ANALYSIS OF SOLID STATE, SOLID OXIDE ELECTROLYTE BASED DIRECT CARBON FUEL CELLS

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Abstract

Efficient use of solid fuels, including coal and biomass, is of paramount importance in light of limited fossil fuel reserves and environmental concerns over pollution and atmospheric carbon dioxide. The solid state, direct carbon fuel cell (DCFC), using an oxide ion conducting electrolyte, is investigated here as a means to efficiently extract work from carbonaceous solids, while producing a capture ready stream of CO_2. The fuel is supplied as a free particle bed in direct contact with an SOFC anode. Thermodynamic performance and behavior, carbon reactivity, and experimental DCFC performance are used as metrics to explore feasibility.

Thermodynamic system analyses are performed in order to estimate DCFC system performance, as compared to systems using separate autothermal or indirect gasification steps upstream of the fuel cell. The maximum work output of the autothermal gasification scheme is lower than the indirect approach, which is lower than the unconstrained case. Detailed calculations, including practical constraints, for the DCFC and indirect gasification plants give nominal system efficiencies between 51 - 58%, compared to the autothermal gasification approach which was found to be 33 - 35% efficient. DCFC efficiencies can be increased to over 60% by an increase in operating voltage and/or inclusion of a bottoming cycle. Open circuit potential measurements agree with equilibrium calculations both for the C/O and C/H/O chemical systems, confirming the governing mechanism and feasibility of the DCFC. The Boudouard reaction ($C + CO_2 \longrightarrow 2\,CO$) governs the cell potential in the presence of solid carbon.

Reactivities (to CO_2) of various carbons and chars were measured and compared using thermogravimetric analysis. Coal and biomass chars created under high heating rates demonstrate the highest reactivities, as compared to slowly heated coals or

highly processed fuels. A five-step heterogeneous reaction mechanism characterizing the Boudouard reaction was employed to model activated carbon gasification in the temperature range of 973 - 1273 K. These experimentally verified kinetic parameters were then included in a transport model to calculate concentrations established in the carbon bed as a consequence of convection, diffusion, and heterogeneous reaction. The model, validated using fixed bed experiments, is used to simulate the effect of an imbedded SOFC, in contact with the carbon bed. The model predicts current densities in the range of 100 - 1000 $mA \cdot cm^{-2}$. Current-voltage measurements performed on a lab scale system demonstrate power densities of 220 $mW \cdot cm^{-2}$ at 0.68 V during operation at 1178 K using activated carbon.

In addition to work production from the DCFC, hydrogen production using a similar SOFC architecture is also considered. The carbon assisted steam electrolysis (CASE) system is analyzed as a means to reduce the losses from conventional electrolysis. Thermodynamic analysis suggests efficiencies of nominally 60% are possible using a coupled CASE-DCFC plant. Measured open circuit voltages agree with theoretical calculations, and gas analysis verifies hydrogen production. Current-voltage measurements were taken of several un-optimized configurations of CASE cells.

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Chapter 1

Introduction

Energy conversion and its associated environmental impacts are of paramount importance in the 21st century. Recoverable petroleum reserves can no longer be considered limitless in a practical or economic sense, and are expected to peak this century if not already [1]. Recoverable coal reserves have the potential to sustain demand for another few hundred years [2]. The steady trend towards global industrialization has significantly increased the demand for these finite, fossil fuel resources. The threat of global warming, caused by the absorptive properties of greenhouse gases, has created increased awareness of CO_2 emissions. Alternative and renewable energy conversion options such as biomass, solar photovoltaics, solar thermal, wind power, wave power, and hydroelectricity, which stem from irradiant solar energy, and tidal energy stemming from the moon's gravitation, are gaining increased attention, although it is unlikely these options will completely overshadow fossil fuel utilization during the next century [2, 3]. In order to mitigate, and possibly reverse, some of the negative impacts of historical and continued fossil fuel use, solutions must address efficiency along with environmental concerns. This study investigates the use of high temperature, solid-oxide electrolyte based fuel cells for power production from solid fuels. Specifically, the Direct Carbon Fuel Cell (DCFC) will be investigated for efficient conversion of solid, carbonaceous fuels. The potential advantages of DCFC systems include higher thermal efficiencies, lower pollutant emissions, and a capture ready exhaust stream of CO_2, as compared to contemporary arrangements.

1.1 Solid Fuel Resources and Utilization

1.1.1 Coal and Biomass as Resources

Biomass, in the form of terrestrial vegetation and wood, has been utilized by man for heat for roughly a million years [4, 5], coincident with early man's mastery of fire. Ease in recovery, ease in utilization for heating, relative local abundance, and renewability on the order of year(s) have made biomass the clear fuel of historical convenience. Coal, with a heating value larger than that of wood, was first utilized about 3000 years ago in China, and then in Europe about 700 years ago [4, 6]. Depletion of forests in England eventually led to widespread use of coal for industry, and circa 1800 brought the introduction of large scale gasification in England, with lighting being the primary end use [7]. Heat engines were first developed during the 1700s and 1800s, in order to produce work from the combustion of these resources. This ability to produce mechanical and electrical work allowed for ever greater feats of engineering, construction, and transportation while simultaneously exponentiating the demand for fossil fuels to drive these heat engines. This dramatic shift in the late 1800s to early 1900s marked the Industrial Revolution, forever changing how fossil fuels were utilized. Contemporary concern about this continued use of limited fossil fuels has created cause for reexamining our energy technologies.

As long as man has burned solids, he has been aware of the acute health hazards of direct combustion of solid fuels. With increasing population densities, and industrial use of fossil fuels, these effects would be magnified. As early as 1306, attempts were made to ban coal burning near populated regions in England [8]. In the 1600s, John Evelyn documented the air pollution in London caused by combustion of coal in the densely populated city [9]. In an especially relevant play on words, he reversed the proverb "there is no smoke without fire" to "there is no fire without smoke", describing the noxious combination of tar, soot, and gas phase irritants found in typical combustion exhaust of the period. Aside from pollution in this explicit sense, Arrhenius published as early as the late 1800s regarding carbon compounds in the atmosphere and their effects on climate [10]. In the contemporary United States, air pollution of many forms is closely monitored by the federal government. The

Environmental Protection Agency (EPA) was formed in 1971 following the passing the of the Clean Air Act (CAA) of 1970 [11], with the primary aim of reducing air pollutants from industrial and transportation sources. Amendments in 1977 and then 1990 further enforced emission of toxic substances to the atmosphere. United States Code Title 42, put into effect in early 2006, mandates a well rounded approach to energy production. Specific to coal, Subchapter VI, Part A, Section 13331 [12] mandates the Secretary of Energy to conduct research into increased efficiency, pollutant control and mitigation, and new technologies. Technologies, such as the DCFC, have the potential to address these concerns by exhibiting increased conversion efficiencies and reduced greenhouse gas emissions.

Currently, about 50% of the electricity generated in the United States comes from coal, with about a quarter of the worlds recoverable coal reserves located in the United States [13]. On a global basis, China and India are emerging coal consumers, and together account for another quarter of the worlds recoverable reserves. These countries have more than tripled their combined use since 1980, and are projected to account for 72% of the increase in coal consumption through 2030 [13]. This signifies that the infrastructure for locating, mining, transporting, processing and using coal are well established, and consumption is projected to increase in the decades to come. In terms of cost, coal (1.60 $dollars/millionBTU$) supplies low cost competition to fuel oil (7.40 - 12.60 $dollars/millionBTU$), natural gas (8.40 $dollars/millionBTU$) and renewables in the electricity generation sector where most coal is consumed [13]. These prices are projected to remain relatively flat through 2030, preserving the current trends and signifying that coal will be economically viable as a fuel for decades to come.

1.1.2 Solid Fuel Use Strategies

Carbonaceous solid fuels have an inherent heating value and availability associated with their inherent chemical bonds and composition, relative to the Earth's terrestrial atmosphere. There exists a wide variety of specific work extraction techniques from these resources, each with specific limitations, costs, complexity, and material

technology requirements. A general description of the major approaches is given, in conjunction with a discussion on the advantages and disadvantages of each.

In practical and historical terms, the most basic method for electrical work extraction from solid hydrocarbon fuels is to burn these fuels in air and use the sensible heat production to drive a heat engine, which can in turn drive an electric generator. The simplest arrangement of solid fuel in a combustor is a fixed bed or moving grate fixed bed. Fixed bed and moving grate combustion systems, typically employ large particle sizes with limited air solid mixing. A boiler and superheater are located in, or directly after, the combustor. Systems based on pulverized coal have been used since the early 1900s [14] and are similar, except that coal is fed by air entrainment or slurry spray in to the combustor. This necessitates the use of smaller sized particles, with a typical feed having 80% of particles between 10 - 100 μm in size [6]. The use of smaller paricles with better mixing allows more efficient char burnout. The presence of a flame zone, with associated high temperatures (above $1000°C$), can induce slagging of molten ash which can have dentrimental effects on boiler tubes and surfaces. This high temperature can also cause increased NO_x emissions, which are a prerequisite of photochemical smog and acid rain [15]. Fluidized bed coal combustion (FBC), introduced rougly 40 years ago [3], uses an inert fluidizing media such as sand to maintain a lower, more spatially homogeneous combustion temperature field. With typical FBC operating temperatures ranging from 750 - $950°C$, slagging is mitigated, thermal NO_x production is reduced and sulfur sorbents can be introduced directly into the bed for sulfur capture [3, 6, 14]. Circulation of particles is sometimes imparted to FBC systems, via cyclone, when high levels of particulates are carried into the freeboard section of the reactor. Particles are caught, cooled, and recycled to the combustor, with some minimum size particles escaping the cyclone. This allows for ash removal without large amounts of unreacted char, and another means of extracting heat from the combustor.

Systems of this type are inherently limited by the Carnot efficiency of the work producing cycle. Irreversibilities include the combustion process itself, the difference in combustion temperature (2000 K) with that of the working fluid during heat transfer, and mechanical/flow losses in the closed cycle [16]. Material constraints generally

rule out external heat transfer to a high temperature gas turbine cycle, although systems have been proposed based on ceramic heat exchangers [17, 3]. Direct firing of coal in gas turbine systems has been considered [18], but is not currently a practical option due to materials issues [14]. This currently relegates the work producer for these direct combustion approaches to a Rankine-type steam cycle. Efficiencies for these plants range from roughly 34% for subcritical plants to 43% for state of the art supercritical plants equipped with flue gas cleanup [2, 7, 14, 17].

Magneto-Hydrodynamic (MHD) power plants utilize the electrical and ionization properties of a working fluid (usually heated by direct combustion) to produce work at efficiencies above 50% [3]. Unlike aerodynamic-based turbine systems, the MHD work producer, or electromagnetic turbine, produces work directly from the working fluid without moving parts, although the material constraints are nontrivial. A magnetic field is applied transverse to the flow direction, which causes a flux of charge carriers, thus producing current (Faraday's Law). There are two primary approaches to MHD power generation from coal. The first method is create a high temperature (3000 K), ionized flue gas by burning fossil fuels in oxygen. The mixture can be seeded with alkali metals for increased conductivity. The second method is to employ fossil fuels as a heat source to maintain a liquid metal working fluid at a temperature of roughly 1000 K. As a method of energy conversion, MHD has been demonstrated and will continue to be the focus of future research targeted at commerialization [3].

Coal gasification refers to the reaction of solid carbonaceous fuels with heat and/or oxygen containing species [14], to create a synthetic fuel gas (syngas), which can then be cleaned and used much the same as natural gas. Gasification improves the handling of fuels, and allows direct use in conventional heat engines or other work producers. The simplest methods of gasification are pyrolysis, and partial oxidation in air. This was the primary technique used until the 1920s, preceding Linde's invention of the cryogenic air separation unit (ASU) [7]. During World War II, Germany used the Fischer-Tropsch process to supply liquid fuels, setting the stage for South Africa's SASOL plants, which convert coal and natural gas to syngas, and then liquid fuels [7]. The following discussion of gasification will be specific to electricity generation, and thus will focus on conversion of solid fuels to syngas. The configuration of solid

fuel gasifiers are myriad, although, similar to combustion systems, the main types are fixed bed, fluidized bed and entrained flow reactors [7]. The benefit of the fluidized bed reactor over the fixed bed arrangement is increased mixing, increased transport and reaction rates, and more uniform spatial fields of temperature and composition. Entrained flow reactors allow shorter residence times, but have more stringent oxygen and particle size requirements.

Integrated gasification combined cycle (IGCC) plants are expected to be one of the primary approaches to coal utilization in the future. The benefits of the IGCC over direct combustion include syngas that can be fed into a high efficiency gas turbine combustor (with a steam cycle acting as a bottoming cycle for the gas turbine heat rejection), and a CO_2 rich effluent stream, undiluted by N_2 from air as in combustion systems (making carbon sequestration possible with this arrangement). The projected efficiencies of such cascaded systems ranges from 40 - 50% [7, 3]. The benefit of using steam as the gasification modifier in IGCC is a higher H_2 concentration in the syngas, allowing for faster reaction rates and greater stability in the gas turbine combustor.

The Department of Energy's Vision 21 concept entails a fuel-flexible, modular, minimally polluting power plant, utilizing advanced technologies in gas turbines, fuel cells, fuels processing and auxiliary equipment [19]. The FutureGen Industrial Alliance, Inc. in partnership with the U.S. Department of Energy were planning construction of a power plant fueled by hydrogen created via coal gasification. The FutureGen project also included plans for flue gas cleanup and underground carbon dioxide sequestration thus producing very low emissions [20]. These programs rely on coal gasification or combustion with complete flue gas cleaning and carbon capture, and demonstrate the need for clean coal technologies.

1.2 Work Production from Fuel Cells

Fuel cells represent a collection of devices that produce electrical work from differences in electrochemical potential of reactant streams. The combination of hydrocarbon fuels and oxygen are generally used to create this electrochemical driving force across

an ionically conducting electrolyte. The relatively low oxygen partial pressure maintained by chemical means in the fuel chamber acts a sink for the oxygen of the air chamber at a larger partial pressure. The benefit of electrochemical conversion over direct combustion lies in the reversibility of the fuel-air reaction process. Direct combustion allows uncontrolled mixing and reaction (flame) of reactants and then uses the sensible energy of the products to produce work. Essentially, the electron rearrangement (bond breaking and forming) is allowed to occur irreversibly. Electrochemical conversion bypasses irreversible mixing, and allows a more controlled reaction of fuel and air by requiring electrons to traverse an external circuit prior to rearrangement forming products. The fact that electrolysis (product conversion to fuel) can be facilitated with similar fuel cell architecture demonstrates the increased reversibility of the fuel cell over direct combustion.

Fuel cell systems operate at temperatures ranging from room temperature up to $1000°C$, utilizing a variety of fuels including hydrogen, carbon monoxide, carbon, syngas, natural gas, hydrogen disulfide, methanol, methane and higher alkanes. The most widely used fuel cell types are discussed here, with the solid oxide based cells being the most relevant to this study. All consist of three basic components; the cathode, electrolyte and anode which collectively comprise the membrane-electrode assembly (MEA). Figure 1.1 shows a general fuel cell assembly, that represents schematically any fuel cell system. Air and fuel are generally introduced separately in order to maintain high and low oxygen activities, respectively. Although single-chambered or premixed arrangements (i.e. premixed air and fuel) have been investigated [21], only separate air and fuel streams will be considered here. The electrodes allow electron flux, or current, to flow through an external circuit thus producing work. Electrodes are generally porous to allow for air and fuel species to diffuse to the electrode/electrolyte interface while still providing good electronic conductivity. According to electrochemical convention, the cathode is designated as the electrode at which reduction occurs (electron transfer from the electrode), while the anode facilitates the oxidation reaction (electron transfer to the electrode). The region in which electrochemical reactions occur, located at the intersection of the gas phase, electron-conducting phase and the ion-conducting phase, is termed the triple phase boundary

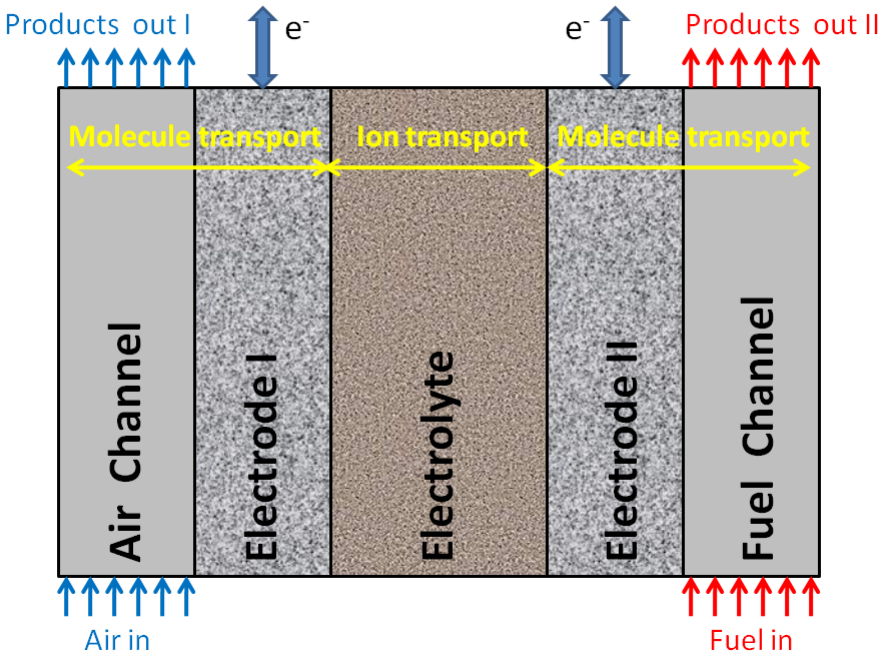


Figure 1.1: General fuel cell schematic (not to scale) showing the electrolyte, electrodes and transport regimes.

(TPB).

1.2.1 Fuel Cell Types and Applicability

The proton exchange membrane fuel cell (PEMFC) represents the default system in most texts and general discussions regarding fuel cells. These cells utilize a proton conducting polymer, usually Nafion, to extract work from gaseous hydrogen and oxygen (from air typically). The PEMFC is limited to temperatures below about $100°C$ since the electrolyte requires liquid water for ion conductivity. Due to this low temperature, expensive catalyst materials are required. Also utilizing the polymer membrane is the direct methanol fuel cell (DMFC). Chemical reactions occur in situ to decompose methanol into protons, electrons and CO_2. These types of cells are generally not appropriate for carbon conversion due to low temperature operation and susceptibility to CO poisoning of electrode catalysts, although the PEMFC does produce the highest power density of all fuel cells [22]. Hydrogen production from coal is one option for PEMFC work production, although the cleaning requirements to remove contaminants and CO are nontrivial. Nonetheless, direct carbon conversion is not a present possibility for the PEMFC.

Phosphoric acid fuel cells (PAFCs), utilize H_3PO_4 suspended in a carbide matrix as the electrolyte. PAFCs are limited to roughly $200°C$ due to properties of the H_3PO_4, and employ Pt catalyst which are susceptible to CO poisoning at these low temperatures. Although commercially mature for hydrogen fuel [22, 3], these low temperature cells are also not a current possibility for direct carbon conversion.

Alkaline fuel cells (AFCs) are based on hydroxide ion (OH^-) transport in aqueous solution. Historically, the AFC has been used in space applications due to high power density and efficient operation on hydrogen and oxygen. Generally, a KOH solution is used, with system temperatures up to $250°C$. Aside from the low operating temperature unsuitable for CO-containing streams, AFCs cannot tolerate even 300 ppm of CO_2 due to carbonate formation [22, 3]. However, this does not rule out the AFC for direct carbon or fossil fuel conversion. Systems employing molten hydroxide (i.e. NaOH or KOH) electrolytes operate at temperatures up to $700°C$. Operation at

these higher temperatures allows for operation on hydrocarbon fuels or CO containing streams, although the formation of carbonates must be mitigated [23].

A considerable amount of DCFC work has been reported using molten carbonate fuel cells (MCFC). The MCFC electrolyte generally consist of molten Li_2CO_3, K_2CO_3 or Na_2CO_3 salts. Carbon dioxide and oxygen undergo a reduction reaction to form carbonate ions, which then participate in the oxidation reaction of the fuel. Cells of this type can operate at temperatures in the range of 600 - 800°C permitting internal reformation, and thus can tolerate the presence of CO and hydrocarbons. The presence of CO_2 is not a hindrance to the MCFC as in the AFC, and a supply of CO_2 is required at the cathode. The electrochemical reaction within the MCFC is limited to solid-liquid interfaces, which generally facilitate lower reaction rates than gas phase reactants. One benefit of the MCFC is the possibility of direct participation of carbon in the electrochemical reaction, allowing for larger fuel utilizations as the activity of carbon is conversion-independent.

The final type of fuel cell to be discussed is the solid oxide fuel cell (SOFC). This family of fuel cells utilizes a solid, oxide ion conducting membrane, generally in conjunction with solid state electrodes. The cathode is located at the air source (Electrode I in Figure 1.1), while the anode is in contact with the fuel (Electrode II in Figure 1.1). SOFC's typically operate at temperatures ranging from 800 - 1000°C, due to conductivity requirements of the oxide electrolyte. The first solid oxide conductors were documented by Nernst in 1899 [24, 25]. Nernst used zirconia (ZrO_2) rods stabilized by yttria (Y_2O_3) as light sources, termed glowers, which conducted a current when externally biased, thus emitting photons from ohmic heating. Stabilized zirconia (YSZ) is still used in modern SOFC systems, owing to its high ion conductivity, low cost, minimized crystal-phase transitions, and chemically inert character under oxidizing and reducing conditions. Ideally, the electrodes of the SOFC provide good catalysis of surface reactions, good electrical and ionic conductivity, with a porous character allowing large active areas for gas-solid interaction. Typical SOFC cathodes are comprised of doped perovskites (ABO_3), such as strontium-doped lanthanum manganite (LSMO) which function as mixed conductors. Typical SOFC anodes consist of composite metal-ceramic (cermet) structures that have both high

electronic and ionic conductivity.

1.2.2 Basic SOFC Processes

Typical steady state fuel cell behavior is generally characterized by the relationship between cell potential and cell current, while operating at a fixed temperature and total pressure. This function V(I) is termed the I-V curve and represents the electrochemical potential as a function of electron flux. Under zero current (or open circuit) conditions, the reversible cell potential is achieved, as calculated from thermodynamic properties. Nonzero currents produce finite ion and molecular fluxes, introducing activation losses, ohmic losses and concentration gradients.

Consider a solid state SOFC, operating with an ambient air supply at the cathode and a stream of equimolar CO/CO_2 (as shown in Figure 1.2). As the co-flowing (in this example) air and fuel streams axially convect through the air and fuel channels, O_2, CO and CO_2 undergo transport in the transverse direction. Oxygen diffuses through the porous cathode and is reduced at the cathode-electrolyte TPB given by the global Reaction R1.1, while CO diffuses through the porous anode and is oxidized at the anode-electrolyte TPB as shown by Reaction R1.2. The overall chemical reaction is given by Reaction R1.3. The net work output arises from the net motion of electrons (or current) through a potential difference, and is given by Equation 1.1.

$$\mathrm{O_2 + 4\,e^{'} + 2\,V_o^{\cdot\cdot} \longrightarrow 2\,O_o^x} \qquad \text{(R 1.1)}$$

$$\mathrm{2\,CO + 2\,O_o^x \longrightarrow 2\,CO_2 + 4\,e^{'} + 2\,V_o^{\cdot\cdot}} \qquad \text{(R 1.2)}$$

$$\mathrm{2\,CO + O_2 \longrightarrow 2\,CO_2} \qquad \text{(R 1.3)}$$

$$\dot{W}_{out} = I * (E_{cath} - E_{an}) \qquad (1.1)$$

The reactants in these electrochemical interactions include charged species such as conduction-band electrons ($e^{'}$) in the metal electrodes, electrolyte oxygen vacancies ($V_o^{\cdot\cdot}$), and mobile oxygen ions (O_o^x) within the electrolyte. These eactions, as written, represent global reactions and do not resolve adsorption or desorption processes,

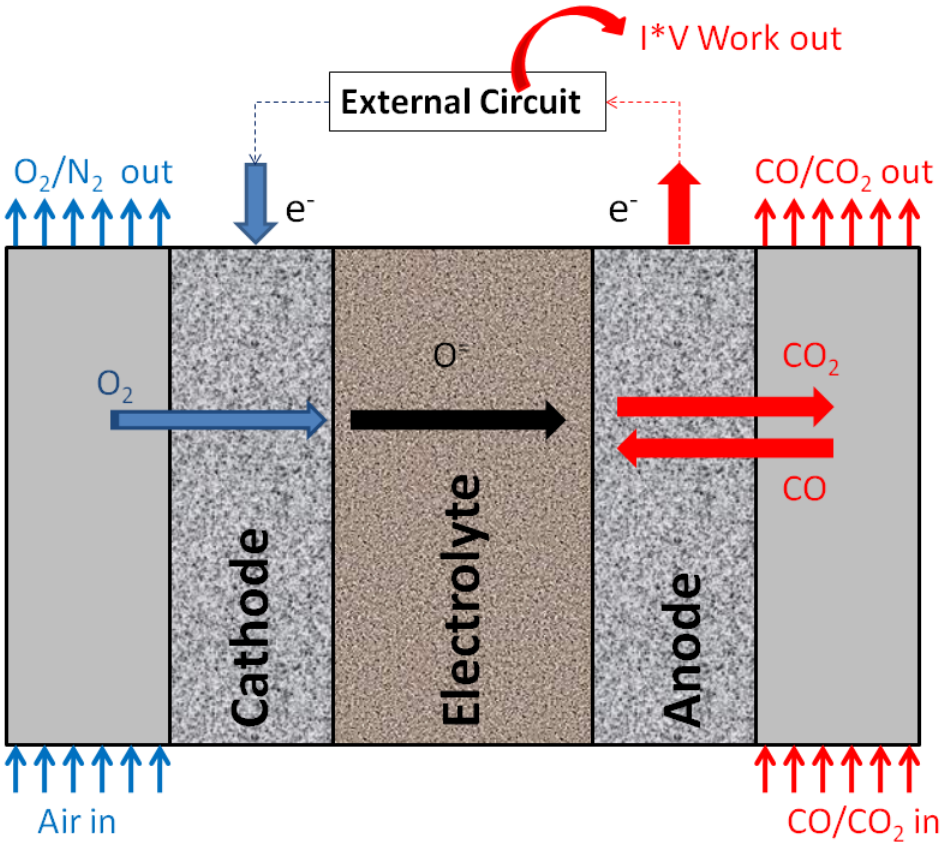


Figure 1.2: SOFC schematic (not to scale) showing the electrolyte, electrodes and transport regimes for inputs of air and CO/CO_2.

migration of surface species, or transport of solid phase species. As many as 9 steps have been formulated to describe the tranport and reactions entailed in the reduction process alone [26]. The reversible cell potential can be formulated as shown in Equation 1.2, where ΔG is the Gibbs energy for Reaction R1.3, n is the number of electrons transferred (4), F is Faradays constant, R is the gas constant, and T is the temperature. Typical values for hydrocarbon fuels are near 1 V, noting that the partial pressure of O_2 is determined by the cathode side composition, while the those of CO and CO_2 are determined by the anode side composition. This reversible cell potential will be revisited in further detail in the following chapter during the

discussion on the thermodynamic expectations of the DCFC.

$$E_{rev} = -\frac{\Delta G(T, P, X_i)}{nF} = E^0 + \frac{RT}{nF} ln \left(\frac{P_{O2} P_{CO}^2}{P_{CO2}^2} \right) \tag{1.2}$$

A simple model is used here to describe typical SOFC behavior for a general system as shown in Figure 1.2. Using the Butler-Volmer model of electrochemical reactions, the current and electrode potential can be related. This model assumes that the electrochemical reaction occurs in one step, termed the rate determining reaction. The reactions considered here involve 4 electrons, and thus an additional simplification is made by assuming a global form for multiple electrons [27]. This will not necessarily give the correct functional form, but allows for simplification here. The Butler-Volmer results, based on Reactions R1.1 and R1.3 are given by Equations 1.3 and 1.4, respectively, where f is defined as F/RT, and n is the number of electrons involved in the reaction. The overpotential η_j is defined as the electrode potential minus the equilibrim potential ($E - E_{eq}$), and the exchange currents I_0 are functions of species concentrations (at the respective TPB) and temperature. The electrode currents I_c and I_a are equal, and under equilibrium conditions ($\eta_j = 0$) no current flows, with the reversible cell potential being achieved. As current is allowed to flow, the cell incurs losses thus lowering the cell potential.

$$I_c = I_{0,c} \left(e^{-n\alpha_c f \eta_c} - e^{n(1-\alpha_c) f \eta_c} \right) \tag{1.3}$$

$$I_a = I_{0,a} \left(e^{n(1-\alpha_a) f \eta_a} - e^{-n\alpha_a f \eta_a} \right) \tag{1.4}$$

During nonzero current flow, concentration gradients arise in the porous electrodes, lowering the available activites of the reactants and raising that of the products. Utilizing Ficks Law, the concentration change ($C_{i,channel} - C_{i,TPB}$) due to diffusion is given by Equation 1.5, where l is the electrode thickness, D is the effective diffusivity, and A represents the cross section area. This represents a loss in reversible cell potential (or overpotential η_{conc}) due to transport-induced gradients, as shown in

Equations 1.6 and 1.7 for the cathode and anode, respectively.

$$C_{i,channel} - C_{i,TPB} = \frac{il}{nFDA} \tag{1.5}$$

$$\eta_{conc,c} = \frac{RT}{nF} ln \frac{P_{O2,channel}}{P_{O2,TPB}} \tag{1.6}$$

$$\eta_{conc,a} = \frac{2RT}{nF} ln \frac{P_{CO,channel} P_{CO2,TPB}}{P_{CO,TPB} P_{CO2,channel}} \tag{1.7}$$

The cell potential may then be formulated as a function of current as shown in Equation 1.8, where Ω_{cell} is the ohmic resistance of the cell and E_{rev} is calculated from reactant concentrations within each channel. If E_{rev} were defined at the TPBs then the explicit inclusion of the concentration overpotential is unnecessary. Note that the cathode overpotential is negative by definition, and thus the absolute value is included. The operating cell potential is then the reversible cell potential minus the various overpotentials due to activation losses, concentration losses, and ohmic losses. The reactant concentrations at each TPB will be less than that in the channel due to concentration gradients within the porous electrodes.

$$E_{cell}(i) = E_{rev} - \Omega_{cell} * I - |\eta_c(I)| - \eta_a(I) - \eta_{conc,c}(I) - \eta_{conc,a}(I) \tag{1.8}$$

Shown in Figure 1.3 are data taken from an SOFC displaying these various limitations. Considering Equation 1.8, at low currents the activation losses dominate the departure from the reversible cell potential. As current increases, linear ohmic losses dominate the cell behavior, and as currents are further increased the concentrations gradients arising from transport dominate this departure. The work output, defined as $I * E_{cell}(I)$, achieves a maximum for some value of current, representing the peak power, although cell efficiency monotonically decreases with current as losses increase. While the data shown in Figure 1.3 display each regime clearly, this is not necessarily so. Fast electrode kinetics may significantly reduce the activation losses, as a high ohmic resistance would give the V(I) function essentially linear character.

The schematic for the DCFC is similar, except that the fuel (CO in this case) is created within the channel (carbon bed) via the Boudouard reaction ($C + CO_2 \longrightarrow 2\,CO$),

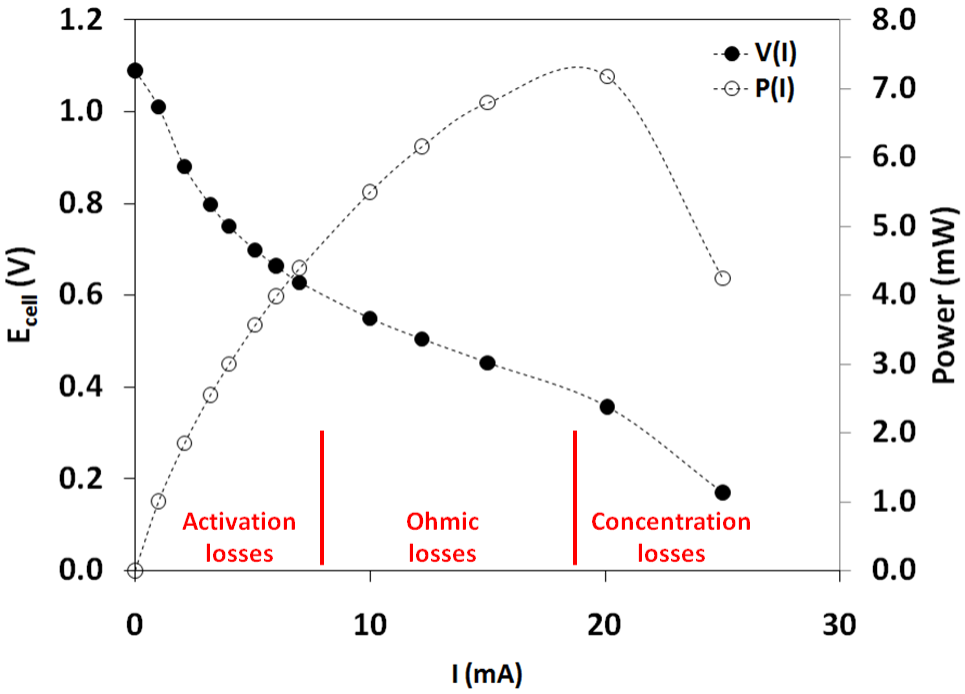


Figure 1.3: Measured potential V(I) and power output P(I) of an SOFC, showing regimes in which the various loss mechanisms dominate performance. The tube cell used for these measurements had relatively large resistances due to electrolyte thickness and un-optimized Pt electrodes.

rather than advectively supplied. The concentrations within the anode channel and electrode, and hence the cell potential, now depend on chemical reactions and transport within the carbon bed. This chemical coupling is the primary complication in studying the DCFC system over the conventional SOFC system. The thermodynamic consequences of this coupling are investigated in Chapter 2, while a study on the rate of this chemical interaction is investigated in Chapter 3.

1.3 DCFC Background and History

Although hydrogen dominates most of the fuel cell efforts to date, the conversion of naturally occurring fuels containing carbon has been evaluated extensively. Direct,

electrochemical work extraction from solid, carbonaceous fuels has been considered for over 150 years. K. Williams provides a detailed, early history of carbon based fuel cells up through the 1930s, given in summary here along with references contained in Chapter 1 of that text [25]. In 1855, Becquerel used a carbon rod inserted in a molten nitrate salt electrolyte within a platinum vessel acting as the cathode to produce electrical work. A similar arrangement, with an iron vessel in lieu of platinum was employed by Jablockoff in 1877 [28], and also by Jacques in 1896 [29, 30], with a sodium hydroxide electrolyte and iron vessel. Taitelbaum in 1910 [31] and Reed in 1918 [32] chose to use solid, ceramic separators containing molten salt electrolytes between the anode and cathode. It is probable that all of these early attempts did not produce actual carbon-air fuel cell behavior, but rather were governed by side reactions involving the electrolyte or other net-consumed material. In 1937, Baur and Pries used a carbon electrode in contact with a solid oxide conducting electrolyte and molten silver cathode [33]. Silver has been shown to dissolve oxygen [34].

In addition to these early investigations of solid carbon were several experimental studies focusing on carbon monoxide. Since coal was the prevalent fuel during this time, gasification to CO/H_2 and subsequent use of this syngas in a fuel cell was of practical concern. In 1904, Haber and Moser used a completely solid state MEA, consisting of platinum electrodes and glass electrolyte to study carbon monoxide [35]. Baur and Ehrenburgh used a nickel anode with a borax electrolyte and molten silver cathode around 1910 to convert CO [36], while 19 years later Greger used a carbonate electrolyte with a continuous oxygen supply to the cathode [37]. Low current densities and sometimes ambiguous governing electrochemical interactions marked these early works, however the results paved the way for future research in this field.

An extensive review of carbon's diverse uses in fuel cells is given by Dicks [38] in 2006, including the potential for high system efficiencies of DCFCs. In 2007, Cao and Wang gave an overview of ongoing DCFC efforts, including some background on the various fuel cell types [39]. These recent and unique efforts will be discussed and summarized in the following sections.

1.3.1 Recent Solid State DCFC work

A comprehensive review of carbon electrochemistry was prepared by Park in 1984 [40], with a focus on aqueous systems and slurries. Although the primary focus is aqueous electrochemical conversion of carbon and coal (at temperatures far below SOFC applications), higher temperature processes involving direct and indirect utilization are mentioned. References within are made of two DCFC experiments, both of practical interest. The first involved a solid state SOFC with a graphite anode, although the current density was too low by practical standards as was the estimated energy conversion efficiency of 2% [41]. The second referenced chemical looping of Pb/PbO in a molten carbonate medium, reacting coal char with oxygen from air. This system was reported to produce 108 $mA{\cdot}cm^{-2}$ at 0.76 V, giving a voltage efficiency of 75% [42].

During the late 1980s to mid 1990s, several groups studied completely solid state MEAs with solid carbon fuels. An exergy analysis was performed on a solid state DCFC system, at current densities up to 8 $mA{\cdot}cm^{-2}$, by Nakagawa and Ishida [43]. The examined system consisted of a YSZ electrolyte disk, painted with Pt electrodes on both sides. Charcoal was place in the anode chamber, in close proximity to the Pt anode, with the anode gas environment purged with nitrogen prior to testing under no-flow conditions. The loss mechanisms and exergy destruction are discussed in terms of ohmic resistance, chemical reactions, and electrode polarization. Carbon gasification to carbon monoxide supplied the fuel for the anode. The carbon monoxide is converted to carbon dioxide at the anode-electrolyte boundary, which can subsequently undergo the Boudouard reaction. Any CO leaving the cell represents a loss of work potential since it was not converted to CO_2. Operation at higher temperature was shown to reduce system losses, by reducing ohmic resistance and increasing reaction rates. Although the current density was quite low as this system was un-optimized, key DCFC behavior is touched upon.

In the early 1990s, Gür and Huggins performed experiments with solid carbon fuel and a solid state electrochemical cell [44]. A one-chamber, two section reaction vessel was used such that the carbon section and MEA-containing section could be maintained at separate temperatures. The vessel was purged with He prior to experimentation, to evacuate oxygen containing species from the gas phase environment

at the anode. The results demonstrated fuel cell behavior at an MEA temperature of $932°C$, with I-V behavior limited most likely by electrode kinetics. The carbon section temperature was controlled to $725°C$ and $955°C$, giving open circuit voltages of 0.9 and 1.1 V, respectively. Again, current densities were low, and were likely driven by the Boudouard reaction of product CO_2 with the solid fuel.

During the mid 1990s, Horita et al. investigated anode materials for DCFC operation [45]. Graphite, as well as metal carbides such as VC, WC , ZrC and TiC were included as anode materials. The cells consisted of a tubular YSZ electrolyte, with a painted Pt cathode on the outside of the tube. The inside of the electrolyte tube was partially filled with a mixture of graphite and metal carbide powders, with a solid carbon rod used as the current collector. This experimental setup allowed for gas flow inside of the electrolyte tube, thus providing a means of controlling anode-side gas composition in the fixed carbon bed. The gas environment in the anode chamber will be discussed further in the next chapter. The results showed metal carbides to be superior to graphite during polarization, although the metal carbide anodes are believed to oxidize, and then recarbidize at a slow rate. The possibility of a rechargeable anode was presented, for recarbidizing the oxidized metal. Gas composition measurements of product CO showed Faradaic conversion of solid carbon to carbon monoxide up to current densities of 30 $mA{\cdot}cm^{-2}$. This arrangement has the advantage of extending the effective anode into the fuel bed, although this approach entails additional materials concerns.

A rechargeable, solid state DCFC was investigated by Hasegawa and Ihara [46, 47]. The electrochemical cell consisted of a YSZ electrolyte disk, with an LSMO cathode and Ni/GDC (gadolinium doped ceria) anode. A gaseous fuel, methane or propane, flows past the anode surface at $900°C$, decomposing and depositing solid carbon throughout the porous anode. The resulting gaseous pyrolysis products are purged with Ar, and then the fuel cell was operated galvanostatically under Ar flow. When the accumulated carbon is consumed, the cell potential dropped, and the power generation interval is concluded. This interesting type of arrangement could produce gaseous hydrogen from methane, or natural gas, and then generate electric power on the accumulated carbon. Power densities of 52 $mW{\cdot}cm^{-2}$ were obtained with type of

cell operating on carbon. This process is an effective means of partitioning and separating hydrogen and carbon species, but does not directly consume a solid feedstock. This is the fundamental difference between this approach and that considered in this study, in which solid fuels are utilized.

1.3.2 Recent DCFC Work with Molten or Liquid Phases

In addition to solid state systems, fuel cells employing liquid phases are another DCFC possibility. Although the management of liquid phases can present challenges, there are several benefits such as ability of solid carbon to come in direct contact with the electrolyte or TPB, and recycling of anode material for cleaning or restoration in a separate process. The performances of the following MCFC systems (up to 800°C)are impressive and relatively consistent, providing a convenient performance target in this study.

In the late 1980s, the MCFC was employed for direct carbon conversion in the temperature range of 500 - 800°C by Vutetakis et al. [48]. Solid fuels including activated carbon, bituminous coal, lignite and anthracite were tested in the stirred melt at solids loadings of up to 0.15 g_{solid}/g_{melt}. Current densities of over 250 $mA{\cdot}cm^{-2}$ were achieved at about 0.3 V, operating at 800 °C. As expected, graphite and anthracite produced lower current densities due to their lower surface reactivities (or more ordered structure). In addition, essentially inert diamond is mentioned briefly as supporting no current at all when loaded into the melt, as would be expected for this material. Open circuit potentials characteristic of carbon equilibrium were reached at temperatures of 700 °C and above.

Peelen and coworkers tested a MCFC fueled with solid carbon up to current densities of 100 $mA{\cdot}cm^{-2}$ [49]. A Li/K melt held between 650 - 750°C was used to perform impedance and voltammetric tests on a graphite anode with an initial gas enviornment of pure CO_2. Note is made that the charge transfer resistance for carbon conversion is three orders of magnitude larger than that of hydrogen at 650°C over a Ni anode. Thermodynamic analysis of this type of MCFC-based plant showed a calculated system efficiency of about 78% [50]. The system under analysis operated

at 100 $mA \cdot cm^{-2}$ and 0.8 V, with 100% fuel utilization. Exergy analysis showed that about 65% of the total loss occurs in the preheating subsystems and hot air rejection to the environment, compared to only about 9.5% of the total loss occurring in the MCFC. Operation at these temperatures, where CO_2 is the preferred product, mitigates the Boudouard reaction between product CO_2 and carbon fuel. The use of a graphite anode inherently necessitates preprocessing of the fuel, and may or may not represent a practical feedstock.

Cherepy et al. at Lawrence Livermore National Laboratory (LLNL) are also investigating molten carbonate based DCFC systems, and have achieved current densities of around 200 $mA{\cdot}cm^{-2}$ at a power density of 100 $mW{\cdot}cm^{-2}$ [51]. One of the unique aspects of this work is a focus on derived carbon fuels that present less concern for current fuel cell materials. Activated carbons and carbon black particles smaller than 5 μm are used. Operation on petroleum coke, with a significant sulfur content of 2.5 - 3 wt% shows a time increasing resistance of the cell, likely from sulphidation of the Ni anode. Also, mention is made that the cathode limits cell performance, implying that optimization of this electrode could significantly improve the power density.

Interesting work by Pointon et al. with a carbon-air fuel cell combines a solid oxide ion conductor with molten carbonate melt to extract electrical work from carbon [52]. A solid oxide electrolyte is used which allows for more diverse cathode materials to be considered since the cathode does not contact the carbonate melt. Also, N_2 is kept from the molten carbonate, thus preventing dilution within the carbonate. The focus in this study was man-portable applications, so energy density and simplicity were the primary constraints. Three simple electrochemical cells, based on hydroxide, carbonate and oxide electrolytes, are compared with the hydroxide being ruled out of consideration due to the water requirement and associated complexities. Based on the stated Ni anode current collector area, the measured current density of 22 $mA{\cdot}cm^{-2}$ at $700°C$ is too low for practical application, but research is ongoing into reducing system losses. The benefits of the use of standard cathode materials for the SOFC in the DCFC is also one of the benefits of the present study.

The use of solid oxide conducting electrolytes with molten salt anodes is investigated by Balachov et al. [53] at SRI International. This system employs a YSZ

electrolyte with an LSMO cathode characteristic of typical SOFCs, although the solid state anode is replaced with a circulating molten anode into which fuel particles have been seeded. Power densities of up to 110 $mW \cdot cm^{-2}$ at a voltage of 0.7 V were demonstrated on a variety of fuels including coal, biomass, tar and plastics. This is an especially relevant study in terms of DCFC commercialization, as a variety of raw fuels are considered.

Currently, CellTech Power LLC is developing SOFCs with molten anodes for fuel flexibility [54]. Target fuels include coal, plastics, bioimass, jet fuel, diesel fuel, gasoline, natural gas and hydrogen. A preliminary stack was operated on carbon/coal for 2 weeks [55]. This cell type allows for accumulation of oxygen, in the form of oxidized anode material, typical of battery-type behavior. The fuel can thus be directly utilized in the cell, or used to reduce the oxidized anode material.

Zecevic and coworkers at Scientific Applications and Research Associates, Inc. (SARA) investigated a DCFC system with molten sodium hydroxide electrolyte that produced current densities of up to 250 $mA \cdot cm^{-2}$ [23]. Graphite rods were used as the anode and current collector, while the melt container served as the cathode. Humidified air was bubbled into the melt to supply oxygen to the cathode. A measured efficiency of 60% is given, and although not explicitly defined appears to refer to a voltage efficiency. An efficiency estimate of 75% is stated for a practical, optimized DCFC system based on a molten hydroxide electrolyte. Systems were tested up to 540 hours, ranging in temperature from $400°C$ to $650°C$. The use of graphite rods again necessitates significant fuel preprocessing.

Another research group at West Virginia University has also investigated molten salt-based DCFCs. A sodium hydroxide melt with immersed carbon rods was tested by Hackett et al. to investigate the effect of carbon type on DCFC performance [56]. Current densities of up to 200 $mA \cdot cm^{-2}$ were obtained at operating temperatures in the range of 600 - 700 $°C$. The carbon fuels included graphite and pressed petroleum coke, coal extract, and mixtures bound with tar pitch. The graphite rods showed higher conductivity but lower reactivity, consistent with differences in carbon structure and the reactivity of the active sites.

Aside from oxide-ion conducting membranes, Cocks and Laviers patented a DCFC

system relying on a carbon-ion conducting electrolyte for coal conversion [57]. The eletrolyte consisted of a carbon containing fluorite or perovskite, immersed in a molten metal bath serving as the anode. A solid state metal cathode was in contact with the carbon ion conducting membrane. This type of arrangement would allow for carbon conversion, although the hydrogen present in coal would not be converted. Nonetheless, this arrangement would allow for carbon capture if the O_2 provided to the cathode were separated from N_2.

1.3.3 Other Works of Interest

In addition to the direct use of solid fuels, there has been a wide range of studies on direct use of other hydrocarbons, and/or the use of solid fuel derived fuel gases. Elimination of external reforming would be a huge advantage in terms of system efficiency and process simplicity, just as bypassing the separate gasification of solids is advantageous. Anodes capable of reforming hydrocarbons in situ are an area of active research. Researchers at the University of Pennsylvania have been investigating direct oxidation of dry hydrocarbons in SOFCs [58]. Fuels including CH_4, C_2H_6, C_4H_{10}, C_4H_8, and C_7H_8 were shown to give stable operation at 973 K [59]. The use of energy-dense oxygenated fuels, such as dimethyl ether (CH_3OCH_3), to reduce carbon deposition has also been investigated with promising results [60].

Researchers at Northwestern University have also been investigating the resilience and performance of anodes under direct hydrocarbon inputs. Conventional fuels such as CH_4 and natural gas [61], along with the higher alkane C_8H_{18} [62]were investigated, with power densities well over 500 $mW \cdot cm^{-2}$.

Panapoulous and coworkers investigated the integration of biomass gasification with SOFC work production [63, 64]. This system consisted of an allothermal gasifier, with heat supplied from combustion of the SOFC exhaust flue stream. A post gasification cleaning step to remove particulates, HCl and H_2S removal prior to combustion. Sensible energy is transferred as heat from the fluidized bed combustor to the gasifier by means of sodium heat pipes. Although the solid fuel stream is not in direct contact with the SOFC, this arrangement is especially relevant to this study.

Allothermal gasification arrangements, or indirect gasification as it will be called in this study, will be shown to provide some of the same benefits of DCFC systems. The efficiency of the investigated system was reported as 32% for a fuel utilization of 0.7 and a current density of 250 $mA{\cdot}cm^{-2}$. This thermally-integrated approach will be shown to be theoretically equivalent to the DCFC in many aspects, as the reaction oxygen must still traverse an ion conducting membrane.

Gopalan et al. modeled hydrogen production via electrolysis using carbon and solid oxide electrolytes [65]. The carbonaceous fuel, in the form of carbon rods that also serve as current collectors, is immersed in a molten Ag anode in contact with the electrolyte. A H_2/H_2O mixture flows over a Ni/YSZ electrode in contact with the electrolyte. The steam is reduced to H_2, as the oxygen is transferred to the carbon side. Under certain gas phase species activities, this arrangement may transfer oxygen spontaneously, although some work input may be used to overcome losses. Another benefit is that the hydrogen is produced from clean water, chemically separated from the solid fuel, rather than a coal/water mixture that requires extensive cleaning of the syngas. This arrangement can also be run in fuel cell mode by introducing oxygen to the molten Ag and H_2 to the Ni cermet. The potential for clean hydrogen production using the DCFC arrangement will be considered in this study since the governing physics and chemistry are essentially the same.

1.4 Objective of the Present Study

1.4.1 Motivation of the Present Study

The detailed account of previous work regarding the DCFC uncovers several areas that require further study. Solid state DCFC systems have not been demonstrated with power densities over 100 $mA{\cdot}cm^{-2}$, which would merit further practical interest. Verification of Faradaic behavior would also demonstrate steady state operation of the DCFC. In addition, efficiency calculations generally consist of the voltage efficiency, but make no mention of the complete system. Even the maximum work output and theoretical efficiency of the DCFC are generally not determined for the system, with

reference to environmental air and proper accounting of heat flows.

Kinetics of solid-liquid and solid-solid interactions are expected to proceed at a lower rate than gas-gas or gas-solid interactions. The solid state DCFC relies on gas-solid interactions during the electrochemical step, as opposed to a solid-liquid interaction as in the MCFC. The higher rate of kinetics translates into smaller reactors with more efficient conversion. Current densities well over 100 $mA \cdot cm^{-2}$ have been achieved with the MCFC, requiring similar performance from solid state systems to warrant comparison and consideration. Solid fuel handling is another issue directly affecting DCFC implementation. The solid state system considered here, physically coupled to a particle bed, allows for convenient feed and ash removal during steady operation. Also, in situ means of contaminant capture may be possible with such a configuration. Systems with preformed fuel beds or pressed rods require preprocessing and must be replaced in a discreet process.

Lastly, the Boudouard reaction is mentioned in the solid state DCFC literature, but is typically refered to as a peripheral process, although it governs cell behavior. Kinetics of the Boudouard reaction have been widely studied, but the rate of gasification is unique to a given feedstock. In addition, gas phase production and transport within the carbon bed must be incorporated into any model of the solid state DCFC. A model capable of predicting gas phase scalar concentration fields is required to determine the work output potential of the DCFC given a particular fuel.

1.4.2 Scope of the Present Study

The present study will focus on the theoretical and practical fundamentals of the direct carbon fuel cell based on solid state membrane-electrolyte assemblies (MEA). In analyzing the solid state DCFC for feasible implementation, the following questions and issues are addressed here.

- What efficiency is expected for the solid state DCFC?
- What process concerns exist for DCFC operation?
- How does the gasification rate of the fuel impact DCFC performance?

- What power densities can be achieved with a solid state DCFC?

In order to address these issues, this work will be organized as follows. The thermodynamic basis for work extraction from carbon is investigated, along with the loss mechanisms within the DCFC. Thermodynamic and process analyses are conducted to determine the expected efficiency of the solid state DCFC. Thermodynamic results are compared with systems utilizing separate indirect and autothermal gasification processes for context. A methodic experimental verification of the governing thermodynamic behavior is included to confirm thermodynamic calculations.

The chemical reactivity of carbon and char to carbon dioxide via the Boudouard reaction is reviewed. A model is presented that incorporates multi-step heterogeneous chemistry along with particle morphology. The char gasification model is experimentally verified using thermogravimetric analysis. Reaction rate parameters are then used within a transport model to model carbon bed dynamics rather than single particle interactions. The transport model is verified experimentally for a fixed bed configuration, and is then used to estimate the supportable current densities of an SOFC in contact with the carbon bed. A schematic of a possible DCFC arrangement is shown in Figure 1.4, with the areas of interest in this study highlighted in red.

Another desirable output, aside from direct electrical work, is gaseous hydrogen for the purpose of energy storage and transportation. The possibility of clean hydrogen production using carbon as a chemical input is also investigated in Chapter 5. Carbon is able to maintain a sufficiently low oxygen activity to reduce H_2O under certain conditions. A simple system will be analyzed to determine the expected exergy recovery in the form of H_2 from the carbon input. The possibility of oxygen transfer from steam to carbon, in the context of energy storage, is verified experimentally using a DCFC-derived system.

A significant portion of the measurements and calculations presented in this dissertation are also available in the published literature [66, 67, 68, 69].

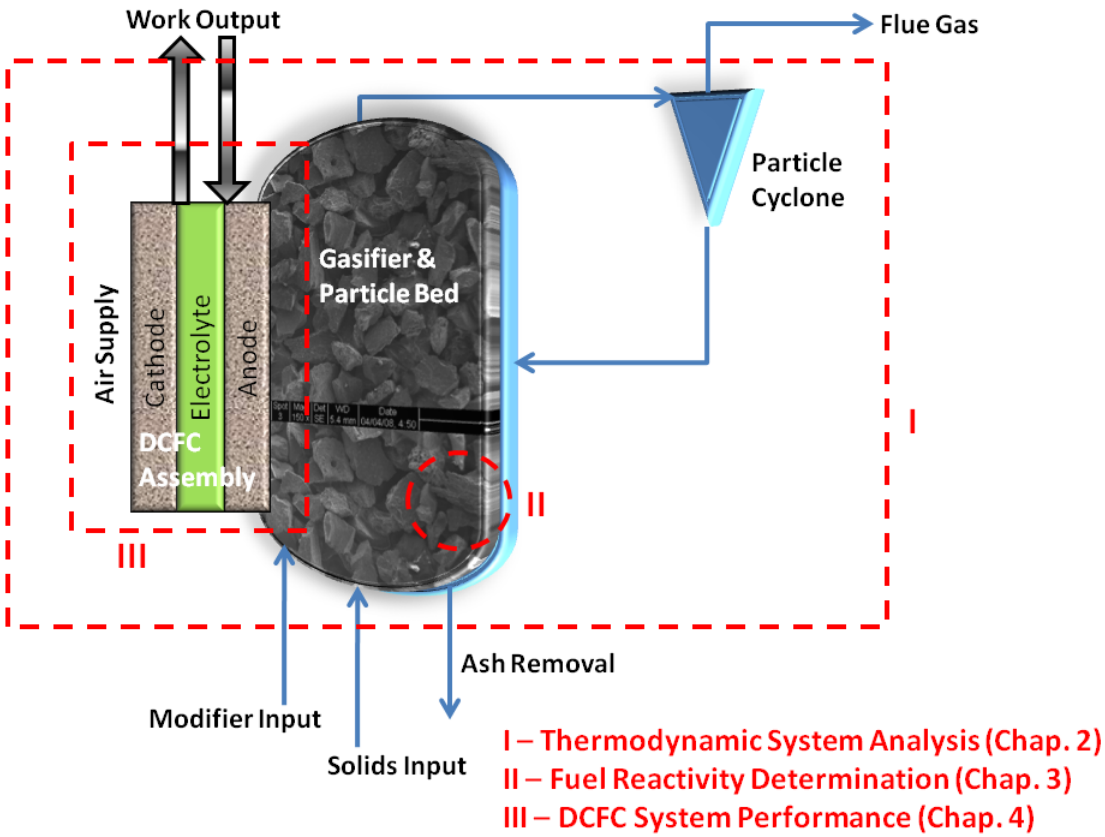


Figure 1.4: General DCFC system, with the specific issues addressed in this study labeled by the dotted red boundaries.

Chapter 2

Thermodynamic and Process Evaluation

Carbon, coal, and biomass are carbonaceous solids with distinct physical and thermochemical properties. There exist limitations on the work that can be produced from systems operating on these fuels. There are several process approaches to solid fuel utilization, and thermodynamic analysis allows direct comparison of various systems. The work potential and efficiency of DCFC systems, along with several other technologies, will be investigated here along with some experimental validation of the DCFC thermodynamic behavior.

2.1 Carbon, Coal, and Biomass Properties

Carbon represents the twelfth element in the periodic table, and generally refers (in this context) to the pure substance in its solid, graphitic phase. Graphite consists of layered graphene sheets, although the variability in crystalline arrangement of carbon powders is significant. This deviation from a perfectly ordered structure gives way to more reactive solids that are of practical use. Biomass represents a large group of complex, C-H-O containing solids, including plants, wood, animal wastes and algae that were formed from biological processes. Biomass fuels are renewable and naturally occurring in large quantities unlike graphitic carbon. Biomass fuels typically have

large moisture and oxygen content relative to coals. Coal derives from biomass, but is accumulated and chemically transformed over tens to hundreds of millions of years under geological flows and stresses [14]. Coals are classified as lignite, sub bituminous coal, bituminous coal, or anthracite, in order of age and fixed carbon content. Lignites will have properties more similar to biomass, while anthracites will have properties more characteristic of graphite. Solid wastes, both municipal and industrial, are also potential energy resources, although to a lesser extent.

Hydrocarbon fuels react exothermically with oxygen, and can be characterized by a heating value. This represents the heat that is released and removed from complete combustion with air for reactant and product temperatures of 298 K. Tables 2.1 and 2.2 show comparison of heating values (HHV in $MJ{\cdot}kg^{-1}$ solid) and composition for various solid fuels and carbon. Work potential generally follows heating value closely for solid, carbonaceous fuels [70, 71]. Wyodak and Illinois #6 coals are included in the Penn State University Coal Sample Bank and Database [72]. Almond shells and switchgrass are included as representative biomass fuels, with properties taken from the Energy Research Center of the Netherlands (ECN) database [73] along with municipal solid waste.

Table 2.1: Various solid fuel proximate analyses (wt%) and heating values ($MJ{\cdot}kg^{-1}$) for comparison.

Fuel	Volatiles	Fixed C	Moisture	Ash	HHV
Carbon(s)	0	100	0	0	32.79
Illinois #6 coal	35.44	39.74	13.2	11.49	24.55
Wyodak coal	33.06	35.06	26.3	5.58	20.82
Switchgrass (dry)	NA	NA	0	10.07	18.02
Almond shell	70.1	NA	7.8	2.78	17.55
Municipal waste	NA	NA	38.5	27.12	7.50

The data presented in Table 2.2 show that solid carbon has the largest heating value, followed by coals, biomass and finally municipal waste. The ultimate analysis is presented on an as received basis, so dried fuels or chars deriving from the different sources will have separate properties.

Table 2.2: Various solid fuel ultimate analyses (wt% as recieved) for comparison.

Fuel	C	H	O	N	S	Cl
Carbon(s)	100	0	0	0	0	0
Illinois #6 coal	57.33	3.98	8.07	0.99	4.8	0.14
Wyodak coal	51.42	4.16	11.53	0.69	0.32	0.001
Switchgrass	47.8	5.76	35.1	1.17	0.1	0.001
Almond shell	46.2	5.5	37	0.68	0.03	0.01
Municipal waste	18.9	0.6	13.2	0.67	0.48	0.437

Reaction vessels employ varied solids feeding mechanisms for steady flow operation. These include lock hoppers, screw feeds, and gas entrainment feeds. Depending upon the friability of the fuel, and the nominal size of fuel particles, each method has its merit. Small particles are easily entrained, and fibrous biomass may be fed more easily using a screw. The particular feeding mechanism will not be specified in this study, although it is assumed that the prevailing technology for that fuel type will be utilized. Ash removal from the gasification reactor will be treated in the same manner.

2.2 C-H-O Thermodynamic System

The compositions considered within a carbon bed are outside the regime typically encountered in combustion products or even gasification products. Shown in Figure 2.1 is the equilibrium composition of $C + nO_2$ at 1173 K as a function of n. Products from lean combustion lie above n values of one, while gasification or partial oxidation products nominally lie between n values of 0.5 and 1. The region characterized by n values of less than nominally 0.5 represent conditions with solid carbon present, typical of pyrolysis regions, or the gas composition within a bed of solid particles. The solid carbon phase is present in equilibrium when there is insufficient oxygen available. This behavior dictates the requirements for full solid carbon transfer to the gas phase, and the conditions under which solid carbon will precipitate or deposit onto surfaces from the gas phase.

The general behavior shown in Figure 2.1 for 1173 K specifically is characteristic

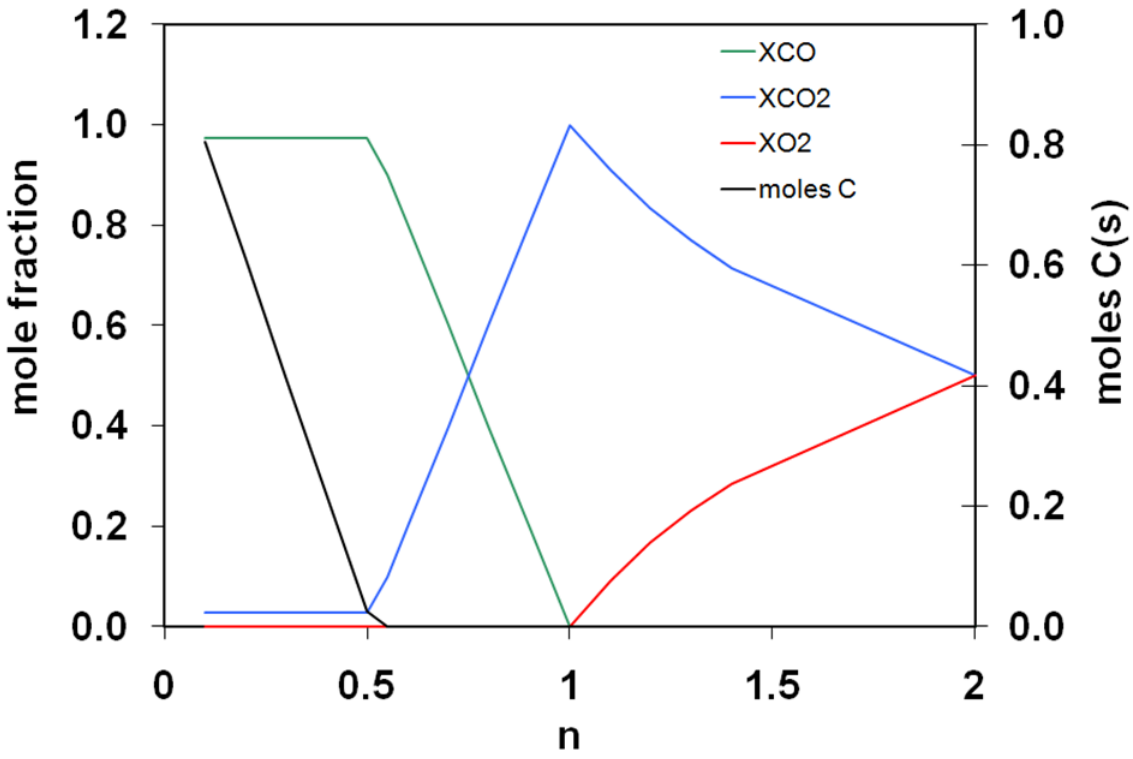


Figure 2.1: Equilibrium composition for the C-O system for $C + nO_2$ at 1173 K for values of n spanning the gasification, partial oxidation and lean combustion regimes.

of temperatures above about 1000 K, where CO is preferred to CO_2 in the presence of excess carbon.

The C-H-O ternary diagram shown in Figure 2.2 graphically represents atomic content of several fuel and product species. Coals typically have C/H ratios of roughly one, by mole. Biomass fuels typically have C/H ratio slightly lower than one, and significant oxygen content. As these fuels react with oxygen, the fuel/product mixture tends to the region between the CO_2 and H_2O points on the ternary diagram. The equilibrium formation of solid carbon occurs on the carbon rich side of a temperature dependent boundary, as shown in Figure 2.3 (calculated using STANJAN [74] using NIST-JANAF thermochemical data [75]).

Studies have been conducted regarding both solid carbon formation and cell open circuit voltage (OCV), using the C-H-O ternary diagram [76, 77, 78]. At the temperatures of interest, near 1273 K, the boundary of carbon deposition is essentially a straight line between the H vertex and the CO point. As the temperature is lowered, this boundary curves with the endpoints moving from the H vertex to CH_4, and from CO to CO_2 as the equilibrium products shift. Almost all solid fuels of interest have compositions falling within the deposition region, signifying that a gasification modifier is required for full conversion to the gas phase. Carbon deposition is a concern with DCFC operation, especially with the presence of solid state electrodes. The fuel gas can deposit carbon on anode surfaces or within the porous anode structure, blocking active sites and reducing catalytic effects.

Prior to determining the thermodynamic potentials of work producing systems, several quantities will be defined. The mole specific enthalpy $h_i(T)$, entropy $s_i(T, P_i)$ and Gibbs functions $g_i(T, P_i)$, for species i at some temperature T, partial pressure P_i, or activity a_i are given by Equations 2.1-2.4, respectively. The specific heat of species i ($C_{P,i}$) is known as a function of temperature in the ranges of interest, and is used to evaluate the given quantities. Thermochemical properties are evaluated using NIST-JANAF data [75].

$$h_i(T) = h_{f,i}(T_{ref}) + \int_{T_{ref}}^{T} C_{P,i}(T')dT' \tag{2.1}$$

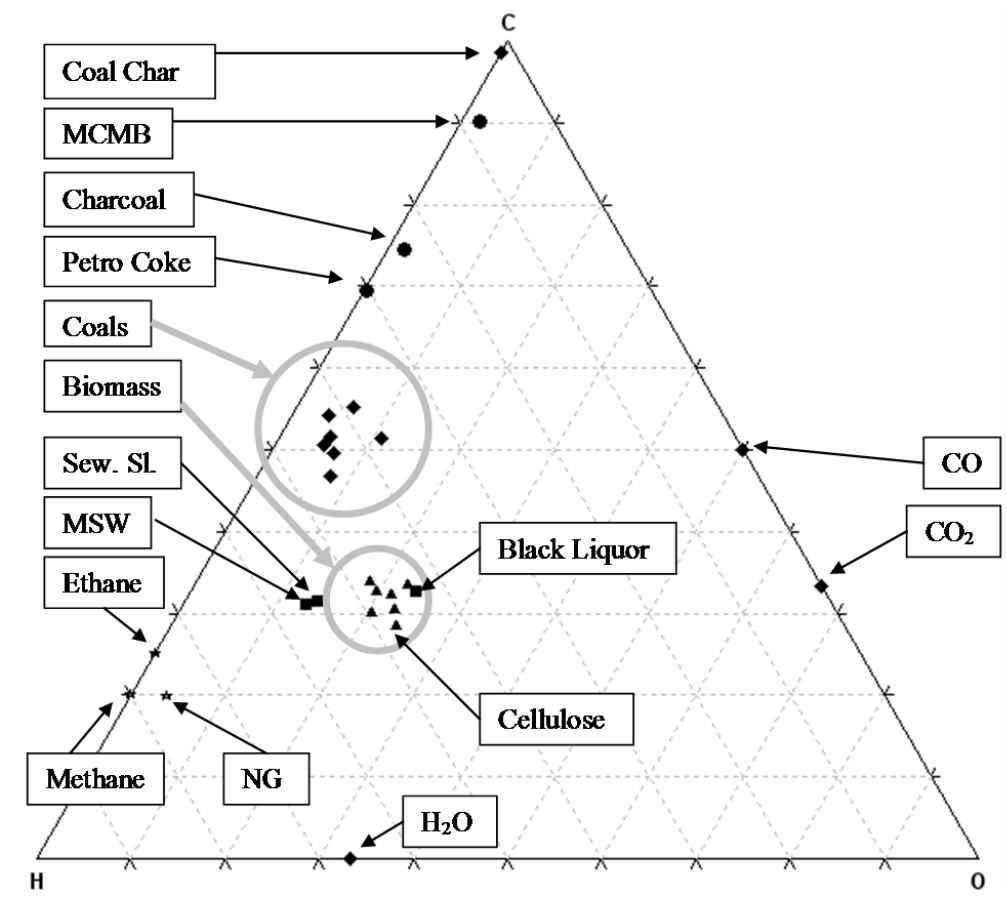


Figure 2.2: C-H-O ternary diagram for representative solid and gaseous hydrocarbon feedstocks, fuels and products.

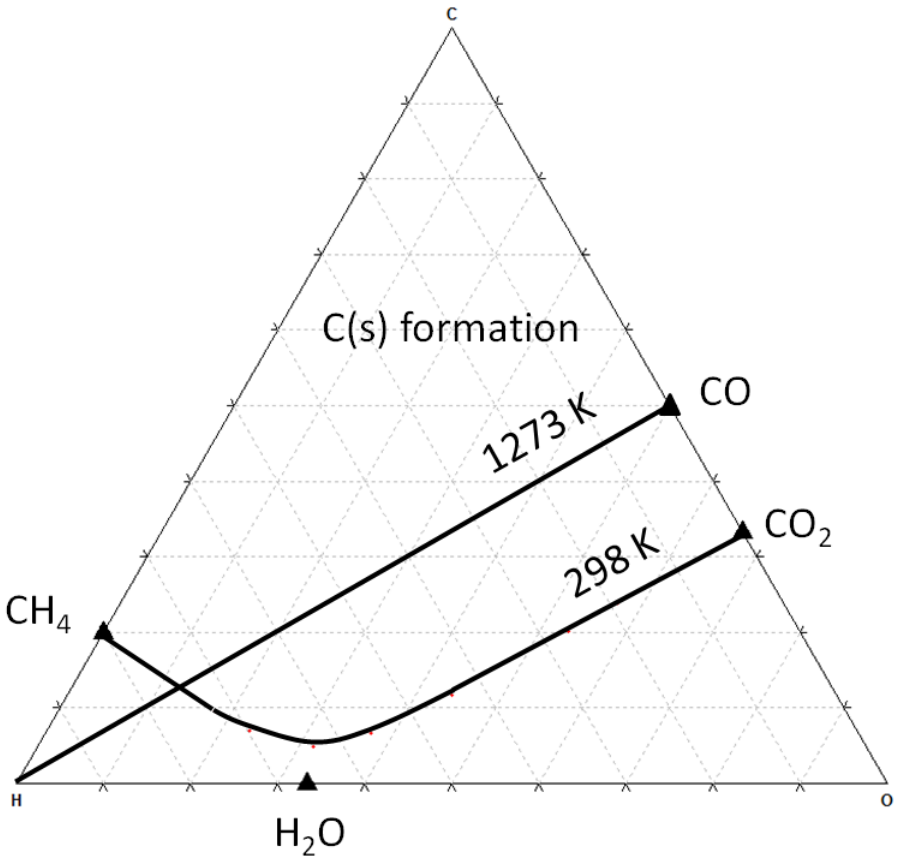


Figure 2.3: C-H-O ternary diagram showing the calculated boundary of carbon deposition, at temperatures of 298 K and 1273 K.

$$s_{i,gas}(T, P_i) = \int_{T_{ref}}^{T} \frac{C_{P,i}(T')}{T'} dT' - Rln\frac{P_i}{P_0} \tag{2.2}$$

$$s_{i,solid}(T, A_i) = \int_{T_{ref}}^{T} \frac{C_i(T')}{T'} dT' - Rlna_i \tag{2.3}$$

$$g_i(T, P_i) = h_i(T) - Ts_i(T, P_i) \tag{2.4}$$

Thermodynamically, the C-O chemical system is the keystone to DCFC operation, and thus will be used for illustration. There are a multitude of possible species to be considered, although only C(s), O_2(g), CO(g) and CO_2(g) are thermodynamically relevant here. When considering the work-producing potential of a fuel, there exists an upper limit that can be calculated based on properties of the fuel, the environment, and any constraints imposed on the system. This maximum work potential, or exergy, is the work that could be done by a system operating reversibly, with the products exiting in the dead state (298 K and 1 atm). A system in which carbon interacts with environmental air, and produces unmixed CO_2 and N_2, with all streams at 298 K (T_0) and 1 atm (P_0), is shown in Figure 2.4. The products are constrained to remain unmixed, characteristic of systems with capture-ready CO_2, as the fuel cell systems considered in the following section will necessarily exhaust unmixed CO_2 and N_2 due to the selective electrolyte membrane. The possibility of extracting work during a concentration relaxation to the unrestricted dead state (or environmental values [79]) will not be considered, as the stream of pure CO_2 requires capture while the remaining stream is rejected directly.

Equations 2.5 and 2.6 display the first and second laws of thermodynamics, respectively applied to the simple control volume in Figure 2.4. The activity of solid carbon is taken as unity, implying a pure solid. The activity of gases is given by the partial pressure of that species in an ideal solution of ideal gases. Note the the addition of gaseous CO_2 does not algebraically alter the first or second laws (unless the input stream is not at T_0 and P_0) and thus is completely arbitrary. This fact suggests

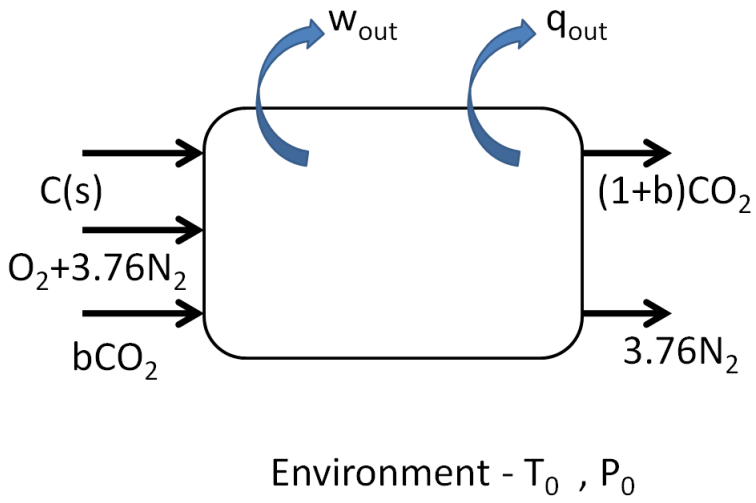


Figure 2.4: Single stage work extraction from a carbon stream, with input and output streams at 298 K and 1 atm.

that input of a gasifying agent is not necessarily required for DCFC operation.

$$\begin{aligned} 0 &= w_{out} + q_{out} + h_{CO2}(T_0) + 3.76h_{N2}(T_0) \\ &\quad - h_{C(s)}(T_0) - h_{O2}(T_0) - 3.76h_{N2}(T_0) \end{aligned} \tag{2.5}$$

$$\begin{aligned} \sigma &= q_{out}/T_0 + s_{CO2}(T_0, P_0) + 3.76s_{N2}(T_0, P_0) \\ &\quad - s_{C(s)}(T_0) - s_{O2}(T_0, 0.21P_0) - 3.76s_{N2}(T_0, 0.79P_0) \end{aligned} \tag{2.6}$$

These two equations can be combined algebraically by eliminating the heat interaction q_{out}, and solving for w_{out}. The work output is maximum when the irreversibility of the system is identically zero, given by Equation 2.7.

$$\begin{aligned} w_{out,max} &= h_{C(s)}(T_0) + h_{O2}(T_0) - h_{CO2}(T_0) \\ &\quad - T_0[s_{C(s)}(T_0) + s_{O2}(T_0, 0.21P_0) + 3.76s_{N2}(T_0, 0.79P_0) \\ &\quad - s_{CO2}(T_0, P_0) - 3.76s_{N2}(T_0, P_0)] \end{aligned} \tag{2.7}$$

The maximum work potential for this system is 388.3 $kJ{\cdot}mole_C^{-1}$, representing the maximum work output that could be realized utilizing solid carbon and the terrestrial environment, while giving a capture-ready stream of CO_2. This is roughly equivalent to the heating value of 393.5 $kJ{\cdot}mole_C^{-1}$ for carbon and air at 298 K. A work potential value of 394.4 $kJ \cdot mole_C^{-1}$ is often reported in the literature, but only applicable if the environment contains pure O_2 and the product is pure CO_2 (unmixed with the environment). The difference in these values is equivalent to the work required to separate the reactant O_2 from air at 1 atm total pressure. The N_2 exiting the system is not in chemical equilibrium with the environment, however it is assumed here that no practical mechanism exists to extract this potential work.

Real processes introduce irreversibility and/or additional operating constraints, producing less work than 388.3 $kJ \cdot mole_C^{-1}$. This value is independent of pressure, since the environment and fuel stream are initially at 1 atm, and the product states are specified as 1 atm. The work requirement for CO_2 compression is not included, although it will be similar between systems per mole of CO_2. Conventional approaches to solid fuel utilization rely on separate gasification or combustion processes prior to work extraction. These approaches represent irreversible chemical interactions which necessarily lower the available work output. Several approaches will be analyzed for comparison with the direct approach.

Indirect gasification (IDG) refers to the reaction of solid fuels with a gasifying agent, such as H_2O or CO_2, and energy transfer as heat. The agent supplies oxygen to convert solid carbon to gas phase CO and CO_2 via endothermic reactions. The absence of direct oxygen addition into the gasifier is the key feature of the IDG process. A simple, ideal IDG system is portrayed in Figure 2.5, incorporating two separate control volumes, constrained independently. The gasifier, represented by control volume I, is constrained to produce an equilibrium stream of $CO/CO_2/O_2$ at a specified temperature T_{gas} and P_0. The gasifying agent (CO_2) input is constrained here such that the output stream has a maximum CO/CO_2 ratio without solid carbon present. The work producing process, depicted by control volume II, is only constrained to produce separate streams of CO_2 and N_2, each at T_0 and P_0.

The first and second laws of thermodynamics, applied to the gasification control

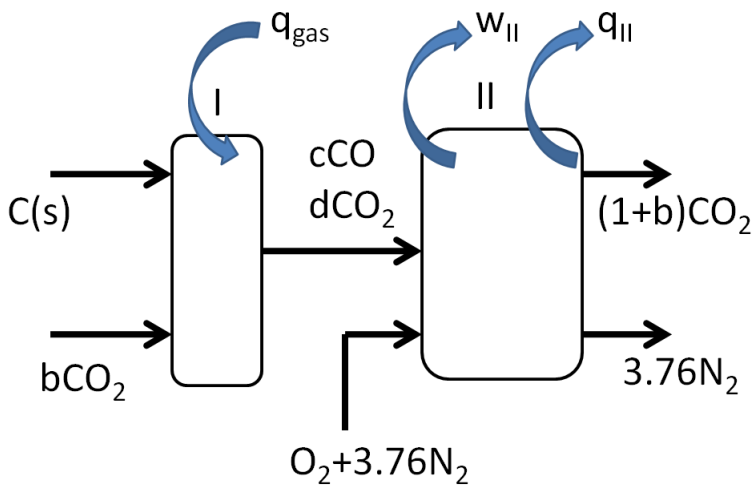


Figure 2.5: Two stage IDG process for work extraction from a carbon stream, with the input and output streams at 298 K and 1 atm.

volume, allow the heat requirement (q_{gas} in Equation 2.8) and irreversibility (σ_I in Equation 2.9) to be calculated. Considering T_{gas} values of 1073, 1173, and 1273 K, the maximum work outputs for this system are 370.1, 367.5, and 363.6 $kJ \cdot mole_C^{-1}$, respectively, as calculated using Equation 2.10. Note that the irreversibility of control volume II is identically zero during this calculation.

$$q_{gas} = c \cdot h_{CO}\left(T_{gas}\right) + d \cdot h_{CO2}\left(T_{gas}\right) - h_C\left(T_0\right) - b \cdot h_{CO2}\left(T_0\right) \tag{2.8}$$

$$\begin{aligned} \sigma_I &= c \cdot s_{CO}\left(T_{gas}, P_{CO,gas}\right) + d \cdot s_{CO2}\left(T_{gas}, P_{CO2,gas}\right) \\ &\quad -s_C\left(T_0\right) - b \cdot s_{CO2}\left(T_0, P_0\right) - \frac{q_{gas}}{T_{gas}} \end{aligned} \tag{2.9}$$

$$\begin{aligned} w_{II} &= g_C\left(T_0\right) + g_{O2}\left(T_0, 0.21P_0\right) + 3.76 g_{N2}\left(T_0, 0.79P_0\right) \\ &\quad -g_{CO2}\left(T_0, P_0\right) - 3.76 g_{N2}\left(T_0, P_0\right) - T_0\left(\sigma_I + \sigma_{II}\right) \end{aligned} \tag{2.10}$$

The heat inputs q_{gas} also have an associated work potential associated with the elevated temperature, as given in Equation 2.11. These work potentials are 166, 171, and 180 $kJ{\cdot}mole_C^{-1}$ at 1073, 1173, and 1273 K, respectively. The second law efficiency can then be determined using Equation 2.12, giving values of 67, 66, and 64% at 1073, 1173, and 1273 K, respectively.

$$w_{gas,max} = q_{gas}\left(1 - \frac{T_0}{T_{gas}}\right) \tag{2.11}$$

$$\eta_{II} = \frac{w_{out}}{\psi_C + w_{gas,max}} \tag{2.12}$$

Although the work output of the IDG systems is only 4.7 - 6.4% lower than that for the unconstrained case, the second law efficiency is significantly lower than the unconstrained case. This is due to the additional heat input requirement. No work is extracted during the gasification process, and the process irreversibility results in lost work potential. Systems operating at elevated temperature may be able to supply energy as heat to the gasification reactor from heat rejection from downstream processes. The use of system heat rejection allows for higher efficiencies, as will be shown in the following analysis.

Autothermal gasification (ATG) refers to reaction of solid fuels with O_2 (from air) and a modifier, such as H_2O or CO_2, without sensible heat addition. The key feature of ATG is the need for pure oxygen, to produce a syngas stream with high heating value by preventing N_2 dilution of the products. The exclusion of N_2 also provides that the effluent stream will be more suitable for CO_2 capture. A simple, ideal ATG system is portrayed in Figure 2.6, incorporating a distinct gasification control volume and work producing control volume. The gasifier is constrained to operate adiabatically and isobarically, with the gaseous output again constrained such that the output stream has a maximum CO/CO_2 ratio without solid carbon present. The work producing process, depicted by control volume II, is again constrained to produce separate streams of CO_2 and N_2 at T_0 and P_0.

The work output (w_{II}) after algebraic combination, is shown in Equation 2.13, and is a maximum when the irreversibility of control volume II (σ_{II}) is identically

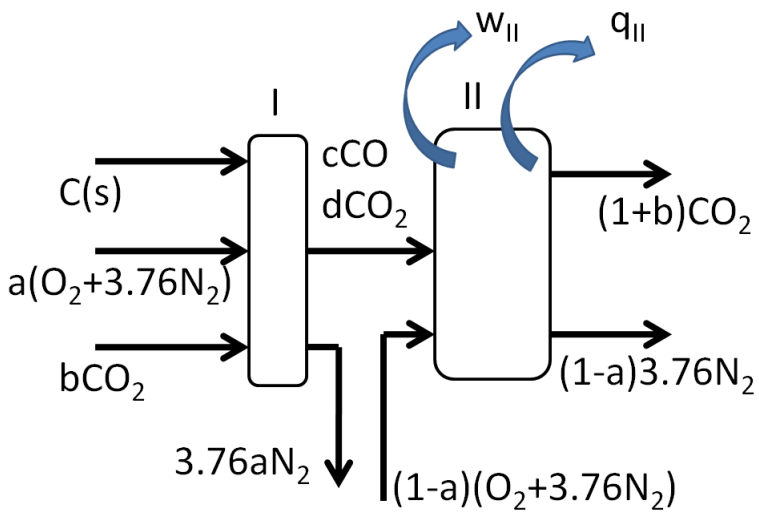


Figure 2.6: Two stage ATG process for work extraction from a carbon stream, with the input and output streams at 298 K and 1 atm.

zero. The irreversibility of the gasification control volume (σ_I) is calculated using Equation 2.14.

$$\begin{aligned} w_{II} &= g_C(T_0) + g_{O2}(T_0, 0.21P_0) + 3.76 g_{N2}(T_0, 0.79P_0) \\ &\quad - g_{CO2}(T_0, P_0) - 3.76 g_{N2}(T_0, P_0) - T_0(\sigma_I + \sigma_{II}) \end{aligned} \tag{2.13}$$

$$\begin{aligned} \sigma_I &= c \cdot s_{CO}(T_{gas}, P_{CO,gas}) + d \cdot s_{CO2}(T_{gas}, P_{CO2,gas}) \\ &\quad + 3.76a\left(s_{N2}(T_0, P_0) - s_{N2}(T_0, 0.79P_0)\right) \\ &\quad - s_C(T_0) - a \cdot s_{O2}(T_0, 0.21P_0) - b \cdot s_{CO2}(T_0, P_0) \end{aligned} \tag{2.14}$$

Since the input and output states, and the work and heat interactions are specified for the gasifier, the irreversibility of the gasifier is now uniquely set. At gasification temperatures of 1073, 1173, and 1273 K, the maximum work outputs of control volume II are 338.4, 340.2 and 340.4 $kJ \cdot mole_C^{-1}$ respectively, or 12 - 13% less than the unconstrained approach shown in Figure 2.4. The work output of the ATG system is

less than 388.3 $kJ \cdot mole_C^{-1}$ due to the irreversible chemical gasification process, with second law efficiencies of 87% for all three temperatures.

This general analysis shows clearly that gasification constraints lead to lower work potentials, with approaches using separate gasification steps incurring higher losses than the arrangement without separate gasification. The primary benefit of the DCFC approach is that a separate gasifier is not required.

Pyrolysis (PYR) refers to the addition of sensible heat to solid fuel, and the subsequent thermochemistry that occurs in a reducing atmosphere in the absence of input gas streams. The PYR process produces char and fuel gas species such as H_2 and CH_4 as the major products, along with larger volatile hydrocarbons and tars. Oxygen containing species are also present, but to an extent dictated by the oxygen content of the fuel. One benefit of this process is the bifurcation of the fuel stream into a gas phase, and a solid phase that would normally require large reaction times, allowing the use of fuel gas separate from the slow-reacting char. Coals typically have lower oxygen content than biomass, and as such are more prone to carbon deposition than biomass during pyrolysis. This process is not applicable to solid carbon, and would not be well modeled using equilibrium methods [7], and thus will not be considered here.

The addition of hydrogen to the C-O system accounts for such key species as H_2, H_2O, CH_4, larger hydrocarbon molecules and smaller O-H compounds. The C-H-O system behaves similarly to the C-O system, although the H/O molar ratio presents an additional degree of freedom. There are several important reactions involving the C-H-O system, as given by Reactions R2.1 - R2.7, that will be discussed here briefly in order to frame the following discussions.

$$\mathrm{C(s) + O_2(g) \longrightarrow CO_2(g)} \qquad \text{(R 2.1)}$$

$$\mathrm{C(s) + \tfrac{1}{2}O_2(g) \longrightarrow CO(g)} \qquad \text{(R 2.2)}$$

$$\mathrm{C(s) + CO_2(g) \longrightarrow 2\,CO(g)} \qquad \text{(R 2.3)}$$

$$\mathrm{C(s) + H_2O(g) \longrightarrow CO(g) + H_2(g)} \qquad \text{(R 2.4)}$$

$$CO(g) + \tfrac{1}{2}O_2(g) \longrightarrow CO_2(g) \quad \text{(R 2.5)}$$

$$H_2(g) + \tfrac{1}{2}O_2(g) \longrightarrow H_2O(g) \quad \text{(R 2.6)}$$

$$CO(g) + H_2O(g) \longrightarrow CO_2(g) + H_2(g) \quad \text{(R 2.7)}$$

Shown in Figure 2.7 are the heats of reaction and standard Gibbs free energy changes of reaction for exothermic, combustion reactions. Shown in Figure 2.8 are the heats of reaction and standard Gibbs free energy changes of reaction for endothermic, gasification reactions and the water gas shift (WGS) reaction (Reaction R2.7). It is evident from these two figures that the endothermic steam (Reaction R2.4) and Boudouard (Reaction R2.3) gasification reactions behave similarly thermodynamically, as do the exothermic H_2 (Reaction R2.6) and CO (Reaction R2.5) combustion reactions. Essentially the presence of solid carbon dictates the thermodynamic behavior, with the presence of hydrogen playing a secondary role. Another important characteristic of the curves shown in Figure 2.8, is the crossover of ΔG_R^0 for the carbon gasification reactions from positive to negative as temperature is increased near 1000 K. This fact will be shown to limit the lower bound of operating temperature for the gasification-driven DCFC.

2.3 Fuel Cell Potential

2.3.1 Theoretical Cell Potentials

As discussed earlier, solid oxide fuel cells operate by extracting work from differences in oxygen electrochemical potential. Generally air is supplied to the cathode, maintaining a nominal oxygen activity of $0.21P_{cathode}$. Note that for all systems considered here, both experimentally and theoretically, oxygen will be supplied to the cathode directly from air at a total pressure of 1 atm. A hydrocarbon fuel is used at the anode to maintain a low oxygen activity, and thus providing the driving force for oxygen transfer. The theoretical cell work potential can be calculated by knowledge of the free energy change per oxygen atom between cathode and anode. The implicit assumptions involved in this theoretical procession are listed below;

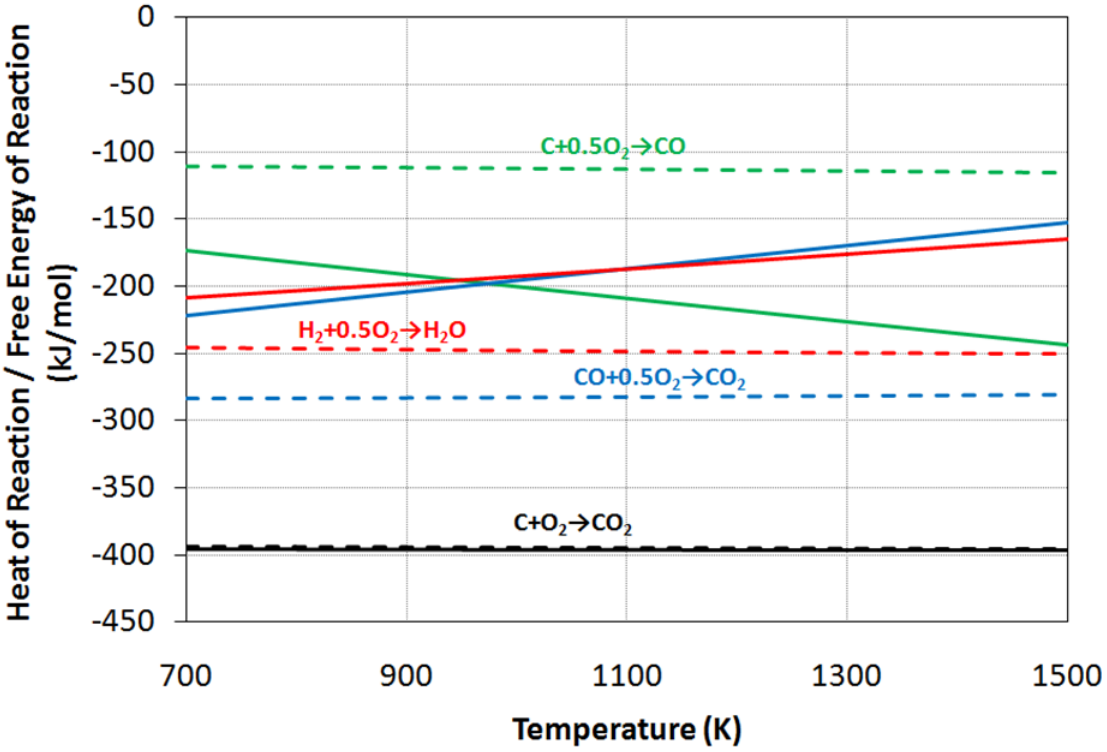


Figure 2.7: ΔH_R (dashed) and ΔG_R^0 (solid) for relevant combustion reactions.

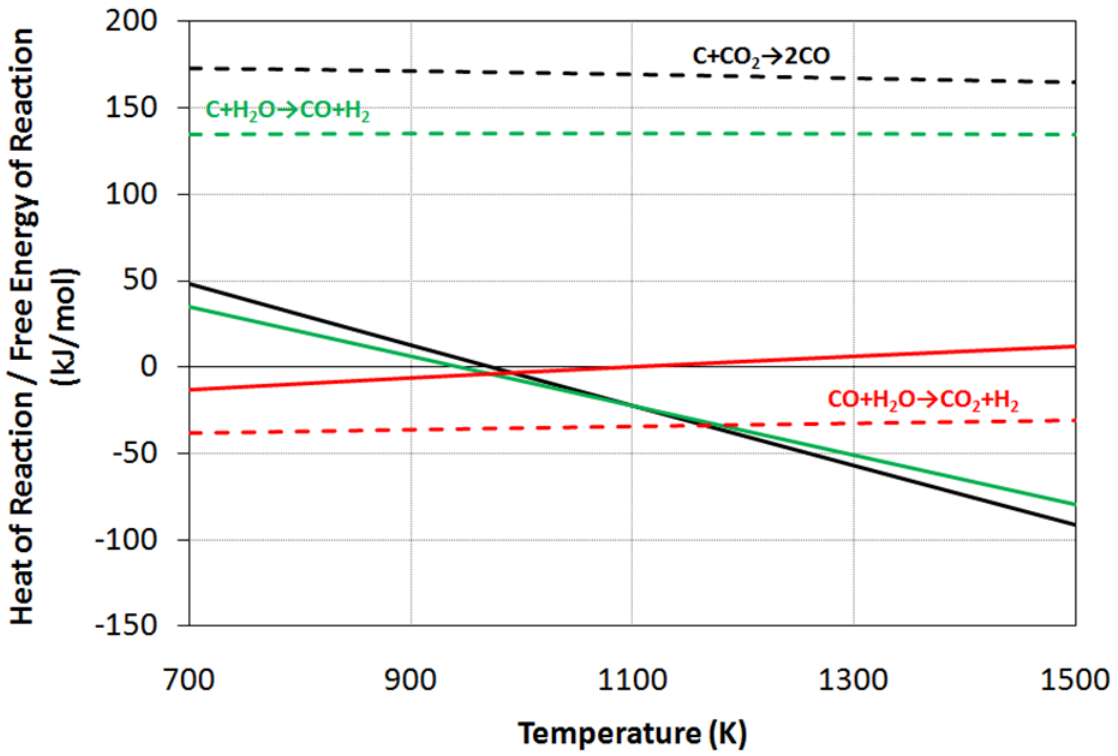


Figure 2.8: ΔH_R (dashed) and ΔG_R^0 (solid) for relevant gasification reactions and the WGS reaction.

- system is isothermal, isobarric, and steady
- gaseous and surface species are in equilibrium (termed half cell equilibrium) at each triple phase boundary (TPB)
- electrolyte conducts only oxygen ions, and has zero electron conductivity
- electrolyte is impermeable to gases or solids
- electrode and electrolyte materials do not influence reversible cell potential
- gaseous and solid mixtures are ideal

Electrochemical systems are generally characterized by electronic potential difference or voltage rather than free energy to describe potential differences. These two quantities are related by a factor $-nF$, where n denotes the stoichiometric number of electrons transferred per reaction (n = 2 per oxygen atom) and F denotes Faraday's constant. At open circuit conditions (i.e. infinite resistive load) there is zero electronic current, and the charge transfer reactions at the electrode/electrolyte interfaces are assumed to be in equilibrium. The adsorption/desorption reactions and heterogeneous reactions are also assumed to be in equilibrium in the these formulations. The open circuit voltage (OCV) is then the electrostatic potential difference that exactly balances the chemical potential difference of the species that traverses the electrolyte. For the SOFC, the charge transfer reactions at the cathode (c in Reaction R2.8) and anode (a in Reaction R2.9) can be formulated as follows, assuming chemical equilibrium at the anode.

$$\mathrm{O_2(g)_c} + 4\,\mathrm{e_c^{'}} + 2\,\mathrm{V_o^{\cdot\cdot}} \longrightarrow 2\,\mathrm{O_o^x} \qquad \text{(R 2.8)}$$

$$2\,\mathrm{O_o^x} \longrightarrow \mathrm{O_2(g)_a} + 4\,\mathrm{e_a^{'}} + 2\,\mathrm{V_o^{\cdot\cdot}} \qquad \text{(R 2.9)}$$

Considering the following global reaction (Reaction R2.10) for oxygen from summing the above reactions, Equations 2.15 and 2.16 express the expected free energy and electronic potential differences (referred to as the Nernst equation), respectively.

$$\mathrm{O_2(g)_{cathode}} \longrightarrow \mathrm{O_2(g)_{anode}} \qquad \text{(R 2.10)}$$

$$\Delta G = g_{O2,a}(T, P_{O2,a}) - g_{O2,c}(T, P_{O2,c}) \tag{2.15}$$

$$E_{rev} = \frac{RT}{nF} ln \left(\frac{P_{O_2,cath}}{P_{O_2,an}} \right) \tag{2.16}$$

Interactions with $\Delta G < 0$ can proceed spontaneously given a mechanistic pathway, and produce work during progression. If the oxygen activity is fixed at the anode by a fuel, the Nernst potential can be formulated in the following manner, in which ν_r represents the molar stoichiometric coefficient of the reactants and ν_p the products. The Nernst potential is then given by Equation 2.17 for such reactions,

$$E_{rev} = \frac{\Delta G_0}{-nF} + \frac{RT}{nF} ln \left(\frac{\prod_{r=1}^{N_r} P_r^{\nu_r}}{\prod_{p=1}^{N_p} P_p^{\nu_p}} P_0^{(\sum_{p=1}^{N_p} \nu_p - \sum_{r=1}^{N_r} \nu_r)} \right) \tag{2.17}$$

resulting in unique temperature and activity dependencies. The first term represents the standard potential E_0, with all reactants and products having unit activities. The second term contains the activity dependencies, and for ideal components, can be rearranged as shown in Equation 2.18.

$$E_{rev} = E_0 + \frac{RT}{nF} ln \left(\frac{\prod_{r=1}^{N_r} x_r^{\nu_r}}{\prod_{p=1}^{N_p} x_p^{\nu_p}} \left(\frac{P_0}{P} \right)^{(\sum_{p=1}^{N_p} \nu_p - \sum_{r=1}^{N_r} \nu_r)} \right) \tag{2.18}$$

Increases in the activity of reactants, or decreases in the activity of products, will produce larger theoretical cell potentials. As demonstrated by Equation 2.18, the cell potential is dependent upon the anode-side composition assuming that the cathode-side oxygen activity is fixed. There are two unique and convenient sets of anode side compositions that can be used for comparison; standard activities and equilibrium activities. The standard potential, E_0 here, gives a nominal cell potential for a given global interaction with all anode side species having unit activities, while the equilibrium potential is based on activities in chemical equilibrium composition at the anode. In terms of fuel cell theoretical potentials, the C-O system gives three key global reactions;

$$\mathrm{C(s)_{anode}} + \mathrm{O_2(g)_{cathode}} \longleftrightarrow \mathrm{CO_2(g)_{anode}} \tag{R 2.11}$$

$$\mathrm{C(s)_{anode}} + \tfrac{1}{2}\mathrm{O_2(g)_{cathode}} \longleftrightarrow \mathrm{CO(g)_{anode}} \quad \text{(R 2.12)}$$

$$\mathrm{CO(g)_{anode}} + \tfrac{1}{2}\mathrm{O_2(g)_{cathode}} \longleftrightarrow \mathrm{CO_2(g)_{anode}} \quad \text{(R 2.13)}$$

while the H-O system includes the following reaction.

$$\mathrm{H_2(g)_{anode}} + \tfrac{1}{2}\mathrm{O_2(g)_{cathode}} \longleftrightarrow \mathrm{H_2O(g)_{anode}} \quad \text{(R 2.14)}$$

The first and second reactions represent direct carbon conversion mechanistically, while the third represents conversion of gaseous CO arising from carbon (via the Boudouard reaction). The fourth reaction represents hydrogen oxidation. The standard potentials for these relevant reactions are shown in Figure 2.9a, for an oxygen activity of 0.21 (air), and standard unit activities of other species. If the C-O system or C-H-O system (H/O ratio of 2) are allowed to equilibrate at the anode side in the presence of solid carbon ($x > 1$) according to the following reactions, then the composition at the anode is uniquely fixed by chemical equilibrium.

$$\mathrm{xC(s) + O_2(g)} \longleftrightarrow \sum_{i=1}^{N_p} a_i A_i \quad \text{(R 2.15)}$$

$$\mathrm{xC(s) + H_2O(g)} \longleftrightarrow \sum_{i=1}^{N_p} a_i A_i \quad \text{(R 2.16)}$$

The $\mathrm{CO/CO_2}$ ratios governed by the Boudouard reaction are shown in Figure 2.10a. The $\mathrm{CO/CO_2}$ ratio increases with increasing temperature, and decreases with increasing pressure in accordance with the Le' Chatelier principle. The OCV for the electrochemical reaction of CO with $\mathrm{O_2}$ from air is shown in Figure 2.10b, for various temperatures and $\mathrm{CO/CO_2}$ ratios. Cell OCV increases with $\mathrm{CO/CO_2}$ ratio at a given temperature, and decreases with temperature for a given ratio.

The equilibrium potentials for the above reactions at various temperatures are shown in Figure 2.9b as solid black curves, along with the standard potentials included as gray curves for reference. The equilibrium potential for $CO_{an} + \frac{1}{2}O_{2,cath} \rightarrow CO_{2,an}$ is coincident with that for $C_{an} + \frac{1}{2}O_{2,cath} \rightarrow CO_{an}$ at temperatures above 1100 K, and

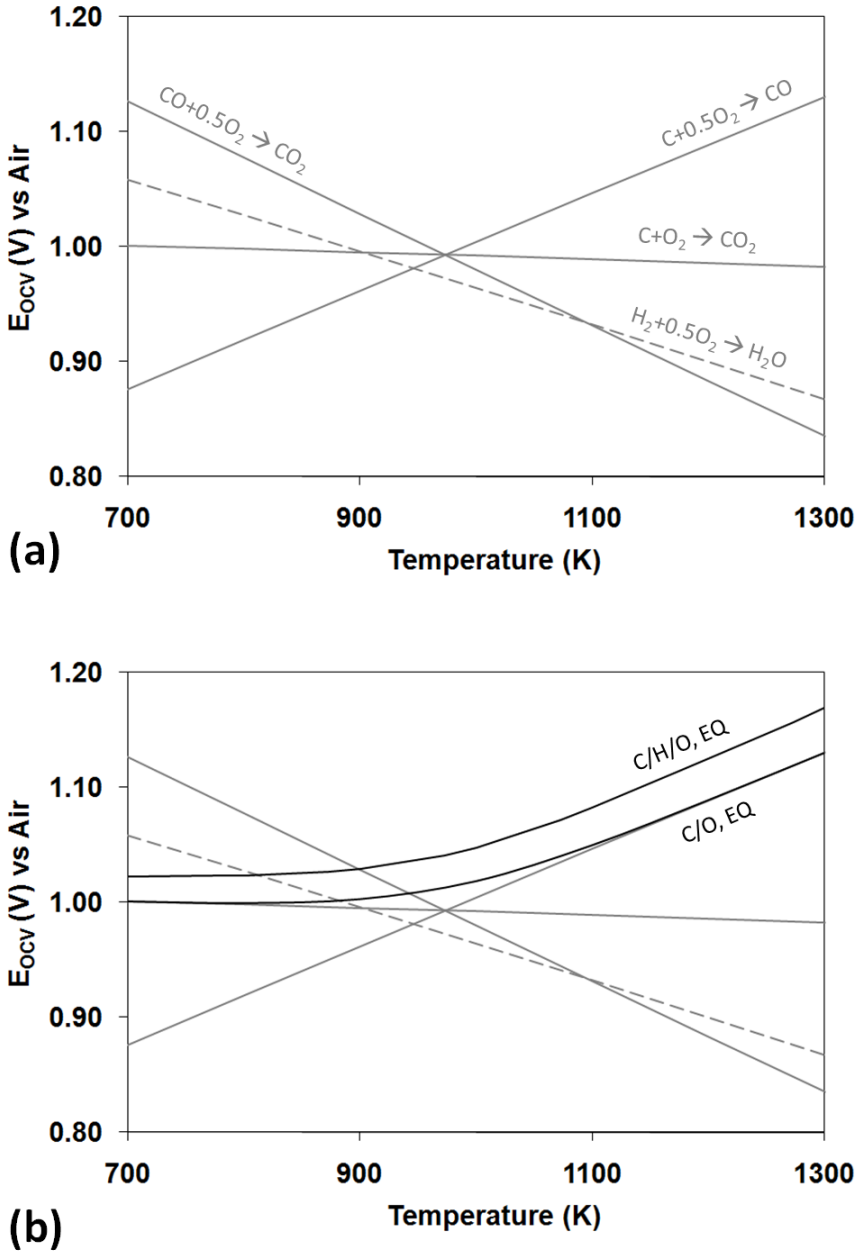


Figure 2.9: (a) Standard and (b) equilibrium potentials calculated for both C-O and H-O interactions as a function of temperature.

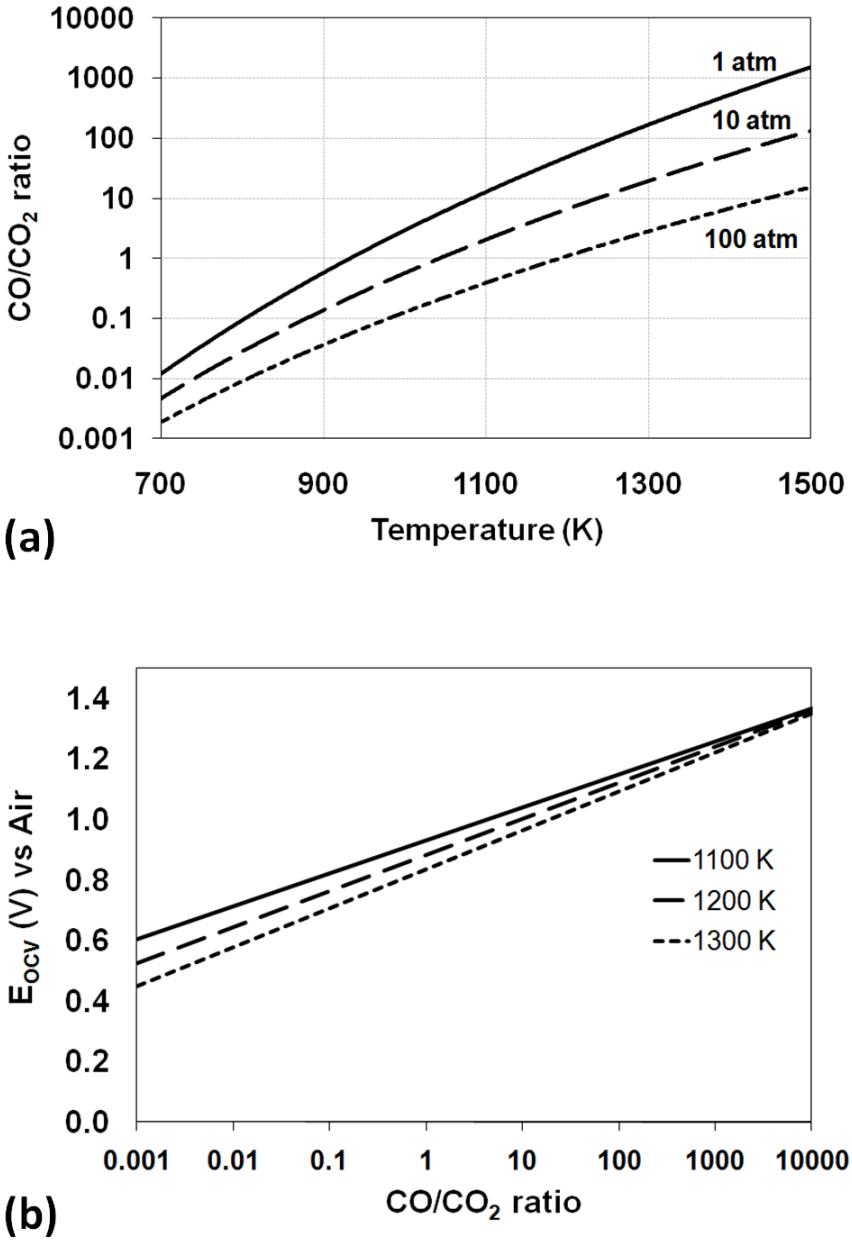


Figure 2.10: (a) Equilibrium CO/CO_2 ratios calculated for various total pressures and temperatures, and (b) OCV values calculated for various CO/CO_2 ratios and temperatures, based on the Boudouard reaction.

is coincident with $C_{an} + O_{2,cath} \rightarrow CO_{2,an}$ below 800 K. This is due to the prevailing CO/CO_2 ratio at the anode, dictated by Boudouard equilibrium. At high temperatures CO is preferred, while at low temperatures CO_2 is preferred. At intermediate temperatures between 800 and 1100 K, the CO/CO_2 ratio transitions from $O(0.1)$ - $O(10)$, causing the observed transition of the equilibrium potential. The equilibrium potential curve for the specified C-H-O system behaves similarly, although shifted 20 - 40 mV higher than that of the C-O system.

The equilibrium potentials arise from decreased oxygen activity at the anode due to the presence of solid carbon. The bed reactions allowing this composition shift are endothermic, and thus energy input is required as heat. Actual DCFC operation, under a nonzero current flux would raise the oxygen activity at the anode, lowering the operating voltage of the cell in accordance with typical SOFC operation. The theoretical OCV values presented here are based on the assumptions outlined at the beginning of this section.

2.3.2 Experimental Verification of Cell Potentials

In order to validate the expected OCV values, experiments were conducted with simple SOFC arrangements. Details of the experimental setup, along with component information, can be found in the Appendices. Gas tight tube cells, with Pt electrodes, were used to measure cell potentials. The tube cells used were made from commercially available, partially stabilized zirconia (PSZ), closed-end tubes. The 45 cm long electrolyte tubes were about 750 microns thick, with a 12.7 mm outer diameter. Although large current densities could not be supported with these thick tubes, they provided a leak-free, mechanically resilient support from which to conduct voltage measurements. Both the cathode and anode were formed by applying Pt ink (available from Englehard) to both sides of the PSZ tube, followed by heating in air to 1173 K, at a nominal rate of 5 $K \cdot min^{-1}$, to remove the organic components. Shown in Figures 2.11 and 2.12 are SEM images of face and cross section of the porous electrodes, respectively. An SEM image of the electrolyte (dark grains underneath) and electrode (lighter network on top) interface are shown in Figure 2.13. Platinum

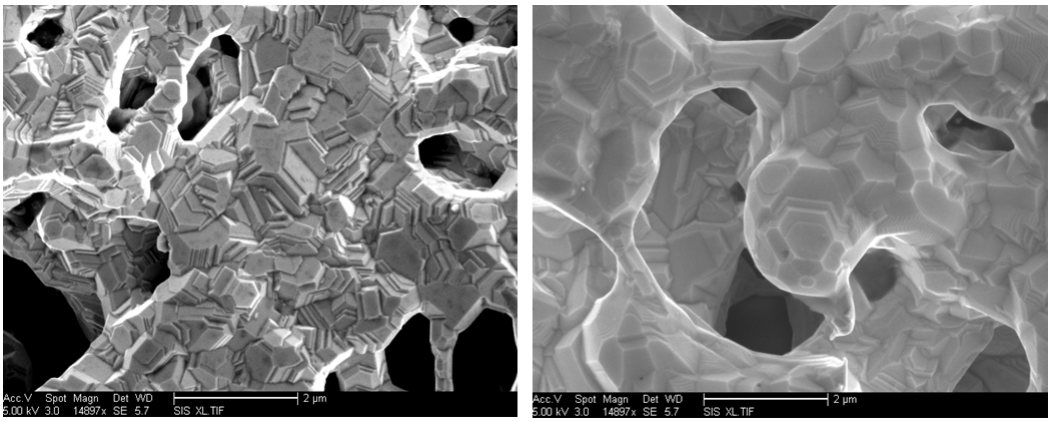


Figure 2.11: SEM image of the tube cell platinum electrode displaying porous structure (face view with 2 μm scale bar).

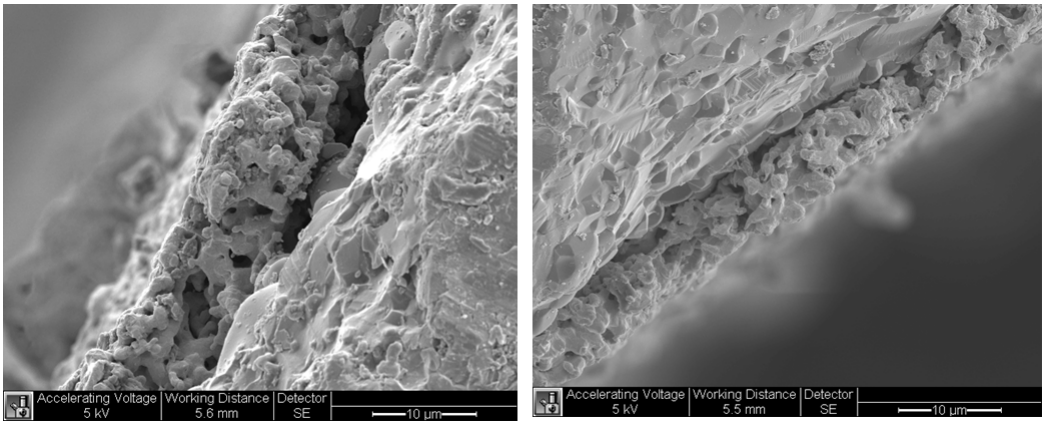


Figure 2.12: SEM image of the tube cell cross section showing dense PSZ and porous Pt (section view with 10 μm scale bar).

wires (0.5 mm diameter) were used as current collectors for both electrodes, and were mechanically bonded with the dried Pt ink.

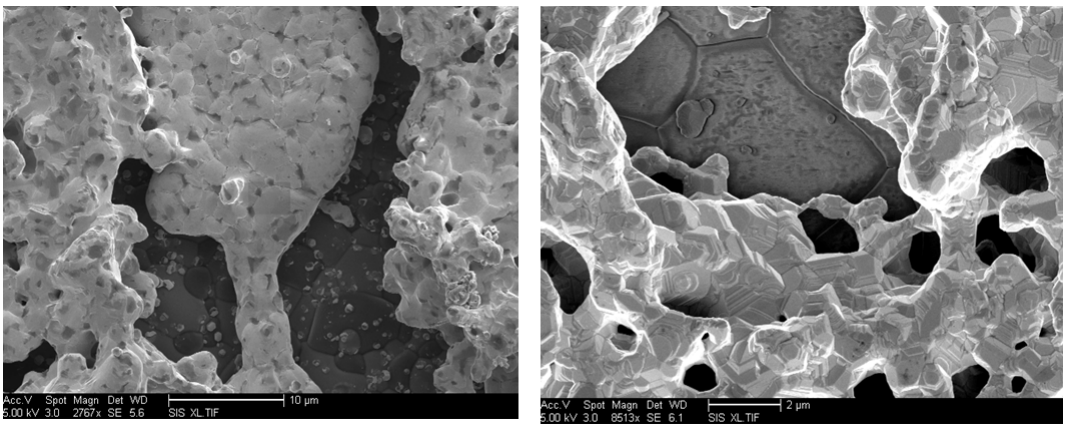


Figure 2.13: SEM image of the tube cell electrolyte (large dark grains) - electrode (lighter network) interface (face view).

The tube assembly was then placed in a fabricated quartz housing so that the electrode environment could be actively controlled. Carbon particles (activated carbon available from Fisher Scientific) were loaded into the quartz reactor, completely covering the active anode region of the tube cell. Gases were then introduced from the bottom of the housing to traverse the bed and exit at the top of the housing through a liquid seal. The anode-side volume in the housing was purged with He prior to heating and measurement. A 3 mm OD quartz tube was inserted into the interior of the PSZ tube to supply environmental air to the cathode via an impeller pump. An insulated K-type thermocouple was inserted adjacent to this quartz tube in order to measure cell temperature at the cathode. The quartz housing was placed in a temperature-controlled tube furnace, with the controlling K-type thermocouple positioned in the region between the quartz housing and furnace wall. A cross section of the entire reactor assembly is depicted in Figure 2.14. This experimental configuration provides a direct means of measuring the effective oxygen activity at the DCFC anode. The open circuit voltages of cells were verified experimentally for both the C-O and C-H-O (H/O molar ratio of 2) systems as shown in Figures 2.15 - 2.17, respectively. The data

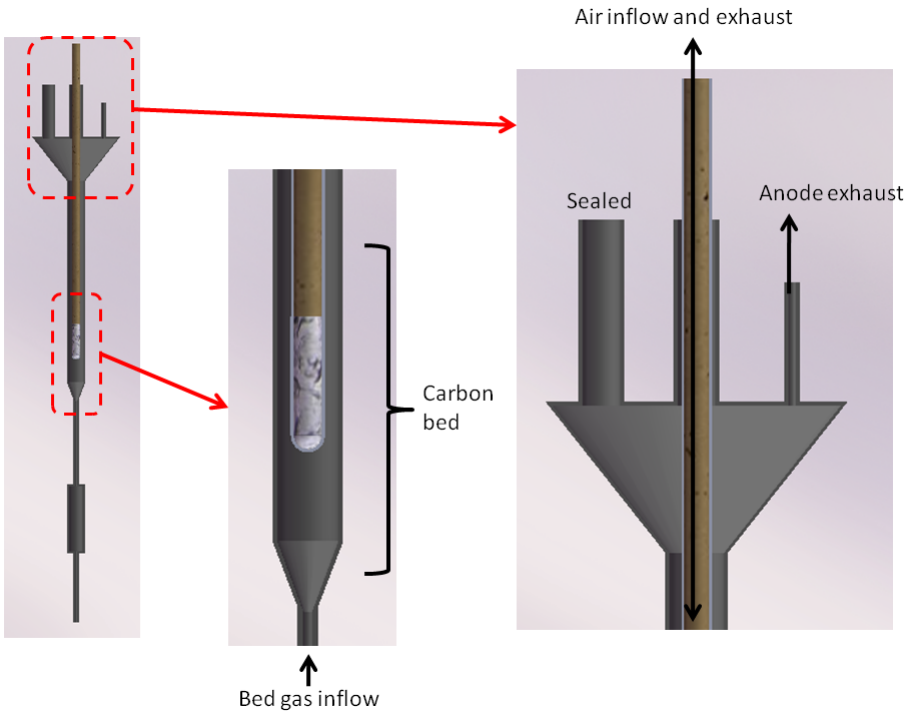


Figure 2.14: Cross section schematic of the tube cell assembly used for OCV measurements.

presented in Fig 7 were taken during temperature ramps between nominally 773 and 1253 K. The ramp rates, in units of $K{\cdot}min^{-1}$, are given numerically in the legend of the figure. The OCV data for the C-O system are not strongly dependent on heating rate in this configuration. This is likely due to the large thermal inertia of carbon particles in the bed, which would exhibit a time lag. The interior of particles, where most of the active surface area resides, would influence the gas phase composition in accordance with the actual interior surface temperature. The data labeled *amb*

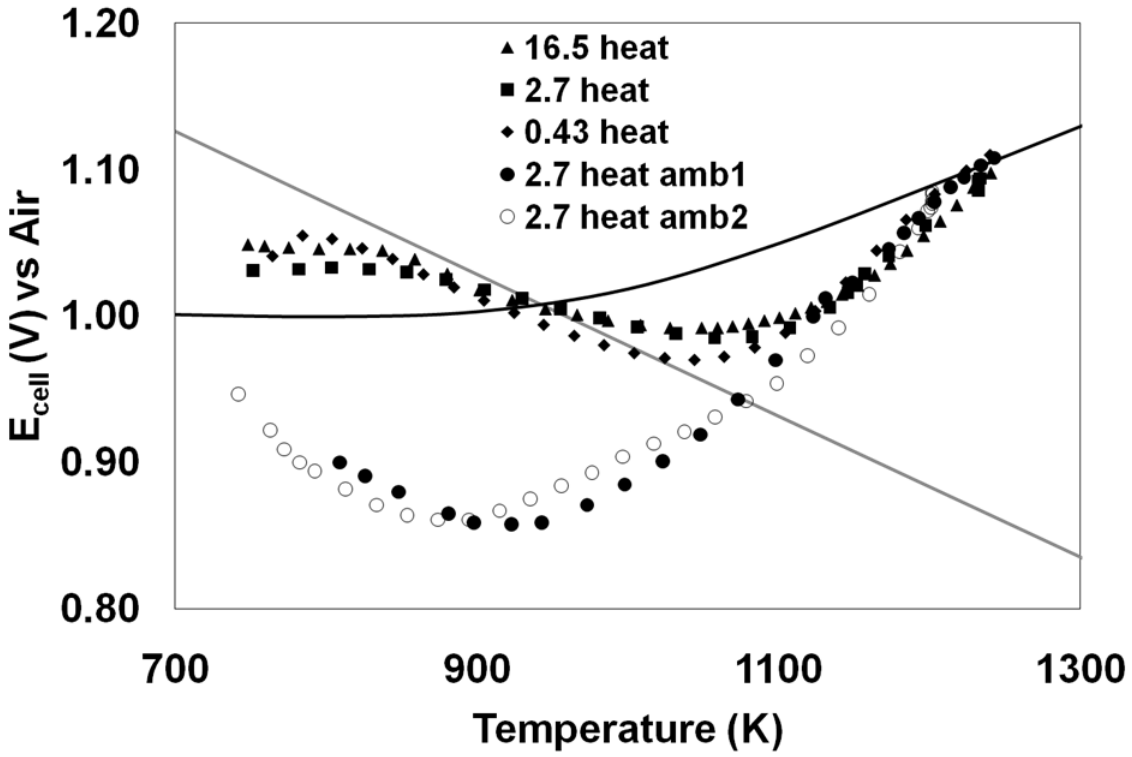


Figure 2.15: Experimental OCV values for the C-air cell during heating ramps, as compared to calculations. The heating rate is given in the figure legend in $K{\cdot}min^{-1}$.

were taken during initial temperature ramps with raw activated carbon. The FAC particles initially have a large amount of adsorbed oxides arising from storage in air, and desorption of these complexes as CO and CO_2 skew the gas phase composition

during initial heating. The lower OCV value is characteristic of larger CO_2/CO ratios, caused by the large oxygen content of air in the storage environment. The data labeled *amb1* was taken with the anode submersed in a particle bed similar to the other data sets. The data labeled *amb2* was taken with a packed bed of carbon in contact with an anode located on the PSZ tube interior and stagnant air at the tube exterior. The agreement in results between experimental configurations is expected since a similar batch of raw particles were used. At temperatures near 1100 K, the potential approaches that of previously heated carbon, indicating that desorption of the initially adsorbed oxides is complete.

During cooling ramps, the experimental data, with the carbon bed present, tracked the equilibrium curves until the temperature decreased sufficiently to slow the chemical reaction rates in the bed (near 1100 K at these ramp rates). The increase in OCV during cooling below about 1100 K is due to effective freezing of the gas phase composition, which causes the OCV to change slope in accordance with a fixed CO/CO_2 ratio. During cooling ramps to temperatures below 900 K, the OCV decreases with decreasing temperature. This is likely due to the reverse Boudouard reaction ($2CO \rightarrow C + CO_2$) on the Pt anode since CO_2 and C(s) are thermodynamically preferred at lower temperature (see Figure 2.8).

Open circuit potential data for an equimolar CO/CO_2 mixture flowing past the anode, with no carbon bed present, is included Figure 2.16 for comparison. The data for this gas mixture match with theoretical calculations above 900 K, depicting a scenario with frozen chemical composition (i.e. a forced CO/CO_2 ratio). At lower temperatures, the reverse Boudouard reaction is thermodynamically favored for the equimolar mixture of CO and CO_2, without a carbon bed. Thus the composition is no longer fixed due to chemical reaction at the electrode, denoted by the curvature of the measured OCV away from the standard potential curve indicative of lower CO/CO_2 ratios.

The data presented in Figure 2.17 depict the inclusion of hydrogen-containing species into the model. The OCV data at temperatures above about 1100 K reach equilibrium values shown by the solid black curve, in which steam-carbon gasification fixes the oxygen activity at the anode. The C-H-O system, as shown by the data

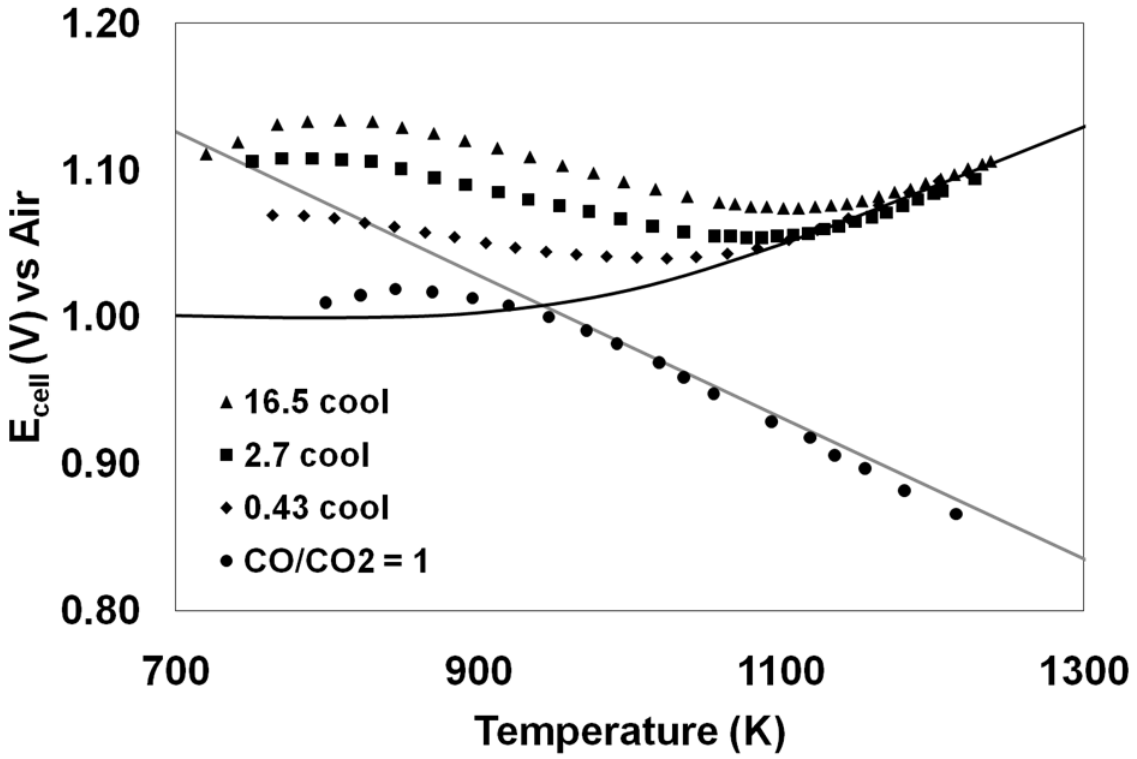


Figure 2.16: Experimental OCV values for the C-air cell during cooling ramps, as compared to calculations. The cooling rate is given in the figure legend in $K \cdot min^{-1}$.

presented in Figures 2.15-2.17, behaves similarly to the C-O system.

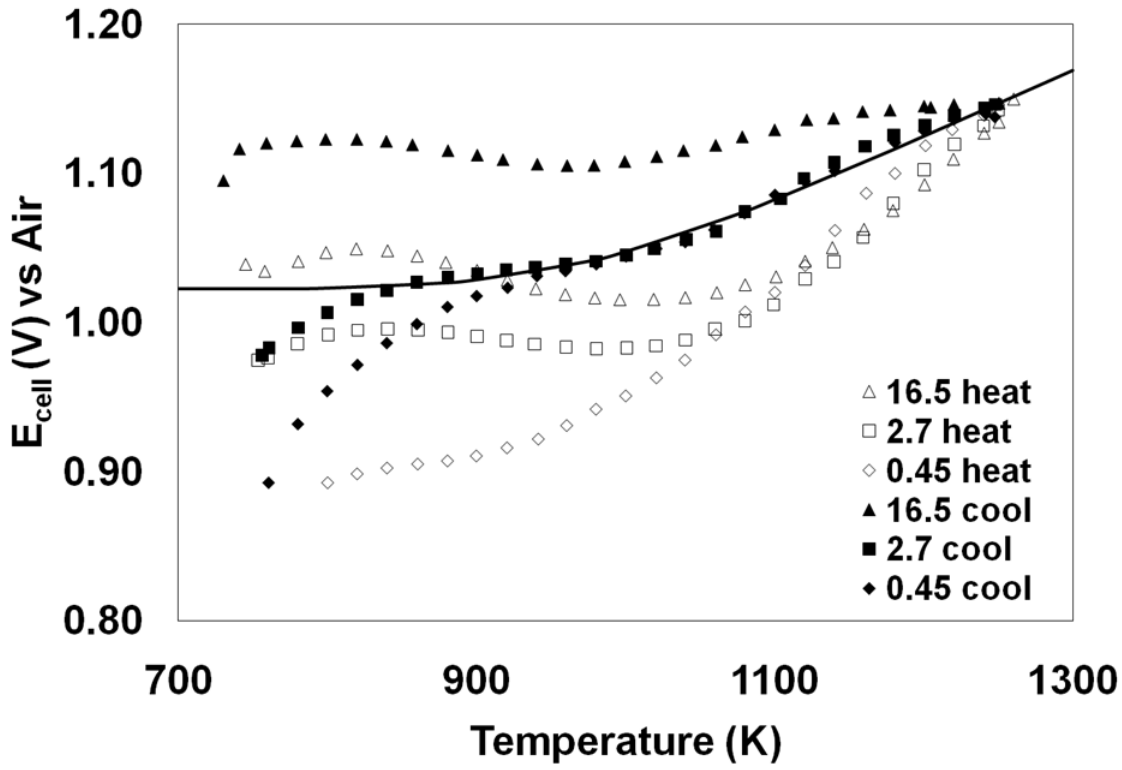


Figure 2.17: Experimental OCV values for the C/H/O-air cell during heating and cooling ramps, as compared to calculations. The heating (open symbols) and cooling (closed symbols) rates are given in the figure legend in units of $K \cdot min^{-1}$.

The transient traces of OCV data are shown in Figure 2.18 for the C-O system under isothermal conditions. Measurements were taken immediately after purging an equimolar mixture of CO/CO_2 through a carbon bed, thus causing the OCV to tend towards the standard potential. The flow rate was then stopped abruptly, and the OCV was immediately recorded as a function of time at isothermal conditions (1196 K and 1224 K). As expected, the cell potentials tend to their equilibrium values in the presence of excess carbon, with the higher temperature case exhibiting a faster approach. This result also signifies that given sufficient time, the gases within the

carbon bed will tend to equilibrium under open circuit conditions in the absence of gas inflow. Introduction of CO_2, for instance, into the bed would result in a steady state OCV value below the equilibrium potential, dependent upon the rates of gas inflow and chemical reaction within the bed.

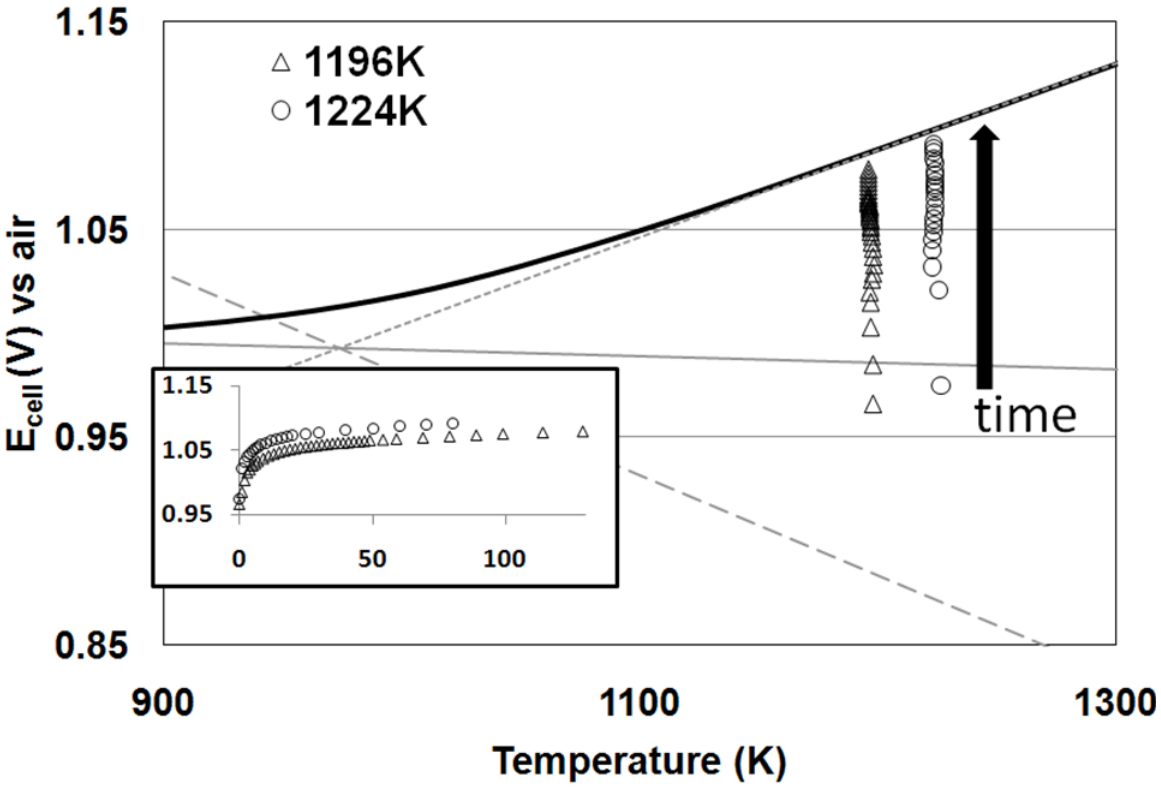


Figure 2.18: Transient OCV traces recorded at 1196 and 1224 K for the C-O system, and (inset) asymptotic approach to equilibrium OCV in time (minutes).

The analysis presented here demonstrates that the Boudouard reaction governs the electrochemical potential of the solid state DCFC, with the concentrations of gas phase CO and CO_2 being the critical parameters. The temperature dependence of OCV measurements match with theory, for systems relaxing towards Boudouard equilibrium and systems whose composition is fixed. The inclusion of hydrogen produces similar results, which agree with theoretical values for steam gasification. Further

analysis is based on this result, that the gasification process and products drive performance regardless of whether the solid fuel is in contact with the anode or located upstream.

2.4 Simple System Analysis

The previous discussion focused on the fundamental thermodynamic potentials for the DCFC, with experimental validation of these potentials. Practical operating constraints and irreversibilities will necessarily limit the efficiency of the DCFC plant, and thus should be included in evaluating potential performance. The DCFC, IDG and ATG plants will be revisited here under specified operating constraints, in order to compare performance. Only total currents will be used, leaving the current density and power density unspecified here as they tend to be hardware/geometry specific, and full-scale design is not addressed in detail here.

2.4.1 Thermodynamic System Selection

The DCFC plant includes gasification and work extraction steps that are mechanically and physically coincident. The gasification process is essentially based on converting solid carbon to gaseous CO by oxygen addition. When the oxygen is added via oxide conducting electrolyte, work and heat are extracted, in contrast to the completely irreversible introduction of gas phase oxygen into the gasifier. This subtle difference allows for higher work outputs as the solid fuel is further converted to CO_2 and H_2O. The DCFC gasifier system does not necessarily satisfy the thermal neutral constraint, and requires a heat interaction. The fuel utilization, air utilization, operating voltage, temperature and total pressure are specified in this model. All internal irreversibilities, or overpotentials, of the fuel cell are included by specifying an operating voltage less than the reversible cell potential. No ASU is required since the solid state electrolyte is effectively impermeable to N_2. The air introduced into the SOFC is preheated via counter-flow heat exchanger (HX) with a specified effectiveness ϵ. Since the fuel bed reactions are endothermic and the fuel cell assembly rejects energy as

heat, there will exist an internal, adverse temperature gradient ($T_{SOFC} > T_{gas}$), creating a similar practical disadvantage as the indirect gasification method. The DCFC is shown schematically in Figure 2.19, without a specified geometry but with the anode in direct contact with the fuel bed. The variables $\dot{Q}_{out,SOFC}$ and $\dot{W}_{out}$ refer to the

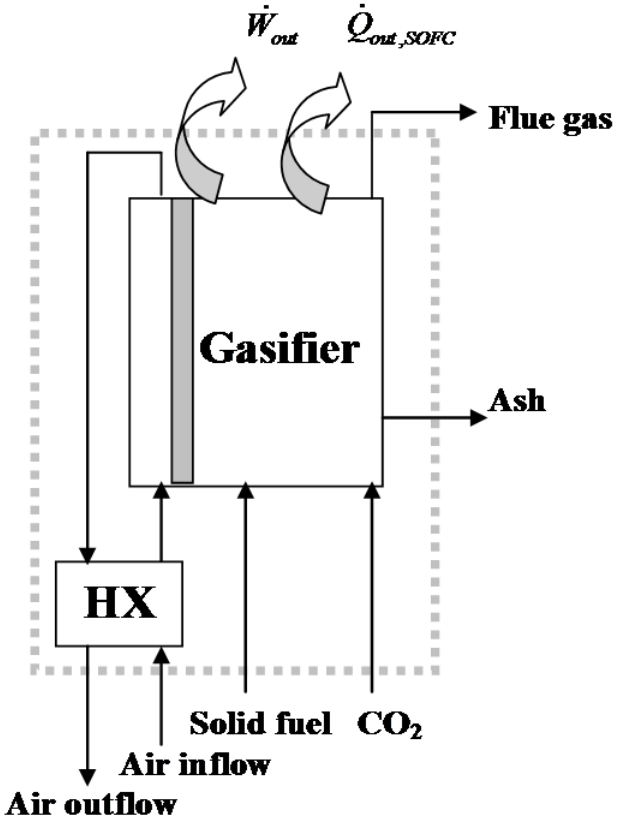


Figure 2.19: Schematic of the constrained DCFC plant. The regions denoted by gray dotted lines reference the control volume in Figure 2.4.

net heat and work outputs of the SOFC. The DCFC system is constrained such that there is not a unique value of CO_2 inflow, so this stream is arbitrary. The products leaving the SOFC are termed flue gas (similarly for each system), with an associated heating value and work potential. Results are presented without CO_2 addition for simplicity, while addition of CO_2 does not change the work output or efficiency, only the flue gas composition and heating value. Note that some gas flow to the gasifier

is generally desired for mechanical agitation to segregate ash, fluidization to enhance mixing, and/or entrained fuel delivery. The DCFC arrangement shown in Figure 2.19 is a further constrained example of the system shown in Figure 2.4, with associated internal irreversibilities.

The second system examined is similar to the DCFC plant, except that the gasifier is physically separate from the SOFC (see Figure 2.20). Sensible energy is transferred as heat into the gasifier to drive the endothermic reactions, termed indirect gasification (IDG). The required modifier input, CO_2 or H_2O here, is uniquely determined such that no solid carbon is present in the gasifier effluent. The syngas leaving the gasifier is converted via SOFC with specified fuel utilization, air utilization and operating voltage. Similar indirect gasification systems, operating at 1173 K, have been analyzed with an SOFC as the work producer [63, 64]. In that study, heat pipes were used to transfer heat into the endothermic gasifier from combustion of syngas. One concern with the type of arrangement considered here is the source for heat transfer into the gasifier. Reaction of the flue gas with oxygen could supply some heat using burner tubes within the carbon bed. The fuel cell heat rejection could plausibly supply some heat, although this opposes the practical advantage of conducting gasification at higher temperatures (faster reaction rates) and fuel cell operation at lower temperatures (sealing and materials concerns). Again, the variables $\dot{Q}_{out,SOFC}$ and $\dot{W}_{out}$ refer to the net heat and work outputs of the SOFC, while $\dot{Q}_{in,gas}$ refers to energy transfer to the gasifier as heat at 1173 K. The IDG arrangement shown in Figure 2.20 is a further constrained example of the system shown in Figure 2.5, as denoted by the dotted regions numbered I and II.

The last plant considered here, the autothermal gasification plant (ATG), utilizes O_2 and H_2O, or O_2 and CO_2 to gasify solid fuel. The gasifier is constrained such that no solid carbon is present in equilibrium and the gasifier is adiabatic, operating at 1173 K (see Figure 2.21). The scenario employing oxygen and steam is similar to the FutureGen process [80, 81]. Similar systems have been investigated for coal and biomass utilization with an SOFC as the work producer [82, 83]. Note that if carbon dioxide were replaced with water for gasification, results differ by less than 1% in terms of work output, efficiency and oxygen requirements, although gas composition

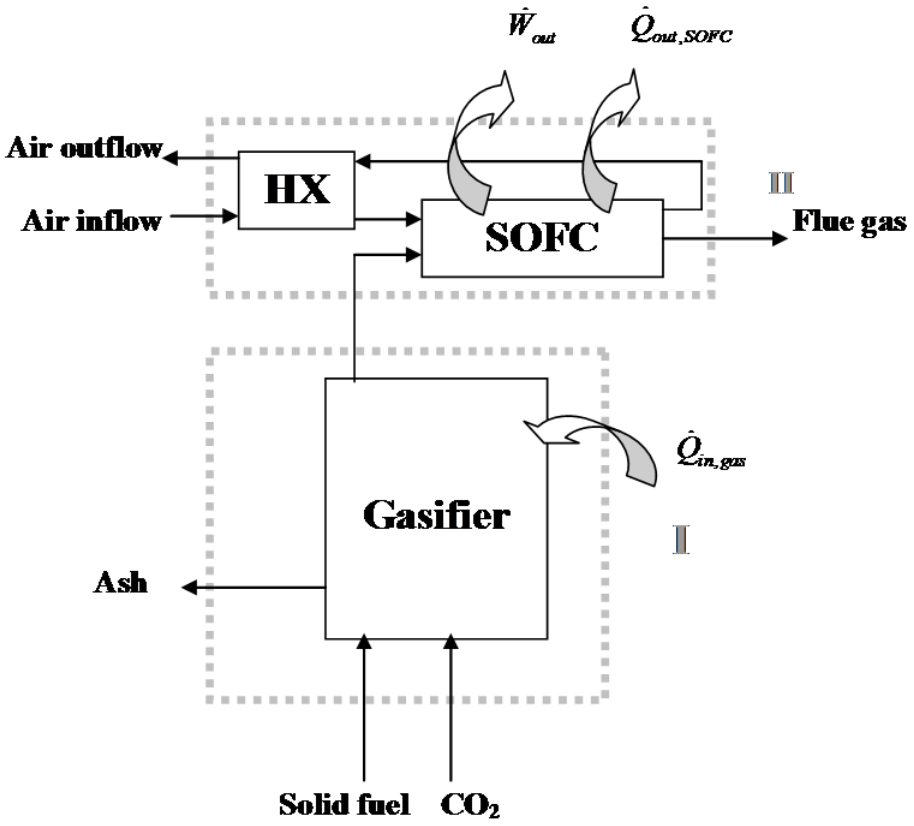


Figure 2.20: Schematic of the constrained IDG plant. The regions denoted by gray dotted lines reference the respective control volumes in Figure 2.5.

is altered. The syngas is again converted via SOFC with fuel utilization, air utilization and operating voltage specified. An ASU is included in the analysis since the gasifier

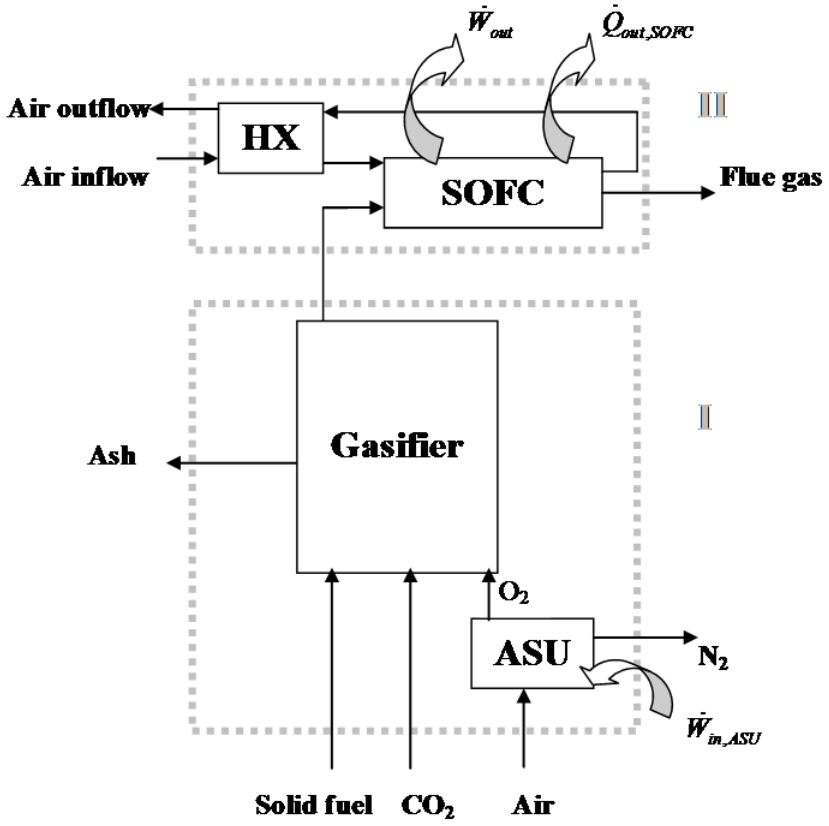


Figure 2.21: Schematic of the constrained ATG plant. The regions denoted by gray dotted lines reference the respective control volumes in Figure 2.6.

is oxy-fired, to prevent dilution of effluent CO_2 by N_2. The work requirement for the ASU is specified as 200 kWh per tonne O_2 (or 720 $kJ \cdot kg_{O2}^{-1}$), consistent with values reported for cryogenic plants delivering 95% pure oxygen at low pressure [84]. The variables $\dot{Q}_{out,SOFC}$ and $\dot{W}_{out}$ refer to the net heat and work outputs of the SOFC, respectively. The gasifier control volume could contain any type of gasifier, including fixed bed, circulating fluidized beds, or entrained flow systems. The ATG arrangement shown in Figure 2.21 is a further constrained example of the system shown in Figure 2.6, as denoted by the dotted regions numbered I and II. Autothermal gasification is

the standard gasification plant employed currently.

A range of solid fuels were investigated for each system as shown in Table 1: solid graphitic carbon, several coals, several biomass fuels, and municipal solid waste (MSW). Graphitic carbon is included as a reference solid. The coals selected include Illinois #6 (Ill6) from Macoupin County in Illinois and Wyodak (WYO) from the Powder River Basin in Wyoming, both of which are included in the Penn State Coal Sample Bank and Database [72] as DECS-24 and DECS-26, respectively. Illinois #6 coal has high-sulfur content while Wyodak coal, on the other hand, has relatively low sulfur content but a large amount of moisture. Switch grass (SG) and almond shells (AS) were selected as representative biomass fuels, although their compositions are similar [73]. The MSW, with a high ash and chlorine content, is included as a representative waste fuel [73]. Except for carbon and switch grass, the data presented in Tables 2.1 and 2.2 for the various fuels are on an as-received basis. Significant moisture can be present in the solid fuels, such as Wyodak coal or MSW, which can alter the amount and composition of gases required for gasification. The drop in heating value between as-received and dry fuels is due to the dilatory effect of water, reducing the H_2O/CO_2 requirement for complete gasification. Heating values listed are in units of $MJ \cdot kg^{-1}$.

2.4.2 Solution Methods

Chemical reactions are assumed to be in equilibrium for simplicity during the following analysis. Equilibrium is used as a reference here, and results will be discussed in the context of this assumption. This simplification allows a unique composition to be determined and used for analysis. Finite rate chemical kinetics slow the relaxation to equilibrium in reactors, especially at lower temperatures or residence times. Equilibrium calculations are performed using the element potential method, as described by Reynolds [74]. Species considered in this study for the equilibrium calculations include C(s), CO, CO_2, CH_4, H_2, H_2O(g), COS, H_2S, SO_2, N_2, NH_3, HCl, Cl_2, and O_2. The element potential method is a mathematical tool used to minimize a thermodynamic variable under atom-balance constraints. The Gibbs function is used here

as the reactors considered operate at specified T and P. The general formulation is given in Equation 2.19, where λ_j represents the element potential of element j, and A_{ij} represents the number of j atoms in species i.

$$g_i(T, P, x_i) = \sum_{j=1}^{N_e} A_{ij}\lambda_j \tag{2.19}$$

This expression can be rearranged for an ideal gas species (Equation 2.20), or solid species (Equation 2.21).

$$g_i(T, P) + RTlnx_i = \sum_{j=1}^{N_e} A_{ij}\lambda_j \tag{2.20}$$

$$g_i(T) = \sum_{j=1}^{N_e} A_{ij}\lambda_j \tag{2.21}$$

The Gibbs function at T and P is a constant for given conditions. The solution method used here consists of the following steps, readily applicable using simple spreadsheets;

- Assume major products, and determine major products mole numbers
- Calculate the element potentials based on the mole numbers
- Using the element potential values, calculate the mole numbers of minor species
- Perform atom balances and redetermine the major products mole numbers
- Repeat calculation until the mole numbers stop changing

This algorithm is used for all equilibrium calculations, and was crosschecked with STANJAN for verification.

Energy balances, given by Equation 2.22, are performed on the gasifier and/or SOFC control volumes in order to relate heat ($\dot{Q}$), work ($\dot{W}$), and flow enthalpy (h_i) interactions. The first term in Equation 2.22 is zero for the ATG and DCFC plants, while the remaining three terms represent the stream enthalpies of the reactants,

products, and ash.

$$\sum_{r=1}^{N_r} \dot{N}_r h_r + \sum_{q=1}^{N_q} \left(\dot{Q}_{q,in} - \dot{Q}_{q,out} \right) = \sum_{p=1}^{N_p} \dot{N}_p h_p + \sum_{w=1}^{N_w} \left(\dot{W}_{w,out} - \dot{W}_{w,in} \right) \tag{2.22}$$

There is no work interaction within the IDG and ATG gasification control volumes. The heat interaction for the IDG gasifier is nonzero, while the ATG gasifier has no heat interaction. The reactants for both gasifiers include the solid fuel and gasification modifier(s), while the products include both gas phase species and ash species. The ash leaving the gasifier is not used for heat transfer in this study and is included for the gasifier energy balance only.

The work outputs from the fuel cells are given by the product of cell operating voltage (E_{cell}) and current (I) as shown in Equation 2.23. The current is directly proportional to the oxygen flux (see Equation 2.24) associated with a fuel utilization (U_f) of 85%. The fuel utilization of the DCFC plant is defined here as the fraction of the supplied oxygen to the total oxygen requirement for full conversion of the carbon and hydrogen in the solid fuel stream to products. The number of oxygen ions that traverse the electrolyte is given by Equation 2.25 for the DCFC plant, and Equation 2.26 for the IDG and ATG plants. The difference in these two equations arises from the different fuel streams, solid as opposed to syngas, that the SOFC experiences. Note that CH_4 is a minor equilibrium product at 1 atm, and is thus not included in the utilization calculation. The ATG plant includes an ASU, requiring a work input proportional to the oxygen flux that must be subtracted from the fuel cell work output.

$$\dot{W}_{SOFC,out} = E * I \tag{2.23}$$

$$I = 2F * \dot{N}_{O,SOFC} \tag{2.24}$$

$$\dot{N}_{O,SOFC} = U_f \left(2\dot{N}_C + \frac{1}{2}\dot{N}_H - \dot{N}_O \right)_{solidfuel} \tag{2.25}$$

$$\dot{N}_{O,SOFC} = U_f \left(\dot{N}_{CO} + \dot{N}_{H2} \right)_{SOFC,in} \tag{2.26}$$

The work output is linearly proportional to the SOFC operating voltage, although there are practical operating ranges over which the voltage can vary. The heat interaction of the SOFC is determined via Equation 2.22, taking into account the work output, reactant input enthalpy and product output enthalpy (including the air side).

The air entering the SOFC is preheated to T_{out} in a counter-flow heat exchanger with an effectiveness (ϵ) of 0.86. In this analysis, the hot stream always has the lower heat capacity ($\dot{m}_{total}C_{P,H}$) due to the deficit of oxygen. The cold side air outlet temperature is determined by solving Equations 2.27 and 2.28, where $\dot{H}_i$ represents the enthalpy of stream i at temperature T, while the subscripts H and C refer to the hot and cold sides of the heat exchanger, respectively [85]. Shown in Equation 2.29 is the expression for the air mass flow rate provided to the SOFC with a utilization U_a defined as the fraction of the oxygen atoms traversing the electrolyte to the total oxygen contained in the air inflow.

$$\dot{H}_H(T_{SOFC}) - \dot{H}_H(T_{H,out}) = \dot{H}_C(T_{out}) - \dot{H}_C(T_0) \tag{2.27}$$

$$\epsilon = \frac{\dot{H}_H(T_{SOFC}) - \dot{H}_H(T_{H,out})}{\dot{H}_H(T_{SOFC}) - \dot{H}_H(T_0)} \tag{2.28}$$

$$\dot{m}_{air} = \left[\frac{MW_{O2} + 3.76MW_{N2}}{U_a}\right]\frac{I}{4F} \tag{2.29}$$

Stream exergies are calculated using Equation 2.30, with the subscript i indicating the steam and subscript *0* indicating the environmental (restricted) dead state [79]. Mixing of the products with the environment is not considered in this analysis. The exergy of the solid fuel stream is estimated as the heating value [70, 71], and a detailed calculation confirms agreement well within 5% for all fuels assuming that the entropy of the solid is negligible compared to that of the gas streams. The exergies of the flue gas and air exhaust streams, denoted by $\dot{\Psi}_j$, are calculated relative to the products reacted completely to a mixutre of SO_2, N_2, Cl_2 and H_2O(L) at 298 K and 1 atm total pressure, and CO_2 exiting as a separate, capture ready stream also at 298 K and 1 atm total pressure. The net heat output ($\dot{Q}_{out,net}$) is equal to the heat rejection of the SOFC minus the energy input requirement for gasification, as shown in Equation

2.31.

$$\dot{\Psi}_j = \sum_{i=1}^{N_r} \dot{N}_i \left((h_i - T_0 s_i)_j - (h_i - T_0 s_i)_0 \right) \tag{2.30}$$

$$\dot{Q}_{out,net} = \dot{Q}_{out,SOFC} - \dot{Q}_{in,gas} \tag{2.31}$$

In order to compare different systems, operating on a variety of fuels, several non-dimensional parameters will be formulated. The system efficiency is given by Equation 2.32 as the ratio of net work output to the heating value of the fuel. The heat rejection (at 1173 K) from the fuel cell, along with the heating value of the flue gas stream can be used in a bottoming process, such as a Rankine cycle, for additional work output. The heating values of the flue gas species contain both the sensible enthalpy change and chemical enthalpy change of complete oxidation to products at 298 K. Assuming the bottoming cycle has a known thermal efficiency (η_B), the total system efficiency can be written as shown in Equation 2.33.

$$\eta' = \frac{\dot{W}_{out,net}}{\dot{m}_{fuel} HHV_{fuel}} \tag{2.32}$$

$$\eta'' = \frac{\dot{W}_{out,net} + \eta_B \left(\dot{Q}_{out,net} + \dot{HV}_{flue} \right)}{\dot{m}_{fuel} HHV_{fuel}} \tag{2.33}$$

The preceding analysis is completed for all three systems for various fuels. The solid fuel input is fixed to 1 $kg \cdot s^{-1}$, thus providing a common basis for comparison amongst systems and fuels. The parameters given in Table 2.3 are used in the thermodynamic model, and represent practical operating conditions [80, 81, 83, 82, 86, 87]. Under the constraints employed, operation at higher pressures does not appreciably affect the system thermodynamic results. The benefits of high pressure operation are primarily in enhancing gasification kinetics and thus reducing reactor size, which is not considered here.

Table 2.3: Operating parameters used for thermodynamic modeling of the fuel cell plants.

Parameter	Value
ϵ	0.86
U_a	0.20
U_f	0.85
E_{cell}	0.70 V
T_{SOFC}	1173 K
T_{gas}	1173 K
P_{SOFC}	1 atm
P_{gas}	1 atm
η_B	0.30
$w_{in,ASU}$	720 $kJ \cdot kg_{O2}^{-1}$

2.4.3 Thermodynamic DCFC Results

Results for the DCFC, IDG, and ATG plants are presented in Tables 2.4 through 2.6 on a basis of 1 $kg{\cdot}s^{-1}$ solid fuel input. The DCFC plant was modeled using fuel data on both the as-received and dry basis for comparison. The raw fuels exhibit lower work outputs than dried fuels, but the thermal efficiencies remain similar to that of raw fuels due to the corresponding drop in HHV. The DCFC is not constrained to require a gasifying agent such as CO_2 or H_2O; the addition of a gasifying agent does not change the work output or efficiency. However, the addition of CO_2 or H_2O alters the energy transfer as heat, along with the composition of the flue gas. Dried solid fuels were used in the IDG (see Table 2.5) and ATG (see Table 2.6) plants, with both CO_2 and H_2O included as gasifying agents, although there is little variation in the work output or efficiency between the agents. The thermal efficiencies of the standalone IDG and DCFC plants operating on carbon, coal or biomass are similar ranging from 52 - 58% (60 - 65% with bottoming cycle), as compared to 33 - 35% (49 - 50% with bottoming cycle) for the ATG plant. Even with the inclusion of a bottoming cycle, the ATG plant exhibits significantly lower efficiencies of around 50%.

Quantitatively, if the operating voltage were increased to between 0.75 and 0.80 V, the work output (and efficiency) of the standalone units would increase by 7 - 14%, respectively. The heat outputwould be reduced accordingly, such that the effect of

Table 2.4: DCFC thermodynamic modeling results for both as received and dry fuels.

Fuel	$\dot{m}_{air}$	$\dot{W}_{out,net}$	$\dot{Q}_{out,net}$	HV_{flue}	η_I	η_{II}
C(s)	49.46	19.14	-2.94	10.27	0.58	0.65
Ill6	32.65	12.64	-1.11	8.84	0.52	0.61
Ill6 d	37.63	14.57	-0.69	9.57	0.52	0.61
Wyo	29.47	11.37	-2.71	8.39	0.53	0.63
Wyo d	39.95	15.43	-2.22	9.93	0.55	0.63
AS	24.11	9.30	-2.03	7.20	0.53	0.62
AS d	26.11	10.08	-1.82	7.45	0.53	0.62
MSW	7.79	3.01	-0.17	3.67	0.40	0.54
MSWd	12.63	4.89	2.34	3.36	0.40	0.54
SG d	25.52	9.89	-2.41	7.28	0.55	0.63

the bottoming cycle would be lessened or even unnecessary. Operation at a larger cell voltage would allow more efficient operation, but may require larger reactor length scales and/or cell active area for the same work output. The similar performance between the DCFC and IDG plants is due to similar oxygen requirements for both plants, thus each requires the same current at the constrained operating voltage. The small deviation in work output between the IDG and DCFC plants is due to the slight difference in the definition of fuel utilization between the two plants. As noted in Tables 2.4 through 2.6, different fuels can give greatly varying results in the DCFC and IDG plants. The lowest system efficiencies are consistently achieved with MSW, which has a large ash content and low heating value. Thermodynamically, pure carbon performs better than all of the heterogeneous fuels, while the coals and biomass fuels perform about equally. The reason for carbons superior performance in the DCFC and IDG plants is that pure carbon has no oxygen content, and thus requires the most oxygen input in the way of electrical current per 1 $kg \cdot s^{-1}$ of solid fuel. The ATG plant does not exhibit significant differences in performance for the different fuels, as oxygen content and requirement track with heating value.

Flue gas compositions calculated for the GD-DCFC exhaust are given in Table 2.7. The major products are CO, CO_2, H_2, and H_2O, as expected. The majority of sulfur is bound in H_2S at the conditions specified, with levels reaching as high as 2% of the flue

Table 2.5: IDG thermodynamic modeling results for dry fuels.

Fuel	$\dot{m}_{air}$	$\dot{Q}_{in,gas}$	$\dot{m}_{CO2}$	$\dot{m}_{H2O}$	$\dot{W}_{out,net}$	$\dot{Q}_{out,net}$	HV_{flue}	η_I	η_{II}
C(s)	48.54	19.15	3.87	0	19.14	-6.59	14.04	0.58	0.65
C(s)	47.67	18.97	0	1.53	18.76	-9.22	17.136	0.57	0.64
Ill6	36.29	11.82	2.29	0	14.34	-3.02	12.32	0.51	0.61
Ill6	35.67	11.65	0	0.90	14.04	-4.6	14.28	0.50	0.60
Wyo	39.15	13.71	2.26	0	15.37	-4.46	12.33	0.54	0.63
Wyo	38.22	13.52	0	0.89	15.04	-5.98	14.30	0.53	0.62
AS	25.42	8.32	0.80	0	10.00	-2.63	8.43	0.53	0.62
AS	25.42	8.42	0	0.32	9.87	-3.23	9.14	0.52	0.61
MSW	12.31	1.29	0.58	0	4.85	1.76	4.02	0.40	0.54
MSW	12.19	1.27	0	0.23	4.80	1.36	4.48	0.39	0.54
SG	25.01	8.88	0.85	0	9.81	-3.29	8.31	0.55	0.63
SG	24.59	8.79	0	0.34	9.68	-3.87	9.07	0.54	0.62

gas stream for Illinois #6 coal. Concentrations of a few hundreds of ppm have been shown to effect SOFC performance [88], and research is ongoing for sulfur resistant anode materials [89]. Chlorine exists mainly as HCl, with concentrations ranging from 3 to 6500 ppm. A study of HCl effects on SOFC performance indicated that as little as 20 ppm can cause degradation [90]. Cell performance was observed to recover, however, with removal of HCl from the syngas. Very little chlorine exists as Cl_2 in equilibrium, which has concentrations (not shown) about 10 orders of magnitude lower than HCl. Nitrogen is primarily bound in N_2, with ammonia concentrations several orders of magnitude lower.

The O_2 concentrations in the DCFC exhaust is sufficiently low to give OCV values above 0.79 V, as calculated using Equation 2.16. This implies that 0.7 V is an achievable operating potential. The flue gas exiting the fuel cell also contains fuel species, which can release further sensible enthalpy by reacting with oxygen, requiring an ASU to prevent N_2 dilution. The work requirement for this oxygen supply amounts to less than 2% of the SOFC work output, since the majority of the oxygen used to convert the fuel traverses the SOFC. The adiabatic combustion temperature from reaction of the hot flue gas with oxygen at 298 K ranges from over 1600 K for wet MSW to over 2000 K for coal and biomass. This sensible enthalpy could be used to

Table 2.6: ATG thermodynamic modeling results for dry fuels.

Fuel	$\dot{m}_{air}$	$\dot{m}_{O2}$	$\dot{m}_{CO2}$	$\dot{m}_{H2O}$	$\dot{W}_{out,net}$	$\dot{Q}_{out,net}$	HV_{flue}	η_I	η_{II}
C(s)	30.81	0.98	1.09	0	11.37	7.91	8.87	0.35	0.50
C(s)	30.57	0.98	0	0.44	11.38	7.23	9.58	0.35	0.50
Ill6	25.31	0.61	0.58	0	9.54	5.93	9.15	0.34	0.50
Ill6	25.31	0.61	0	0.23	9.49	5.49	9.63	0.34	0.50
Wyo	26.09	0.70	0.27	0	9.79	5.95	8.67	0.35	0.50
Wyo	26.11	0.70	0	0.11	9.77	5.70	8.90	0.35	0.50
AS	17.10	0.46	0	0	6.42	3.75	6.35	0.34	0.50
MSW	11.12	0.07	0.40	0	4.32	2.74	3.67	0.35	0.51
MSW	11.01	0.07	0	0.16	4.30	2.46	3.98	0.35	0.51
SG	16.04	0.49	0	0	5.98	3.52	6.11	0.33	0.49

contribute heat into the gasifier similar to recent studies [63, 64], along with some heat transfer from the fuel cell to the gasifier or gasifier inlet streams. The IDG and DCFC plants would benefit from this approach, although all three systems would require a similar gas cleaning step. Catalytically stabilized combustion could possibly be used for conversion of the flue gas containing large amounts of CO_2 and H_2O.

2.5 Gas Cleanup and Materials Issues

2.5.1 Freeboard Region

Generally, the region within a gasifier above the dense particle bed is termed the freeboard. In this freeboard section, the CO/CO_2 ratio would decrease as the anode is traversed under current flowing conditions, as opposed to an immersed anode. Further conversion of the effluent CO to CO_2, prior to exiting the gasifier, would require a freeboard section of the DCFC, without solid carbon. In the absence of carbon, the CO/CO_2 ratio will decrease monotonically as the freeboard section of the anode is traversed under current flow. The decrease in CO/CO_2 ratio will cause a decrease in theoretical cell potential along the anode length in the direction of gas flow. This may necessitate or encourage the use of a separate (and geometrically optimized) clean up SOFC, or burner section, rather than attempting to completely

Table 2.7: Calculated DCFC exhaust gas mole fractions (superscripts i denote x10^i) in equlibrium at 1173 K.

Fuel	CO^1	CO_2^1	H_2^1	H_2O^1	O_2^{15}	N_2^3	NH_3^7	H_2S^3	COS^4	HCl^4
C(s)	3.0	7.0			0.45					
Ill6	1.4	5.0	6.1	2.8	1.1	4.7	2.3	19	13	5.3
Ill6 d	1.7	5.3	5.3	2.2	0.83	5.2	2.0	20	20	5.1
Wyo	1.1	4.3	7.6	3.7	1.2	3.1	2.6	1.2	0.56	0.036
Wyo d	1.7	5.1	6.6	2.6	0.77	3.9	2.3	1.4	1.1	0.033
AS	1.0	4.4	6.9	3.8	1.5	3.5	2.4	0.13	0.059	0.40
AS d	1.2	4.7	6.7	3.4	1.3	3.7	2.3	0.12	0.067	0.43
MSW	0.41	3.5	5.1	5.5	5.9	5.9	1.9	3.6	0.91	31
MSW d	1.7	6.6	2.4	1.2	1.2	13	0.94	6.4	14	65
SG d	1.2	4.6	6.9	3.5	1.3	6.1	3.1	0.43	0.23	0.041

convert the gaseous products in the freeboard. This limitation manifests practically in fuel utilizations of less then 100% in the DCFC, as discussed earlier.

2.5.2 Fuel Impurities

Environmentally, sulfur emissions can form sulfuric acid (H_2SO_4), which contributes to acid rain. Sulfur compounds represent a large portion of the flue gas or syngas cleanup effort. Sulfur occurs in several forms in solid fuels, can form several product compounds. Solid fuels generally contain sulfur in several forms [14]; (i) organically bound sulfur that exists within the carbon matrix and elutes along with char burnout, (ii) sulfur bound in iron pyrites (FeS_2), and (iii) sulfur bound in metal sulfates (M$\cdot$ SO_4). Under oxidizing conditions, sulfur oxides such as SO_2 and SO_3 are formed in the products, while reducing conditions favor hydrogen sulfide (H_2S) production, and carbonyl sulfide (COS) production to a lesser extent. The capture strategies for sulfur vary depending upon which of these compounds the sulfur is bound within.

Conventional coal-fired plants dictate that fuel-bound sulfur is released into an oxidizing environment, giving primary products of SO_2 and SO_3. Since the flue gas is used to transfer heat to a work producing cycle, the cleanup step can potentially be included at several different stages. Fluidized bed combustors can utilize sulfur

sorbents directly in the particle bed, upstream of the boiler or superheater. Alternatively (or in addition), SO_2 scrubbers can be included downstream of the boiler and superheater, consisting of limestone ($CaCO_3$) slurry introduction. The SO_2 reacts with $CaCO_3$ to form calcium sulfate, which as a dihydrate is termed gypsum ($CaSO_4 \cdot 2H_2O$), with a variety of industrial uses. The use of limestone can typically achieve 98% SO_2 capture [91]. In order to apply this approach to the DCFC, cleanup would be limited to the post fuel cell fuel stream and hence the DCFC anode would need to be sulfur resistant.

Gasification processes produce a gaseous fuel mixture that is subsequently oxidized, thus sulfur capture can occur in a reducing atmosphere or an oxidizing atmosphere. The sulfur cleanup could occur after the syngas stream has been oxidized, where the previous discussion would apply. However, for fuel cell applications, generally the sulfur must be removed prior to contacting the anodic surface, necessitating removal or conversion of H_2S. Fluidized bed gasifiers can benefit from direct limestone addition ($>$ 98 % H_2S and COS removal [91]), and possible addition of metal oxides. Hot gas cleanup represents a suite of approaches generally employing metal oxides to chemically bind sulfur, with typical results demonstrating sulfur reduction down to 10 - 100 ppm [92, 93, 94]. These techniques typically occur at temperatures ranging from 773 - 1073 K, and entail chemical looping to regenerate the sorbent in O_2-containing environments.

Another active area of research encompasses studies into the effects of fuel impurities on fuel cell materials and the formulation of chemically resilient anodes. The effects of HCl and H_2S in coal syngas on SOFC performance was investigated to determine cell degradation [88, 90]. In addition to resilience, the possibility of SOFCs that use H_2S as a fuel, converting the sulfur to its elemental form or SO_2, have been investigated [95, 96, 97, 89]. The H_2S can be converted via the Claus process in which H_2S interacts with sub stoichiometric O_2 or air to form H_2O and elemental sulfur. Electrochemical conversion of up to 3000 ppm H_2S from fuel streams using the MCFC to remove sulfide ions from the stream has been investigated with promising results [98]. These studies demonstrate that the materials issues prevalent with sulfur-containing fuels can possibly be addressed by careful anode selection or in situ

capture by sorbents.

2.5.3 Other Concerns

Carbon dioxide capture and sequestration presents a contemporary engineering challenge, but is considered a necessary component of future environmental carbon mitigation [84]. Systems producing a capture-ready stream of CO_2, or unmixed CO_2, have an inherent advantage over systems requiring CO_2 separation from mixtures with significant diluent, gas phase species. The DCFC system produces a concentrated stream of CO_2, as the process O_2 is separated from N_2 in air through the SOFC electrolyte.

Carbon deposition is another major concern for systems employing hydrocarbon fuels directly. Streams of pure hydrocarbons have been shown to deposit carbon on anode surfaces [99]. The conditions within the fuel bed at the DCFC anode however, lie on the opposing side of the carbon deposition boundary if char is used. The CO/CO_2 mixture at the anode is limited to that prescribed by Boudouard equilibrium within the bed, with any oxygen addition from the anode favoring the forward progression of the Boudouard reaction. In addition, fuel preparation in terms of devolatilization may be desired to reduce tar formation and other condensables that could lead to deposits. Other fuel preparations of interest would be de-ashing or de-sulfuring of feedstocks to allow for optimum DCFC operation.

In addition to chemical and deposition resilience, the DCFC anode must also withstand abrasion from the particle bed. The schemes investigated in this study do not specify the gasifier type, although particle contact with the anode is foreseen. Physical separations, such as barrier layers or separate (indirect) gasification, could mitigate these effects and allow impurity removal.

The DCFC system does not require a large ASU in contrast to ATG-based systems. It was shown, however, that a smaller ASU may be desired to supply heat via utilizing flue gas. The presence of an ASU would also help to mitigate tar formation if O_2 were introduced with the solid feed. Introduction of O_2 to the ash prior to removal would also benefit the DCFC by mitigating the output of carbon with ash in addition to

the sensible energy this would provide. Ash, consisting primarily of metals and metal oxides, presents some chemical and mechanical challenges for gasification systems as fly ash must be separated from syngas if entrained. Metal and metal oxide particles could physically block the anode surface or chemically alter anode composition. The gasifiers considered here are generally below slagging temperatures, so ash would be present as solid particulate. Depending upon the gasifier details ash can be removed using current technologies such as cooling screws or lock hoppers.

Chapter 3

Carbon Gasification Chemistry

There are several global reactions that characterize carbon gasification, the process of converting solid, bound carbon atoms to free, carbon containing molecules existing in the gas phase such as CO, CO_2, CH_4, etc. Partial oxidation refers to reaction of solid carbon with oxygen to create a CO and/or H_2 rich stream. Steam gasification refers to reaction of solid carbon with H_2O to form a CO/H_2 rich stream. Gasification using CO_2, referred to here as the Boudouard reaction, refers to reaction of solid carbon to form CO. This last reaction is of concern to this study. The rate at which feedstock react with gas phase species are fuel specific, and can vary significantly between different solids. The structure of the solid particulate also influence gasification rates. The following analysis will include measurements of fuel reactivities, along with some discussion of structure, for both comparison and quantitative insight. The DCFC is dependent upon sufficiently reactive particles to sustain a practical current density.

3.1 Basic Physical Processes

The kinetic processes governing gas-solid interactions and surface kinetics dictate the rate at which the solid fuel is consumed, and the rate at which the gas species are consumed or created. Using the Langmuir-Hinshelwood approach, heterogeneous reaction chemistry involves three main processes; adsorption of gas phase species, surface reaction, and desorption of surface species. These surface reaction processes

must be incorporated into a transport equation for the gas species, both in the bulk gas phase and the gas phase within the porous particle structure. In order to isolate the kinetic rates, and mitigate transport limitations, reactions under kinetically-limited conditions will be considered.

The fuel particles considered here are porous and thus have larger internal surface area arising from the pore network. Shown in Figure 3.1 is an illustration of the different transport regions. In the particle bed, there exists a bulk convective flow and mass diffusion between the particles, with a length scale similar to the particles (of order 100 μm). Surrounding each reacting particle exists a concentration boundary layer through which species interchange between the bulk flow and the particle inner surfaces. Finally, within each particle gas species must undergo adsorption, subsequent surface reaction and desorption, while also convecting and diffusing through the porous structure with length scales of < 1 μm. The assumptions and development of the treatment used to address these effects is included in the following sections.

3.2 Literature Review and Previous Work

There have been extensive experimental and theoretical studies regarding the Boudouard reaction and the surface chemistry of carbons. A comprehensive review and overview was provided by Laurendeau in 1978 [100]. This study includes extensive comparisons amongst measurements regarding char gasification in CO_2 and H_2O, as well as char combustion in O_2. A brief summary of some relevant experimental work is included here.

In 1955, Ergun employed a 2-step mechanism to describe the gasification of graphite, activated graphite and activated carbon by CO_2 at 1 atm in the vast temperature range of 700 - 1400°C. The reaction scheme, given below includes the chemisorption of gaseous CO_2 onto an active carbon surface site, and subsequent desorption of an oxide complex as gaseous CO. The first reaction is reversible while the second reaction is considered not reversible, as carbon deposition was not observed experimentally. Note that the second reaction is not balanced, as the bulk carbon that is uncovered following CO desorption is not included on the left hand side. This

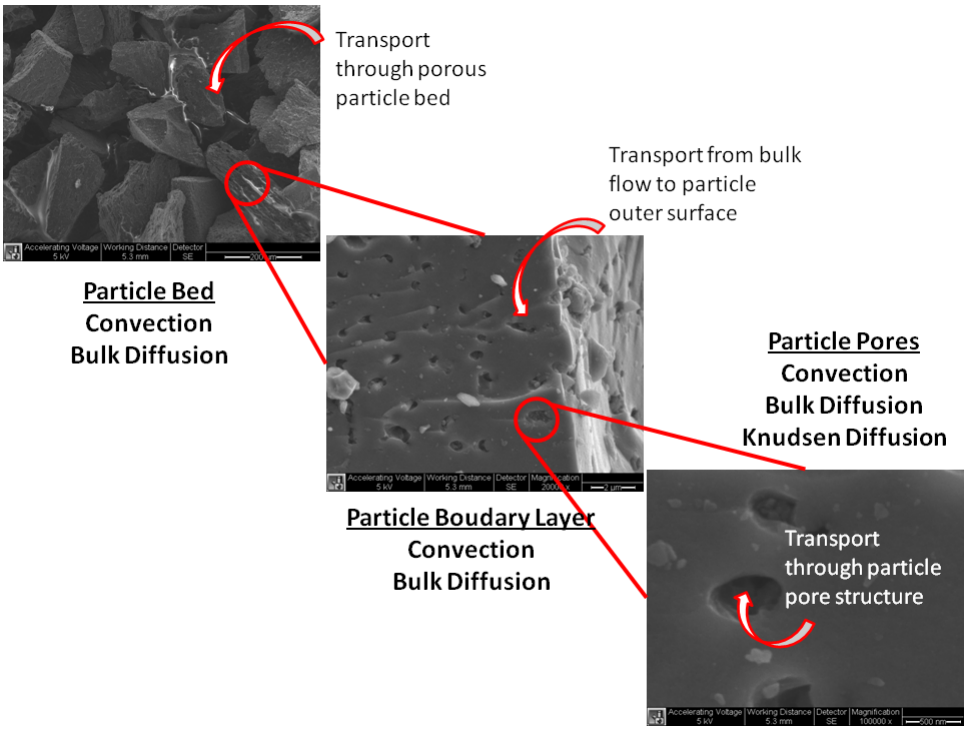


Figure 3.1: Illustration and summary of gas transport within a bed of porous particles. There exists a net mass flow radially outward due to gasification, thus producing a Stefan flow.

reaction imbalance is evident in several mechanisms reviewed here, but does not impact the reaction rate dependencies as the bulk carbon does not participate in the desorption.

$$\mathrm{CO_2(g) + C_f \longleftrightarrow CO(g) + C(O)} \tag{R 3.1}$$

$$\mathrm{C(O) \longrightarrow CO(g) + C_f} \tag{R 3.2}$$

A fluidized bed arrangement was used to determine exhaust gas compositions from which reaction rates were determined. The equilibrium constant for Reaction R3.1 (k_f/k_r) was determined to be independent of carbon fuel, varying only with temperature. The heat of reaction was determined to be 23 $kcal \cdot mol^{-1}$ (96 $kJ \cdot mol^{-1}$), which is equivalent to the difference in activation energy for the forward and reverse reactions [101]. The activation energy of the desorption reaction (Reaction R3.2) was determined to be 59 $kcal{\cdot}mol^{-1}$ (246 $kJ{\cdot}mol^{-1}$) for all three fuels, although the actual rate coefficients varied considerably [101]. The conversion of solid carbon limited to about 5% in this study, giving reactivities of carbons with extents of reaction near zero.

A study by Blackwood and Ingeme in 1959 examined the Boudouard reaction at pressures of up to 40 atm in the narrow temperature range of 790-870°C [102]. The experimental setup was similar to that of Ergun, using an isothermal bed with a volume of 17.6 mL, and a fuel prepared from coconut shell. The char in the reactor experienced less than 10% conversion, indicative of raw char. The results show significant inhibition from CO, and increases in reaction rates with pressure. A 5-step mechanism (Reactions R3.3 - R3.7) is used to describe the experimental results, specifically the effects of high pressure and high CO comcentrations.

$$\mathrm{CO_2(g) + C_f \longrightarrow CO(g) + C(O)} \tag{R 3.3}$$

$$\mathrm{CO(g) + C_f \longleftrightarrow C(CO)} \tag{R 3.4}$$

$$\mathrm{C_f + C(O) \longleftrightarrow CO(g)} \tag{R 3.5}$$

$$\mathrm{CO_2(g) + C(CO) \longleftrightarrow 2\,CO(g) + C(O)} \tag{R 3.6}$$

$$\mathrm{CO(g) + C(CO) \longleftrightarrow CO_2(g) + C_f} \qquad \text{(R 3.7)}$$

Although activation energies are presented, the temperature range is likely too small to accurately quantify the temperature dependencies. The reaction rate coefficient of Reaction R3.3 ranged from 8.7e-8 to 1.1e-6 $mol{\cdot}s^{-1}{\cdot}\,g^{-1}$ corresponding to 790°C and 870°C, respectively at 1 atm [102]. Other constants were also determined, involving algebraic groupings of the five reaction rate coefficients.

Two decades later, in 1976, Strange and Walker conducted thermogravimetric measurements to test the applicability of Langmuir-Hinshelwood kinetics (Reactions R3.8 - R3.10) to the Boudouard reaction [103]. The experimental conditions ranged from 900-1007°C, and 2-20 kPa, using pressed graphite as the carbon source. The fuel had an initial specific surface area of 1.8 $m^2{\cdot}g^{-1}$ and a reported porosity of 0.02-0.06. A 3-step mechanism was proposed to describe chemisorption and desorption, along with physical adsorption of gaseous CO.

$$\mathrm{CO_2(g) + C_f \longleftrightarrow CO(g) + C(O)} \qquad \text{(R 3.8)}$$

$$\mathrm{C(O) \longrightarrow CO(g) + C_f} \qquad \text{(R 3.9)}$$

$$\mathrm{CO(g) + C_f \longleftrightarrow C(CO)} \qquad \text{(R 3.10)}$$

The heat of reaction for Reaction R3.8 was found to be 25 $kcal{\cdot}mol^{-1}$ (105 $kJ{\cdot}mol^{-1}$), in general agreement to within 10% with that found by Ergun [101]. The reaction rate coefficients of Reaction R3.8 were found to be 20.5e-7, 3.2e-7 and 0.65e-7 $g_C{\cdot}s^{-1}{\cdot}\,m^3_{CO2}{\cdot}\,mol^{-1}_{CO2}\cdot g_0^{-1}$ at temperatures of 1007°C, 950°C, and 902°C, respectively [103].

Experiments published by Koenig, et al.. in 1984 [104], utilized the same Fisher activated carbon (FAC) employed in the present study, and thus provides a basis for comparison. A packed bed configuration was used to determine chemical kinetic rates in the temperature range of 724 - 821°C, at a pressure of 1 atm. This low temperature range allowed low CO_2 conversions to CO, and more accurate determination of reaction rates. The surface area of the char was found to be 1020 $m^2\cdot g^{-1}$ at 10% conversion. The same 3-step Languir-Hinshelwood mechanism used in Strange and Walker's study was employed, and found to accurately describe experimental data

for low to high CO concentrations near 1 atm. The experimental reaction rate determined at a temperature of $800°C$ was estimated to be 3.4e-5 s^{-1} (calculated as $-1/m \cdot dm/dt$ for a 20 mg sample) for 99% CO_2 [104].

Research at the High Temperature Gasdynamcis Laboratory at Stanford University has also focused on char reactivity in both oxidizing and CO_2 rich environments at pressures up to 25 atm. Tsai investigated the effects of high CO concentrations on char particle reactivities over a large range of pressures and CO mole fractions [105]. The presence of CO was found to significantly inhibit the Boudouard reaction. Research by Campbell [106], focusing on carbon oxidation, produced a detailed mechanism for char conversion in oxygen-containing environments. The model proposed includes reversible physisorption of gas species, chemisorption of physisorbed species, surface migration, and surface desorption with distributed activation energies. Ma further extended the concepts explored by Campbell to chars formed at high pressure, under both oxygen and carbon dioxide containing environments [107]. The results of these studies are applied here, particularly that of Ma, regarding the theoretical approach to modeling char gasification. The use of surface area measurements, temperature programmed desoprtion experiments, and isothermal reactivity measurements are based in part on the efforts of these researchers.

3.3 Theoretical Approach

The chemical reaction mechanism proposed and validated by Ma is employed, due to its applicability over various operating pressures and CO concentrations [107]. The mechanism is given in Table 3.1, with this numbering convention used throughout. Reaction R.1 represents the chemisorption of CO_2 on a free carbon site, and subsequent desorption of a gas phase CO molecule. The second reaction represents the desorption of an oxide surface complex to a gas phase CO molecule. Reactions R.3 and R.5 represent pathways resulting from surface complex interaction with gas phase species. Surface physisorption of CO is represented by Reaction R.4. The reaction rates resulting from the 5-step mechanism dictate production of gas phase species and surface species. The production rates of gas phase CO and CO_2, along with surface

Table 3.1: 5-step reaction mechanism used to model char reactivity to CO_2. Note that the the reactions here will be referred to as $R.i$ to distinguish from other chemical reactions (of form Rj,k) labeled in the text.

Reaction	Label
$CO_2 + C_f \leftrightarrow CO + C(O)$	R.1f , R.1r
$C_b + C(O) \rightarrow CO + C_f$	R.2
$C_b + CO_2 + C(O) \rightarrow 2CO + C(O)$	R.3
$C_f + CO \leftrightarrow C(CO)$	R.4f , R.4r
$CO + C(CO) \rightarrow CO_2 + 2C_f$	R.5

species C(O) and C(CO) are given by Equations 3.1-3.5, in unit of $moles \cdot m^{-2} \cdot s^{-1}$.

$$R_{CO} = k_{1f}[CO_2]S\theta_f - k_{1r}[CO]S\theta_O + k_2 S\theta_O + 2k_3[CO_2]S\theta_O + k_{4r}\theta_{CO} - k_{4f}[CO]S\theta_f - k_5[CO]S\theta_{CO} \quad (3.1)$$

$$R_{CO2} = -k_{1f}[CO_2]S\theta_f + k_{1r}[CO]S\theta_O - k_3[CO_2]S\theta_O + k_5[CO]S\theta_{CO} \quad (3.2)$$

$$R_{\theta_O} = k_{1f}[CO_2]S\theta_f - k_{1r}[CO]S\theta_O - k_2 S\theta_O \quad (3.3)$$

$$R_{\theta_{CO}} = k_{4f}[CO]S\theta_f - k_{4r}\theta_{CO} - k_5[CO]S\theta_{CO} \quad (3.4)$$

$$\theta_f = 1 - \theta_O - \theta_{CO} \quad (3.5)$$

The gas phase species source equations are on an active surface area basis, but transport equations for the gas phase are not referenced to the active surface. The parameter consisting of active surface area per geometric volume ($S_G \cdot \rho_{app}$) converts the surface-referenced reaction rates to volume-referenced terms suitable for gas transport governing relations. The reaction rates forming surface species must be recast in order to determine the time variation of the surface species, due to the changing surface area available for coverage. As the extent of conversion (x_c defined by Equation 3.6) of a particle proceeds, the active surface area changes thus requiring the surface concentrations to respond. Equation 3.7) presents a differential expression for surface species coverage fraction, rearranged to a form suitable for time integration. The first term on the right hand side represents changes due to chemical reaction, while the

second term represents that due to surface area changes.

$$x_c = 1 - \frac{m_c}{m_{c,0}} \tag{3.6}$$

$$\begin{aligned} \frac{d(S\theta_i S_{gC} M_C)}{dt} &= R_i S_{gC} M_C \\ S_{gC} M_C \frac{d(\theta_i)}{dt} + \theta_i \frac{d(S_{gC} M_C)}{dt} &= \frac{S_{gC} M_C}{S} R_i \\ \frac{d(\theta_i)}{dt} &= \frac{R_i}{S} - \frac{\theta_i}{S_{gC} M_C} \frac{d(S_{gC} M_C)}{dt} \end{aligned} \tag{3.7}$$

The PTGA measures the total mass of sample particles as a function of time, which dictates that the only directly measurable quantities from PTGA analysis are the total particle mass and the time derivative of the total mass. The rate of mass change of a particle includes the effects of char conversion, adsorption and desorption. Equations 3.8 and 3.9 display the expression for char mass loss and particle mass loss, with mass leaving the particle taken as positive.

$$\frac{dM_C}{dt} = (S_{gC} M_C MW_C) \cdot (k_2 S\theta_o + k_3 [CO_2] S\theta_o - k_5 [CO] S\theta_{co}) \tag{3.8}$$

$$\begin{aligned} \frac{dM_P}{dt} &= (S_{gC} M_C) \cdot (16 k_{1r} [CO] S\theta_o - 16 k_{1f} [CO_2] S\theta_f \\ &\quad + 28 k_2 S\theta_o + 12 k_3 [CO_2] S\theta_o \\ &\quad + 28 k_{4r} S\theta_{co} - 28 k_{4f} [CO] S\theta_f + 16 k_5 [CO] S\theta_{co}) \end{aligned} \tag{3.9}$$

The active surface area of porous particles changes as the particles react and pores are opened. A random pore model proposed by Bhatia and Perlmutter [108], accounted for changes in pore size and overlapping of widening pores. The functional dependence of mass specific surface area on extent of conversion is given in Equation 3.10.

$$S_{gC}(x_c) = S_{gC0} \sqrt{1 - \psi ln(1 - x_c)} \tag{3.10}$$

There are two parameters in this formulation, the initial surface area S_{gC0} and the structure parameter ψ. The initial surface area is measured and thus specified, and the structural parameter can be determined from comparison with measurements at various extents of conversion.

3.4 Experimental Approach

The theoretical framework described in the previous section is used along with experimental data to specify the reaction rate parameters for the Boudouard reaction for specific carbon fuels. Surface area measurements, reactivity thermogravimetry, and temperature programmed desorption ramps are obtained experimentally.

3.4.1 Thermogravimetric Analysis of Reactivity

A Cahn model 151 pressurized, thermogravimetric analyzer (PTGA) is used to accurately measure sample weight as a function of time in a prescribed environment. A schematic of the PTGA is given in Figure 3.2, showing the reaction gas, furnace gas, and purge gas inlets and outlets along with the sample pan, balance, furnace, and thermocouple. A detailed description of the PTGA, including the flow control and data acquisition (performed using Labview software), is provided in several previous works [105, 107, 106]. The PTGA is capable of maintaining test temperatures up to 1373 K, and pressure up to 100 atm. Solid samples are placed on a quartz sample pan, suspended from the microbalance by a thin Nichrome wire. The sample pan is located within a vertical, 5 cm diameter quartz tube, surrounded by a wire-wound ceramic heater. This arrangement thus provides a chemically inert reaction vessel (quartz) capable of high temperature operation. A K-type thermocouple probe is inserted just below the sample pan to measure reactor temperature. The reaction gas streams, microbalance purge stream, and furnace purge stream are controlled with thermal mass flow controllers. Reactive environments are prescribed using input method files (included in Appendix B), and the sample mass is recorded as a function of time, at 5 second intervals. Generally, isothermal conditions, in the kinetically limited

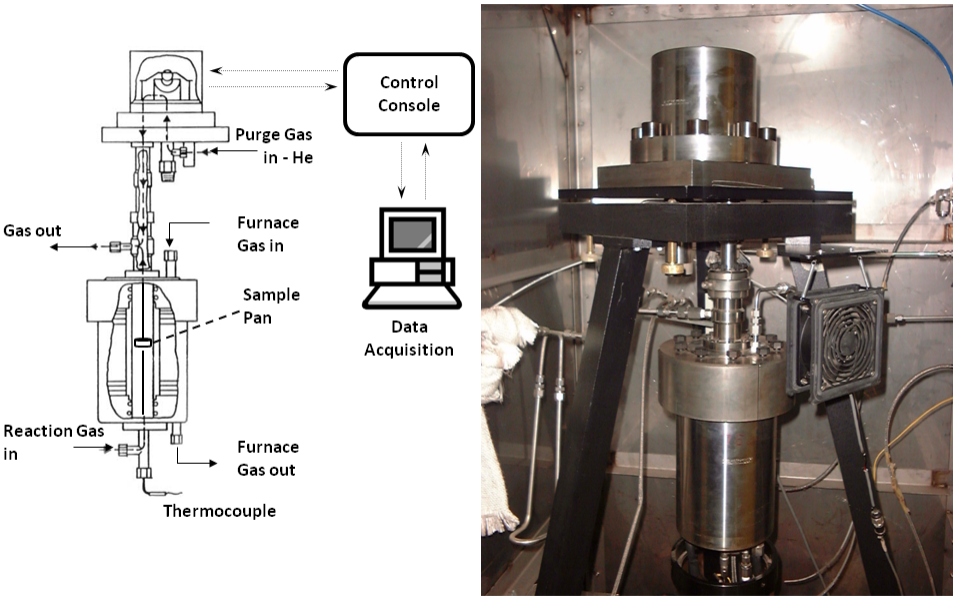


Figure 3.2: Schematic and digital photograph of the PTGA used in this study to determine particle surface area and reactivity.

regime, are used to measure the rate of particle mass loss. Temperature programmed desorption (TPD) experiments can also be conducted with the PTGA to determine the desorption rate of surface complexes to stable gas phase species. The variation in char surface character creates variation in the chemical properties of oxide complexes [109, 110], thus a distribution of desorption activation energy is used. In order to resolve the desorption properties of the surface oxides, C(O) namely, the oxide complex is accumulated onto the active surface of the particles. Reaction in pure CO_2 until 5 - 20% particle conversion is completed, and then followed by a cooldown temperature ramp in pure CO_2 to prevent premature oxide desorption. Once cooled to room temperature (298 K), the gas environment is purged with N_2 to remove all non-adsorbed oxygen species from the reaction chamber. A controlled heating ramp is then initiated (the TPD process), and the CO mole fraction in the effluent stream is measured in time.

3.4.2 Thermogravimetric Analysis of Surface Area

The surface area of a carbonaceous solid includes the large interior surfaces of the porous structure. The total surface area is estimated using the theoretical development of Brunauer, Emmett and Teller (BET) [111]. The BET theory essentially applies Langmuir adsorption theory to multiple layers of adsorbing species from the gas phase onto an active surface. While this theory was developed using kinetic considerations of adsorption processes, the same result can be deduced from statistical thermodynamics of a solid surface under specific conditions [112]. The resulting formulation, relating the volume of adsorbed gas (ν) to the partial pressure (p) and saturation pressure (p_0) of the adsorbing gas, is given by Equation 3.11.

$$\frac{1}{\nu}\frac{p}{(p_0 - p)} = \frac{1}{\nu_m c} + \frac{(c-1)}{\nu_m c}\frac{p}{p_0} \tag{3.11}$$

The quantity ν_m represents the volume of adsorbed gas in a single layer, completely covering the active surface. Assuming that the cross section area of the adsorbed gas is known (0.222 nm^2 for CO_2 [113]), the surface area can be deduced from ν_m. The PTGA is used to conduct BET analysis of solids to determine surface area. Various partial pressures of CO_2, in a N_2 balance, are introduced to the sample pan at 298 K and a total pressure of 10 atm, and the weight change is recorded. The background measurement allows differences in weight due to bouyancy to be separated from the effects of adsorption. A characteristic surface area measurement and background measurement are included in Appendix A.

3.4.3 Particle Sizing

The size distributions of particles and pulverized fuels is determined using a Coulter Multisizer. The basis of operation of the multisizer is a temporal measurement of ionic conductivity of an electrolyte containing the particles of interest. Two electrodes, separated by a pre-selected orifice, measure the ionic conductivity, and as particles are drawn through this orifice, through suction created by a diaphragm pump, the ionic conductivity of the solution at the orifice changes. The change in ionic conductivity

is explicitly related to the length scale of the particle cross-section in the orifice. The electrolyte used in this study consists of 20 % glycerol in balance H_2O, with 1 % wt/vol NaCl (1 g per 100 mL solution) added to increase ionic conductivity. The orifice size used in this study is 280 μm, consistent with nominal particle sizes of 100 μm. The size distributions extracted from the Coulter Multisizer are corroborated using SEM imaging.

3.5 Fuels Considered

Several fuels were used to determine char reactivity to CO_2. Several SEM images are provided for various fuels, showing the structure and size of both raw and processed particles. In addition to SEM images, size distributions are included for the various particles for reference. Table 3.2 contains ultimate analysis of various fuels used in DCFC studies and PTGA analysis. The fuels include two different industrial carbons MCMB and Fisher Scientific activated carbon (FAC), Lower Kittanning coal (LK), Wyodak coal (WYO), and also biomass solids including almond shells (AS) and rice straw (RS). The rice straw is shown for comparative purposes only. Ultimate analyses for MCMB, FAC, LK, LKFC, AS, and RS were performed by Hazen Research, Inc. (of Golden, CO), while that for WYO was taken from the Penn State University Coal Sample Bank and Databasae [72]. Carbon black spherical particles, termed MCMB

Table 3.2: Ultimate analysis (wt%) for various fuels included for comparison.

Fuel	C	H	O	N	S	Ash	Moisture
MCMB	91.23	3.39	3.52	1.09	0.34	0.01	0.43
MCMBpt	95.17	0.69	2.70	0.92	0.29	0.01	0.23
FAC	80.56	0.46	1.33	0.3	0.11	2.59	14.65
WYO	51.42	4.16	11.53	0.69	0.32	5.58	26.3
LK	65.69	3.83	9.50	1.34	1.60	17.03	1.01
LKFC	71.32	0.09	0	0.98	1.35	26.98	1.18
AS	27.16	3.37	26.62	0.93	0.04	36.21	5.67
RS	33.57	3.93	24.23	0.58	0.07	30.48	7.14

(available from Osaka Gas Co.), were selected as a representative carbon black (see

Figure 3.3). An activated carbon, termed FAC here (available from Fisher Scientific), was selected as a representative de-ashed, processed char. A char was formed from heating Lower Kittanning coal at a slow rate of order 10 $K \cdot min^{-1}$, although SEM images of LK char formed under high heating rate are also included for comparison. Another char was formed from heating Wyodak coal under a large heating rate of order 1e5 $K \cdot s^{-1}$. Almond shell char, also heated under a rate of 1e5 $K \cdot s^{-1}$ in the LFR, is included as a representative biomass.

3.5.1 MCMB and Activated Carbon Particles

Both MCMB and FAC particles are non-natural fuels in present form, yet have significantly different properties. Shown in Figure 3.3 are MCMB particles, while FAC particles are depicted in Figure 3.4. The spherical form of the MCMB is due to isotropic formation from pyrolysis of gas phase hydrocarbons, with a nominal diameter of about 20 μm, although the distribution contains particles as large as 100 μm. The primary use of MCMB is in electrodes for Li-ion batteries. The carbon comprising MCMB thus arose from light hydrocarbon species originally rather than solid phase mixtures. These fuels represent pretreated DCFC fuels rather than raw feedstock, thus mitigating the problems posed by electrode contamination, ash removal, or tar condensation. The angular form of the activated carbon, more similar to milled

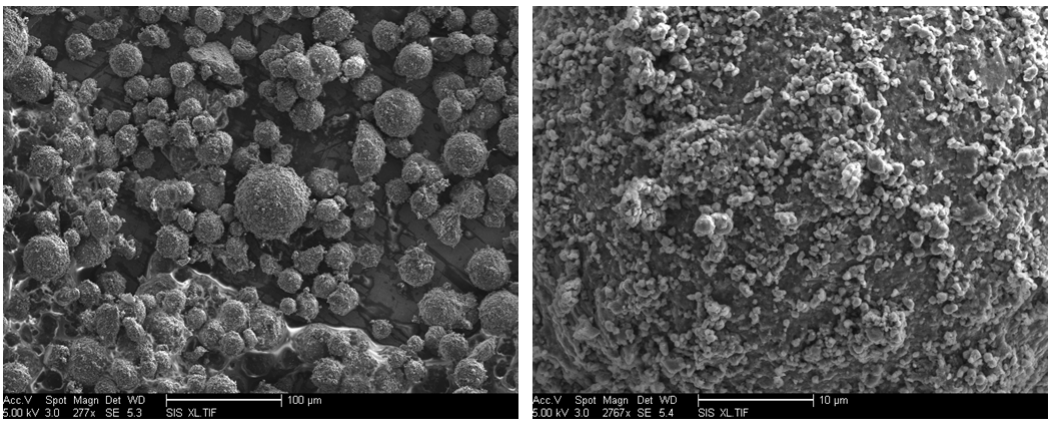


Figure 3.3: SEM image of raw, spherical MCMB particles.

coal, arises from pulverizing after the de-ashing process. The carbon comprising FAC arose from the fixed carbon content in a biomass feedstock, in contrast to MCMB. The size of the FAC ranges from nominally 75-300 μm as received, although particles sieved to less than 100 μm were used for thermogravimetric reactivity analysis. This difference in size and shape between these fuels arises from formation and preprocessing, highlighting this diversity in solid feedstock. The size distributions of MCMB

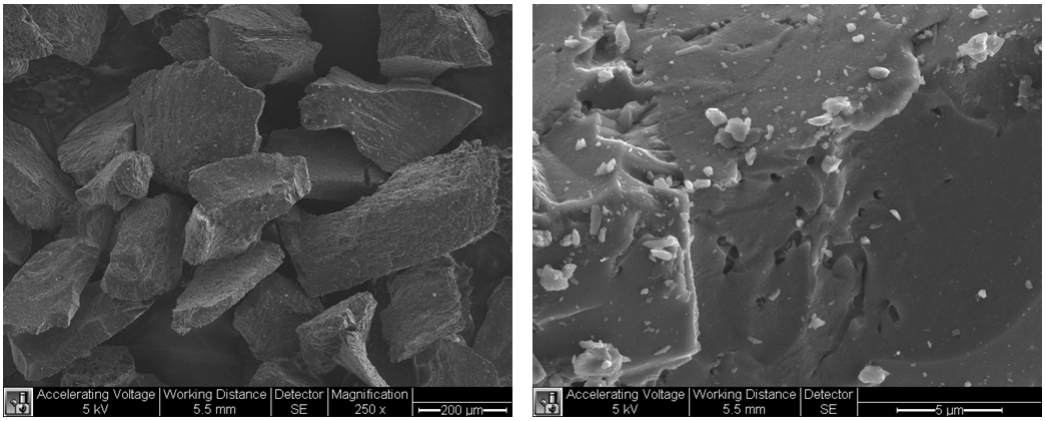


Figure 3.4: SEM image of raw, angular FAC particles.

and FAC are shown in Figures 3.5 and 3.6, measured using the Coulter Multisizer. The distributions are calculated based on total particle volume (mass), so as not to superficially emphasize the small carbon mass contained in the large number of small sized particles. The notation *raw* denotes the unreacted fuel, while *pretreated* denotes a 1173 K temperature hold to remove volatiles, and *FC* denotes post DCFC experiment fuel. The MCMB particles were sieved to provide a distribution of larger particles for improved handling, as a large amount of fines were present. The raw MCMB used for reactivity and DCFC measurements had an initial peak size of just below 50 μm, with a slight decrease in size upon pretreatment at 1173 K for 1 hour. Post-DCFC size measurements show that the particle size has decreased to less than 20 μm due to attrition and reaction as expected in a prolonged gasifying environment. The FAC particles size has a peak at roughly 75 μm, larger than that of MCMB. Aside from their respective origins and directly observable shapes and sizes,

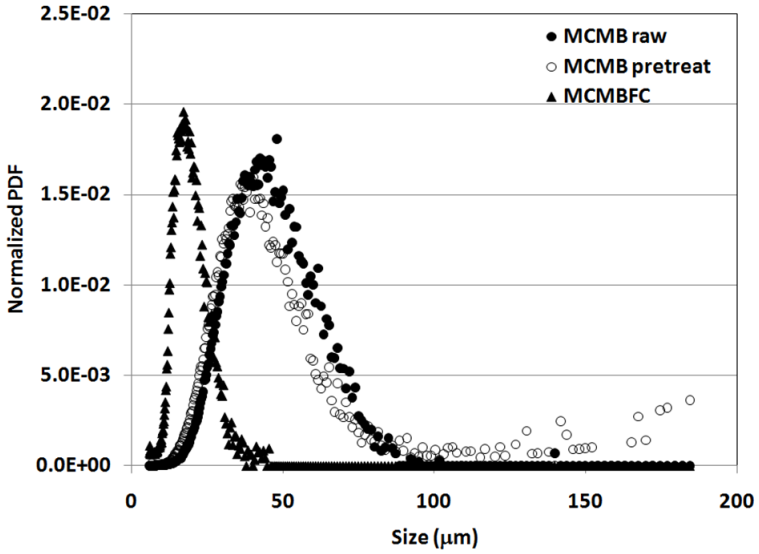


Figure 3.5: Size Distribution (Volume) of MCMB particles.

the primary difference of these carbons lie in their respective active surface areas. The FAC particles have an initial specific surface area of nominally 750 $m^2 \cdot g^{-1}$, as determined from BET analysis. The MCMB particles have specific surface areas of order 1 $m^2 \cdot g^{-1}$, below the accurate measurement capability of the PTGA (representative values presented in the literature range from 0.377 [114] to 1.4 $m^2 \cdot g^{-1}$ [115]). The relatively low internal surface area of MCMB would also cause the large decrease in particle size in the DCFC, as the exterior is primarily gasified thus reducing the particle diameters. The reactivities of MCMB and FAC would be expected to differ due to the differences in structure and active surface area. The rate of mass loss of the solid exposed to a controlled, constant, reactive environment allows for quantitative comparison. The normalized mass loss curves for MCMB and FAC are presented in Figures 3.7 and 3.8, respectively. This metric, in units of s^{-1}, supplies a direct comparison of reactivity between fuels in given environments. The temperatures are given in the figures for pure CO_2 environments, while the notation 50/50 represents an environment of equimolar CO/CO_2. The data presented here was taken at at a total pressure of 1 atm. Note that the variable x_p is the extent of conversion of the

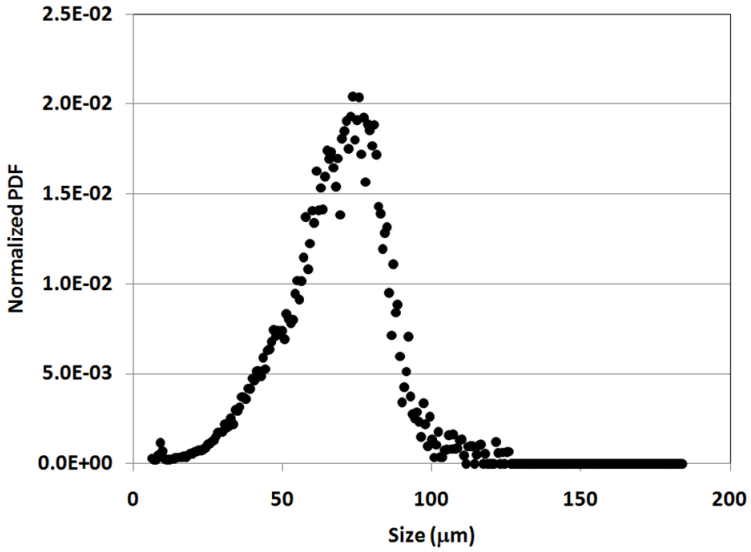


Figure 3.6: Size Distribution (Volume) of FAC particles.

entire particle (including adsorbed species) as defined in Equation 3.12.

$$x_p = 1 - \frac{m_p}{m_{p,0}} \tag{3.12}$$

It is evident from Figures 3.7 and 3.8 that FAC is roughly an order of magnitude more reactive to CO_2 than MCMB. This fact has direct implications for the DCFC, since the rate of the Boudouard reaction drives the cell performance. Higher reactivities are desired to create CO at higher rates within the bed to drive the DCFC under current flow. The large difference in active surface area is one cause for the increased reactivity of FAC. The normalized mass loss for FAC at 1073 K in pure CO_2 was measured to be 4.08e-5 s^{-1} near 10% conversion, within 20% of the value 3.4e-5 s^{-1} found by Koenig et al [104].

A more general observation of the mass loss shows that the time scale for solids conversion, the inverse of the normalized mass loss, lies between 1e2 and 1e6 seconds at the test temperatures. This fact supports the correct assertion that the Boudouard reaction is slow with respect to conversion of the solid at these temperatures. However,

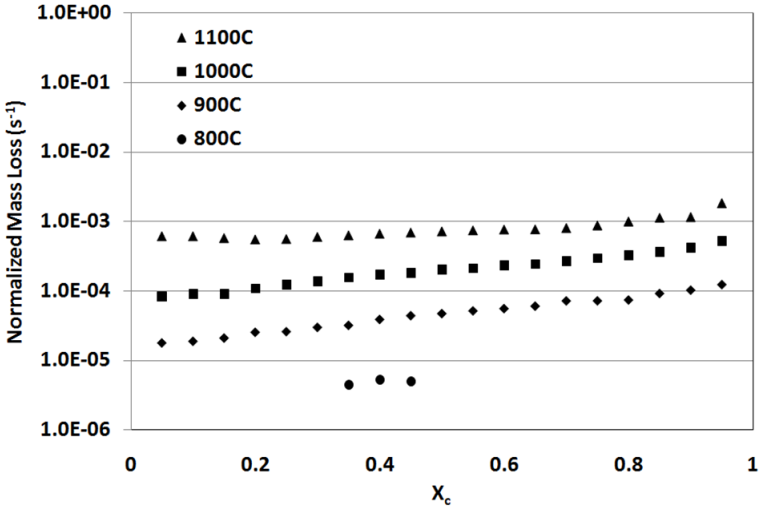


Figure 3.7: Reactivity of MCMB to CO_2 at various temperatures.

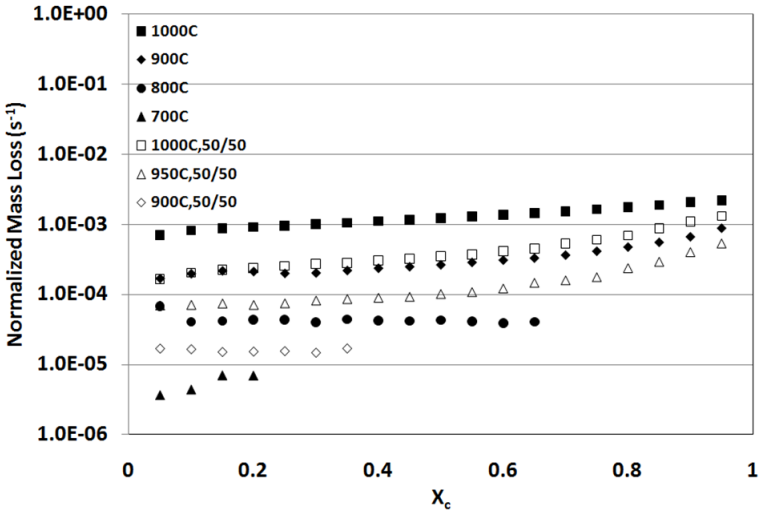


Figure 3.8: Reactivity of FAC to CO_2 and equimolar CO/CO_2 at various temperatures.

it will be shown in subsequent analysis of the DCFC that small changes in the solid conversion can lead to significant changes in gas composition within a fixed bed.

3.5.2 Wyodak and Lower Kittanning Coals

Coal chars represent a more practical feed source than MCMB or FAC, but introduce more complexity in composition, structure and chemical behavior. The reactivities of two coal chars, created under vastly different conditions, to CO_2 were measured in the PTGA. The raw particles are dried and devolatilized to remove water and hydrocarbons, leaving solid, carbonaceous particles of roughly the same size. The first sample, Lower Kittanning (LK) coal char, was created under low heating rates characteristic of radiant furnace temperature ramps (of order 10 $K \cdot min^{-1}$). Images of the raw ruel are included in Figure 3.9, while images of the char created in the LFR are included in Figure 3.10. This fuel has been investigated in other studies and was selected as a representative Eastern USA coal [107]. An image of LK char

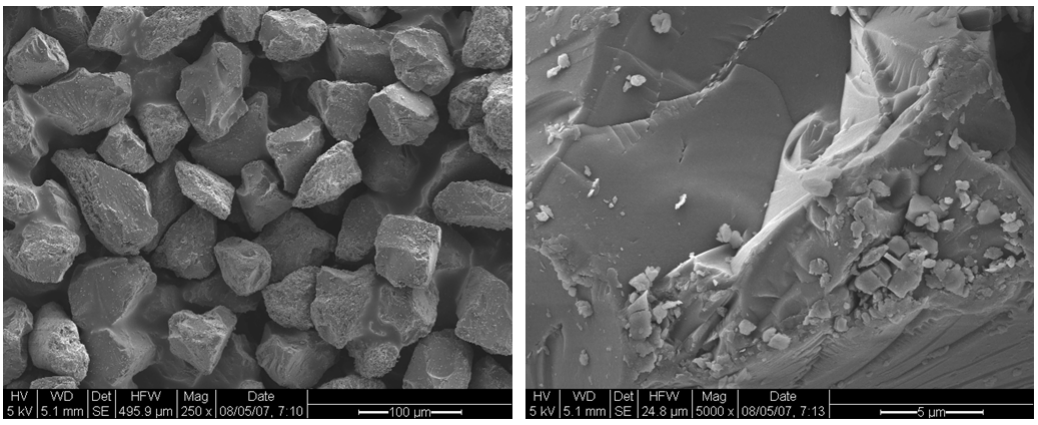


Figure 3.9: SEM image of raw Lower Kittanning coal particles. Char created at low heating rate is visually similar.

taken from post DCFC operation is given in Figure 3.11, however, LK char created in the LFR is included for reference in Figure 3.10. The LK chars created under different heating rates have completely different character visually. The char that underwent slow heating rates resembles the raw fuel, with angular shape and low

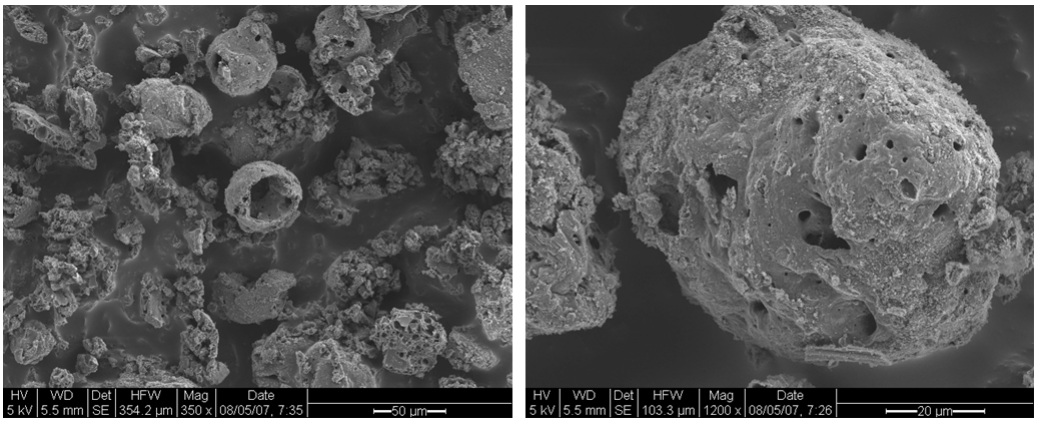


Figure 3.10: SEM image of Lower Kittanning coal particles treated in the LFR, showing a significant change in morphology.

porosity. The char that underwent a high heating rate is rounded in shape with large voids visible due to rapid devolatilization. Also, the char extracted from the LFR has a more varied size range, due to fragmenting from rapid release of volatile species. Hollow particles, termed cenospheres, are also visible along with denser particles and particles of intermediate density. The variation in morphology, and subsequent theoretical treatment of reactivity was investigated by Ma [107]. In this study, the bulk reactivities will be investigated based on a lumped sample. The LK

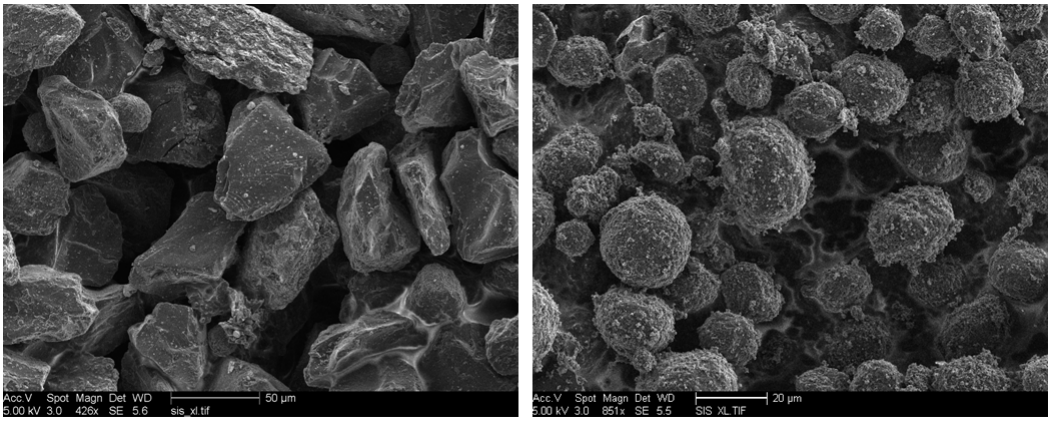


Figure 3.11: SEM image of LKC (l) and mcmb (r) fuels after DCFC operation.

char was sieved prior to use, with the size distributions depicted in Figure 3.12. The raw coal particles have a peak size of 65 microns, reducing to 54 microns as the char is created although the general shape of the distribution is preserved. The LK char particles removed from the DCFC post-operation peak at roughly 39 microns, with a much broader size distribution. The decrease in particle size upon treatment is likely due to fracturing (producing large resulting fragments), as the treated sample still resembles the parent fuel in shape. The decrease in size upon DCFC operation is likely due to the same process of fragmentation. The size distribution of the post-DCFC particles contains particles larger than the original fuel, likely due to pieces of fiber insulation amongst the fuel particle bed. The second coal char, Wyodak (WYO),

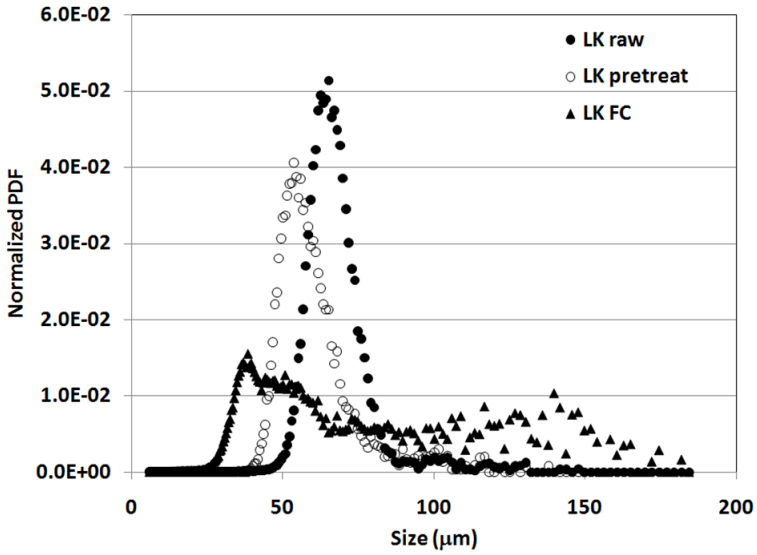


Figure 3.12: Size Distribution (Volume) of LK char particles.

was created under high heating rates characteristic of the LFR. This coal originates in the Powder River Basin of Wyoming. Currently, well over half of the recoverable coal reserves in the United States reside in the western states, with about 40 % of the total recoverable reserves residing in Wyoming [116]. As such, western coals are an important feedstock for future power production in the US. Images of the WYO coal and subsequent char are depicted in Figures 3.13 and 3.14, respectively. The

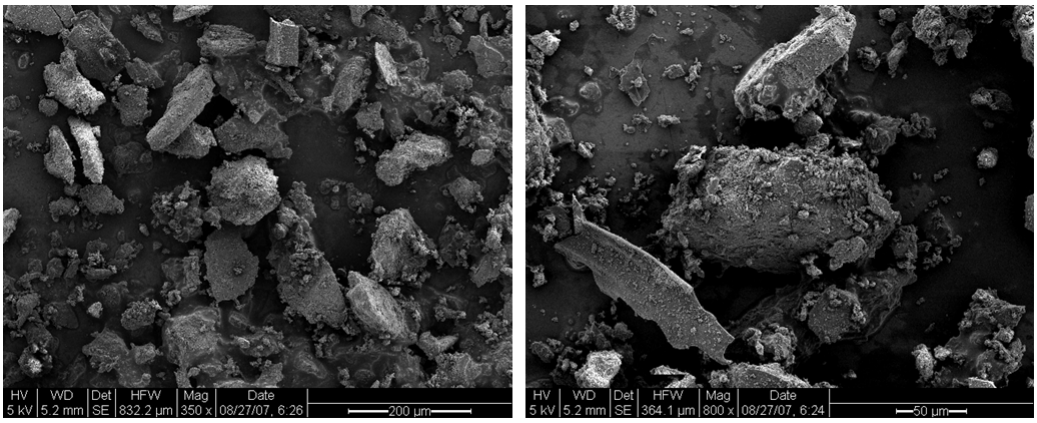


Figure 3.13: SEM image of raw Wyodak coal particles.

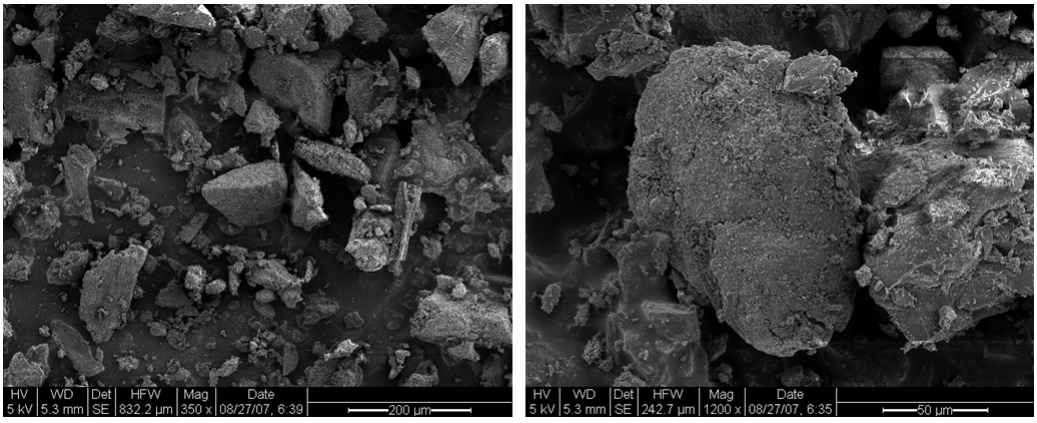


Figure 3.14: SEM image of Wyodak coal particles treated in the LFR.

Wyo coal char more closely resembles a biomass than LK coal in appearance under the SEM, lacking sharp features of pulverized mature coal. This coal is also visibly a brown hue which, along with the large water content, is characteristic of a less mature coal. The size distribution for the WYO char, as measured in the Coulter Multisizer, is given in Figure 3.15, for the raw and post-LFR particles. The particle size shifts to lower values upon introduction to the LFR, due to fragmenting of the particles under the large heating rates. However, the peak size is about 100 μm for both the coal (106 microns) and char (99 microns). The resulting distribution implies that the particles remain rougly the same size, with fragmentation populating the lower-size portion of the distribution. These coal chars, with vastly different morphology created

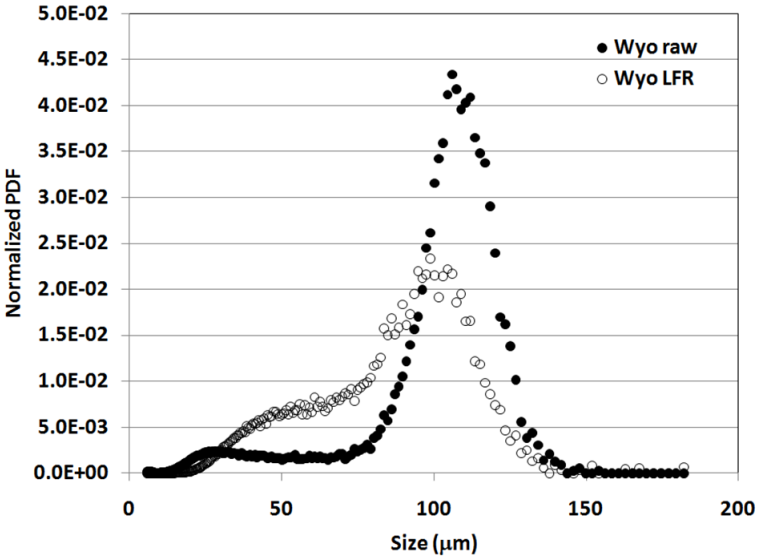


Figure 3.15: Size Distribution (Volume) of Wyo char particles.

under heating rates differing by 5 orders of magnitude would not be expected to have similar reactivities to CO_2. The reactivities of LK and WYO chars are presented as normalized mass loss in Figures 3.16 and 3.17, respectively. The Wyo char has a reactivity to CO_2 that is well over an order of magnitude greater than that of LK char.

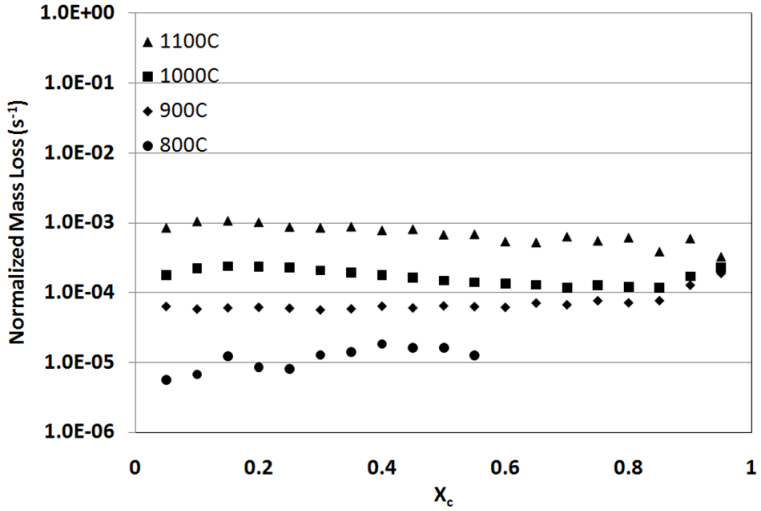


Figure 3.16: Reactivity of LKC to CO_2 at various temperatures.

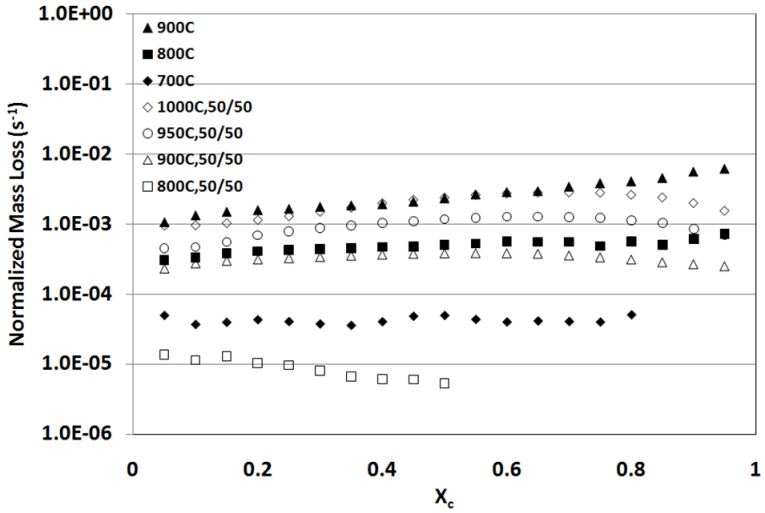


Figure 3.17: Reactivity of WYO to CO_2 and equimolar CO/CO_2 at various temperatures.

3.5.3 Biomass Particles

In addition to carbon and coal char, biomass char is an important fuel for energy production. The primary benefit of biomass fuels reside in their renewable character, thus helping to mitigate carbon release from fossil fuel conversion. The fibrous nature of biomass, along with large moisture and volatile content, set it apart from coals in structure and behavior. Almond shells (AS), treated in the LFR, will be used for reactivity measurements. Shown in Figures 3.18 and 3.19 are the raw particles and char, respectively. The biomass char is similar to the raw fuel in both appearance (under the SEM), and the visible color of the particles (light brown for the raw sample and dark brown for the char). The size distributions of the almond shell particles and

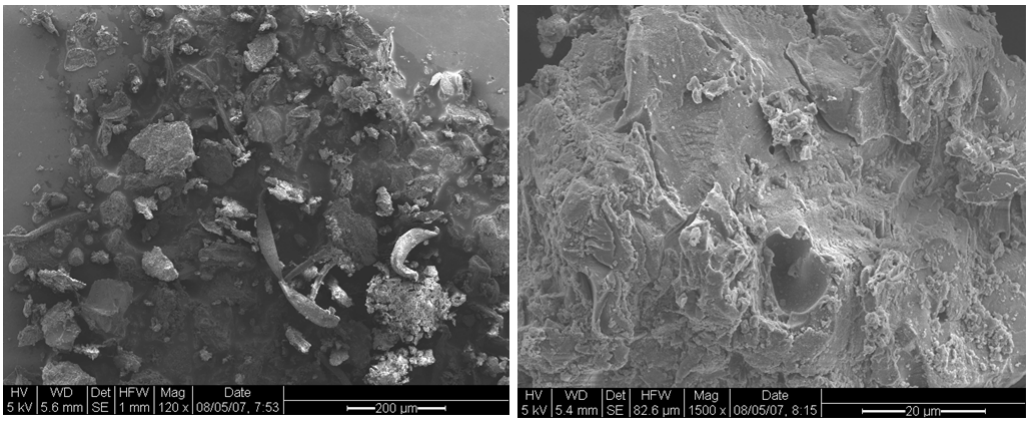


Figure 3.18: SEM image of raw almond shell particles.

char are given in Figure 3.20. The raw particle distribution peaks at 80 μm while the char peaks slightly lower at 74 μm. As can be seen from the SEM images, however, the range in aspect ratio and hence length scale for the particles varies considerably. The chemical analysis of the AS particles show large oxygen (27%wt) and ash (36 %wt) content. The large oxygen content implies a large amount of volatiles that desorb upon heating, thus increasing the rate of mass loss from solid phase to gaseous products. The carbon content is only 27% (compared to over 50% for the coal samples and over 80% for the carbons), which coupled with the large oxygen content causes only about 10 - 20% of the inital char particle mass to be fixed carbon that undergoes

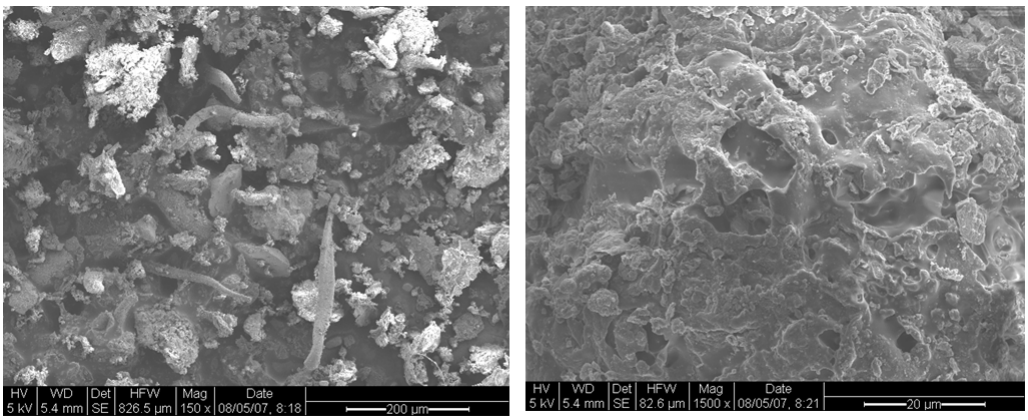


Figure 3.19: SEM image of almond shell particles treated in the LFR.

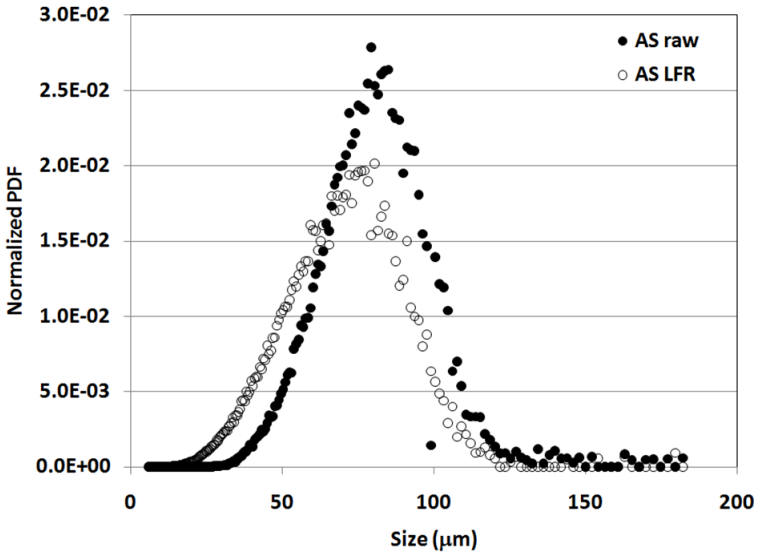


Figure 3.20: Size Distribution (Volume) of AS char particles.

the Boudouard reaction. The measured reativity of the AS char to CO_2 is given in Figure 3.21, for both pure CO_2 and equimolar CO/CO_2. Using BET analysis, the AS

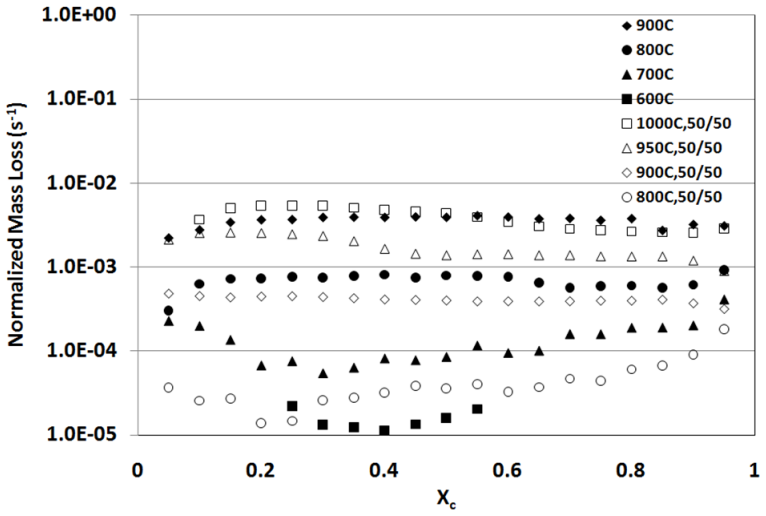


Figure 3.21: Reactivity of ASC to CO_2 and equimolar CO/CO_2 at various temperatures.

char was found to have an initial surface area of 640 $m^2 \cdot g^{-1}$. The AS char, created under high heating rates exhibited the highest reactivity of the fuels tested here.

3.6 Global Comparison of Carbon Reactivities

The data presented in the previous section is summarized and plotted in Figure 3.22. The normalized mass loss values are taken at an extent of conversion of 0.1, near the beginning of the particles lifetime. Particles devolatilized under high heating rates (HHR) are more reactive than those that were heated slowly. The addition of CO inhibits the reaction rate although the hierarchy amongst fuels remains. The Boudouard reaction gives the global progression for CO_2 gasification of carbon. This is the simplest mechanism that can be formulated in order to gain insight into carbon reactivity to CO_2. A global rate equation can be formulated using the law of mass

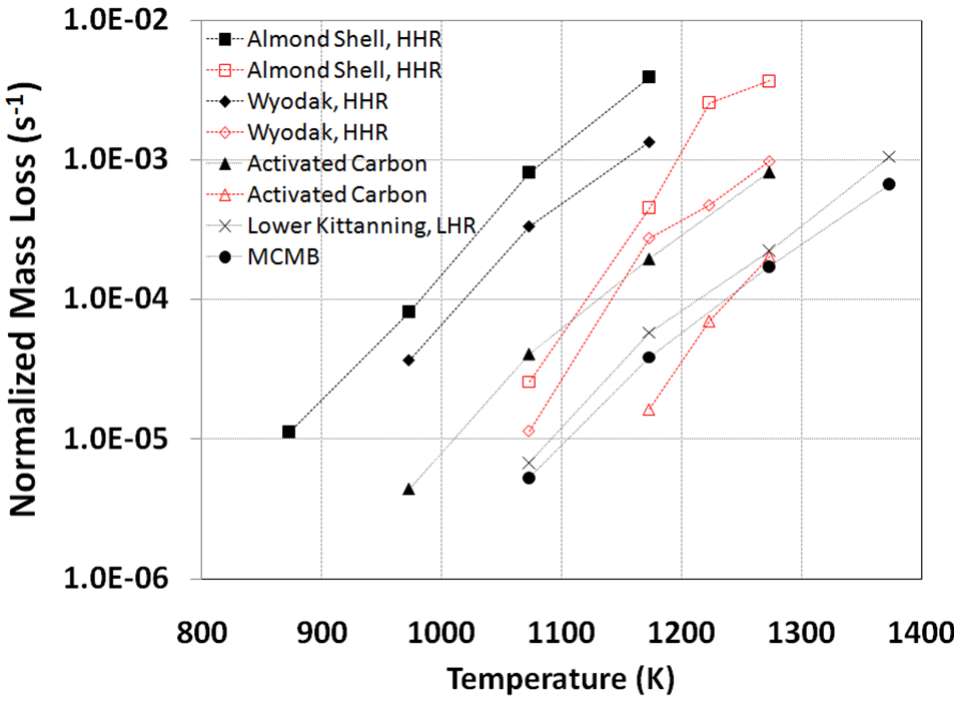


Figure 3.22: Reactivity for the studied fuels taken at an extent of conversion of 0.1. Solid symbols denote pure CO_2 while open symbols denote equimolar CO/CO_2.

action as shown in Equation 3.13, where the equilibrium constant K_p is given by Equation 3.14, and the total gas concentration C_0 is given by Equation 3.15.

$$R = k_{fuel}(T)\left([CO_2] - \frac{[CO]^2}{K_p C_0}\right) \tag{3.13}$$

$$lnK_p = -\frac{\Delta G^0}{RT} \tag{3.14}$$

$$C_0 = \frac{P_0}{RT} \tag{3.15}$$

Using this form, the rate coefficient k is the rate at which a mixture relaxes towards Boudouard equilibrium, or for a fixed gas phase composition, the rate at which carbon is consumed. This form represents the product of two terms, the first representing the relaxation rate to equilibrium, and the second representing the deviation of the mixture from its equilibrium composition. This allows a direct comparison of the reactivity across different fuels, and a means for simple computation. The expression given by Equation 3.13 can be further arranged to match units of normalized mass loss (s^{-1}), as shown in Equation 3.16.

$$\frac{1}{m}\frac{dm}{dt} = MW_c S_{gC}(x_c) k_{fuel}(T)\left([CO_2] - \frac{[CO]^2}{K_p C_0}\right) \tag{3.16}$$

This form is now amenable to comparison with experimental data for determination of rate coefficients. The right hand side of Equation 3.16 can be further arranged, using the functional relationship for the specific surface area and extent of reaction. If Arrhenius behavior is assumed, giving $k_{fuel} = Ae^{-\frac{E_a}{RT}}$ (see Equation 3.17), then there are 3 unknowns for each fuel; ψ, A, and E_a.

$$\frac{1}{m}\frac{dm}{dt} = \sqrt{1 - \psi ln(1 - x_c)} A^0 e^{-\frac{E_a}{RT}}\left([CO_2] - \frac{[CO]^2}{K_p C_0}\right) \tag{3.17}$$

In more general form, this can be written as the product of three functions Ψ, Γ and Θ (see Equation 3.18).

$$\frac{1}{m}\frac{dm}{dt} = \Psi(x_c) \cdot \Gamma(T) \cdot \Theta([CO_2], [CO], K_p, C_0) \tag{3.18}$$

Using experimental data taken at various extent of reactions, a system of nonlinear equations can be constructed. A nonlinear least squares algorithm can now be employed to determine best-fit parameters for these three unknowns. Defining the following matrices, **F** representing the RHS and **D** representing the LHS of Equation 3.17 for N data points taken various temperatures and extents of conversion.

$$\mathbf{F} = \begin{bmatrix} f(T_1, x_{c,1}, [CO_2]_1, [CO]_1, \psi, A, E_a) \\ f(T_2, x_{c,2}, [CO_2]_2, [CO]_2, \psi, A, E_a) \\ \vdots \\ f(T_i, x_{c,i}, [CO_2]_i, [CO]_i, \psi, A, E_a) \\ \vdots \\ f(T_N, x_{c,N}, [CO_2]_N, [CO]_N, \psi, A, E_a) \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} \left(\frac{1}{m}\frac{dm}{dt}\right)_1 \\ \left(\frac{1}{m}\frac{dm}{dt}\right)_2 \\ \vdots \\ \left(\frac{1}{m}\frac{dm}{dt}\right)_i \\ \vdots \\ \left(\frac{1}{m}\frac{dm}{dt}\right)_N \end{bmatrix}$$

This is a nonlinear system in terms of the three unknowns, and there exists more equations than unknowns. There is not necessarily a unique combination of values that will cause the model to exactly coincide with the data points. An iterative approach is used within MATLAB, based on an initial guess, in order to minimize the residual (using the least squares formalism), given by the difference $\mathbf{D} - \mathbf{F}$. The MATLAB code is included in Appendix B.

The results of the LSQ analysis are given in Table 3.3 for five fuels under various test conditions. The global activation energy lies in the range 176 - 205 $kJ \cdot mole^{-1}$ (42 - 49 $kcal \cdot mole^{-1}$) for 100 % CO_2 at 1 atm, in general agreement with values presented in the literature [100]. Also included in Table 3.3 is the rate coefficient Γ at the temperatures examined. Direct comparison amongst the fuels shows a two orders of magnitude span in reactivity. The results imply that fuels undergoing high heating rates are preferred for DCFC fuels in order to create chars exhibiting higher

reactivities. Although processed fuels such as MCMB and FAC exhibit lower reactivities, they are typically cleaner than raw fuels in terms of impurities such as sulfur or chlorine, and contain significantly less ash than coal char. Data taken in pure CO_2 at

Table 3.3: Global parameters and $\Gamma(\mathrm{m}^3 \cdot mole^{-1} \cdot s^{-1})$ from LSQ analysis for various fuels under test conditions X_{CO2} (with CO balance).

Fuel	MCMB	FAC	FAC	LK	WYO	WYO	AS	AS
Test	1.0	1.0	0.5	1.0	1.0	0.5	1.0	0.5
ψ	5.0	3.8	5.0	0.0	5.0	0.8	0.0	0.4
A^0	2.3e3	2.0e4	4.1e6	1.0e3	5.8e4	1.4e4	2.5e4	2.9e4
E_a	205	182	272	186	195	186	176	181
Γ_{873}	-	-	-	-	-	-	6.8e-7	-
Γ_{973}	-	3.4e-7		-	2.1e-6	-	8.3e-6	-
Γ_{1073}	2.5e-7	2.8e-6	-	8.6e-7	2.0e-5	1.2e-5	6.3e-5	4.3e-5
Γ_{1173}	1.8e-6	1.6e-5	3.1e-6	5.1e-6	1.3e-4	6.9e-5	3.4e-4	2.4e-4
Γ_{1223}	-	-	9.8e-6	-	-	1.5e-4	-	5.2e-4
Γ_{1273}	9.1e-6	6.8e-5	2.8e-5	2.3e-5	-	3.1e-4	-	1.0e-3
Γ_{1373}	3.7e-5	-	-	8.2e-5	-	-	-	-

1 atm can be reproduced by the global model. The primary drawback to this global approach arises when trying to fit parameters to data taken under high-CO conditions or at higher pressures. The above analysis breaks down for an equimolar mixture of CO/CO_2, due to the inhibiting effect of CO on the Boudouard reaction, especially at lower temperature. There is no global value of Γ that fits data at these different conditions. The detailed chemical behavior cannot be captured using a single step reaction, indicating the presence of additional rate-limiting processes. In addition to reactivity, adsorption and desorption processes (and hence surface species) are not addressed in the global model, as only bulk carbon is included. Therein lies the need for a more elementary model of the gasification process, allowing for more robust predictive capability. tion and desorption processes (and hence surface species) are not addressed in the global model, as only bulk carbon is included. Therein lies the need for a more elementary model of the gasification process, allowing for more robust predictive capability.

3.7 Detailed Kinetic Modeling of Activated Carbon

The normalized mass loss rates presented previously for the various fuels give a direct comparison of reactivity between different feedstock. In order to generalize the char reactivity over various gas compositions and temperatures, reaction rate coefficients based on the five-step mechanism were determined. The activated carbon FAC is studied in detail, since it was used in fixed bed and DCFC experiments. Some data on WYO char particles will also be presented for comparison.

3.7.1 Surface Area Determination During Conversion

Thermogravimetric measurements were made of the specific surface areas of FAC and WYO (see Figure 3.23), at various extents of conversion. This was accomplished by performing an initial BET test on the raw sample, then performing a partial reactivity test followed by another BET test. The dotted curves on each figure denote the surface area model with the specified structure parameter for reference. The direct measurement of specific surface area allows for direct determination of the structure parameter, independent of simultaneous chemical reaction. The FAC particles have an initial surface area of 750 $m^2 \cdot g^{-1}$, while WYO particles have an initial surface area of 300 $m^2 \cdot g^{-1}$. The surface area of FAC is nearly 900 $m^2 \cdot g^{-1}$, as compared to 1020 $m^2 \cdot g^{-1}$ found in Koenig's study [104]. In addition, FAC has a structure parameter of 3 while WYO has a structure parameter of 5, using one significant figure. The two parameters determining the functional dependence of surface area with the extent of conversion are now explicitly specified.

3.7.2 Temperature Programmed Desorption Measurements

The TPD procedure, enacted within the PTGA, consists of the following processes;

- loading oxide complexes onto the carbon surface using CO_2 at fixed temperature
- cooling the sample under CO_2 flow to mitigate desorption

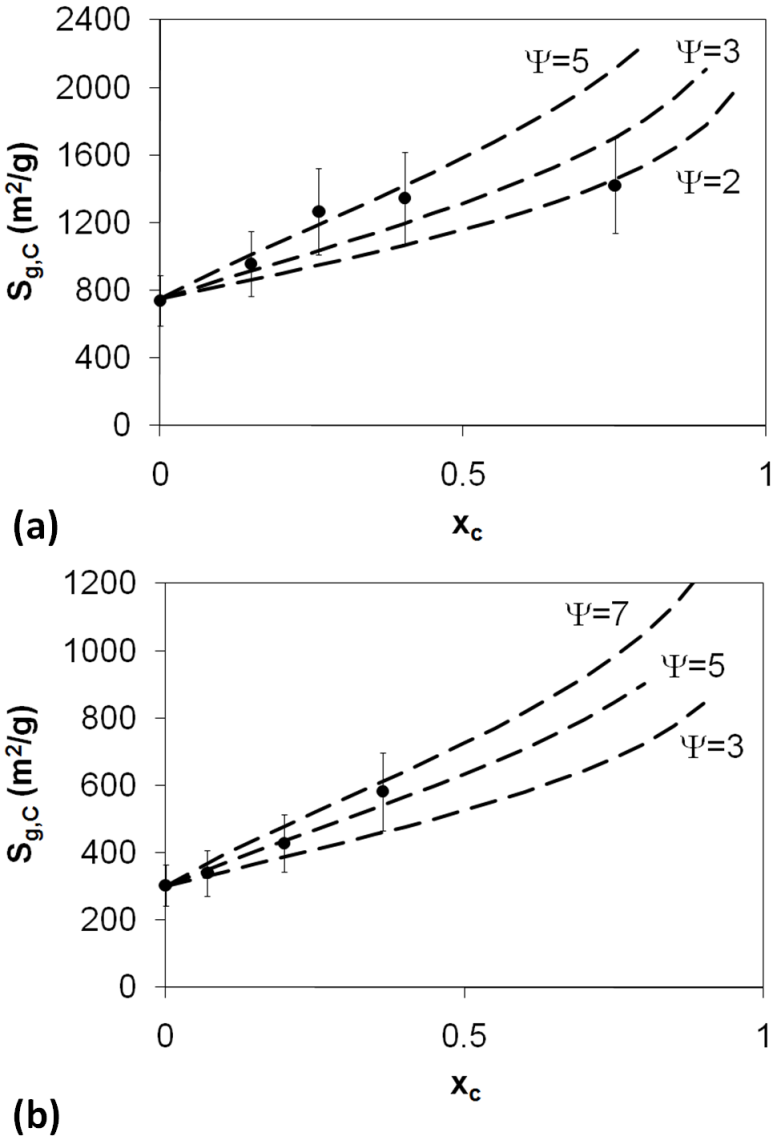


Figure 3.23: Measured specific surface area (with 20% error bars) for (a) FAC and (b WYO at various extents of conversion, along with the Bhatia and Perlmutter model.

- purging of the sample environment with N_2 (inert)
- ramping the temperature at 25 $K \cdot min^{-1}$ under a constant N_2 flow
- measurement of CO and CO_2 ppm as a function of time

The output of TPD experiments are the measured time traces of temperature and mole fractions of CO and CO_2. The flow rates are user specified and hence known, so Equation 3.19 can be used to determine the molar flows of CO and CO_2 as a function of time.

$$\dot{N}_{CO} = \frac{CO_{ppm}}{1e6}(\dot{V}_{purge} + \dot{V}_{furnace} + \dot{V}_{rxn})\frac{P}{RT} \tag{3.19}$$

The rate equation for CO desorption (normalized by the total CO), given by Equation 3.20, is integrated in time to determine the instantaneous and cumulative about of gas desorbed per unit area,

$$\frac{1}{[CO]_0}\frac{d[CO]}{dt} = \int_0^\infty Ae^{-\frac{E}{RT}}e^{g(E,t)}f_0(E)dE \tag{3.20}$$

where $g(E,t) = -A\int_0^t e^{-\frac{E}{RT}}$ and $f(E)$ is a normalized activation energy distribution function (see Equation 3.21). A Gaussian disribution is used here entailing the determination of two parameters, the distribution mean and standard deviation energies. The TPD process allows determination of k_2 directly, aiding in the determination of the complete set of rate parameters.

$$f(E) = \frac{1}{\sigma}\sqrt{\frac{1}{2\pi}}e^{-\left(\frac{(E-E_{avg})^2}{2\sigma^2}\right)} \tag{3.21}$$

The experimental data from the TPD analysis in the PTGA for FAC and WYO are shown in Figure 3.24, respectively. The dotted lines in these figures represent modeling results, using Equation 3.20. The average and standard deviation energies that best-fit the experimental data are determined by integrating Equation 3.20 in time and then comparing the normalized CO evolution as a function of temperature. The average and standard deviation energies of R.2 for FAC were found to be 375 and 28 $kJ \cdot mole^{-1}$, and for WYO were found to be 335 and 17 $kJ \cdot mole^{-1}$, respectively.

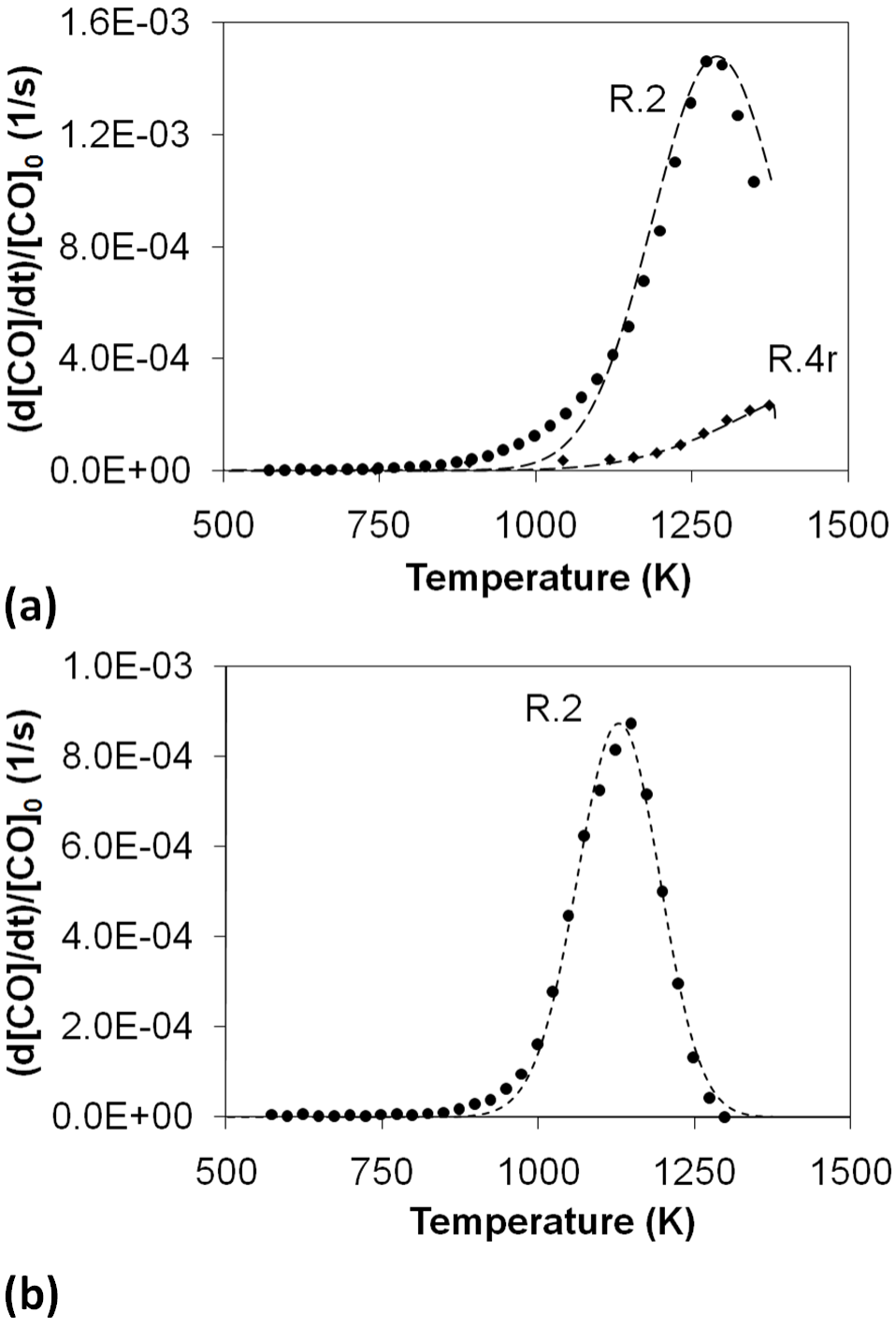


Figure 3.24: Temperature programmed desorption results for (a) FAC and (b) WYO, with a Gaussian-distributed activation energy model fit.

The average and standard deviation energies of R.4r for FAC were found to be 455 and 53 $kJ \cdot mole^{-1}$, respectively. Reactions R.2 and R.4r are now specified for the FAC particles, using an A value of 1.0e13 s^{-1}.

3.7.3 Calculated Reation Rate Parameters

Using the nonlinear least squares formalism, detailed kinetic parameters of the 5-step mechanism were determined for FAC (see Table 3.4). The calculation is performed

Table 3.4: Kinetic parameters determined for FAC particles using thermogravimetric analysis and the nonlinear, least squares formalism.

		A_i	$E_i(kJ/mol)$	$\sigma(kJ/mol)$
$CO_2 + C_f \rightarrow CO + C(O)$	R.1f	5.00e3	185	-
$CO + C(O) \rightarrow CO_2 + C_f$	R.1r	1.08e2	89.7	-
$C_b + C(O) \rightarrow CO + C_f$	R.2	1.00e13	375	28
$C_b + CO_2 + C(O) \rightarrow 2CO + C(O)$	R.3	1.00e4	58	-
$C_f + CO \rightarrow C(CO)$	R.4f	0.89	148	-
$C(CO) \rightarrow C_f + CO$	R.4r	1.00e13	455	53
$CO + C(CO) \rightarrow CO_2 + 2C_f$	R.5	1.01e7	262	-

using Equation 3.9. The surface area at various particle conversions is estimated assuming that $S_{gC}(x_p) \approx S_{gC}(x_c)$. In addition, the concentrations of surface species were assumed as their quasi-steady state values. Using a zero accumulation term, these are calculated using Equations 3.22 and 3.23. These linear equations are solved explicitly for θ_i using 2x2 matrix simplification.

$$\theta_{O,ss} = \frac{k_{1f}[CO_2](1 - \theta_{CO,ss})}{k_{1f}[CO_2] + k_{1r}[CO] + k_2} \tag{3.22}$$

$$\theta_{CO,ss} = \frac{k_{4f}[CO](1 - \theta_{O,ss})}{k_{4f}[CO] + k_{4r} + k_5[CO_2]} \tag{3.23}$$

Reactions R.2 and R.4r were specified previously from TPD analysis. Amongst the remaining steps, Reactions R.1f and R.3 dominate in environments containing CO_2 with minimal CO present. Data measured in pure CO_2 and CO_2/N_2 mixtures was then fit using Reactions R.1f, R.2, and R.3. The remaining Reactions R.1r, R.4f,

R.5 were determined from data taken under an equimolar CO/CO_2 gas environment. The normalized mass loss of FAC particles in these various environments, as modeled using these chemical kinetic parameters are plotted along with the experimental data for comparison (shown in Figures 3.25 - 3.27). The model is capable of predicting the conversion of the FAC particles at temperatures ranging from 973 - 1273 K, under both low-CO and high-CO conditions. The kinetic parameters determined

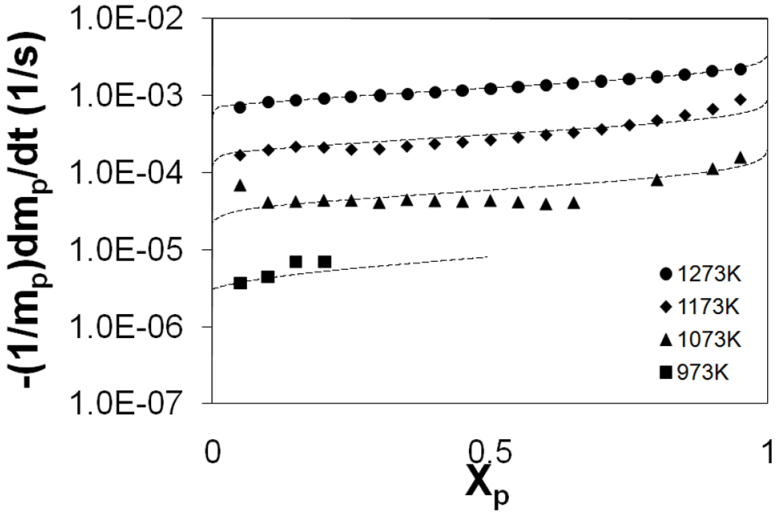


Figure 3.25: Experimental data and detailed kinetic model of FAC in 100% CO_2 at various temperatures.

here will be employed in a transport model in the following chapter, and compared with experimental data measured from a fixed particle bed arrangement. The scalar concentration fields within the DCFC fuel bed can vary widely, requiring a kinetic model that is capable of simulating these conditions.

The sensitivity of intrinsic carbon reactivity to the reaction rate coefficients determined from nonlinear least squares analysis are shown in Figure 3.28. The sensitivity S is defined here as given by Equation 3.24, where Y represents the output ($R_{i,C}$) and X represents an input parameter (k_i). The partial derivative, and subsequent finite difference approximation, are evaluated using a numerical pertubation of 5% in

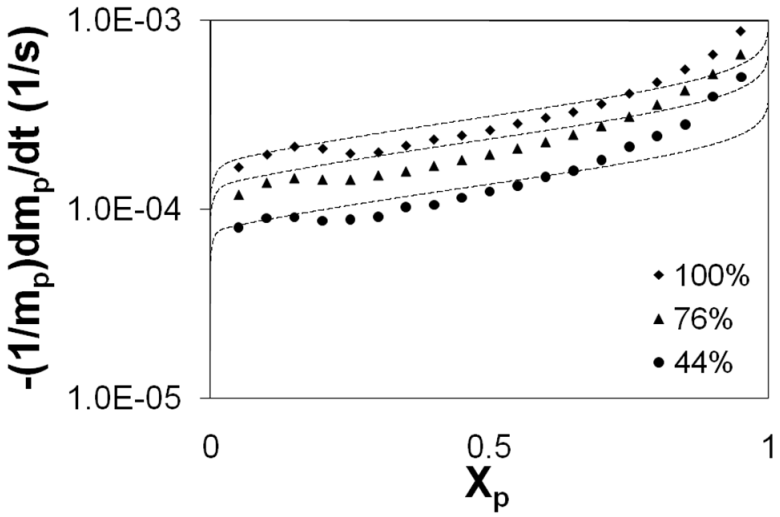


Figure 3.26: Experimental data and detailed kinetic model of FAC in CO_2/N_2 mixtures at 1173 K.

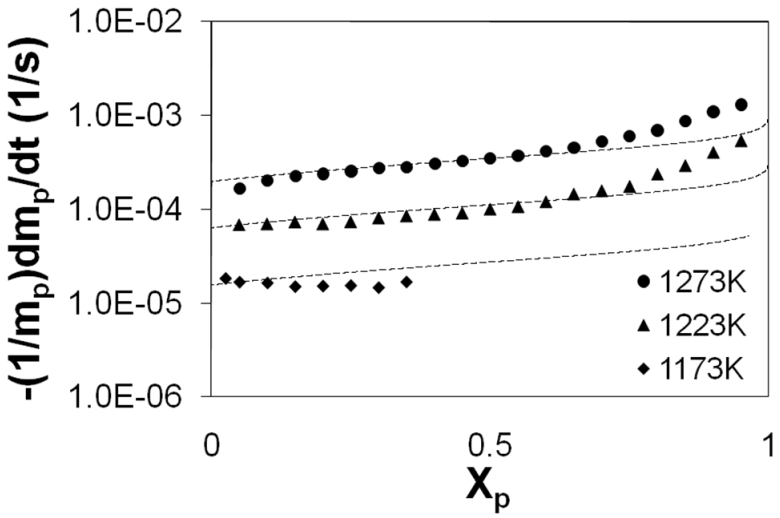


Figure 3.27: Experimental data and detailed kinetic model of FAC in an equimolar CO/CO_2 mixture at various temperatures.

the reaction rate coefficient.

$$S = \left(\frac{\delta R_{i,C}}{\delta k_i}\right)_0 \frac{k_{i,0}}{R_{i,C,0}} \approx \left(\frac{\Delta R_{i,C}}{\Delta k_i}\right)_0 \frac{k_{i,0}}{R_{i,C,0}} \tag{3.24}$$

The sensitivity results show that intrinsic reactivity is most sensitive to reaction

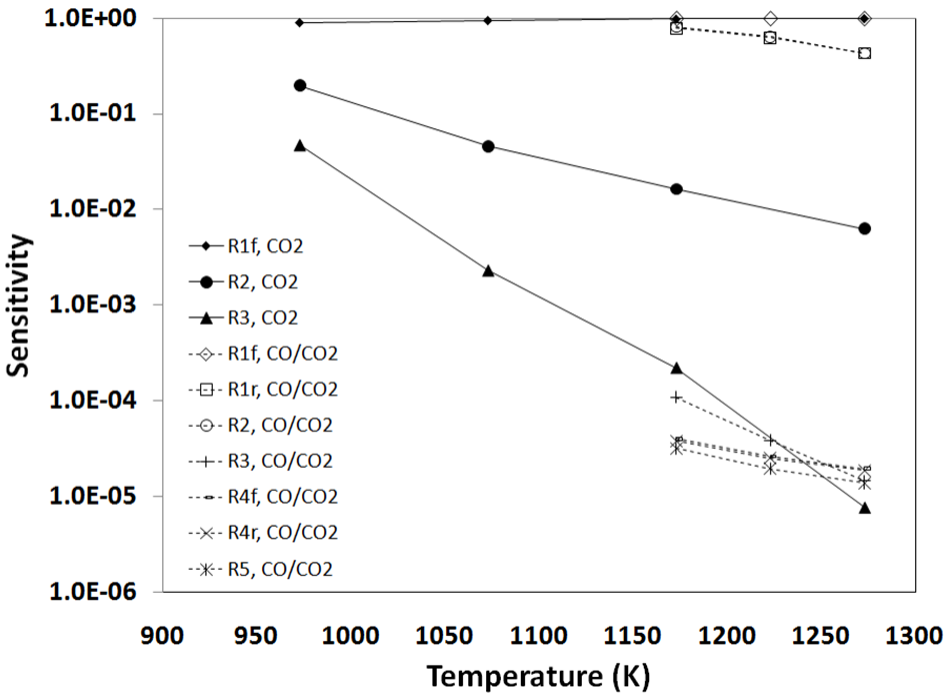


Figure 3.28: Sensitivity of $R_{i,C}$ $(g_C \cdot m^{-2} \cdot s^{-1})$ to reaction rate coefficients for both 100% CO_2 and equimolar CO/CO_2.

R.1f in both 100% CO_2 and an equimolar CO/CO_2 mixture, with S almost equal to unity. At 1 atm, reaction R.1f is the dominating step in the mechanism. In pure CO_2, intrinsic reactivity is also sensitive to Reactions R.2 and R.3 by roughly an order of magnitude, although these sensitivities diminish with temperature. The presence of CO causes sensitivity to Reactions R.1r and R.2, since the C(O) surface complex can proceed along either of these paths. In the equimolar environment, Reactions R.3 - R.5 do not significantly affect reactivity at low total pressure for FAC fuel.

Chapter 4

DCFC Experiments and Modeling

Experiments were performed with laboratory-scale, single electrochemical cells to determine basic behavior and properties of the solid state DCFC. The geometries investigated include electrolyte-supported tube cells, electrolyte-supported button cells, and anode-supported tube cells. The testing facility described in the following section was used for all cells. In addition to DCFC testing, a transport model is developed and verified to model the concentration fields within the fuel bed. This model is then used to predict the expected current densities achievable by the DCFC.

4.1 Fuel Cell Operation and Testing Facility

The fuel cell testing facility used for these experiments includes both electrical and flow control equipment, along with a gas analysis system. Shown in Figure 4.1 is a process schematic of the DCFC, test stand and instrumentation for reference. Clamshell wire wound heaters, with a maximum temperature limit of 1273 K, are used to supply heat to the electrochemical cells. Heaters are controlled with Omega CN8200-DC1 PID process controllers, in conjunction with Omega SSRL240DC25 solid state relays, programmed to maintain or ramp temperature as measured by an immersed K-type thermocouple probe. Gas flow rates were manually controlled by Omega series FL-3102SA rotometers (sapphire float), and verified with a positive displacement Varian Intelligent Digital Flowmeter. The carbon bed is purged with either CO_2 or He, while

air was supplied to the DCFC cathode. The liquid seal was formed by a Pyrex bubbler filled with low-volatility pump oil, helping to mitigate upstream air diffusion, and allowing for a simple means of leak detection. A Varian model 3400 gas chromatograph

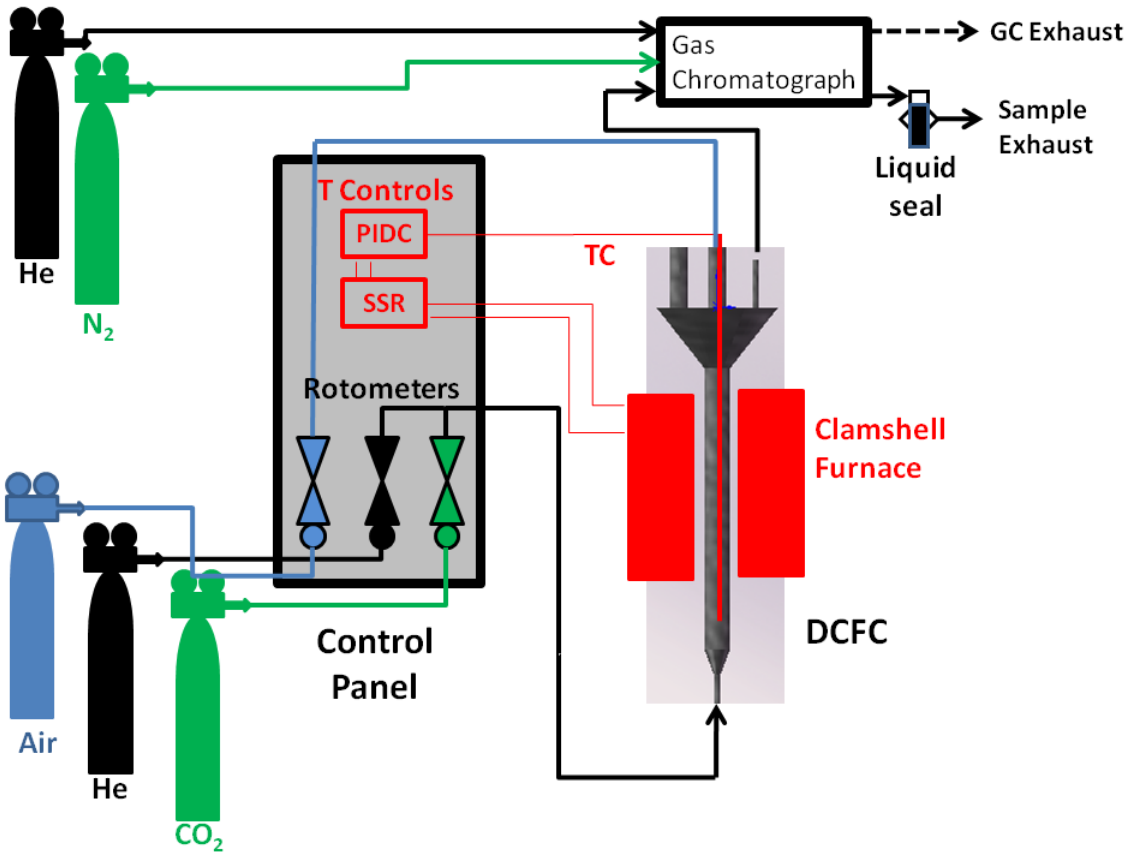


Figure 4.1: Schematic of the DCFC test stand with GC, displaying the major process streams and controls.

(GC), equipped with a porous polymer column (HaySEP), molecular sieve column (5A), and thermal conductivity detector (TCD) is used to quantify gas compositions. The mole fractions of species including H_2, CO, CO_2, O_2, N_2 can be measured. The entire DCFC exhaust sample is directed through the GC sample line, as the flowrates are generally small and large pressures aren't required to overcome pressure drops. Either He or Ar are used as carrier gases in this study, the latter being used when measuring H_2. Compressed N_2 is used to facilitate pneumatic valve switching for the

sample loop. Calibration curves for the GC are included in Appendix A.

4.2 Electrochemical Equipment and Methods

A Princeton Applied Research (PAR) Model 173 Potentiostat/Galvanostat was used to maintain constant DC operation at specified voltage or current (1 A maximum). Open circuit voltage measurements are made with a Hewlett Packard 3455A Digital Voltmeter, while assorted handheld meters are also employed to monitor and compare voltages. An Instek PSP-2010 Programmable Power Supply (PPS) is used to compensate for lead resistance when using simple resistive loads, under high current conditions. The PPS has the capacity to maintain a specified current or voltage, in addition to measuring DC current. Shown in Figure 4.2 is a schematic of the configuration used during DC loading and measurement of cells under high current. The lead resistance (R_L) can be significant, so the PPS is used as a voltage source to match the readings between digital multimeter (DMM) DMM1 measured across non current-carrying leads from the cell, and DMM2 which measures the potential across the resistive load L. This arrangement allows the lead resistance to be exactly compensated and canceled without powering the cell. The voltage across a known (0.1 or 1 ohm) resistor, included in the load L, is measured using DMM3 to corroborate the current reading of the PPS. A Solartron 1250 Frequency Response Analyzer and 1286 Electrochemical Interface are used to perform galvanostatic (DC) and impedance (AC) measurements on the cells. Shown in Figure 4.3 is a schematic of the configuration used during DC and AC-DC coupled measurements using the Solartron 1250 and 1286. The working electrode (WE) is connected to the cathode current-carrying lead, while the counter electrode (CE) is connected to the anode current-carrying lead. The reference electrodes are connected to non current-carrying leads at the cathode (RE2) and anode (RE1). In this arrangement, the cell potential as specified to the 1286 is negative (RE1 - RE2) during normal galvanic DCFC operation, while the current entered is positive.

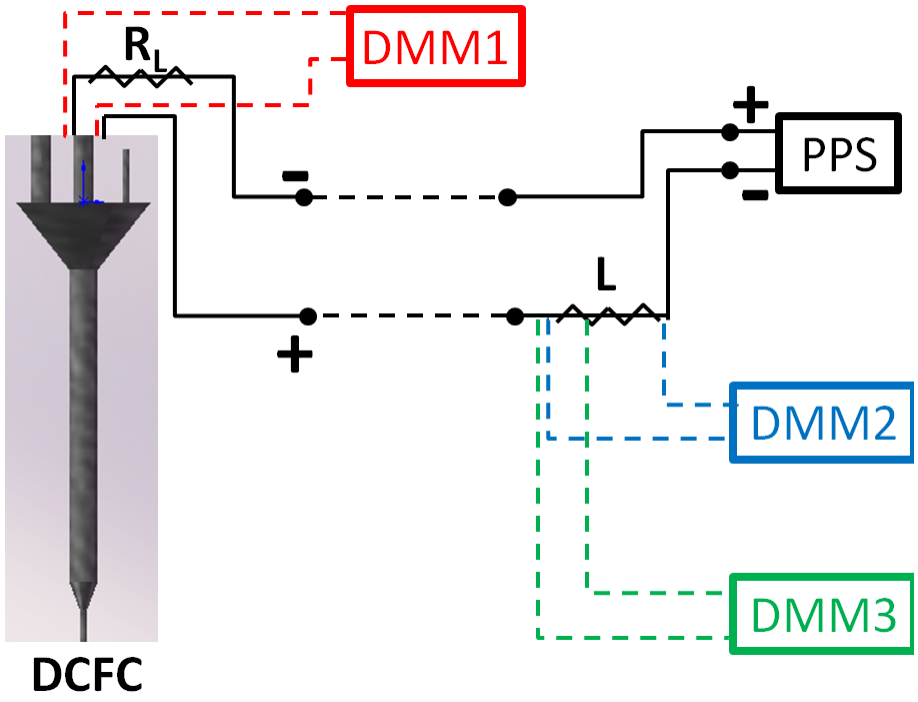


Figure 4.2: Schematic of the PPS wiring during DC measurements under resistive loading.

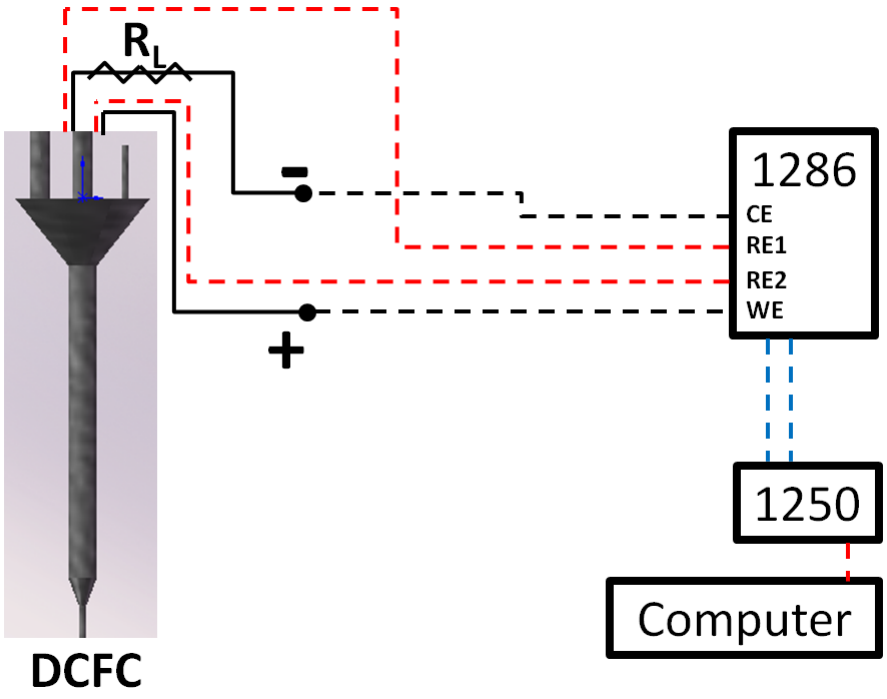


Figure 4.3: Schematic of the Solartron 1286 and 1250 connections during DC-AC measurements.

4.3 Electrolyte-Supported Tube Cells

The electrolyte-supported tube cells (ESTC) described here are identical to those described in Chapter 2, in the context of OCV measurements. The cell resistance, which is significant, is now a factor in the performance. Demonstration of large current or power densities is not possible with such cells, although the verification of cell potentials, general trending, and exhaust gas analysis were completed. Nonetheless, the ESTC provides a convenient, gas tight SOFC assembly that is structurally robust under mechanical and thermal stresses.

4.3.1 ESTC DC Measurements

The ESTCs were first characterized using air at both electrodes under forcing potentials. Potentiostatic air/air experiments were conducted while the cell was maintained at nominally 1173 K. In addition to allowing the resistance of the cells to be estimated, this forcing can increase porosity and active Pt-PSZ interface area [117]. The PAR potentiostat was used to fix cell potential and measure DC current. Shown in Figure 4.4a is a set of current-voltage measurements, with positive current values corresponding to an oxygen flux radially outward. The exterior of the tube was purged with ambient air, and then left open to the atmosphere while the tube interior was purged with a steady air flow during positive current operation and no air flow during negative current operation. Current measurements taken at various air flow rates, during operation at 1 V or 1.5 V, are shown in Figure 4.4b. Following air/air operation the tube cells were immersed in a bed of carbon particles, with air supplied to the interior. Shown in Figure 4.5 are I-V curves of the 20 cm^2 cell as CO_2 flowrate is varied through a bed of MCMB particles. The results show that performance decreases as the CO_2 flowrate is increased. This is due to the reduction in local CO/CO_2 ratio caused by increased convection of CO_2 into the bed relative to the finite destruction rate of CO_2 via the Boudouard reaction. Shown in Figure 4.6 is the CO mole fraction at the MCMB particle bed exhaust, formed via the Boudouard reaction within the carbon bed. The tube cell is in place, but under open circuit conditions with no oxygen flux to the anode. The carbon particles cover the anode

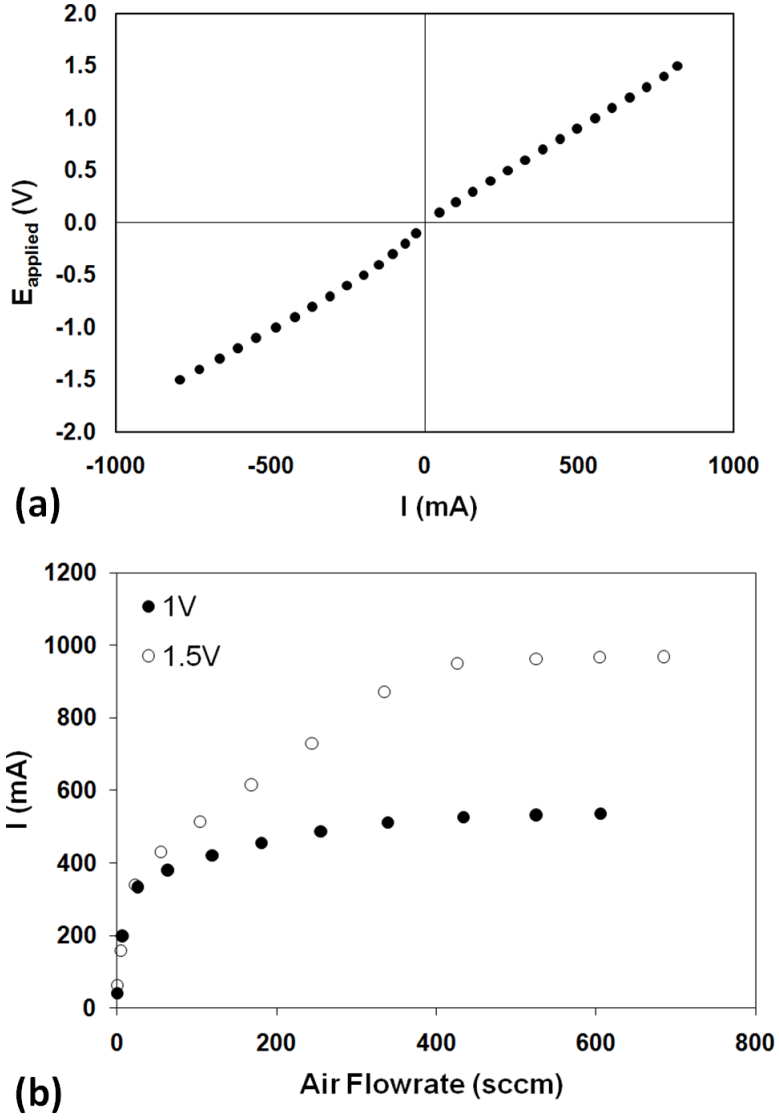


Figure 4.4: (a) DC current-voltage behavior for electrolyte-supported tube cell at 1173 K, with air at both electrodes with positive current corresponding to oxygen flux radially outward. (b) Current as a function of air flow through PSZ tube interior.

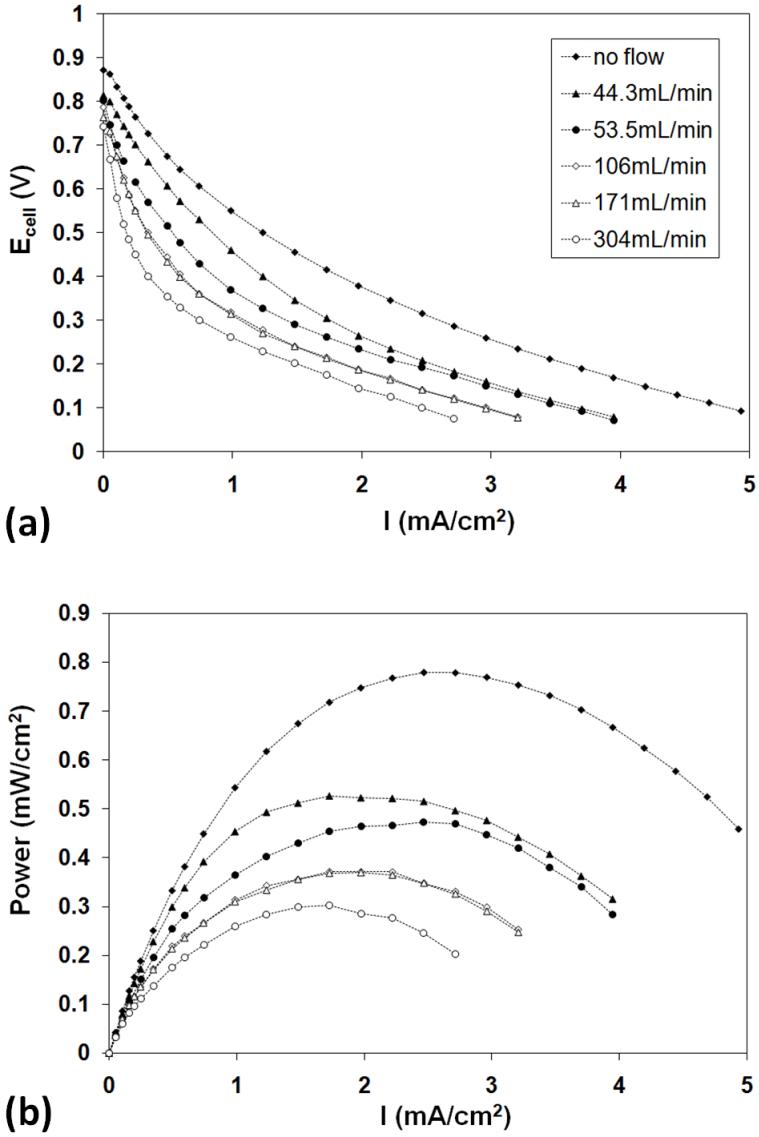


Figure 4.5: Current-voltage measurements and power density for electrolyte-supported tube cell at 1173 K, as CO_2 flow is varied.

surface, forming a bed roughly 10 cm high, with an OD of 2.5 cm and ID of 1.3cm. The fine particle size of MCMB caused observable bubbling and channeling of flow through the bed, thus reducing CO production. This is demonstrated by comparison of the measured and equilibrium CO mole fractions; for example, the measured mole fraction was 0.16 at 1173 K as compared to an equilibrium value of 0.97. Increased temperature is shown to increase the rate of the Boudouard reaction as expected from the previous discussion on gasification reaction rates. Approaches to increase the CO mole fraction exiting the DCFC fuel bed include reduction of the CO_2 flow rate, or use of a more reactive solid fuel. Shown in Figure 4.7 are I-V data for operation

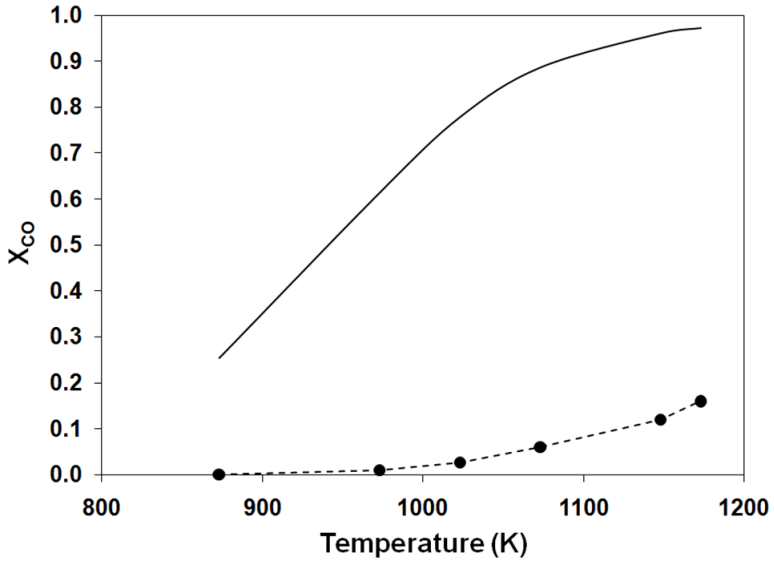


Figure 4.6: CO mole fraction at the MCMB bed exhaust (symbols), and equilibrium values (solid curve), as a function of bed temperature with an inlet CO_2 flowrate of 45 $mL{\cdot}min^{-1}$, without current flow.

on a variety of gaseous fuels, nonfuel gas species, and carbon. The ohmic resistance of the cell and leads ranged from 7-8 ohms as determined by the linear portions of the gaseous fuel I-V data, and remained roughly constant across the different fuels. Significant activation losses were observed for CO fueling, with the 1.6 % stream suffering from larger losses than the pure stream as expected. The data titled *100% CO_2* represents a flow of pure, 3.0 grade CO_2 without carbon present. The OCV for

this case is well below the others, approximately 200 mV, signifying that no chemical oxygen sink is present. The case without a direct chemical sink will be investigated further in the following section. In order to verify Faradaic behavior of the DCFC, gas

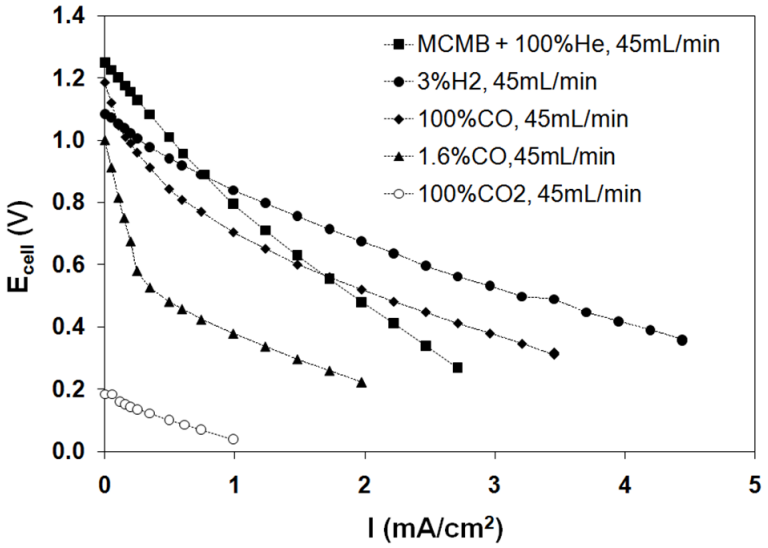


Figure 4.7: Current-voltage measurements for an electrolyte-supported tube cell at 1173 K, with various anode-side environments.

measurements were conducted of the anode chamber exhaust under current flow. The cell was maintained at 1173 K, with an air flow of 46 $mL{\cdot}min^{-1}$ at the cathode. Using MCMB fuel, a He flow of 119 $mL{\cdot}min^{-1}$ was introduced through the particle bed to convect product CO and CO_2 away from the anode chamber. This effluent stream was directed into the GC, and measurements were taken at current values of 10, 25, 30, and 35 mA, as shown by the closed symbols in Figure 4.8. This experiment was repeated for a He flow of 44 $mL{\cdot}min^{-1}$ and current values of 15, 25, 35, and 50 mA, as shown by the open symbols in Figure 4.8. The dashed line in this figure denotes a slope of unity (represented by the functional relationship given in Equation 4.1), indicating that all oxygen traversing the electrolyte (left hand side) leaves the anode chamber bound in CO or CO_2 (right hand side). The pressure and temperature are evaluated as 298 K and 1 atm, giving the molar flow for the volumetric He flow rate

$\dot{V}_{He}$.

$$\frac{I}{4F} = \frac{P}{RT}\dot{V}_{He}\left(\frac{X_{CO}}{2} + X_{CO2}\right) \tag{4.1}$$

The gas measurements agree with the current derived oxygen flux to within 6%,

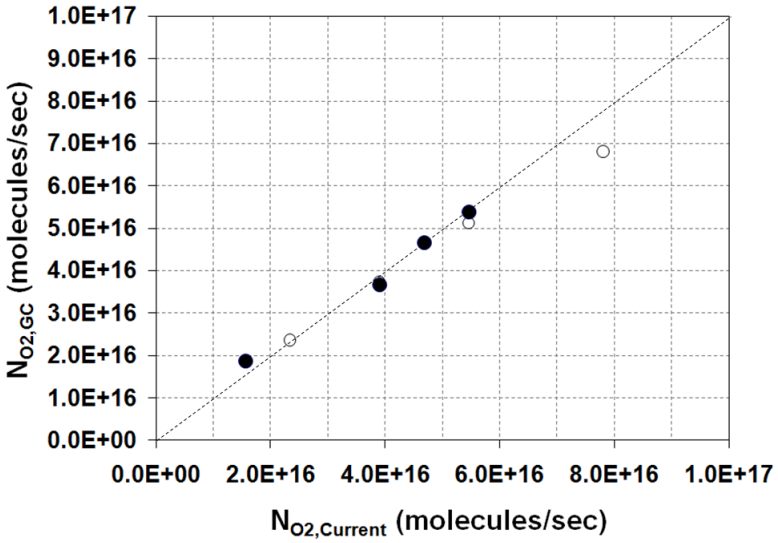


Figure 4.8: Oxygen balance for an ESTC at using MCMB fuel with He fluidization flows of 119 $mL{\cdot}min^{-1}$ (closed symbols) and 44 $mL{\cdot}min^{-1}$ (open symbols), and an air flow of 46 $mL{\cdot}min^{-1}$ at the cathode at 1173 K.

except at the highest current which is within 13%. This agreement between the measured data points and the theoretical oxygen balance suggests Faradaic behavior. The oxygen content of the effluent is thus directly related to the current, but the partitioning of oxygen between CO and CO_2 results from the chemical interactions of the product gases with the particle bed as the bed is traversed. The composition of the exhaust gas thus depends on transport and chemical processes along with bed geometry that are independent of the cell current.

4.3.2 SOFC Operation without Direct Fueling

Hydrocarbon fuels provide a chemical oxygen sink, maintaining low oxygen chemical potentials, with resulting OCV values of nominally 1 V. The oxygen chemical potential at the anode can also be maintained at low values by physical or mechanical means, such as purging with a nonfuel gas species, or pumping the anode chamber to high vacuum. The presence of solid carbon, not in direct contact with the TPB, presents similar complexities. Several experiments were conducted to investigate these possibilities, and their relevance to the DCFC. In the absence of fuel, a nonzero cell OCV can exist if a difference in oxygen chemical potential exists. Purging the anode side gas environment with a steady stream of CO_2 or N_2 results in OCV values of several hundred millivolts, consistent with oxygen impurity contents of 10 ppm or greater.

Another possible anode process, in addition to CO oxidation, is the direct desorption of oxygen at the anode TPD, and subsequent transport and reaction in the bed. Simple experiments were conducted to determine the feasibility of this mechanism without external biasing of the cell. The same experimental test stand was used, with modifications to the carbon bed location. The carbon bed (10 grams raw FAC) was located upstream of the anode, not in direct contact with the Pt surface. This geometry helps to mitigate chemical reactions of the anode products with the carbon bed. A He stream was forced through the carbon bed and, upon traversing the anodic surface, was analyzed in the GC. Shown in Figure 4.9 are the two configurations tested. These arrangements are identical except for the proximity of the carbon bed to the anode; configuration I (left) includes a bed-anode spacing of roughly 50 mm, while configuration II (right) increases this spacing to 250 mm and includes a necked region. The purpose of these arrangements is to determine under what conditions gaseous O_2 would elute from the anode. As such, the carrier gas should contain minimal oxygen or oxygen-containing species to prevent oxidation of the carbon to CO which would act as an oxygen sink. Initial experiments utilizing research grade He produced detectable and persisting amounts of CO and CO_2 leaving the carbon bed, suggesting a significant amount of oxygen impurity in the input He stream. In order to reduce the concentrations of oxygen-containing impurities, a Cu mesh maintained

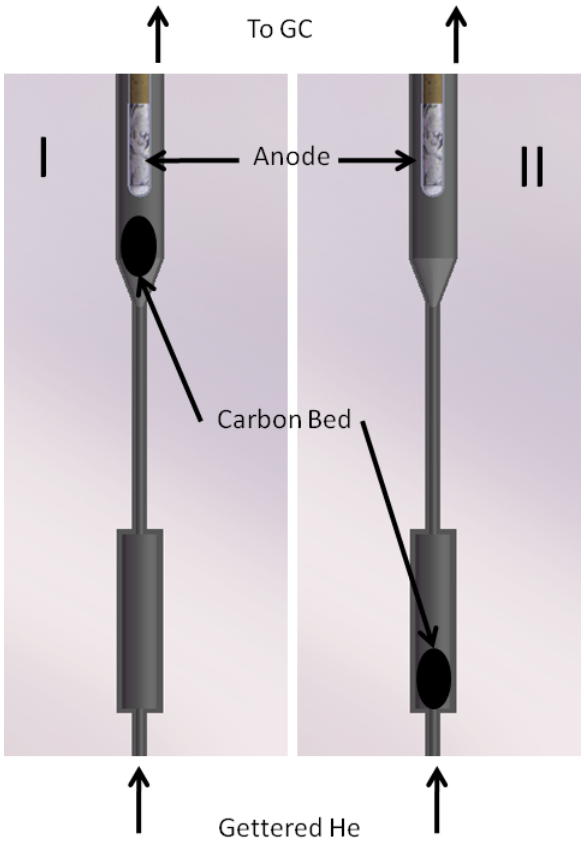


Figure 4.9: Experimental setup used for pure O_2 detection.

at nominally 500 K was introduced upstream of the carbon bed to clean the incoming stream of oxygen.

Under a He purge of 36 $mL{\cdot}min^{-1}$, GC measurements of the anode-side exhaust were taken at roughly 30 minute intervals. The GC traces for current values of 0 and 100 mA are shown in Figure 4.10. The three traces under open circuit conditions show relatively constant CO_2,O_2,N_2, and CO peaks consistent with steady operation. The

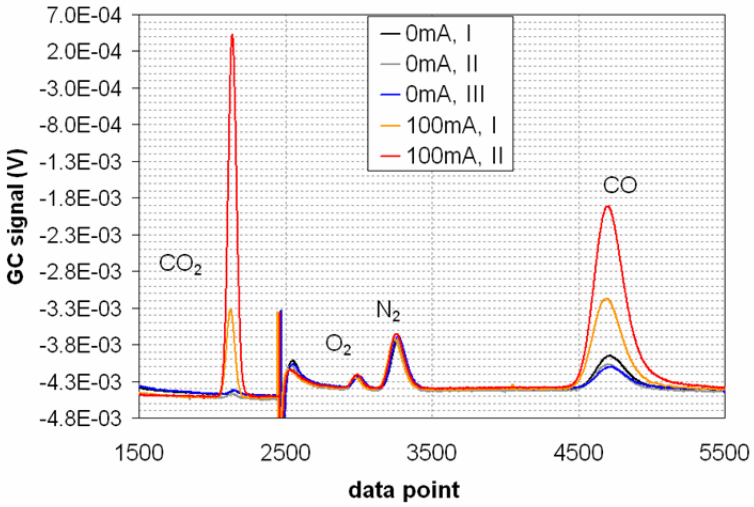


Figure 4.10: GC traces under configuration I, with He purge.

cell potentials for these traces are 1.080, 1.075, and 1.063 V, respectively, indicating low oxygen activities. The O_2 and N_2 are present from a small leak into the GC system from the environment, and are in the proportion of air. When 100 mA of current is allowed to pass, two sequential GC traces show an increase in both the CO_2 and CO peaks, with the O_2 peak remaining constant. The cell potentials for these traces are 0.300 and 0.180 V, respectively. This result implies that CO is likely converted to CO_2 at the anode, and the resulting CO_2 interacts with the carbon bed via the Boudouard reaction. The only mechanism for CO concentration to increase with current is the reaction of product CO_2 with the carbon bed. The residence time of gas between the carbon bed and anode surface due to convection can be estimated

by Equation 4.2, and is on the order of 10 s. The characteristic diffusion time for species diffusing between the anode surface and carbon bed can be estimated using Equation 4.3, also resulting in a time scale on the order of 10 s.

$$\tau_{res} = \frac{V_{void}}{\dot{V}_{anode}} \tag{4.2}$$

$$\tau_{diff} = \frac{L^2}{D_{eff}} \tag{4.3}$$

Axial diffusion upstream in configuration I is non-negligible as shown by the simple time scale estimation. Another drawback of the initial experiments using configuration I is the large background CO concentration, which prevents the appearance of gaseous O_2. The Cu getter was not significantly oxidized from visual inspection, and thus the CO likely did not originate from the He purge, but rather downstream as oxide complexes present on the carbon surface undergoing desorption. Further experiments, enlisting configuration II, addressed these issues by increasing the square of the bed-anode spacing by an order of magnitude, thus mitigating upstream diffusion, and increasing the purge time of He through the bed to reduce the concentration of adsorbed oxide complexes in the carbon bed. The necked region of the reactor between the particle bed and anode also reduces upstream diffusion. Shown in Figure 4.11 are GC traces taken under various current conditions. Note the reduced CO concentration, and achievable current values for this configuration. As the current is increased, CO concentration decreases and ultimately disappears from the GC trace, indicating that oxygen from the anode is not interacting with the carbon bed. The CO_2 peak increases with current, and eventually becomes constant. The O_2/N_2 ratio remains constant as current is increased up to 2 mA, and increases above 5 mA of current, as shown in Figure 4.12. The CO is completely consumed above 5 mA, and thus there is no chemical sink for additional oxygen. Although the presence of air in the background is undesirable, the N_2 peak's standard deviation is about 8%, indicating minimal drifting throughout the experiment. The cell potentials for the five galvanostatic traces are 0.917, 0.803, 0.547, 0.156, and 0.118 V, in order of increasing current. Due to the low fuel concentrations, the cell potentials fall rapidly with

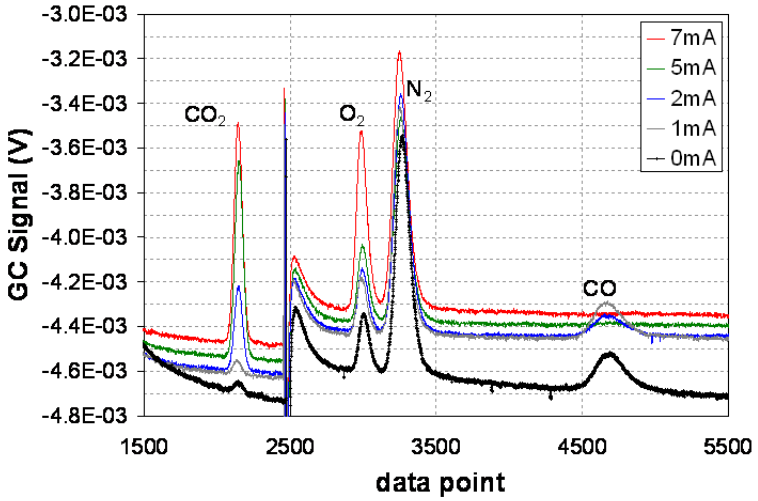


Figure 4.11: GC traces for configuration II for various cell currents, with He purge.

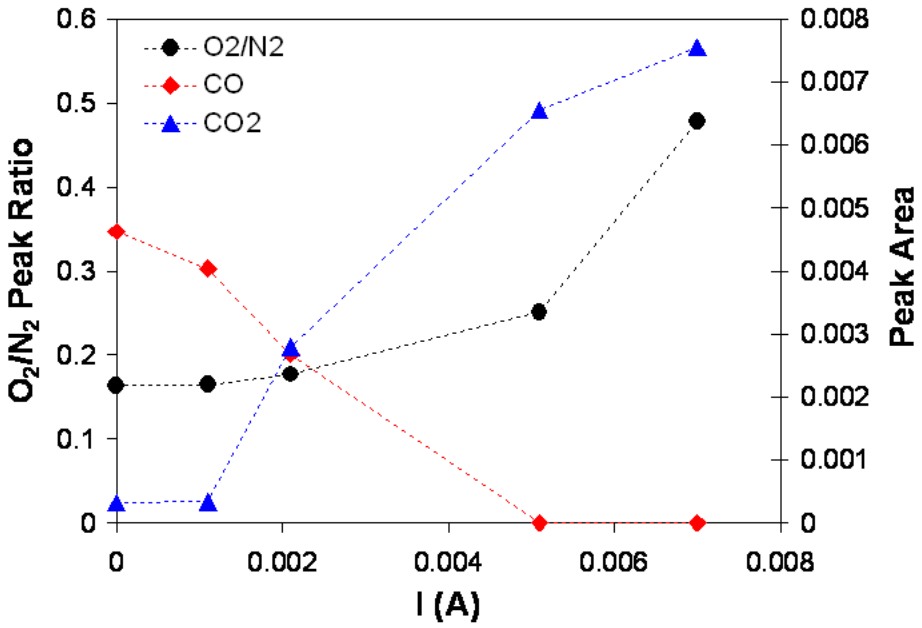


Figure 4.12: Selected peak areas plotted against current for configuration II.

current, even below 10 mA. The same cell was operated at currents of 100 mA when sufficient CO was present, recalling data from Figure 4.10.

The low cell potential during the desorption of gaseous O_2 can be explained by examining oxygen transport in the porous anode. If a steady current is imposed on the SOFC, in the complete absence of gaseous fuel species, O_2 may desorb from the anodic Pt surfaces. Fick's Law is given by Equation 4.4, approximately describing oxygen transport in the porous anode structure, neglecting convective fluxes.

$$J''_{O2} = -D_{eff}\nabla C_{O2} \approx -D_{eff}\frac{\Delta C_{O2}}{\delta} \tag{4.4}$$

Assuming that the anode is thin compared to its radius of curvature gives a linear result, in which the radial flux is constant across the electrode thickness. The concentration difference is evaluated between the anode/channel interface and the TPB interface, respectively and thus the length scale δ is the anode thickness. Upon algebraic rearrangement of Equation 4.4, the oxygen concentration at the TPB can be solved for (see Equation 4.5), with the diffusion resistance of oxygen through the electrode given by the first term. The concentration of oxygen at the bed-anode interface is dependent upon the chemical and physical interactions in the bed, but is assumed to be low ($C_{O2,bed} << 1$). As a best case scenario for performance, this quantity will be taken as zero.

$$C_{O2,TPB} = \frac{J''_{O2}\delta}{D_{eff}} + C_{O2,bed} \tag{4.5}$$

The reversible cell potential against air can be determined using Equation 4.6, neglecting diffusion losses across the cathode layer. Substituting the current for the oxygen molecular flux gives Equation 4.7.

$$E_{rev} = \frac{RT}{nF} ln\left(\frac{0.21\frac{P}{RT}}{\frac{J''_{O2}\delta}{D_{eff}}}\right) \tag{4.6}$$

$$E_{rev} = \frac{RT}{nF} ln\left(\frac{0.21nFD_{eff}P}{IRT\delta}\right) \tag{4.7}$$

The above expression can be evaluated, assuming the following numerical values;

- pressure of 101325 Pa
- temperature of 1173 K
- effective diffusivity of 1e-5 $m^2 \cdot s^{-1}$
- stoichiometric number (n) of 4 for O_2
- current density of 100 to 10000 $A \cdot m^{-2}$
- electrode thickness of 100 microns

giving cell potential values between 0.054 and 0.170 V. This potential is too low to be considered practical, and for Ni containing anodes would result in NiO formation. The diffusion resistance of the electrode is too great to maintain an effective oxygen sink, and even if the thickness were decreased to 10 microns, the cell potential would only increase to 0.228 V at a current of 100 $A \cdot m^{-2}$. The chemical oxygen sink must be brought into the porous anode and to the TPB to maintain useful oxygen concentrations. Direct desorption of O_2 within the anode, although possible, is thus not a practical possibility with the configurations considered.

4.4 Electrolyte-Supported Button Cells

In order to reduce the ohmic losses incurred by the electrolyte-supported tube cells, planar button cells (BC) were tested using the same experimental facility. Button cells (courtesy of Ceramatec, Inc.) consisting of a YSZ disk (25 mm diameter and 200-300 μm thick), with Ni/YSZ cermet anodes and LSMO cathodes were affixed to 25 mm OD alumina tubes using a glass seal or ceramic seal (SEM of BC cross-section provided in Figure 4.13). The electrodes, with imbedded Pt current collecting meshes, were approximately 100 μm thick. The planar button cell geometry, however, necessitates that the cell is not immersed in a particle bed. The carbon particles are rather loaded into the alumina tube, resting on top of the anode. This geometry does not allow bed fluidization (unless the particles are blown away from the anode) or plug flow of an inlet stream through the bed. As such, the flow-pattern used in the

anode chamber will be termed either no-flow (no inlet flow) or purging flow (nonzero inlet flow to counteract leakage into the system). Regardless, under nonzero current flow, there will be a net mass flow out of the anode chamber arising form the oxygen flux through the electrolyte. Current collectors, consisting of silver or platinum leads

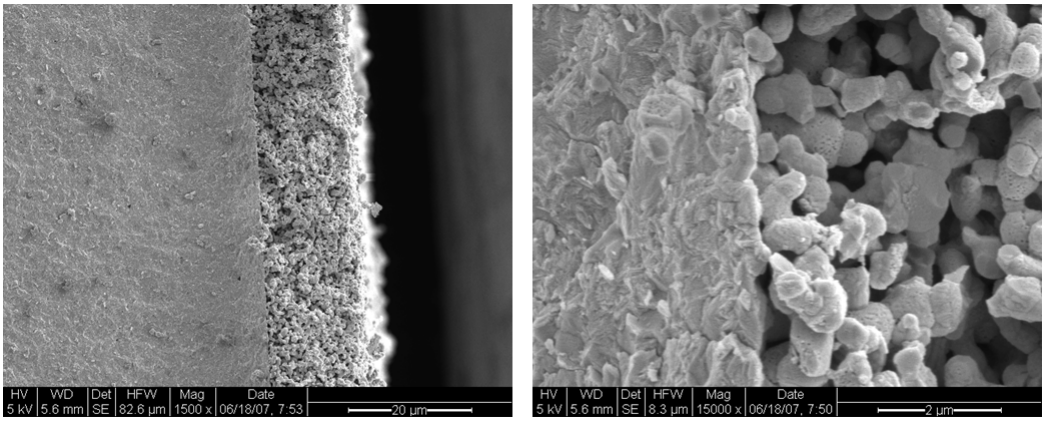


Figure 4.13: SEM image of button cell cross-section showing electrode-electrolyte (porous-dense) interface.

were used to measure and control the cell potential and current. Gas management was achieved using inserted inlet and outlet tubes (6.3 mm diameter), with a liquid seal at the exhaust. A schematic of the button cell assembly, including the temperature controlled furnace, is shown in Figure 4.14.

The Ni/YSZ-YSZ-LSMO BC was first tested using MCMB fuel directly at the anode. The cell cathodes were open to ambient air within the tube furnace, and a small purging flow of CO_2 was used in the anode chamber to prevent air leakage. The initial results show a peak power density of 140 $mW \cdot cm^{-2}$, with a nominal OCV of 0.87 V. This OCV is consistent with CO_2 addition into the bed, yielding CO/CO_2 ratios of O(1). Shown in Figure 4.15 are two sequential I-V curves and power curves. The cell performance degraded in time (on the order of an hour), likely due to electrode oxidation and delamination during operation at lower cell potentials. Both I-V curves exhibit a mass transport limitation at larger currents. Neither data set exhibits a large activation loss at lower current values. The linear portions of these

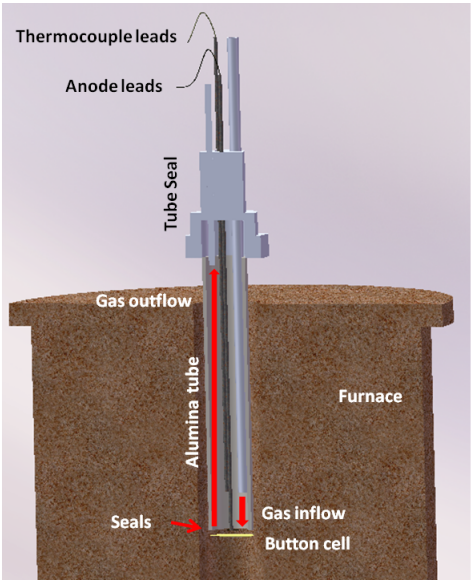


Figure 4.14: Schematic of button cell geometry used for DC measurements.

I-V curves yield ohmic resistances of 0.58 ohms, a significant reduction over the ESTC geometry. For comparative purposes, the BC was then tested in a variety of gaseous fuels, including pure CO, 1.6 % CO (N_2 balance), and 3 % H_2 (N_2 balance), as shown in Figure 4.16. The cell exhibits the largest power densities when fueled with pure CO (maximum of 163 $mW{\cdot}cm^{-2}$ at 0.543 V), as compared to fuels at lower concentration. The performance with pure CO is similar to that using MCMB fuel. Mass transport (fuel starvation) resistances are observed for these N_2 diluted fuels, consistent with the reduced concentrations available for electrochemical conversion. Since the I-V curve for pure CO does not exhibit mass transport limitation, the limitation likely does arise from transport within the cathode or anode, but rather the advected fuel supply. This then implies that the I-V data for MCMB is limited by the carbon bed at larger current values, characteristic of the low reactivity of MCMB fuel. The button cell geometry supplies a different geometry and relative positions of the anode and fuel bed than the ESTC. During galvanostatic operation, gas analysis was performed on the MCMB bed exhaust and again compared with the current. Shown in Figure 4.17 is the oxygen balance performed on the BC geometry at current values of 30 and

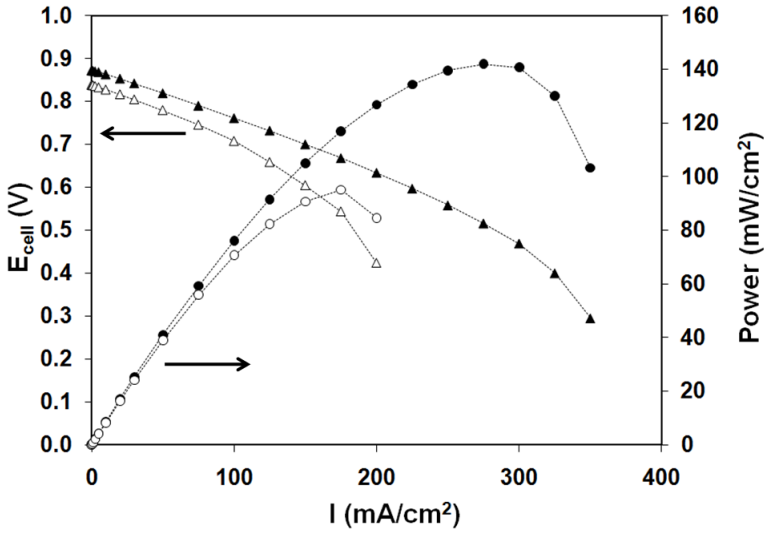


Figure 4.15: Current-voltage measurements (triangles) and cell power (circles) for a BC using MCMB fuel at 1173 K. Comparison of initial measurements (closed symbol) and measurements after about one hour (open symbols) show degradation.

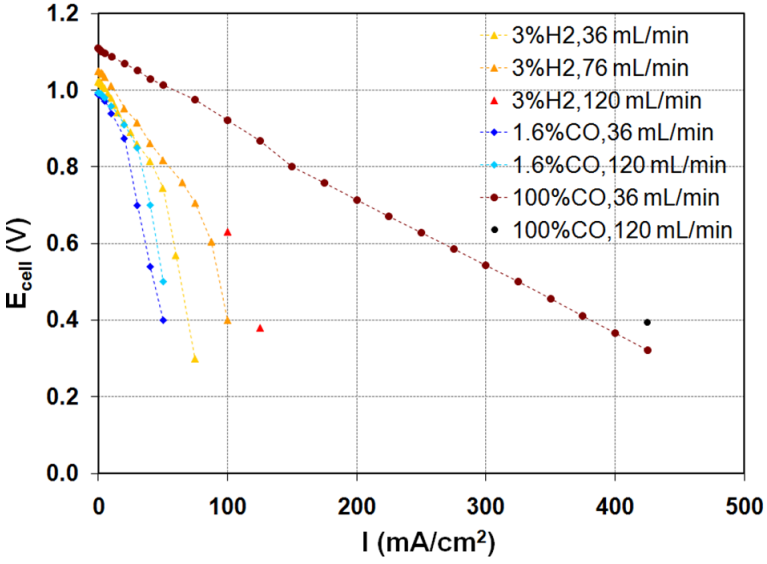


Figure 4.16: Current-voltage measurements for an electrolyte-supported button cell at 1173 K, with various gaseous fuels.

50 mA, with a purging He flow of 30 $mL \cdot min^{-1}$ at 1173 K. Agreement of the data

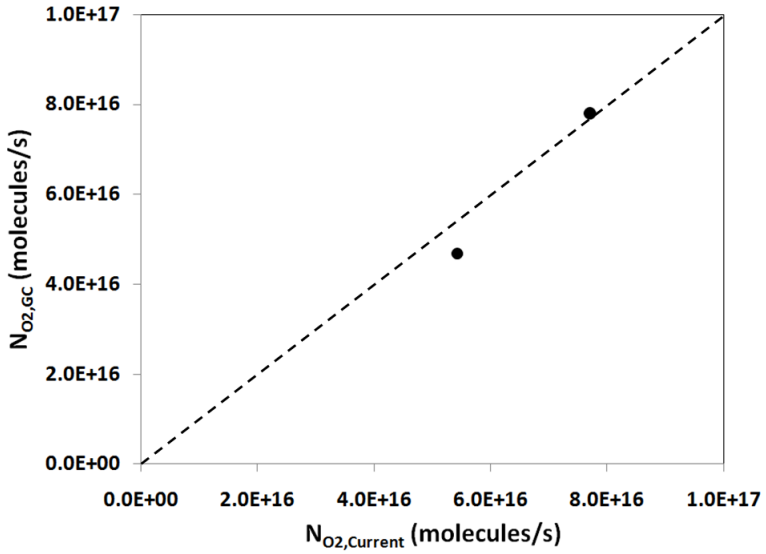


Figure 4.17: Oxygen balance for BC at currents of 30 and 50 mA, using MCMB with 30 $mL \cdot min^{-1}$ He flow at 1173 K.

with the theoretical oxygen balance (within 16 % at 30 mA and 2 % at 50 mA) again confirms Faradaic operation.

Shown in Figure 4.18 are DC and AC-DC coupled data taken with a BC geometry, using Pt electrodes and FAC particles as the fuel. The slope of the I-V data lies between 1.5 and 2 ohms, consistent between both the DC and AC-DC coupled measurements. The general shape of the impedance plot displays two semicircles, indicative of activation processes. The ohmic resistance of 0.5 ohms (1.0 $\Omega \cdot cm^2$ is large for this Pt electrode cell.

4.5 Anode-Supported Tube Cell

In addition to electrolyte-supported cells, an anode-supported tube cell (ASTC) was tested in order to further reduce both ohmic losses and the potential for gas leakage. These cells were provided by Materials and Systems Research, Inc. of Salt Lake

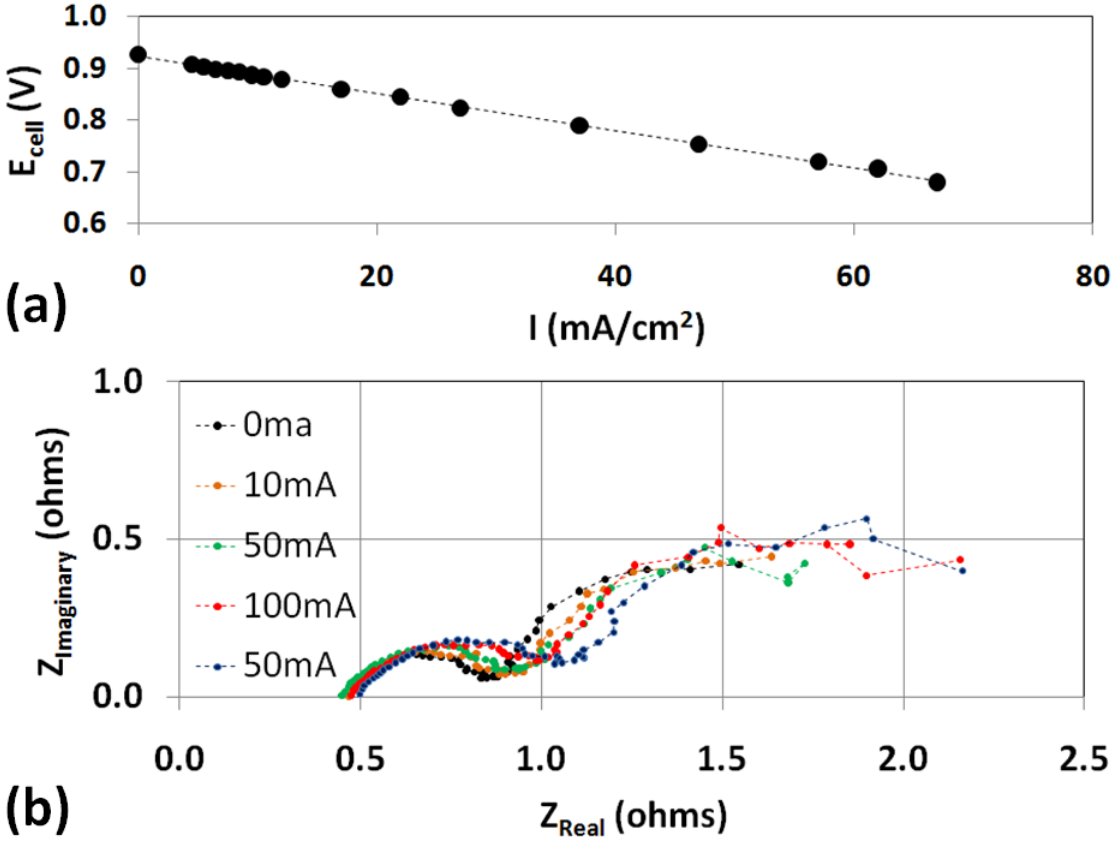


Figure 4.18: (a) DC and (b) AC-DC measurements for Pt/Pt button cell using FAC fuel at 1173 K.

City, UT. The cell consisted of a closed end, 12.7 mm OD Ni/YSZ cermet anode (about 800 μm thick), coated with a 10 μm thick YSZ electrolyte layer. A 100 μm thick layer of LSMO served as the cathode. The cathode had an active area of 5 cm^2, which was used to determine the current and power densities. The SOFC was bonded to a stainless steel tube for structural support and gas management. Four grams of raw, FAC were loaded into the tube interior completely covering the anode, extending approximately 10 cm downstream of the active anode region. A low rate of argon purge (15 $mL{\cdot}min^{-1}$) was introduced above the carbon bed to mitigate leakage. No change in cell potential was observed when the argon flow rate was changed or stopped. The entire DCFC assembly was placed inside a temperature-controlled tube

furnace. High-wattage resistors were used in series to serve as external load, and a DC power supply was used to compensate for the current-collecting lead resistance (roughly 0.1 ohms). The lowest resistive load condition, corresponding to the largest current density of 0.32 $A \cdot cm^{-2}$ was held for 40 minutes, with no noticeable change in current or voltage, indicating stable cell performance. The IV behavior shown in Figure 4.19 demonstrates that it is possible to achieve power densities of 220 $mW \cdot cm^{-2}$ at 0.68 V from this un-optimized cell at 1178 K. The theoretical OCV for a carbon bed at 1178 K is 1.1 V, giving a voltage efficiency of 62 % at this condition.

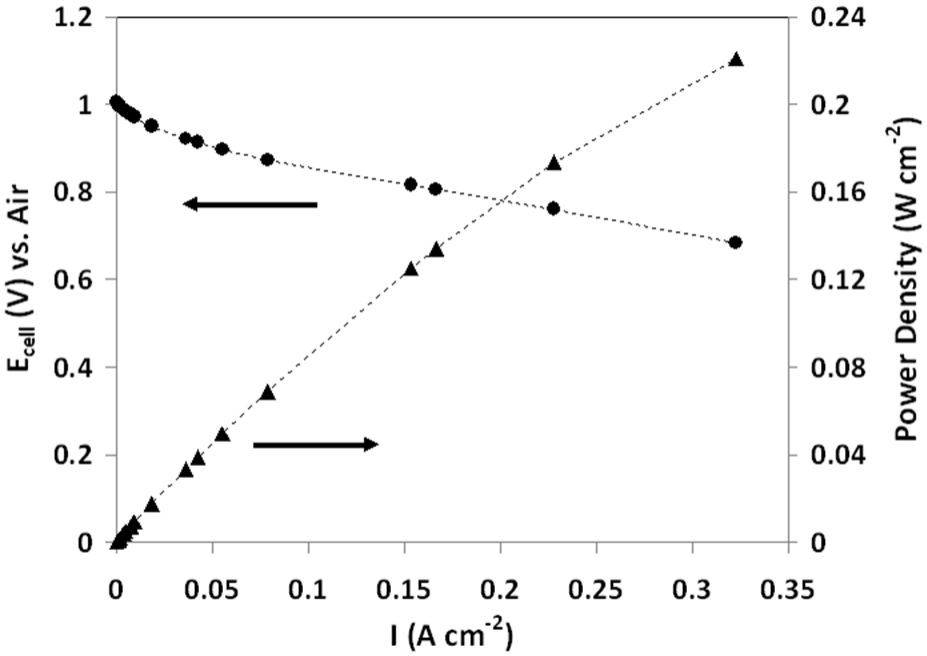


Figure 4.19: Current-voltage measurements taken with an ASTC using FAC fuel at 1178 K.

The ohmic portion of the V(I) curve represents a resistance of 0.15 ohms, or 0.78 $\Omega \cdot cm^2$. The presence of solid carbon causes the product gas CO_2 leaving the anodic surface to undergo chemical reactions in the bed. Thus any gas composition measurement downstream of the bed is dependent upon the extent of the electrochemical and chemical reactions. Gas chromatography measurements on the bed exhaust for this

configuration showed only CO, with CO_2 mole fractions below the detectable limit. This is in accordance with the CO/CO_2 ratio tending to equilibrium (roughly 36:1 at 1173 K) as the carbon bed is traversed.

4.6 Summary of Results and Discussion

4.7 Fuel Bed Modeling

The previous section contained experimental work concerning DCFC operation fueled by the Boudouard reaction. A model will be developed here to predict DCFC performance based on the fuel bed properties. The kinetic rate parameters determined from thermogravimetric analysis describe the rate of production of CO and CO_2, but a transport equation is required to describe these two concentration profiles throughout a spatial region. The amount of gas within the bed at any time is insignificant compared to the mass of carbon present. Stated differently, the carbon particles change over time scales much larger than the residence time of the gas within the bed. The SOFC in contact with a carbon bed is essentially an oxygen source, of strength directly proportional to current density, for the gas volume within the bed. A simple system can be constructed that serves to determine feasibility of the present DCFC configuration.

The velocity field and pressure field within a porous media can be described by Darcy's Law, as given by Equation 4.8 where k is the permeability of the media and μ_{mix} is the viscosity of the gas phase mixture [118]. A fixed particle bed is characteristic of porous media with fixed porosity and particle size. Systems under low flow conditions are approximated by the fixed bed configuration, below the minimum fluidization conditions.

$$\vec{u} = -\frac{k}{\mu_{mix}} \nabla P \tag{4.8}$$

The bed permeability (2.8e-11 m^2 for FAC) is estimated using the Blake-Kozeny equation [118] which is suitable in the low velocity regime, and is based on the mean carbon particle size d_p (155 μm) and the bed porosity ϵ_{fix} (0.4) as given by Equation

4.9.

$$k = \frac{d_p^2 \epsilon_{fix}^3}{150(1-\epsilon_{fix})^2} \tag{4.9}$$

The viscosity of each pure gas species i in the gas mixture is calculated using Equation 4.10, where k_B is Boltzmann's constant, T is temperature, m is mass per molecule, σ_i is the molecular length scale, and $\Omega_{\mu,i}$ is the collision integral (available as a function of temperature in [118]). The effective mixture viscosity is then determined using Chapman-Enskog theory, given by Equation 4.11 where Φ_{jk} is calculated using Equation 4.12 [118].

$$\mu_i = \frac{5}{16} \frac{\sqrt{\pi m_i k_B T}}{\pi \sigma_i^2 \Omega_{\mu,i}} \tag{4.10}$$

$$\mu_{mix} = \sum_{j=1}^{N} \left(\frac{X_j \mu_j}{\sum_{k=1}^{N} X_k \Phi_{jk}} \right) \tag{4.11}$$

$$\Phi_{jk} = \frac{\left(1 + \left(\frac{\mu_j}{\mu_k}\right)^{0.5} \left(\frac{MW_k}{MW_j}\right)^{0.25}\right)^2}{\sqrt{8}\left(1 + \frac{MW_j}{MW_k}\right)^{0.5}} \tag{4.12}$$

The velocity, $\vec{u}$, included in Darcy's Law is the superficial velocity defined per total, geometric cross section area (as opposed to the physical velocity in the voids), and derives from averaging the pressure drop due to viscous laminar flow within pores [119]. The time-independent continuity equation can then be used to determine the pressure and velocity fields, as given by 4.13.

$$\nabla \cdot (\rho \vec{u}) = \dot{S}_{gas} \tag{4.13}$$

The term $\dot{S}_{gas}$ in Equation 4.13 is the net production of gas phase matter due to chemical reaction, in units of $kg \cdot m^{-3} \cdot s^{-1}$.

A species transport equation can be written for each gas phase species i using the velocity defined by Darcy's Law, as shown in Equation 4.14. The divergence of the species flux is equal to the volumetric source of that species, in units of $mole \cdot m^{-3} \cdot s^{-1}$. The active surface area to volume ratio, a_v, and the effective diffusivity, $D_{i,e}$, contain information about the carbon particles in the bed. The effective diffusivity of species

i in porous media can be related to its bulk diffusivity as shown in Equation 4.15, where ϵ_{fix} is bed porosity and τ is the bed tortuosity. The bulk diffusivity is given by Equation 4.16, simplified for a binary mixture where R is the gas constant, C is total gas concentration, N_{av} is Avogadro's number, σ_{ij} is an effective molecular length scale, and $\Omega_{D,ij}$ is the collision integral for diffusion (with numerical values available in [118]). Since the mole fractions of the gaseous species sum to one, only N-1 transport equations must be solved.

$$\nabla \cdot (C_i \vec{u} - D_{i,e} \nabla C_i) = a_v \sum_{j=1}^{N_R} \dot{R}_{ij} \tag{4.14}$$

$$D_{i,e} = \frac{\epsilon}{\tau} D_{ij} \tag{4.15}$$

$$D_{ij} = \frac{3}{16} \sqrt{\frac{2RT}{\pi} \left(\frac{1}{MW_i} + \frac{1}{MW_j} \right)} \frac{1}{C N_{av} \sigma_{ij}^2 \Omega_{D,ij}} \tag{4.16}$$

Since surface species s, do not require a transport equation, the net production of these species (such as θ_O for example) in steady state must be zero as shown in Equation 4.17. This gives the same functional result as imposing the quasi-steady state assumption for these surface species. Again, only N-1 surface species equations must be solved, since in steady state the total surface species are fixed by the site density.

$$0 = \sum_{j=1}^{N_R} \dot{R}_{sj} \tag{4.17}$$

The concentration at any spatial location in the bed can also be assumed to apply to the interior of particles if reactions proceed at kinetically limited rates. To assure that the heterogeneous reactions are within the kinetically limited regime, the Thiele modulus ϕ_s of the carbon particles are estimated [120]. The Thiele modulus, given by Equation 4.18, provides a quantitative comparison of the relative time scales of pore diffusion and chemical reaction within a particle.

$$\phi_s = r_p \sqrt{\frac{R_{CO2} S_g \rho_p}{[CO_2] \varepsilon_p D_{eff}}} \tag{4.18}$$

In this formulation, R_{CO2} represents the net production of CO_2 within the particle ($mol{\cdot}m^{-2}{\cdot}s^{-1}$), r_p is the mean particle radius, ρ_p/ε_p is the mass of carbon per volume of gas within a particle ($g{\cdot}m^{-3}_{fluid}$), $[CO_2]_0$ is the concentration in the bulk gas phase outside of the particle ($mol{\cdot}m^{-3}$), and D_{eff} is the effective pore diffusivity within the particle ($m^2 \cdot s^{-1}$). For the 100 to 125 μm diameter activated carbon particles, the effective pore diffusivity (Knudsen and continuum diffusion) is estimated to be in the range of 1e-7 to 1e-6 $m^2 \cdot s^{-1}$ at temperatures of interest. For the conditions tested in the PTGA, the Thiele modulus ranged from about 0.05 to 0.6. Values of $\phi_s < 1$ suggest that reactions for the gas species occur under kinetically limited conditions.

4.8 Model Discussion and Results

The annular, axisymmetric geometry used for the fuel bed modeling is shown in Figure 4.20. The boundary conditions for this annular flow system depend upon the fluxes at the boundaries. At the inlet (z = 0), the CO_2 concentration and inlet velocity are specified from the assigned inflow. At the outlet (z = L), the pressure is fixed to 1 atm. The diffusive fluxes are assumed to be negligible on this boundary, with convection being the only mode of transport. The outer bed wall ($r = R_o$) is assumed to be impermeable to all species, with all molar fluxes set to zero. The inner bed wall ($r = R_i$) boundary condition depends upon the configuration modeled. The inner radial boundary condition is changed from the insulation constraint of the thermocouple to a specified flux condition to simulate fuel cell conditions. The inner bed radius is also increased from 1.6 mm (thermocouple) to 6.4 mm which is typical of fuel cells used in this study. Along the anodic surface, beyond 2.54 cm impermeable entry region, the atomic oxygen flux is now related to the current density via the relation $I'' = nFJ''_O$, where n = 2.

In order to verify that the transport model captures the particle bed behavior, a thermocouple ($R_i = 1.5$ mm) was placed on the centerline of a 25 mm OD bed of FAC particles. The height and temperature of the bed were varied, and GC measurements were conducted of the bed exhaust. The transport model was implemented using COMSOL, a multi-physics finite element software package. Model output for both

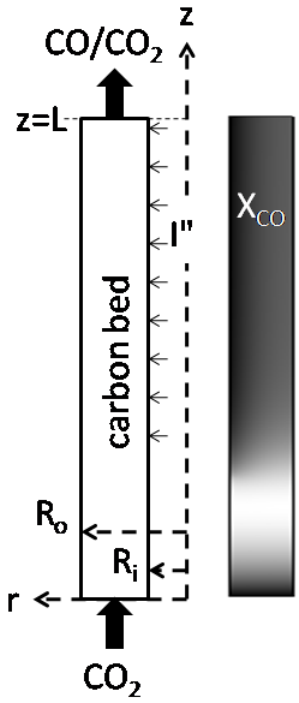


Figure 4.20: Split view showing carbon bed geometry and characteristic COMSOL output.

CO and CO_2 mole fractions are shown in Figure 4.21, indicating good agreement with experimental carbon bed data. The dashed lines correspond to the fixed bed condition, and the thin solid lines correspond to the minimum fluidization condition. The fixed and minimally fluidized configurations give almost identical results, signifying that the effects of bed expansion are not significant for this system. The inlet flow rates ranged from 7.6 to 14.2 $mL{\cdot}min^{-1}$ (maintaining a 15 $mL{\cdot}min^{-1}$ effluent flow rate), as calculated from a steady-state oxygen balance. The inlet velocity ranges from 0.1 to 0.2 $cm{\cdot}s^{-1}$, while the minimum fluidization velocity is estimated to range from 0.05 to 0.2 $cm{\cdot}s^{-1}$ based on Ergun's equation [118, 121]. By observation, the bed is below the bubbling regime (bubbling was observed above about 1.0 $cm{\cdot}s^{-1}$); there

is no large-scale movement of the solid particles in the bed although some agitation is observed.

At each temperature, the CO concentration increases with bed height, as expected. At the highest gas temperature employed, equilibrium values are reached at a bed height of 7.6 cm. The model agrees with the experimental bed data, indicating that bed dynamics are adequately captured. The kinetic model does slightly overpredict the CO mole fraction at temperatures near 1273 K, as shown in Figure 4.22 using the same data as included in Figure 4.21. The experimentally verified transport model can now be employed for predictive calculations with an imbedded fuel cell. A full simulation using the transport model determines the spatial mole fraction (see Figure 4.23) and pressure scalar fields. At 1173 K and 10 $mLmin^{-1}$ of CO_2 inlet flow, the CO mole fraction increases with entry length, then decreases as the active anode region is traversed under current flow. This result illustrates the importance of geometry when discussing the DCFC as opposed to purely fuel-convection driven cells.

This large amount of scalar field data can be reduced by calculation of the mixed mean CO mole fraction exiting the particle bed, under various temperature and current density conditions, as given in Figure 4.24. The mixed mean mole fraction of species i is calculated using Equation 4.19 from the scalar field values, with the pressure fixed as a constant 1 atm across the entire exit plane (indicative of atmospheric operation). The mixed mean value represents the mole fraction if the bed exhaust mixes completely, eliminating radial gradients in concentration, and is the appropriate bulk value for the composition of reactor exhaust.

$$X_{i,mm} = \frac{\int_{Ri}^{Ro} u(r) X_i(r) 2\pi r dr}{\int_{Ri}^{Ro} u(r) 2\pi r dr} \tag{4.19}$$

The results shown in Figure 4.24 indicate that operation at larger temperatures allows for greater current densities. Also, maintaining high CO mole fractions requires higher temperatures and limited current densities. Adjustment of the cell geometry could also enhance performance, such as limiting the active anode area relative to the carbon bed.

Figures 4.25 and 4.26 include model results predicting achievable current densities

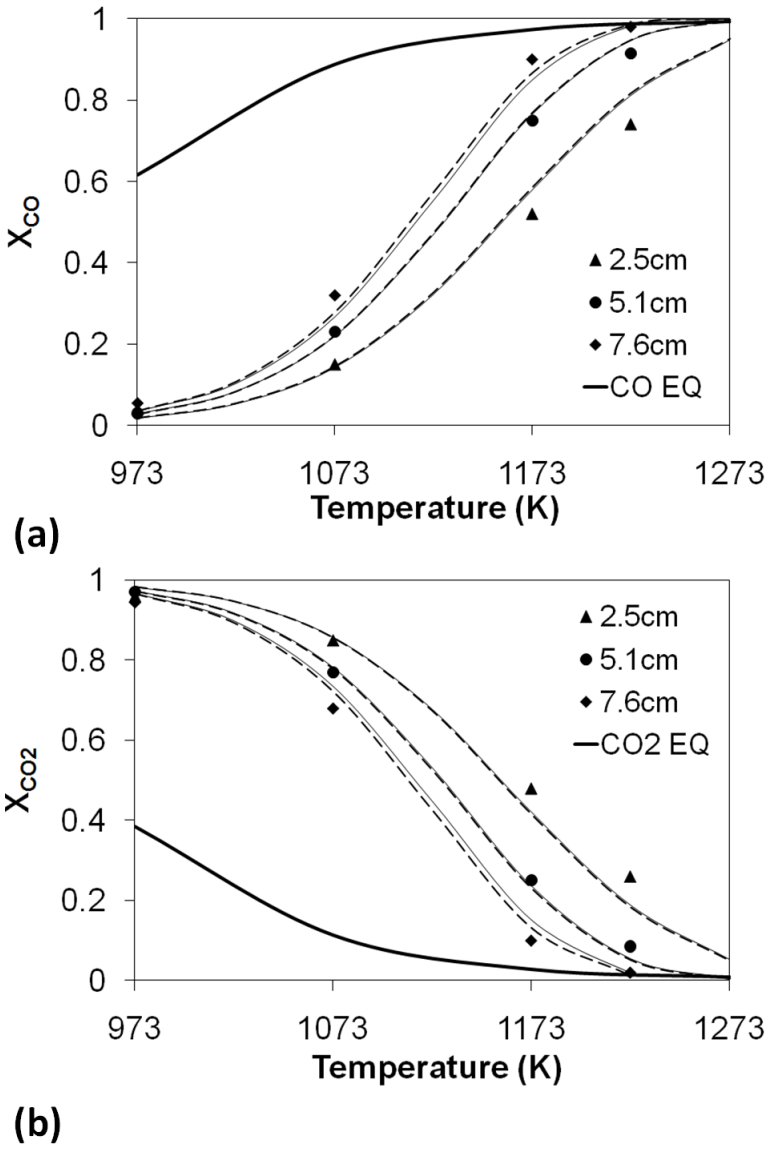


Figure 4.21: Measured and calculated mole fractions of (a) CO and (b) CO_2.

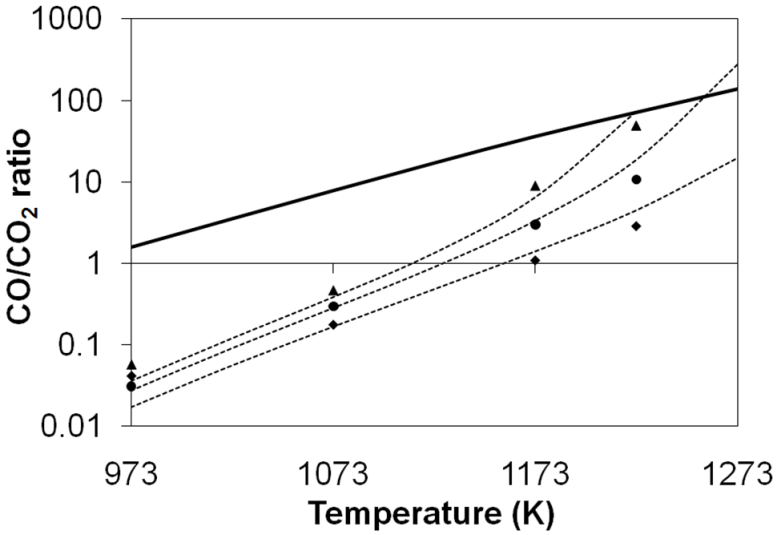


Figure 4.22: Measured and calculated CO/CO_2 ratio as a function of temperatures at bed heights of 2.5 cm (bottom curve), 5.1 cm (middle curve), and 7.6 cm (top curve). Again, the equilibrium CO/CO_2 ratio is given by the solid black curve.

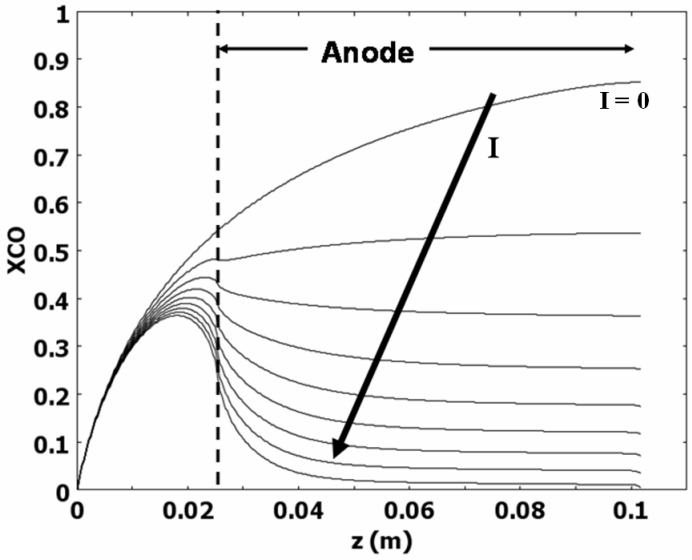


Figure 4.23: Calculated CO mole fraction field in the particle bed at 1173 K and 10 $mL{\cdot}min^{-1}$, for current densities of 0 to 800 $mA{\cdot}cm^{-2}$ in 100 $mA{\cdot}cm^{-2}$ intervals.

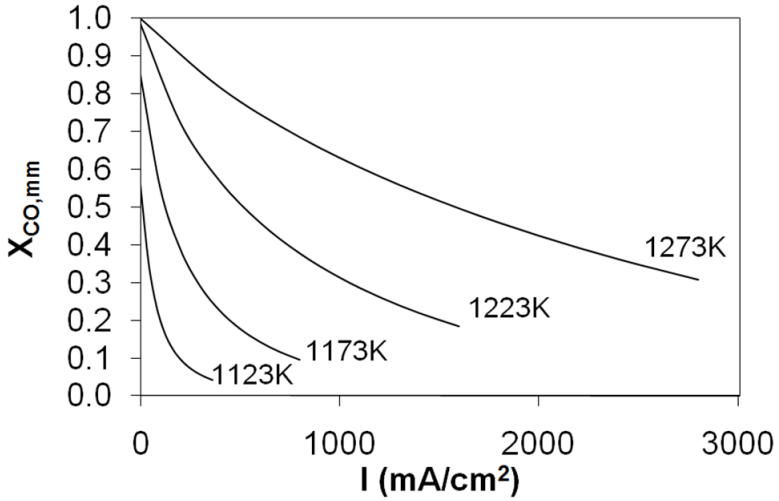


Figure 4.24: Calculated mixed mean CO mole fraction exiting the particle bed at various temperatures and current densities.

parameterized by CO terminal mole fraction, temperature and inlet flowrate. The terminal CO mole fraction is defined here as the mole fraction at the exit plane directly at the anode surface. This location represents the minimum CO mole fraction along the active SOFC area, and thus lends a unique quantification. The corresponding current density is then termed the *terminal current density.*

The dashed lines in Figure 4.25 represent the range of practical interest in current density. At 1173 K, current densities above 100 $mA{\cdot}cm^{-2}$ are achieved while maintaining an equimolar mixture of CO/CO_2. Temperature strongly effects the terminal current density, as expected since CO production occurs under kinetically limited conditions. The impact of inlet CO_2 flow rate on the terminal current density at 1173 K is shown in Figure 4.26. Terminal current density is not a strong function of inlet CO_2 flow rate except at terminal CO values above about 0.5, thus low inlet CO_2 flow rates are required if high CO mole fractions are desired. Large CO mole fractions are beneficial for large cell potentials and power densities. There is a maximum in terminal current density at zero inlet flow rate for all cases and a monotonic decrease

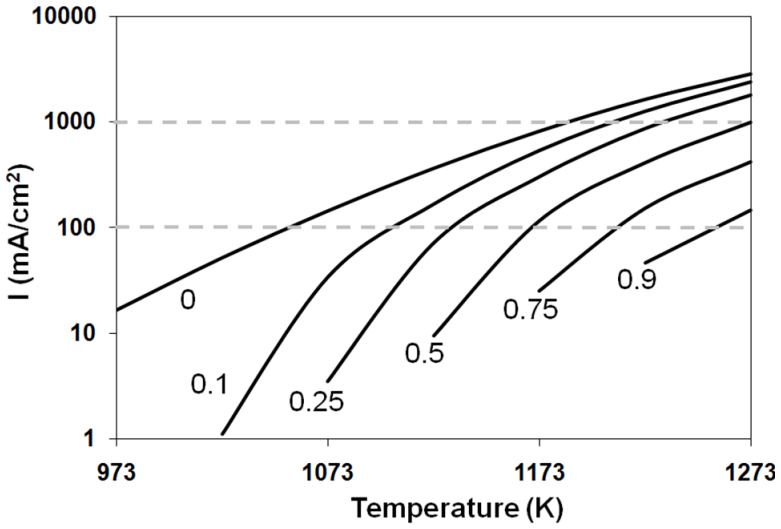


Figure 4.25: Calculated limiting current density vs. temperature parameterized by terminal CO mole fraction.

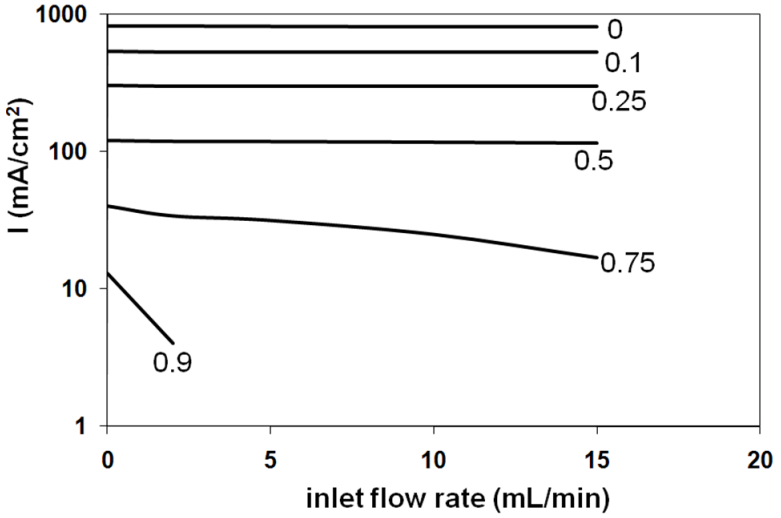


Figure 4.26: Calculated limiting current density vs. inlet flowrate, parameterized by terminal CO mole fraction, at 1173.

in terminal current density as inlet flow rate is increased. In general, higher bed temperatures further reduce the effect of flow rate, whereas lower bed temperatures show increased sensitivity due to kinetic limitations.

These results imply that the use of activated carbon or a more reactive fuel, and optimized DCFC geometry, can provide current densities in the range of practical interest.

Chapter 5

Steam-Carbon Cell System Analysis

Distributed hydrogen production is one solution considered for next generation low-temperature fuel cell based transportation. The steam-carbon cell (SCC), based on the DCFC configuration, allows for clean hydrogen production. Conventional hydrogen production approaches rely on steam reforming of hydrocarbon, and catalysis of the water gas shift reaction to create an H_2 rich stream. Separation is required to remove carbon containing species, especially CO which can poison catalysts at low temperature.

5.1 Approaches to Electrolysis

The production of H_2 gas via electrolysis is generally the method considered to create clean hydrogen. Thermolysis, or the thermal splitting of steam and subsequent separation of the resulting H_2 and O_2, is another approach. This latter method was shown however, to be less efficient and much more equipment-demanding than the former due to the high temperatures and associated energy transfers required. As such thermolysis is not considered here as a practical alternative [122]. In considering water electrolysis, there still remains great variation in the specific approach.

Low-temperature electrolysis, utilizing a proton exchange membrane, is a widely researched approach for distributed generation and has been demonstrated at current densities of well over 1 $A \cdot cm^{-2}$ [123]. This approach relies on large work inputs to electrolyze water, and typically operate above 1.5 V applied. The standard potential at 298 K is roughly 1.23 V for hydrogen conversion to water, with higher applied potentials representing additional work input to overcome losses. The primary drawback to this approach is the electrical work input, which is generally created using heat engines or chemical engines with efficiencies ranging from 30 - 40%. Thus the overall efficiency of the electrolysis process is bounded by this value. The combination of inefficient work input and high applied potentials limit PEM electrolyzer performance.

The work input can be reduced by increasing the operating temperature (thus lowering the required work input and activation losses), readily realized using the SOFC configuration. This approach was investigated for commercialization by Dornier System GmbH during the 1980s [124, 125]. The process, termed HOTElly (High Operating Temperature Electrolysis), employed YSZ as the electrolyte and was investigated at temperatures near 1273 K. Due in part to work input from inefficient centralized plants, the overall efficiency of the HOTElly process was estimated at 40 - 50% when combined with the heat engine power plant, well above the 25 - 28% efficiencies estimated for conventional means. The efficient use of work and heat outputs from power plants can allow for more efficient hydrogen production, allowing use of both the work and heat output of the plant. Another study comparing the greenhouse gas emissions and acidification potential of different H_2 production techniques found high temperature electrolysis to compete favorably [126]. The specific process entailed using a coupled nuclear power plant to supply heat and work to the solid oxide-based electrolyzer. The only technologies that had lower environmental impacts were hydroelectric and wind coupled electrolysis systems, both with energy densities far below those for fossil fuels. Steam reforming was found to emit large amounts of greenhouse gases per kg of H_2.

The previously discussed electrolysis approaches involve coupling a thermal power

plant and electrolyzer, with varying degrees of thermodynamic intimacy. If this approach is taken a step further towards integration, the power plant can be removed and the hydrocarbon fuel can be used to electrolyze water in a single process step rather than intermediate work production. This approach has been considered using natural gas and a solid oxide electrolyte, yielding an efficiency of 70% [127], and also using carbon in a solid oxide electrolyte based electrolyzer with a molten Ag anode [65]. It is this coupled approach, using a solid state architecture and free particle bed, that will be analyzed here.

5.2 Thermodynamic Analysis of the SCC System

In addition to DCFC operation, the reducing atmosphere in the presence of solid carbon can be used to remove oxygen from H_2O without physical mixing, creating gaseous H_2. If the air present at the DCFC cathode were replaced with steam, the SCC system results. The fundamental operation of the SCC is similar to the DCFC, and is thus a natural extension of the previous analysis. The hydrogen stream in this arrangement is completely separated from carbon-containing species by an oxide ion conducting membrane, thus no gas cleanup is required on the hydrogen side other than initial water treatment.

5.2.1 Thermodynamic Basis for the SCC

The DCFC arrangement has been shown to be an efficient means of extracting work from carbon. Consider the simple systems shown in Figure 5.1, for producing clean, unmixed H_2. In this figure, A represents a fuel while D and E represent partial or complete fuel products upon oxygen addition, with the molar coefficients given by a, b, c, d, and e. These represent chemical/electrochemical systems that remove oxygen from steam using a fuel or work input. These arrangements differ considerably from steam reformation in that the H_2 stream is carbon-free. The WGS reaction has essentially been partitioned between H_2/H_2O and CO/CO_2, allowing for the use of carbon to maintain a low oxygen activity sink at the anode. If the systems shown

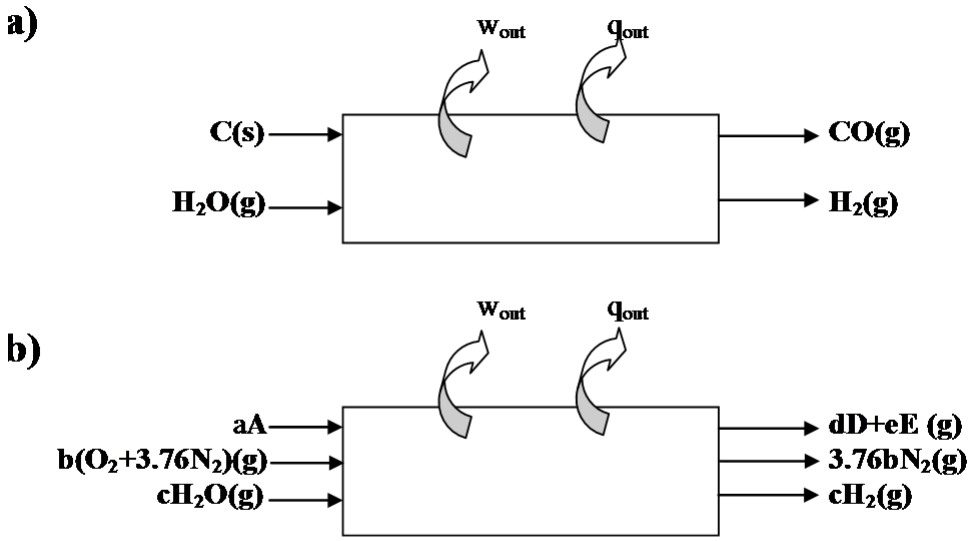


Figure 5.1: Schematic of (a) a simple carbon-assisted hydrogen producing system, and (b) a generalized hydrogen production system.

in Figure 5.1 are assumed to operate either reversibly, or without explicit work input then the heat and work interactions can be uniquely determined. Shown in Table 5.1 are the species and molar coefficients for several hydrogen production schemes. The variables s_i and h_i refer to the stream entropies and enthalpies for the input and output streams in units of $kJ \cdot mol^{-1}$ per the given reaction. Table 5.2 contains the thermodynamic results for the systems considered. The work (w_{out}) and heat (q_{out}) interactions for each arrangement are tabulated (in units of $kJ \cdot mol^{-1}$) for reversible operation (zero entropy generation). This represents the best performance possible although work interaction may be required. The variables HHV_{flue} and Ψ_{flue} refers to the higher heating value and stream exergy of the flue gas (non-H_2) stream. The exergy is based on a restricted dead state of a separate CO_2 stream at 298 K and 1atm, a separate H_2O(L) stream at 298 K and 1atm, or an O_2 stream at 298 K and 0.21 atm.

Table 5.1: Simple hydrogen-producing systems considered, as shown in Figure 5.1b. Stream entropies (s) and enthapies (h) are given in units of $kJ \cdot mol^{-1}$.

case	A	a	b	c	d	e	D	E	s_{in}	s_{out}	h_{in}	h_{out}
1	C(s)	1	0	1	1	0	CO	/	75.7	328.3	-285.8	-110.5
2	C(s)	1	0	2	1	0	CO_2	/	145.6	475.2	-571.7	-393.5
3	C(s)	1	0.5	1	1	0	CO_2	/	548.7	704.7	-285.8	-393.5
4	CO	1	0	1	1	0	CO_2	/	267.6	344.5	-396.4	-393.5
5	/	0	0	1	0.5	0	O_2	/	70.0	233.3	-285.8	0
6	CH_4	1	0	1	1	2	CO	H_2	256.2	605.6	-360.7	-110.5
7	CH_4	1	1	2	1	2	CO_2	H_2O	1272.1	1335.5	-646.5	-965.2

Table 5.2: Heat (q) and work (w) ouputs of the reversible hydrogen-producing systems as shown in Figure 5.1b, along with the higher heating value (HHV_{flue}) and stream exergy (Ψ_{flue}) of the non-H_2) streams, are given in units of $kJ \cdot mol^{-1}$.

case	A	a	b	c	d	e	D	E	$q_{out,rev}$	$w_{out,rev}$	HHV_{flue}	Ψ_{flue}
1	C(s)	1	0	1	1	0	CO	/	-75.3	-100.0	283.0	254.2
2	C(s)	1	0	2	1	0	CO_2	/	-98.2	-79.9	0.0	0.0
3	C(s)	1	0.5	1	1	0	CO_2	/	-46.5	154.2	0.0	0.0
4	CO	1	0	1	1	0	CO_2	/	-22.9	20.1	0.0	0.0
5	/	0	0	1	0.5	0	O_2	/	-48.7	-237.2	0.0	3.9
6	CH_4	1	0	1	1	2	CO	H_2	-104.1	-146.1	854.7	717.7
7	CH_4	1	1	2	1	2	CO_2	H_2O	-18.9	337.5	0.0	0.0

There are several main conclusions that can be drawn from this simple analysis. Operation without external oxygen (air) input, requires a net heat and/or work input. Stated differently, a standalone SCC plant without external heat or work interaction would need to utilize some fraction of the heating value and/or exergy of the flue gas stream by using environmental air. This process of oxygen addition could take the form of an SOFC fueled by the flue gas. The flue gas heating value and exergy of *case 1* in Table 5.2 demonstrate that the electrolysis heat and work requirements can be met by utilization of this stream.

Also, it can be seen that the use of methane as the oxygen sink necessitates that the hydrogen originating in this stream remains mixed with carbon-containing species. The standalone methane plant (*case 7*) provides similar performance to the standalone SCC system (*case 3*), considering that methane has an HHV of more than

twice that for carbon (890.3 vs. 393.5 $kJ \cdot mol^{-1}$).

Another observation is that the effective water-gas shift WGS reaction (*case 4*) is practically stand-alone, as the work output and heat input requirements are roughly the same. This is due to the low ΔG^0 and ΔH_R for the WGS reaction. The performance of the WGS system (*case 4*) and the standalone CASE system (*case 3*) are similar in that both have a net work output and heat input, although the use of C(s) as the fuel allows for more oxygen transfer per mole and hence more work output.

5.2.2 Cell Potentials for the SCC

Shown in Figure 5.2 are the OCV values for a fixed carbon bed in equilibrium (both C-O and C-H-O), and streams of varying H_2O/H_2 ratio, all referenced to air. The absolute OCV values are arbitrary in this discussion, as the difference between the curves gives the driving force for hydrogen production. This difference increases with temperature, assuming that the carbon bed is maintained near equilibrium with excess carbon. The standard potential curves for both CO and H_2 conversion differ by several hundred millivolts at temperatures of interest, indicating that oxygen transfer could be accomplished with minimal work input for this system.

Tube cells (ESTC) were used to verify the OCV characteristics of the steam-carbon cell. As discussed previously, PSZ tubes were used as the SOFC electrolyte and for structural support. Although the thick PSZ tube has poor oxide ion conductivity, it is well suited for OCV measurements. Both the cathode and anode consisted of porous Pt, to ensure chemical inert character at experimental conditions. Electrodes were applied in identical fashion to those used for the DCFC experiments discussed in Chapter 2. Carbon particles were added to the tube interior (anode side), forming a fixed fuel bed. The exterior the PSZ tube (cathode side) was encased in a quartz housing for gas management and control of the reactive environment, as shown in Figure 5.3. This arrangement allowed for independent control of the anode and cathode chambers, and mitigated the appearance of liquid water within the reactor. The cathode side gas mixture is created by flowing metered water (Eldex HPLC pump) into the base of the quartz reactor. The liquid water boils at a fixed location,

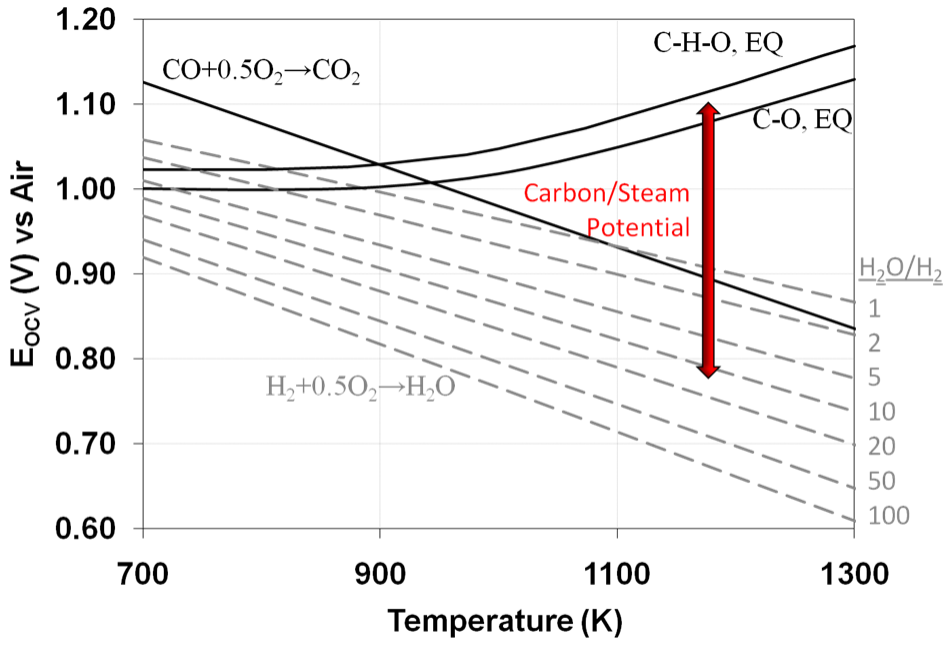


Figure 5.2: Theoretical C-O, C-H-O (H/O = 2/1), and H-O OCV curves

roughly 1 cm from the base of the furnace, and cathode-side gaseous diluents are injected above this phase-change point in order to enhance mixing. Either pure H_2 or a mixture of 3% H_2 in N_2 balance is used as a steam diluent, and to actively fix the H_2/H_2O ratio. The cell is heated initially with ambient air at the cathode, and

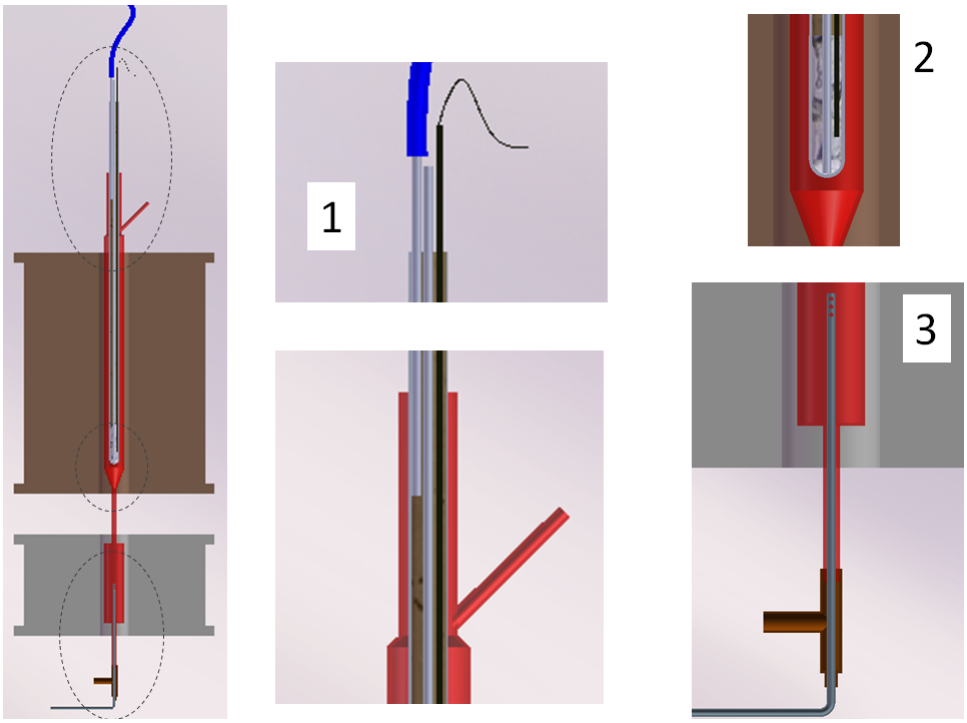


Figure 5.3: Cross-section view of the steam-carbon tube cell reactor.

a small He purge above the carbon bed. Once the system has reached its operating temperature, the air-referenced OCV is noted (near 1 V), giving an accurate measure of the carbon-side oxygen activity. The air is then displaced with the specified H_2/H_2O mixture. This method allows for the impact of the small He purge on the anode oxygen activity to be included in the OCV calculation.

Shown in Figure 5.4, are theoretical and measured OCV values for the steam-carbon system at 1073 K and 1173 K, with carbon-air OCV values of 1.155 V and 1.070 V, respectively for these temperatures. The exceptional agreement between the two data sets taken at 1073 and 1173 K, spanning 3 orders of magnitude in H_2/H_2O

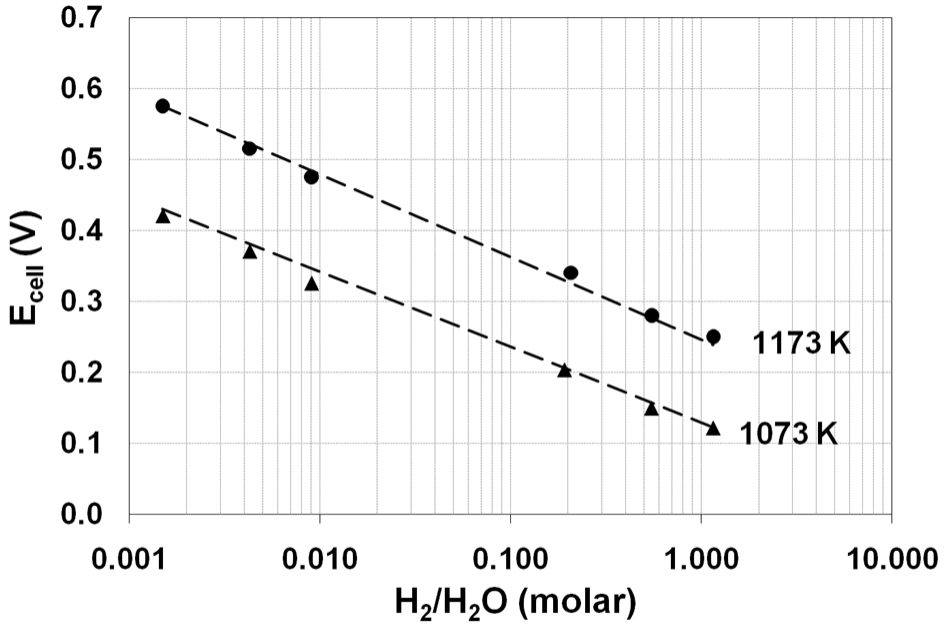


Figure 5.4: Measured OCV differences between a bed of FAC particles and steam (0.1 $mL{\cdot}min^{-1}$ water flow).

ratio, demonstrates again that the steam-carbon cell is governed by the Boudouard reaction at the anode, and the H_2O/H_2 ratio at the cathode. The oxygen potential at the cathode is larger than that at the anode, implying that steam electrolysis can proceed spontaneously (without work input). The previous thermodynamic analyses have shown that Boudouard equilibrium is reached and maintained in a particle bed at temperatures near 1173 K. High temperature operation results in larger driving forces as the carbon bed causes the anode oxygen activity to decrease while the oxygen activity in a fixed H_2/H_2O mixture increases with temperature.

5.2.3 Simple Process Model for the SCC

A simple model constructed here will be used to illuminate some of the expected behavior of the SCC system at 1173 K. The previous OCV measurements validate the

thermodynamic expectations of the SCC system. Shown in Figure 5.5 is a system in which solid carbon and steam are introduced to the anode and cathode, respectively. The resulting CO/CO_2 (anode) and H_2/H_2O (cathode) streams are also chemically and physically separate. The streams interact via oxygen transfer through the SOFC

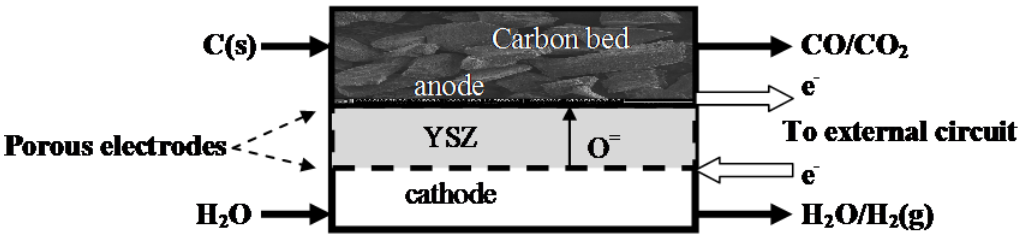


Figure 5.5: Schematic diagram of the SCC system used for simple process analysis at 1 atm, with inlet streams at 298 K.

architecture, which is directly related to the cell current. Using a basis of 1 $mole \cdot s^{-1}$ carbon input, and equilibrium CO/CO_2 ratio (36.25) exiting the particle bed, a specified H_2O/C ratio (S), the system was explored for an estimate of efficiency. The molar outflows are given by Equations 5.1 - 5.4, and the current is given by Equation 5.5. The first law applied to the electrolyzer control volume, given by Equation 5.6, yields the heat input requirement $\dot{Q}_{CASE,in}$ from a specified work input (see Equation 5.7).

$$\dot{N}_{CO,out} = \frac{R}{R+1} \tag{5.1}$$

$$\dot{N}_{CO2,out} = \frac{1}{R+1} \tag{5.2}$$

$$\dot{N}_{H2,out} = \dot{N}_O \tag{5.3}$$

$$\dot{N}_{H2O,out} = S - \dot{N}_O \tag{5.4}$$

$$\frac{I}{2F} = \dot{N}_O = \dot{N}_{CO} + 2\dot{N}_{CO2} = \frac{R+2}{R+1} \tag{5.5}$$

$$\begin{aligned} &\dot{N}_C h_C + \dot{N}_{H2O} h_{H2O,in} + \dot{Q}_{CASE,in} + \dot{W}_{CASE,in} = \\ &\dot{N}_{CO} h_{CO} + \dot{N}_{CO2} h_{CO2} + \dot{N}_{H2O,out} h_{H2O} + \dot{N}_{H2,out} h_{H2} \end{aligned} \tag{5.6}$$

$$\dot{W}_{CASE,in} = -I \cdot E_{cell} \tag{5.7}$$

The hot CO/CO_2 stream has an associated heating value (Q_{HVCO}) of 303.3 $kJ{\cdot}s^{-1}$ at 1173 K (including both sensible energy and chemical energy). The hot H_2/H_2O stream also has an associated heating value ($HV_{H2O/H2}$ including only the sensible portion). If the CO rich stream is used to provide the work input, either through an SOFC or combustion-coupled heat engine, then the remaining heating value can be used to supplement the SCC input heat requirement along with the steam side sensible energy. Included in Table 5.3 are the results of this thermal analysis with heat and work interactions. In Table 5.3, the column labeled η_{CO} represents the required efficiency of the work extraction from the flue gas to provide the SCC work input, and is below 33% at these conditions.

The second law efficiency of this constrained system is 62%, defined as the ratio of cold H_2 exergy ($\dot{N}_{H2,out}\cdot$ 235.2 $kJ{\cdot}mol_{H2}^{-1}$ [16]) to the exergy of input carbon (388.3 $kJ \cdot s^{-1}$). The first law efficency of this system is 75%, defined as the ratio of the cold H_2 heating value (294 $kJ{\cdot}s^{-1}$) to that of the input carbon (393.5 $kJ{\cdot}s^{-1}$). This result agrees with the other studies considered, and demonstrate the benefit of the coupled SCC-power plant system. The use of work and heat inputs from the power plant allow for more efficient operation.

5.3 Experimental Validation of the SCC System

Several measurements were conducted of simple SCC systems under varying experimental conditions. Measurements were taken in both the tube cell and button cell configurations. Activated carbon is used as the feedstock, and the geometries tested are identical to those used for the DCFC studies presetnted earlier.

Table 5.3: Results for simple analysis of the SCC system (in units of V and $kJ \cdot s^{-1}$.

S	E_{cell}	$\dot{W}_{in,CASE}$	$\dot{Q}_{in,CASE}$	$HV_{H2O/H2}$	Q_{HVCO}	Q_{avail}	η_{CO}
	(V)	(kJ/s)	(kJ/s)	kJ/mole	kJ/s	kJ/s	
1.2	-0.5	99.07	144.28	40.04	204.25	244.29	0.33
1.2	-0.4	79.26	164.09	40.04	224.07	264.11	0.26
1.2	-0.3	59.44	183.91	40.04	243.88	283.92	0.20
1.2	-0.2	39.63	203.72	40.04	263.70	303.74	0.13
1.2	-0.1	19.81	223.54	40.04	283.51	323.55	0.07
1.2	0.0	0.00	243.35	40.04	303.33	343.37	0.00
1.5	-0.5	99.07	167.48	63.24	204.25	267.49	0.33
1.5	-0.4	79.26	187.29	63.24	224.07	287.31	0.26
1.5	-0.3	59.44	207.11	63.24	243.88	307.12	0.20
1.5	-0.2	39.63	226.92	63.24	263.70	326.94	0.13
1.5	-0.1	19.81	246.74	63.24	283.51	346.75	0.07
1.5	0.0	0.00	266.55	63.24	303.33	366.57	0.00
2.0	-0.5	99.07	206.14	101.91	204.25	306.16	0.33
2.0	-0.4	79.26	225.96	101.91	224.07	325.98	0.26
2.0	-0.3	59.44	245.77	101.91	243.88	345.79	0.20
2.0	-0.2	39.63	265.59	101.91	263.70	365.61	0.13
2.0	-0.1	19.81	285.40	101.91	283.51	385.42	0.07
2.0	0.0	0.00	305.22	101.91	303.33	405.24	0.00

5.3.1 Tube Cell Results

The ESTC configuration used to measure SCC OCV values was also employed to measure the DC characteristics of the cell under potentiostatic operation. Shown in Figure 5.6 are I-V data for the SCC, with notations of the three regimes of interest. The region characterized by $I > 0$ and $E > 0$ includes spontaneous oxygen transfer from cathode to anode with positive work output. Work input is required for operation in the region characterized by $I < 0$ and $E > 0$ (representing forced oxygen transfer from anode to cathode) or the region characterized by $I > 0$ and $E < 0$ (representing forced oxygen transfer from cathode to anode). In addition to DC measurements, gas analysis was performed on the ESTC during galvanostatic operation at 1173 K. The molar ratio of H_2/CO was found to range from 0.87 to 1.00, suggesting Faradaic behavior. Every mole of oxygen traversing the electrolyte forms one mole of H_2 at the cathode, and roughly one mole of CO at the anode if the

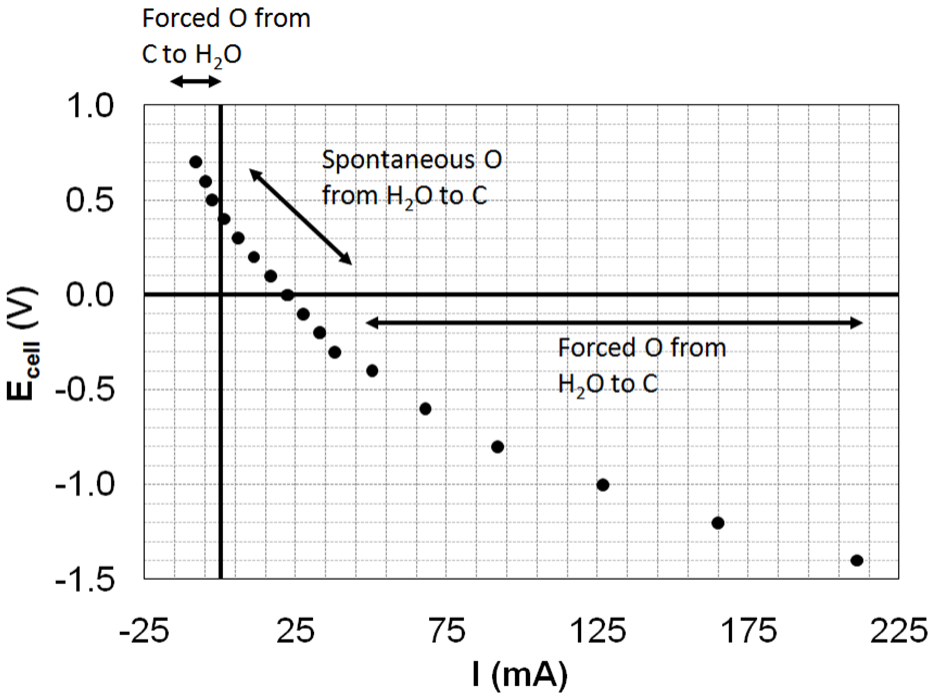


Figure 5.6: Current-voltage DC measurements of an ESTC under SCC conditions.

CO/CO_2 ratio tends to its equilibrium value. The CO_2 was below detectable limits, indicating that Boudouard equilibrium was established by the effluent stream. The current density performance of the ESTC is marginal as expected, although oxygen transfer is demonstrated.

5.3.2 Button Cell Results

In order to increase the achievable current density, a BC with Pt electrodes was tested under SCC conditions. The 25 mm BC was bonded to a 25 mm diameter alumina support tube, with the anode located on the interior. The interior of the alumina tube was loaded with 1 gram of raw FAC particles, forming a fixed bed. The anode chamber was maintained under a slight He purge to mitigate ambient air diffusion, with a liquid seal at the exhaust allowing for gaseous outflow during nonzero current flow such that the cell maintained 1 atm operation. Shown in Figure 5.7 are I-V data for the BC and the previous ESTC, normalized by their respective active areas. Similar behavior is observed for both types of cells, although the BC achieves about an order of magnitude increase in current density, with a resistance of about 7 ohms.

A similar BC was tested under both C-air and C-steam conditions at 1173 K (data shown in Figure 5.8), with a bed of FAC particles (1 gram) in contact with the anode. The anode chamber was maintained under no-inflow conditions (with no He purge), although the liquid seal was still in place to allow for gas outflow. The ohmic resistance was about 7 ohms for the DCFC case at currents above about 20 $mA \cdot cm^{-2}$. The Pt electrodes allow for this flexibility in cathode composition, an the possibility of convenient comparison. The difference in OCV between the DCFC and SCC systems is directly evident from the data presented in Figure 5.8. Again, ohmic losses appear to dominate the cell performance, suggesting that there remains significant room for improvement.

In addition, coupled AC-DC measurements were made using this BC under both DCFC and SCC conditions (with frequency ranging from 0.1 Hz to 60 kHz). The data, shown in Figure 5.9, suggest that the performance is limited by the cathode,

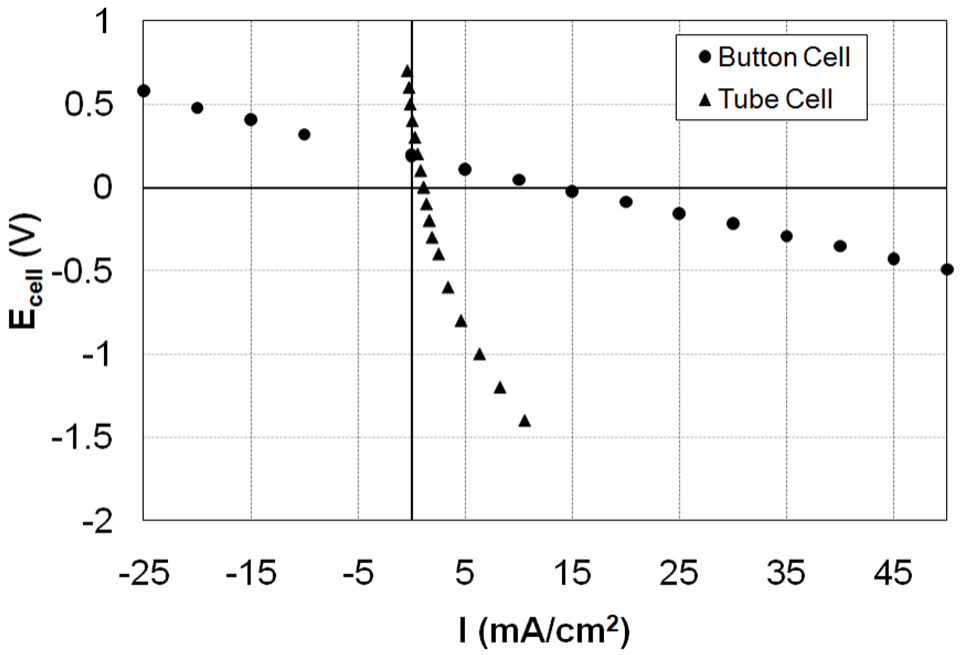


Figure 5.7: comparison of current-voltage DC measurements of an ESTC and BC under SCC conditions at 1173 K, indicating relative cell resistances.

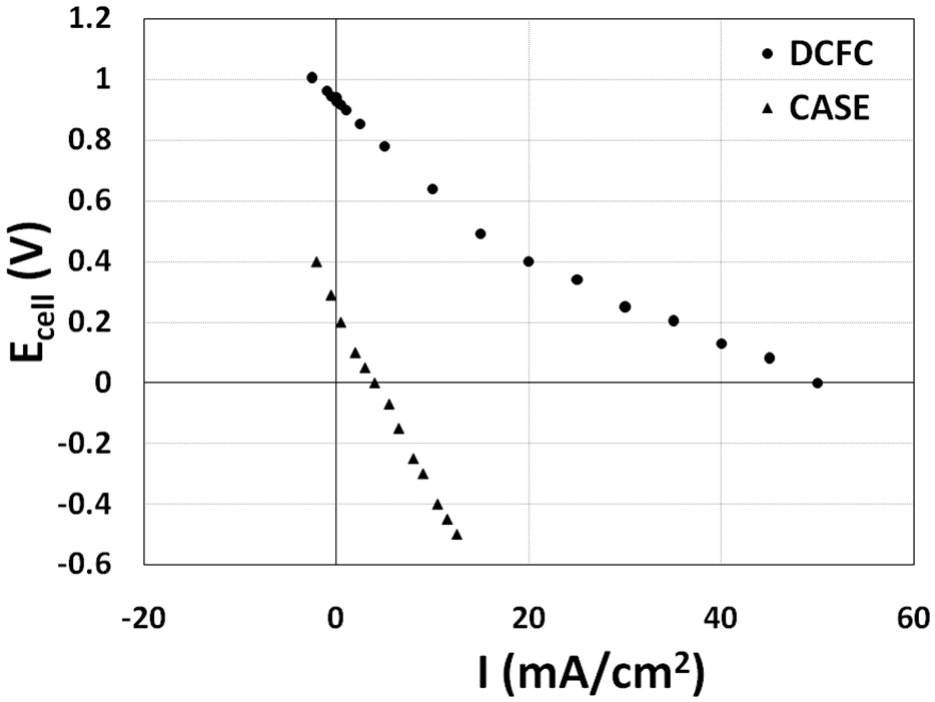


Figure 5.8: Current-voltage DC measurements of a BC under DCFC and SCC conditions at 1173 K.

as the electrode impedance (semicircle) changes significantly as the air is replaced with steam, for the same carbon bed. During DCFC operation, the semicircle peaks occurred at 600, 755 and 1510 Hz for 0, 5, and 25 $mA{\cdot}cm^{-2}$, respectively. During SCC operation, the semicircle peaks occurred at 301 and 755 Hz for 3 and 5 $mA \cdot cm^{-2}$, respectively. The low-frequency impedance agrees with the slope of the I-V curve as expected.

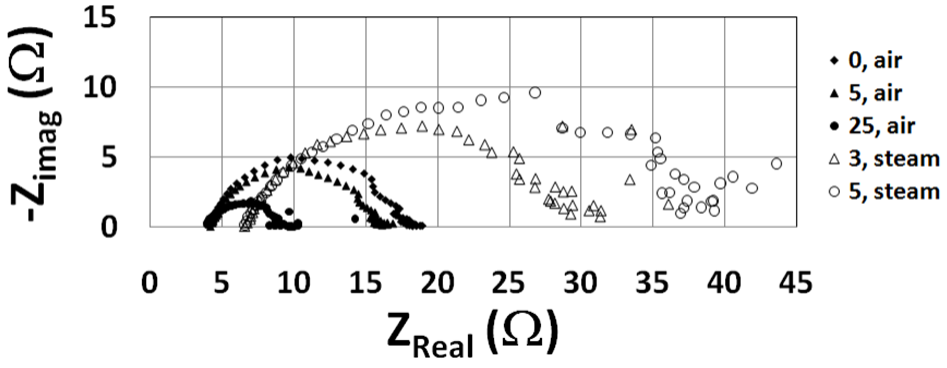


Figure 5.9: Current-voltage DC measurements of a BC under DCFC and SCC conditions at 1173 K.

Chapter 6

Conclusion

6.1 DCFC Feasibility

The thermodynamic analysis demonstrated that the DCFC approach to coal conversion is potentially more efficient than conventional approaches. Removal of the separate gasification constraint has the potential to reduce system losses and increase work output. Simple thermodynamic analysis of reversible work production showed that gasification under autothermal or indirect constraints necessarily introduce losses and reduce the maximum work output of systems utilizing carbon. Essentially the DCFC sidesteps the need for gasification, and thus removes one source of system losses. A major finding was that the maximum theoretical DCFC efficiency can not be greater than 100%, and with environmental air and separate CO_2 effluent is 99%. Subsequent detailed modeling, including practical constraints and irreversibilities, estimated that the DCFC arrangement achieved thermal efficiencies of over 50%, as compared to the less than 40% efficiency of the ATG process. Using the IDG approach, with sensible energy transfer to the gasifier as heat, efficiencies comparable to the DCFC were achieved. Use of the fuel side effluent in order to boost cell operating potential or as a heat source for a bottoming cycle increase the possible efficiency to around 60%. Efficiencies of 50 - 60%, along with fuel flexibility (via internal reforming or gasification) and a capture ready effluent stream make the DCFC competitive with conventional approaches. In addition to high thermal efficiency, the coupling of a free

particle bed and solid state fuel cell offers process advantages. The thermally and chemically integrated SOFC and particle bed can reduce the number of process steps and associated losses. Direct contact between the anode and fuel bed also mitigates the need for a carrier gas or fluidizing agent. Particle feeding and ash separation and removal can be be achieved via conventional means. Sorbents of various types could be incorporated to reduce contaminants in situ.

6.2 Kinetic Model and Particle Bed Model

6.2.1 Fuels Comparison

It was shown experimentally that the Boudouard reaction dominates solid-state DCFC performance. The fuel supply of the solid state DCFC is not supplied by in-flow as is typical of SOFC systems, but rather chemically-driven by in situ gasification. Reactivity measurements confirm that solid fuel reactivities vary significantly (up to several orders of magnitude), with the most reactive fuels being chars created under high heating rates. Pretreated fuels demonstrated lower reactivities, as did Lower Kittanning coal devolatilized under a slow heating ramp. The Boudouard reaction is slow when referenced to the carbon particles, but changes in gas phase concentrations are significant (even reaching equilibrium) within the arrangements considered here.

6.2.2 Detailed Kinetic Modeling of Activated Carbon

Detailed modeling of the gasification of activated carbon, using the 5-step mechanism proposed by Ma [107], was completed and compared with thermogravimetric experimental results. Agreement with measurements under kinetically-controlled conditions for both low-CO and high-CO environments were demonstrated. Direct determination of the desorption reaction rate parameters was achieved using temperature-programmed desorption measurements in the PTGA, and the specific surface area was also measured as a function of particle conversion.

6.2.3 Verification of Transport Model for a Particle Bed

The kinetic parameters determined for an activated carbon from thermogravimetric analysis were incorporated into a gas transport model within a particle bed, similar to the conditions at the DCFC anode. The model incorporated porous convective flow (Darcy's Law), mass transport (convective and diffusive) and surface chemistry to predict the scalar fields of pressure and gas concentration within a particle bed. Using a fixed bed geometry, gas chromatographic measurements of exhaust gas composition were carried out at temperatures of 973 to 1223 K, and compared with the model. This transport model was then used to simulate the effects of an embedded SOFC, and predict the expected performance of such systems. The results showed that practical current densities in the range of 100 - 1000 $mA{\cdot}cm^{-2}$ are achievable with FAC, a fuel of intermediate reactivity.

6.3 DCFC Behavior

Open circuit measurements of the DCFC demonstrated that the Boudouard reaction governs the cell potential for the C-O system, and steam gasification governs the cell potential for the C-H-O system. These two chemical systems behave similarly in the presence of solid carbon. At temperatures above about 1100 K, chemical equilibrium is achieved in the particle bed, verified by comparison with theoretical cell potential calculations. The gas composition within the particle bed was also shown to asymptotically relax towards an equilibrium composition, with a time scale of O(10) minutes near 1200 K. Experimental verification of the DCFC and practical power densities were demonstrated, with output well above that previously documented in the literature for solid state systems. An anode supported tube cell produced a power density of 220 $mW{\cdot}cm^{-2}$ at a cell voltage of 0.68 V during operation at 1178K using activated carbon. An electrolyte-supported button cell achieved a power density of 140 $mW \cdot cm^{-2}$ at 0.56 V, using MCMB. These values were achieved with fuels of intermediate reactivity with relatively large ohmic losses, suggesting that better performance is achievable. Another important conclusion to be drawn is that the

solid state DCFC performs as well as systems based on the MCFC. Gas composition analysis and oxygen balance confirmed Faradaic operation of the DCFC. Oxygen ions traversing the solid electrolyte become bound in gas phase molecules leaving the anode chamber during steady state operation. This verification was one of the major problems with the first DCFC experiments during the 1800s and early 1900s. Although an oxygen balance is generally assumed, this verification is rarely performed in contemporary experiments.

6.4 SCC System Conclusions

The possibility of generating a clean stream of H_2 was demonstrated using a DCFC-type apparatus. Replacement of air by steam at the cell cathode allows the electrolysis of water to commence, with a minimized work input as compared to conventional electrolysis. This arrangement also prevents mixing between with product hydrogen stream and the carbon-containing fuel input stream, maintaining complete chemical separation. Through maintaining a sufficiently low oxygen activity at the cell anode, the carbon bed is able to assist in the oxygen removal from steam, yet remain completely unmixed with the H_2 stream. An estimated exergy efficiency of the standalone SCC system was calculated to be nominally 60%, superior to that of conventional electrolysis via uncoupled power-electrolysis plants. The possibility of a standalone SCC plant is important, as it includes the production of work to drive the electrolysis cell.

Measurements of the open circuit potential for a characteristic SCC system were consistent with theoretical values for the Boudouard reaction and the H_2/H_2O system, as calculated using the Nernst equation. Current-voltage measurements were conducted on an un-optimized system, demonstrating the possibility of oxygen transfer. Gas analysis demonstrated H_2 production from a Faradaic process, in agreement with expectations.

6.5 Recommendations for Related Future Work

6.5.1 Thermal Analysis of the DCFC

While this study has provided a detailed account of thermodynamic expectations, char reactivities and DCFC performance, there remain pertinent issues to be addressed. Temperature gradients within the fuel bed and SOFC were not addressed in detail, yet are critical in commercializing the DCFC. The use of flue gas burning to supply sensible heat for the endothermic particle bed was proposed, although optimization is required to create a favorable temperature field within the DCFC gasifier. A likely arrangement, building upon the systems considered here, would entail immersed combustion tubes within the carbon bed that supply localized sensible energy to drive the gasifier. In addition, if indirect gasification were used in a separate process, an immersed tube arrangement could again be employed for thermal coupling. A small amount of O_2, either pure or diluted with CO_2, would be required to facilitate this heat release, and could be supplied by either cryogenic air separation means or oxygen permeable membrane.

In addition to heat addition to the bed, a fluidizing or gasifying agent may be introduced to help segregate ash, facilitate mixing of the bed, or to act as an particle entraining media for feed. Suitable candidates for this stream include steam, CO_2, exhaust gas recirculation (EGR), of some combination of these. One benefit of EGR is that the fuel concentrations within the bed could be maintained at larger levels, as compared to the use of H_2O or CO_2. Another benefit of EGR stream is that temperature of the EGR stream matches that of the gasifier. In addition, the EGR stream does not require treatment, but does allow for the possibility of contaminant removal. The addition of gaseous O_2 directly into the bed causes a decrease in the thermodynamic efficiency as shown in Chapter 2, but there are benefits to small amounts of O_2 addition. These include the mitigation of tar formation (and subsequent deposition), increased control of heat release locally to complement endothermic bed reactions, and a mechanism of utilizing any char contained in the ash prior to removal. Particle bed temperature would then be a key consideration; large amounts of EGR could supplement the relatively slow Boudouard reaction during operation at lower bed

temperatures, or help to homogenize the spatial temperature field during operation at higher temperatures. The effects of using practical gasifying and fluidizing agents in the DCFC gasifier require investigation and characterization. There likely exists some optimum combination of gasifying agent that preserves efficiency yet assists in chemical and thermal interactions within the bed.

6.5.2 Fuel Reactivities

Aside from thermal analysis of the DCFC and SCC systems, reactivity measurements of solid fuels under varying environments are equally critical. Ideally a library of practical fuel reactivities could be generated, or perhaps a tabulation of reactivity comparisons at relevant conditions. The Boudouard reaction depends upon fuel characteristics, necessitating that some metric is required for comparison between fuels or systems employing different fuels. The fuels included here, especially Wyodak coal char and almond shell biomass, could be devolatilized under varying heating rates, allowing comparison of the resulting reactivities. A practical means of delivery solid particles into the gasifier must be specified, allowing the heating rate of raw particles to be specified. This then allows for detailed thermogravimetric analysis of chars created under these conditions. The kinetic parameters can then be verified in a fixed bed experiment similar to that used in this study. In addition to CO_2 containing environments, the rate of steam gasification also should be addressed for these solid fuels. The use of steam as a gasifying agent has the benefit of local H_2 production, which can improve DCFC performance. A quantification of this benefit should also be completed, as this would have implications for practical system layouts.

In addition to further characterizing the Boudouard reaction for various fuels, fuel pretreatment is another topic of interest for DCFC. Raw fuels contain volatile matter and moisture, that are released upon heating of the particles. One approach that may warrant investigation is pyrolyzing of the solid fuels prior to introduction into the DCFC. The volatile matter could be handled separately via reforming for work production, or a chemical process such as the Fischer-Tropsch process for liquid fuel production. This type of dual-use scenario is in line with the Department of Energy's

Vision 21 power plant concept.

6.5.3 Fuel Impurities

Lastly, the impact of fuel impurities, such as sulfur and chlorine, requires quantification under DCFC and SCC conditions. In addition, techniques for mitigating these effects such as fuel preprocessing, in situ capture, and materials resilience all require further investigation. It is widely known that Ni based anodes suffer from sulfur poisoning due to sulfide formation, however these cermet anodes have a relatively low cost and generate good performance. The preferred method of preventing poisoning of Ni would be to pretreat fuels thus eliminating a majority of sulfur and other contaminants from entering the gasifier. In addition, sulfur capture via solid sorbents within the gasifier is another option, possibly undergoing an external chemical looping process to regenerate. If an IDG process were used to create syngas, gasification products could be cleaned prior to contact with the SOFC anode. This would allow for the prevailing syngas cleaning technologies to be employed. Thermal coupling would then be critical, as the heat output from the SOFC and the heating value of the flue gas (released upon combustion) require transfer to the gasifier. Immersed SOFC tubes and flue gas burner tubes (within the carbon bed) would allow for a compact, thermally coupled system.

Some pertinent experiments might include similar fixed bed GC measurements as those discussed in Chapter 4. Using sulfur as an example here, sulfur containing species (H_2S, COS, or SO_2) could be passed through the bed and the exhaust gas composition could be measured. This experiment could be performed at various bed temperatures, using various bed heights. The sulfur containing stream could be further diluted with H_2O, CO_2, or an inert gas. Measurements of the exhaust gas content of H_2S, COS, or SO_2 would provide insight into the fate of sulfur in the DCFC bed. This approach would allow global kinetic rates to be extracted, or if coupled with PTGA measurements, would allow for more detailed mechanistic studies. If the same experiment were performed with an embedded SOCF, sulfur tolerance could be determined directly. Using this same approach, sulfur sorbents could be added

to the bed or included separately, and the sulfur reduction benefit could be directly determined.

Appendix A

Experimental System Details

The experimental systems used to control inputs and measure data were characterized prior to use. Included in this appendix are characterizations for the PTGA, fuel cell test stand, preparation equipment, and gas chromatograph.

A.1 Gas Chromatograph

Gas analysis was performed with a Varian 3400 gas chromatograph. A Haysep Q porous polymer was used, in conjunction with an MS 5A molecular sieve to separate gas species prior to a thermal conductivity detector (TCD). Both helium and argon were used as the carrier gas during different analyses, set at constant flowrates of 30 $mL \cdot min^{-1}$ for both the carrier and reference. Streams containing H_2 benefit from the use of an argon carrier gas due to the large difference in thermal conductivity, while streams containing CO and CO_2 (without H_2) benefit from the use of a helium carrier gas. The calibration curves for N_2 and O_2 are given in Figure A.1, while the calibration curves for CO_2 and CO are given in Figure A.2, both using a He carrier.

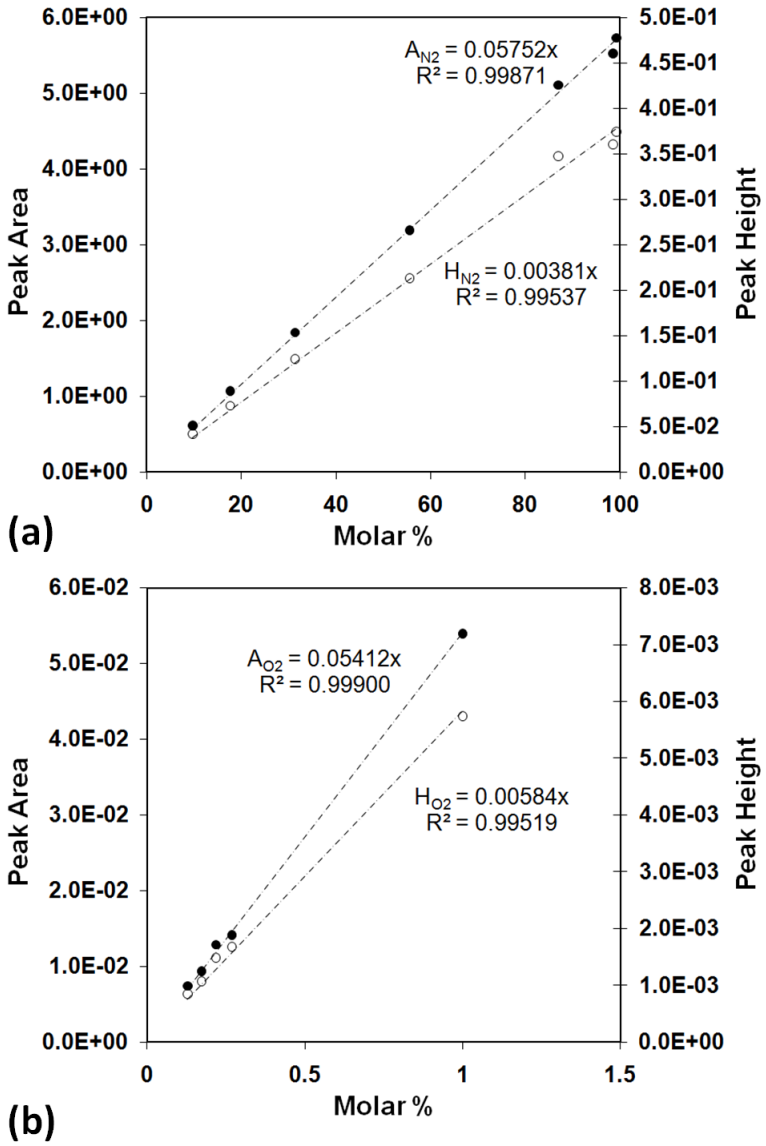


Figure A.1: GC peak area (closed symbols) and height (open symbols) calibration for N_2 and O_2, with linear regression.

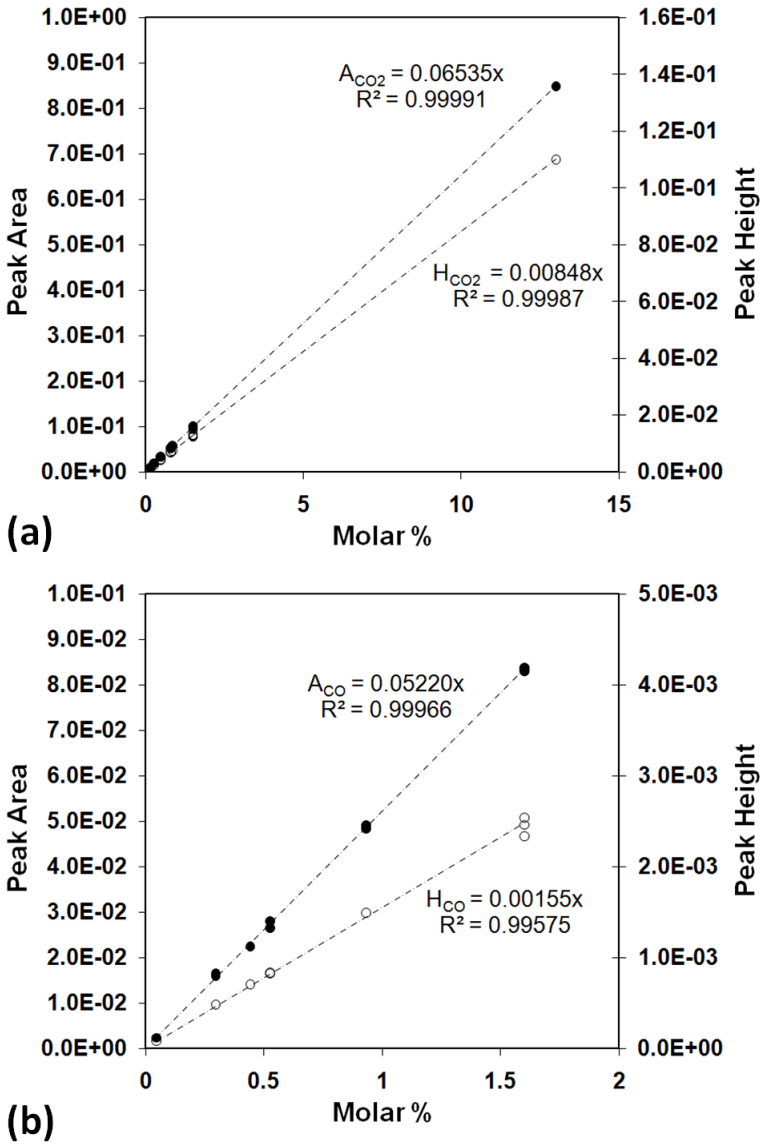


Figure A.2: GC peak area (closed symbols) and height (open symbols) calibration for CO_2 and CO, with linear regression.

A.2 PTGA

A.2.1 Gas Sensor Calibration and Balance Check

The balance and gas sensors are checked for operation and calibrated using calcium oxalate (monohydrate), with chemical notation $CaC_2O_4(H_2O)$. Upon heating, one mole of this compound releases one mole of water, one mole of CO and one mole of CO_2 in order of temperature. This repeatable decomposition, leaving only CaO on the sample pan, allows for independent checking of the PTGA balance and gas sensors output. Shown in Figure A.3 are the gas sensor readings and sample pan mass as functions of time for 730 sccm purge gas flow, 495 sccm RXN N_2 flow, and 290 sccm furnace gas flow, with all streams at 1 atm total pressure. As shown in

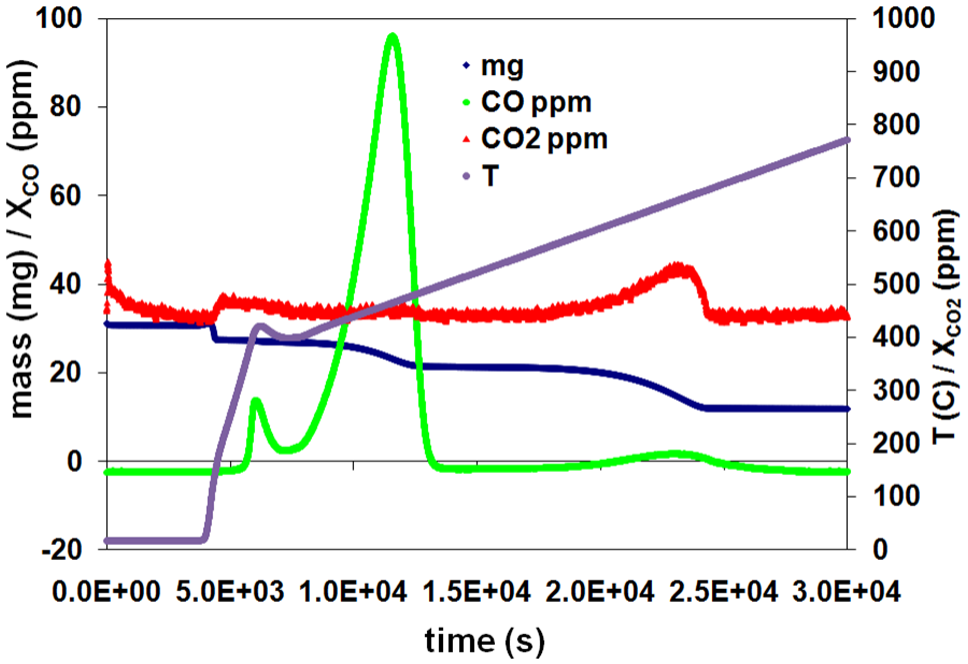


Figure A.3: Measured time traces of mass and mole fractions during calcium oxalate (27.45 mg dry) decomposition.

Figure A.3, the calcium oxalate releases CO near 723 K, and CO_2 near 873 K. The

measured, post-decomposition masses of CO, CO_2, and CaO agree with calculated values, differing by only 2.6 %, 0.4 %, and 1.7 %, respectively (see Table A.1). The measured CO and CO_2 released using the gas sensors give values slightly larger than expected, with calibration constants of 0.753 and 0.719 respectively.

Table A.1: PTGA balance calibration using the decomposition of calcium oxalate (27.45 mg dry).

	CO	CO_2	CaO
Measured (M)	6.16	9.39	11.78
Calculated	6.00	9.43	12.01
Difference	2.6 %	0.4 %	1.7 %
Measured (gas)	8.17	13.06	
Calibration Constant	0.753	0.719	

The total mass of gas species i (M_i), as measured by the gas sensors, is calculated as the time integral (Equation A.1) of the instantaneous molar flow rate (Equation A.2). The concentration in Equation A.2 is determined at STP. The measured mole fractions were zeroed upon post-processing, and were then used in the time integration.

$$M_i = MW_i \int_0^t \dot{N}_i dt \tag{A.1}$$

$$\dot{N}_i = \frac{X_{i,ppm}}{1e6}(\dot{V}_{purge} + \dot{V}_{furnace} + \dot{V}_{rxn})\frac{P}{RT} \tag{A.2}$$

A.2.2 Surface Area Check Using Carbosphere

The specific surface area is determined for the carbonaceous solids using BET analysis. The results from BET analysis are explicit, without corrections or correlations, so a check is performed to ensure accurate results. Commercially available Carbosphere is used for this purpose as it has a stated surface area of 1000 $m^2 \cdot g^{-1}$. The measured value, using the PTGA and BET approach, was found to be 1007 $m^2 \cdot g^{-1}$, differing by less than 1 % from the manufacturers specification. The BET test was performed at 10 atm total pressure, for CO_2 mole fractions of 0.34, 0.64, 0.75, and 1.00 (N_2 balance). The measured mass, as a function of time, during a blank BET test and Carbosphere BET test is included in Figure A.4. Note the reduced drops in

mass for the Carbosphere case as compared to the blank case, indicating increased adsorbed CO_2 as the mole fraction is increased. The vertical shift of the data is of no consequence, as only the changes in mass are required. Shown in Figure A.5a

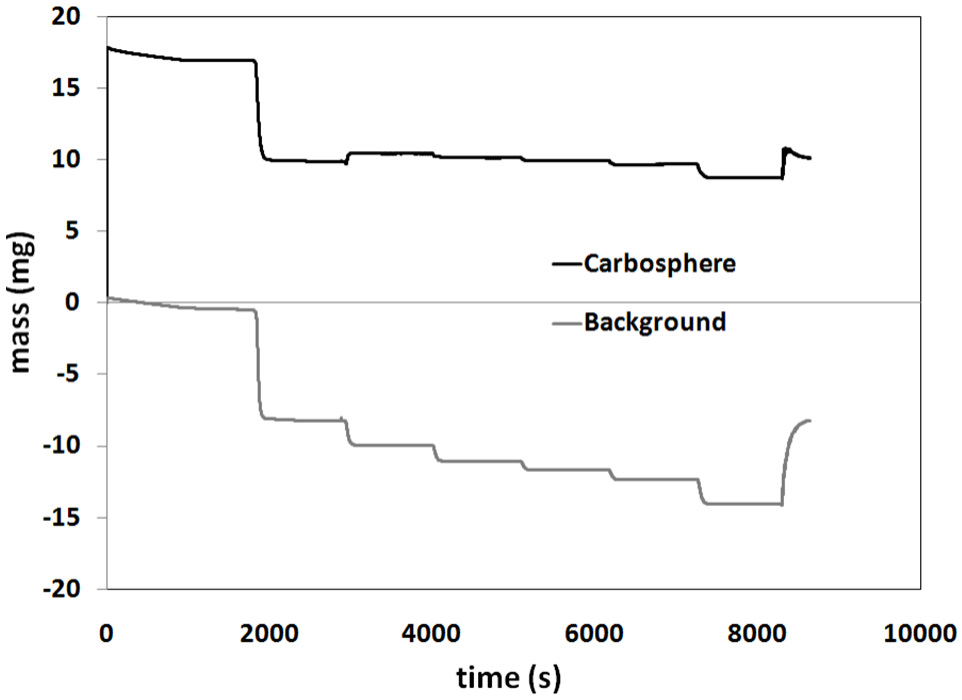


Figure A.4: Time traces measured during a background BET test and with Carbosphere loaded in the PTGA.

is the background drag and buoyancy corrections to the sample pan mass measured during the background test. Shown in Figure A.5b are the measured results from Carbosphere, plotted in BET-derived variables, demonstrating linear behavior.

A repeat of this test using a Ni mesh pan gave a specific surface area of 981 $m^2 \cdot g^{-1}$, also in excellent agreement with the manufacturer specification.

A.3 Fuel Cell Test Stand and Preparation

The fuel cell test stands used employ a large variety of equipment to monitor and control process streams. Included in Table A.2 are the part/component names and

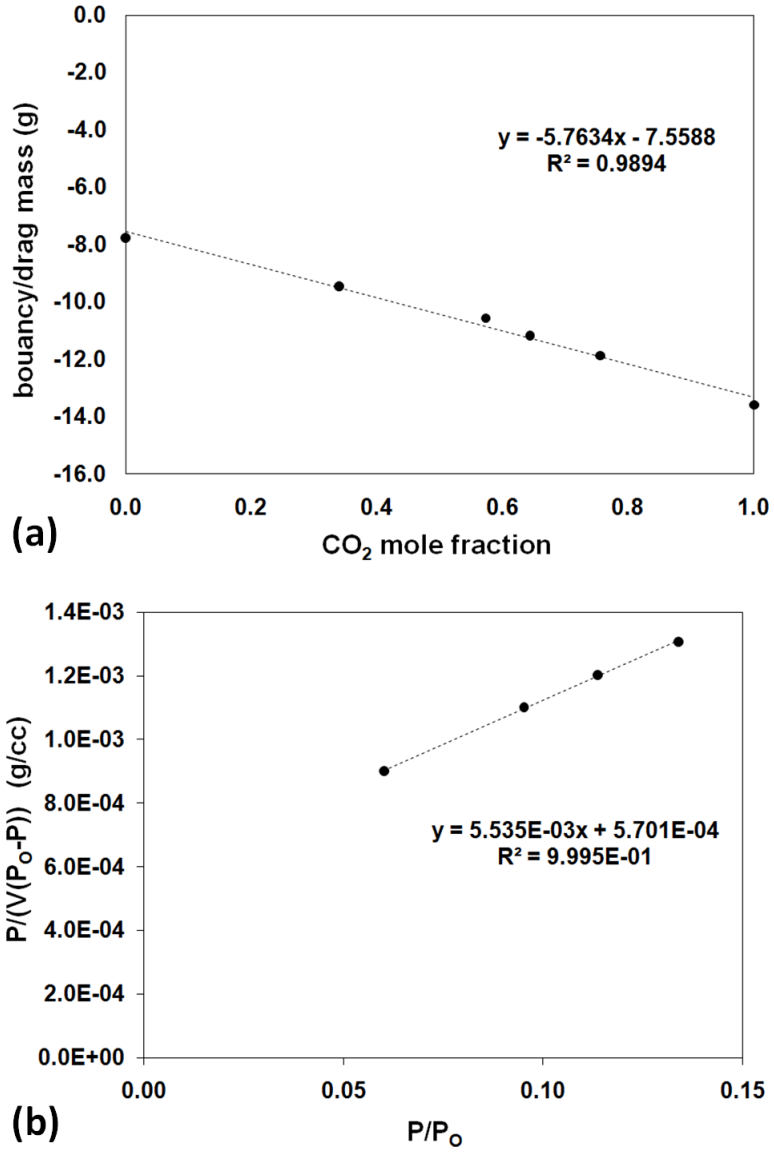


Figure A.5: (a) Background drag/buoyancy correction, and (b) BET correlation for Carbosphere.

vendors for reference. Temperature measurements were taken with K-type (alumel-

Table A.2: Parts list and details for the DCFC test stand and equipment

Component	Part/Model	Vendor C	Details/Notes
Top Rotometer	FL-3102SA	Omega	Sapphire float,150mm
Bottom Rotometers	FL-3035SA	Omega	Sapphire float,75mm
Digital flowmeter	Intelligent	Varian	0.1-1000 sccm
GC	3400	Varian	MS5A + Haysep Q
GC DAQ	OMB-DAQ-54	Omega	0-10Vanalog inputs
T controllers	CN8201	Omega	DC1 output
Relays	SSRL240DC	Omega	10-20 Amp
SSR Fins	FHS-1	Omega	
TC wire	XT-K-24	Omega	ceramic insulated
Clamshell furnace		Multiple Unit	115VAC,300-1000W
Tube Furnace	21100	Thermolyne	1373 K max
Hi-T furnace	Blue	Lindberg	1773 K max
FRA	1250	Solartron	Rear Connections
ECI	1286	Solartron	
Load Resistors		Newark	5W
DC Power Supply	PSP-2010	Instek	30V/10A
Water Pump	Optos 1LM	Eldex	2.5 $mL{\cdot}min^{-1}$ max
Housings	custom	MSGB	408-946-4838

chromel) thermocouple probes. The thermocouples junctions were formed by spot welding, and the thermocouple meters and controllers contained built-in cold junctions (thermistors). Alumina, double-bore tubes were used as electrical insulation and structural support for the thermocouple probes' (24 gage) wires.

A.3.1 Rotometer Calibration

The rotometers used to control and measure gas flow rates were calibrated using the Varian Intelligent Digital flowmeter in-line, with the results included in Tables A.3 and A.4. The measured flowrate values agree with those stated by the manufacturer for air.

A.3.2 Electrochemical Test Equipment

Prior to use the EIS facility (Solartron 1286 and 1250) were tested using the 1186ECI testbox module, and another simple RC circuit (100 ohm resistor in series with parallel 1800 ohm resistor or 104e-9 F capacitor). The results for the testbox are presented in Table A.5, while the results for the RC circuit are given in Figure A.6. The impedance for the RC circuit is calculated using Equation A.3. The semicircle peak occurs as a measured frequency of 867 Hz, as compared to a a calculated frequency of 850 Hz (2 % difference). The results demonstrate that the 1286/1250 arrangement is operational and agrees with calculated values.

$$Z(\omega) = R_{100} + \frac{1}{\frac{1}{R_{1800}} + j\omega C} \tag{A.3}$$

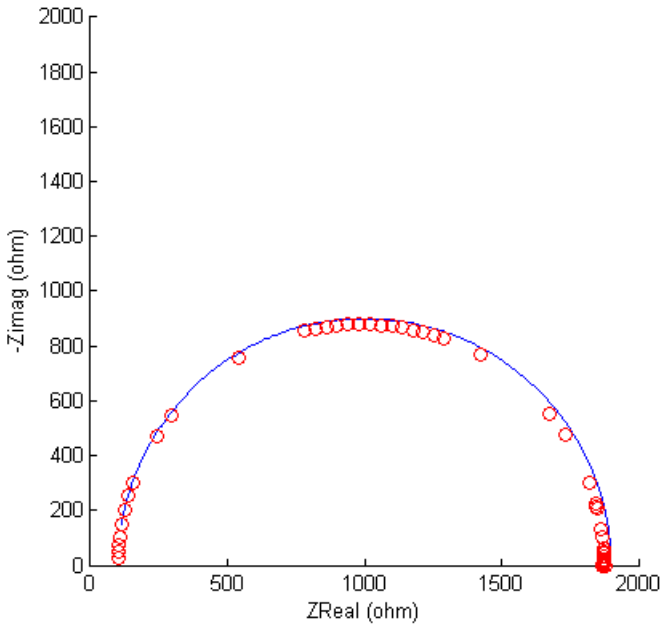


Figure A.6: Measured (symbols) and calculated (curve) impedances for RC test circuit (100 Ω in series with parallel 1800 Ω and 104e-9 F).

A.3.3 Water pump

The Eldex 1LM pump used to accurately meter water flow was calibrated using a graduated beaker and stopwatch. The results of the volume measurements, converted to time rates, is included in Figure A.7. The slope of the regression line differs from unity by less than 1 %, indicating that the user inputted flowrate is accurately acheived by the syringe pump. The CCF (compressibility compensation factor) value is set to 12, and a refill stroke of 50 % is used to reduce the strength of fluctuations during the refill syringe draw.

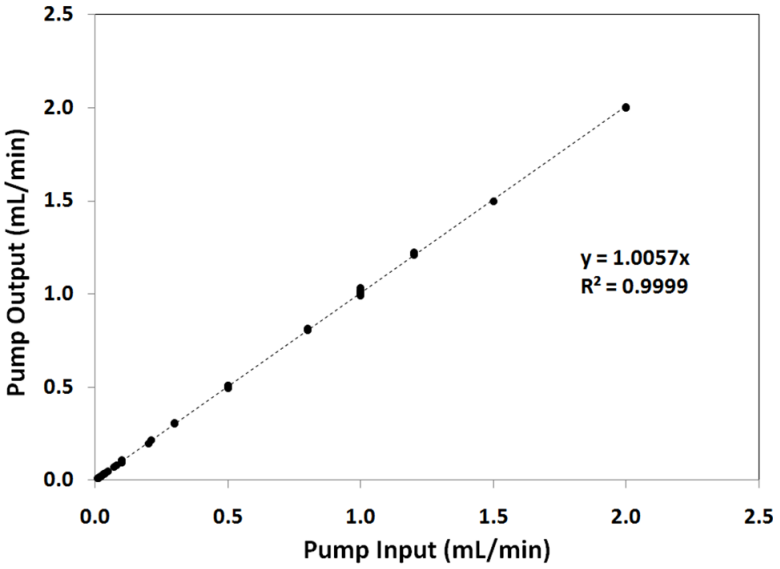


Figure A.7: Eldex Optos pump water output vs. user input ($mL{\cdot}min^{-1}$), with CCF = 12 and 50 % refill.

Table A.3: Rotometer (FL3102SA) calibration data ($mL \cdot min^{-1}$) for gases at 1 atm and 298 K.

Gas	Scale	Flowrate	Gas	Scale	Flowrate
air	10	46.3	1.6CO	5	36
air	20	67.5	1.6CO	10	45
air	30	89.6	1.6CO	20	62
air	50	153	1.6CO	30	90
air	70	216	1.6CO	40	123
air	100	301	1.6CO	50	150
air	130	387	1.6CO	60	189
air	150	457	1.6CO	70	220
			1.6CO	80	256
O2	2	27.7	1.6CO	90	286
O2	5	31.3	1.6CO	100	322
O2	10	40.3	1.6CO	110	350
O2	15	46.3			
O2	20	56.2	CO2	10	45
			CO2	20	75
He	10	44.1	CO2	30	101
He	20	64.1	CO2	50	167
He	30	89.2	CO2	70	223
He	35	104	CO2	100	277
He	40	119	CO2	130	343
He	45	138	CO2	150	383
He	50	151			
He	70	226			
He	100	353			
He	130	524			
He	150	730			

Table A.4: Rotometer (FL3035SA) calibration data ($mL \cdot min^{-1}$) for gases at 1 atm and 298 K.

Gas	Scale	Flowrate
N2	5	6.25
N2	10	11.3
N2	13	12.0
N2	20	17.5
N2	25	23.5
N2	35	30.1
N2	45	43.5
N2	50	52.0
He	10	9.2
He	20	16.0
He	30	24.3
He	40	35.5
He	50	47.5
He	60	63
CO	5	3.4
CO	10	5.3
CO	19	9.6
CO	30	15.0
CO	45	26.3
CO	55	37.0
CO	65	48.1

Table A.5: Measured impedance for Solarton 1186ECI testbox.

Freq.	Measured	Actual	error (Re/Im)%
10000	1.82+0.16i	1.82+0.173i	0.1/7.6
1522	2.32+0.52i	2.33+0.523i	0.3/0.5
240	2.78+0.29i	2.78+0.28i	0.08/1.8
5	6.18+3.40i	6.15+3.38i	0.5/0.7
1	9.34+1.31i	9.28+1.30i	0.6/0.6

Appendix B

Computer Codes and Programs

Computer software was used to collect, analyze and present experimental data. The following sections provide some samples of software used for this study, and a general description of operation.

B.1 MATLAB Programs

The various MATLAB script files used for numerical analysis are included in this section. The codes are included verbatim, and any data present in the codes is representative. The codes include the following;

- TPD - Determination of TPD kinetic parameters
- Global Reactivities - Determination of global kinetic parameters (REACTglobalcalc.m calls globalFACfwd.m)
- Particle Sizing - Formatting and plotting of size distribution from CM
- GC Peak Determination - Peak height and area determination from GC output signal

B.1.1 TPD

```
%Andrew Campbell Lee
```

```
%This m file determines the best fit parameters for gasification reaction
%2, Cb+C(O)-->CO+Cf         where C(O) loaded with CO2
%An initial guess is made and then iteration begins near this guess,
%and refinement can be made.
%-------------------------------------------------------------------------
clear all;
format long;
clc;
tic
sites = 3.8e19;             %carbon sites/m2, calculated from graphite
sitemoles = sites/6.02e23;  %moles carbon sites/m2, from graphite
mass0 = 24.8/1000;          %grams, initial mass char
Sg0 = 750;                  %m2/g, inital surface area
psi = 3;                    %surface area structure parameter
R = 8.314;                  %J/moleK, universal gas constant
molesCO = 1.68e-5;         %moles, C(O) at t=0 from experiment
N2flow = 148;               %sccm, flow of reactant N2
Heflow = 280;               %sccm, flow of purge He
furnflow = 289;             %sccm, furnace gas flow
totalgas = (N2flow+Heflow+furnflow)*(1/60000000)*(101325/(8.314*273));%moles/s, total gas flow for CO(gas)
repeatloop = 1;             %loop logic, repeat or break
T0 = 300;                   %K, starting T
ramp = 0.39;                %K/s, temperature ramp from experiment

%Parametric values for error sweep
%f(E) constants for f(E)=A(E-Emax)^2*(E-Emin)^2
Emin = 308000;%linspace(130000,150000,4);          %J, lower limit of f(E) 4th order
Emax = 430000;%linspace(700000,900000,4);          %J, upper limit of f(E) 4th order
a = 1;
b = 1;

Ebar = 375000;  %J, Mean energy peak
sigma = 28000;  %J, standard deviation

EL = 143200;%linspace(143500,144000,4);                                    %J, lower limit of f(E) Beta
ER = 398000;%linspace(395000,398000,4);                                   %J, upper limit of f(E) Beta
alpha = 6;                                          %Beta PDF constant
beta = 1;                                           %Beta PDF constant
prefac = (gamma(alpha + beta)) / ((gamma(alpha))*(gamma(beta)));

ll = 0;%1e4;     %J, lower limit of integration for gaussian
A = 1.0e13;%linspace(5e8,1e11,4);                   %1/s, preexponential 4th order
Ab = 1.0e13;                                          %1/s for gaussian
Ac = 1.0e13;                                          %1/s for Beta
Nmin = length(EL);
Nmax = length(ER);
NA = length(A);
error = zeros(Nmin,Nmax,NA);        %initialize error matrix
%-------------------------------------------------------------------------
%Experimental results for comparison (081407-fishercarbon-24.8mg)
%T in Kelvins and CO moles/s
XPMNT = [573 3.1806E-11;
        598     3.86644E-11;
        623     7.20853E-11;
        648     3.90519E-11;
        673     5.4797E-11;
        698     7.00949E-11;
        723     7.00964E-11;
        748     1.33373E-10;
        773     1.6312E-10;
        798     2.25519E-10;
        823     2.56568E-10;
        848     3.5286E-10;
        873     4.96072E-10;
```

```
        898     6.90594E-10;
        923     8.84075E-10;
        948     1.22422E-09;
        973     1.61269E-09;
        998     2.0961E-09;
        1023 2.68156E-09;
        1048 3.4373E-09;
        1073 4.37978E-09;
        1098 5.49717E-09;
        1123 6.92846E-09;
        1148 8.64214E-09;
        1173 1.13792E-08;
        1198 1.43893E-08;
        1223 1.8516E-08;
        1248 2.20709E-08;
        1273 2.45735E-08;
        1298 2.43547E-08;
        1323 2.13257E-08;
        1349 1.73468E-08];
n = length(XPMNT(:,1));
%-------------------------------------------------------------------------
%begin loop
total = 0;
DE = 10000;
dt = 15;                    %seconds, time step of iteration
for d = 1:Nmin
    for e = 1:Nmax
        for c = 1:NA

AA = (Emax(b) - Emin(a))^5 / quad(@(x)NORMfE(x,Emin(a),Emax(b)),Emin(a),Emax(b));             %Coefficient for f(E), normalizing
AAb = 1 / quad(@(x)NORMfEgauss(x,Ebar,sigma),ll,1e6);             %Coefficient for f(E), normalizing
CHECK = quad(@(x)NORMfEbeta(x,alpha,beta),0,1);
t = zeros(150,1);
t(1) = 1;                     %initialize time
T = zeros(150,1);
T(1) = T0;
i = 1;                      %initialize index
Oo = molesCO/(sitemoles*Sg0*mass0);
dCOdt = 0;                  %moles/s, gas CO evolution from carbon
Oob = molesCO/(sitemoles*Sg0*mass0);
dCOdtb = 0;                 %moles/s, gas CO evolution from carbon
Ooc = molesCO/(sitemoles*Sg0*mass0);
dCOdtc = 0;                 %moles/s, gas CO evolution from carbon
while repeatloop == 1
%Begin Runge-Kutta time march
i = i+1;
t(i) = t(i-1) + dt;
T(i) = T0 + t(i)*ramp;

Ttemp = T(i-1);
sumtemp = 0;
sumtempb = 0;
sumtempc = 0;
sumtemp2 = 0;
sumtemp2b = 0;
sumtemp2c = 0;
sumtemp4 = 0;
sumtemp4b = 0;
sumtemp4c = 0;

for Etemp = ll:DE:1e6

    if Etemp <=Emin | Etemp >= Emax
    Innertemp = quad(@(x)KERNEL1TPD(Etemp,R,T0,ramp,x),1,t(i-1));
```

```
KERNEL1 = 0;
KERNEL1b = exp(-Ab*Innertemp);
KERNEL2 = 0; %4th order
KERNEL2b = exp(-Etemp/(R*Ttemp)) * KERNEL1b * (1/sigma)*((1/(2*pi))^0.5)*exp(-0.5*((Etemp-Ebar)^2)/(sigma^2)); %Gaussian

Ttemp2 = Ttemp + ramp*dt/2;
Innertemp2 = Innertemp + quad(@(x)KERNEL1TPD(Etemp,R,T0,ramp,x),t(i-1),t(i-1)+dt/2);
KERNEL12 = 0;
KERNEL12b = exp(-Ab*Innertemp2);
KERNEL22 = 0; %4th order
KERNEL22b = exp(-Etemp/(R*Ttemp2)) * KERNEL12b * (1/sigma)*((1/(2*pi))^0.5)*exp(-0.5*((Etemp-Ebar)^2)/(sigma^2)); %Gaussian

Ttemp4 = Ttemp + ramp*dt;
Innertemp4 = Innertemp + quad(@(x)KERNEL1TPD(Etemp,R,T0,ramp,x),t(i-1),t(i-1)+dt);
KERNEL14 = 0;
KERNEL14b = exp(-Ab*Innertemp4);
KERNEL24 = 0; %4th order
KERNEL24b = exp(-Etemp/(R*Ttemp4)) * KERNEL14b * (1/sigma)*((1/(2*pi))^0.5)*exp(-0.5*((Etemp-Ebar)^2)/(sigma^2)); %Gaussian
    else
Innertemp = quad(@(x)KERNEL1TPD(Etemp,R,T0,ramp,x),1,t(i-1));
KERNEL1 = exp(-A*Innertemp);
KERNEL1b = exp(-Ab*Innertemp);
KERNEL2 = exp(-Etemp/(R*Ttemp)) * KERNEL1 * (AA*((Etemp - Emin(a))^2)*((Etemp - Emax(b))^2)/((Emax(b) - Emin(a))^5)); %4th order
KERNEL2b = exp(-Etemp/(R*Ttemp)) * KERNEL1b * (1/sigma)*((1/(2*pi))^0.5)*exp(-0.5*((Etemp-Ebar)^2)/(sigma^2)); %Gaussian

Ttemp2 = Ttemp + ramp*dt/2;
Innertemp2 = Innertemp + quad(@(x)KERNEL1TPD(Etemp,R,T0,ramp,x),t(i-1),t(i-1)+dt/2);
KERNEL12 = exp(-A*Innertemp2);
KERNEL12b = exp(-Ab*Innertemp2);
KERNEL22 = exp(-Etemp/(R*Ttemp2)) * KERNEL12 * (AA*((Etemp - Emin(a))^2)*((Etemp - Emax(b))^2)/((Emax(b) - Emin(a))^5)); %4th order
KERNEL22b = exp(-Etemp/(R*Ttemp2)) * KERNEL12b * (1/sigma)*((1/(2*pi))^0.5)*exp(-0.5*((Etemp-Ebar)^2)/(sigma^2)); %Gaussian

Ttemp4 = Ttemp + ramp*dt;
Innertemp4 = Innertemp + quad(@(x)KERNEL1TPD(Etemp,R,T0,ramp,x),t(i-1),t(i-1)+dt);
KERNEL14 = exp(-A*Innertemp4);
KERNEL14b = exp(-Ab*Innertemp4);
KERNEL24 = exp(-Etemp/(R*Ttemp4)) * KERNEL14 * (AA*((Etemp - Emin(a))^2)*((Etemp - Emax(b))^2)/((Emax(b) - Emin(a))^5)); %4th order
KERNEL24b = exp(-Etemp/(R*Ttemp4)) * KERNEL14b * (1/sigma)*((1/(2*pi))^0.5)*exp(-0.5*((Etemp-Ebar)^2)/(sigma^2)); %Gaussian
    end

X = (Etemp - EL(d))/(ER(e) - EL(d));
    if Etemp <=EL(d) | Etemp >= ER(e)
KERNEL2c = 0; %Beta
KERNEL22c = 0; %Beta
KERNEL24c = 0; %Beta
    else
KERNEL1c = exp(-Ac*Innertemp);
KERNEL2c = exp(-Etemp/(R*Ttemp)) * KERNEL1c * prefac *((X)^(alpha-1)) * ((1-X)^(beta-1))/CHECK; %Beta

KERNEL12c = exp(-Ac*Innertemp2);
KERNEL22c = exp(-Etemp/(R*Ttemp2)) * KERNEL12c * prefac *((X)^(alpha-1)) * ((1-X)^(beta-1))/CHECK; %Beta

KERNEL14c = exp(-Ac*Innertemp4);
KERNEL24c = exp(-Etemp/(R*Ttemp4)) * KERNEL14c * prefac *((X)^(alpha-1)) * ((1-X)^(beta-1))/CHECK; %Beta
    end

sumtemp = sumtemp + KERNEL2*DE;
sumtempb = sumtempb + KERNEL2b*DE;
sumtempc = sumtempc + KERNEL2c*DE/(ER(e)-EL(d));
sumtemp2 = sumtemp2 + KERNEL22*DE;
sumtemp2b = sumtemp2b + KERNEL22b*DE;
sumtemp2c = sumtemp2c + KERNEL22c*DE/(ER(e)-EL(d));
sumtemp4 = sumtemp4 + KERNEL24*DE;
sumtemp4b = sumtemp4b + KERNEL24b*DE;
```

```
sumtemp4c = sumtemp4c + KERNEL24c*DE/(ER(e)-EL(d));
end

K1 = -Oo(1)*A*sumtemp*dt;
K1b = -Oob(1)*Ab*sumtempb*dt;
K1c = -Ooc(1)*Ac*sumtempc*dt;
K2 = -Oo(1)*A*sumtemp2*dt;
K2b = -Oob(1)*Ab*sumtemp2b*dt;
K2c = -Ooc(1)*Ac*sumtemp2c*dt;
K4 = -Oo(1)*A*sumtemp4*dt;
K4b = -Oob(1)*Ab*sumtemp4b*dt;
K4c = -Ooc(1)*Ac*sumtemp4c*dt;
Oo(i) = Oo(i-1) + (1/6)*(K1+4*K2+K4);
Oob(i) = Oob(i-1) + (1/6)*(K1b+4*K2b+K4b);
Ooc(i) = Ooc(i-1) + (1/6)*(K1c+4*K2c+K4c);
dCOdt(i) = -sitemoles * Sg0 * mass0 * (Oo(i) - Oo(i-1)) / dt;
dCOdtb(i) = -sitemoles * Sg0 * mass0 * (Oob(i) - Oob(i-1)) / dt;
dCOdtc(i) = -sitemoles * Sg0 * mass0 * (Ooc(i) - Ooc(i-1)) / dt;

if Ttemp >= 1100+273
    repeatloop = 0;
end
%repeat loop update logic for breaking loop

end

repeatloop = 1; %reset while loop logic after breaking loop
%Compare model with experimental data at T points
for ii = 1:n
temp = min(abs(T-XPMNT(ii,1)));
MINUS = find(T == XPMNT(ii,1) - temp);
PLUS = find(T == XPMNT(ii,1) + temp);
    if MINUS <= i & MINUS >= 0
    errort(ii) = (XPMNT(ii,2) - dCOdtc(MINUS))^2;
    else
    errort(ii) = (XPMNT(ii,2) - dCOdtc(PLUS))^2;
    end
end
error(d,e,c) = sum(errort);
if error(d,e,c) == NaN
    error(d,e,c) = 0;
elseif abs(error(d,e,c)) >= 1e-10
    error(d,e,c) = 0;
end
        total = total + 1
        end
    end
end

%Now find minimum entry in error matrix and rescale parameters
error
EL
ER
plot(T,dCOdt,T,dCOdtb,T,dCOdtc,XPMNT(:,1),XPMNT(:,2),'o');title('f(E) comparison with TPD');
xlabel('Temperature (K)');ylabel('Gas Evolution (CO moles/s)');
legend('4th order','gaussian','Beta','data','location','southwest');
toc

%output k2eff for 973,1072,1173,1273 for input into Mech worksheet.
Teff=973;
keff973 = quad(@(x)keff(x,Emin,Emax,Teff,A,AA),Emin,Emax);
Teff=1073;
keff1073 = quad(@(x)keff(x,Emin,Emax,Teff,A,AA),Emin,Emax);
Teff=1173;
```

```
keff1173 = quad(@(x)keff(x,Emin,Emax,Teff,A,AA),Emin,Emax);
Teff=1273;
keff1273 = quad(@(x)keff(x,Emin,Emax,Teff,A,AA),Emin,Emax);
```

B.1.2 REACTglobalcalc

```
%Least squares determination global rate parameters
%090909 Andrew Campbell Lee
clear all
clc
%Initialize data matrices for comparison
REACTdataFAC;
xc = linspace(0.05,0.95,19);    %extent of conversion matrix
%Specify Kp for Boudouard reaction as a function of temperature
%Data from interpolation of JANAF data
DG1100 = -56.841 + (-74.058+56.841)*(1100-1027)/(100); %kJ/mol
DG1000 = -39.54 + (-56.841+39.54)*(1000-927)/(100);%kJ/mol
DG950 = -39.54 + (-56.841+39.54)*(950-927)/(100);%kJ/mol
DG900 = -22.149 + (-39.54+22.149)*(900-827)/(100);%kJ/mol
DG800 = -4.664 + (-22.149+4.664)*(800-727)/(100);%kJ/mol
DG700 = 12.916 + (-4.664-12.916)*(700-627)/(100);%kJ/mol
DG600 = 30.592 + (12.916-30.592)*(600-527)/(100);%kJ/mol
KP1100 = exp(-DG1100*1000/(8.314*1373));
KP1000 = exp(-DG1000*1000/(8.314*1273));
KP950 = exp(-DG950*1000/(8.314*1223));
KP900 = exp(-DG900*1000/(8.314*1173));
KP800 = exp(-DG800*1000/(8.314*1073));
KP700 = exp(-DG700*1000/(8.314*973));
KP600 = exp(-DG600*1000/(8.314*873));
C600 = 101325/(8.314*873); %mole/m3 CO2
C700 = 101325/(8.314*973); %mole/m3 CO2
C800 = 101325/(8.314*1073); %mole/m3 CO2
C900 = 101325/(8.314*1173); %mole/m3 CO2
C950 = 101325/(8.314*1223); %mole/m3 CO2
C1000 = 101325/(8.314*1273); %mole/m3 CO2
C1100 = 101325/(8.314*1373); %mole/m3 CO2
%-------------------------------------------------------
%Initialization of nonlinear least squares
%unknowns are psi, A and E
options = optimset('tolfun',1e-15,'maxiter',10000)
%-------------------------------------------------
%FAC FAC FAC FAC FAC FAC FAC FAC FAC FAC FAC FAC FAC FAC FAC
X0 = [2 1000 200000]';
LB = [0 1e-5 50000]';
UB = [5 1e5 500000]';
FAC = lsqnonlin(@globalFACfwd,X0,LB,UB,options)
%calculate model values
    for j = 1:19
        FAC700C(j) = FAC(2)*exp(-FAC(3)/(8.314*973))*((1-FAC(1)*log(1-xc(j)))^0.5) * (C700);
        FAC800C(j) = FAC(2)*exp(-FAC(3)/(8.314*1073))*((1-FAC(1)*log(1-xc(j)))^0.5) * (C800);
        FAC900C(j) = FAC(2)*exp(-FAC(3)/(8.314*1173))*((1-FAC(1)*log(1-xc(j)))^0.5) * (C900);
        FAC1000C(j) = FAC(2)*exp(-FAC(3)/(8.314*1273))*((1-FAC(1)*log(1-xc(j)))^0.5) * (C1000);
    end
%plotting function to show agreement with model
figure(1);
semilogy(xc(1:4),FACCO2700C,'ob',xc(1:13),FACCO2800C,'og',xc,FACCO2900C,'or',xc,FACCO21000C,'ok');
hold on;
semilogy(xc,FAC700C,'b',xc,FAC800C,'g',xc,FAC900C,'r',xc,FAC1000C,'k');
%Gamma calculations for Table of Global values
GammaFAC973 = FAC(2)*exp(-FAC(3)/(8.314*973))
GammaFAC1073 = FAC(2)*exp(-FAC(3)/(8.314*1073))
```

```
GammaFAC1173 = FAC(2)*exp(-FAC(3)/(8.314*1173))
GammaFAC1273 = FAC(2)*exp(-FAC(3)/(8.314*1273))
```

B.1.3 globalFACfwd

```
function F = globalFACfwd(x)

psi = x(1);
A = x(2);
E = x(3);
%-----------------------------------------------------------
%Calling experimental data file and calculated KP
REACTdataFAC;
DG1100 = -56.841 + (-74.058+56.841)*(1100-1027)/(100);
DG1000 = -39.54 + (-56.841+39.54)*(1000-927)/(100);
DG950 = -39.54 + (-56.841+39.54)*(950-927)/(100);
DG900 = -22.149 + (-39.54+22.149)*(900-827)/(100);
DG800 = -4.664 + (-22.149+4.664)*(800-727)/(100);
DG700 = 12.916 + (-4.664-12.916)*(700-627)/(100);
DG600 = 30.592 + (12.916-30.592)*(600-527)/(100);
global KP1100 KP1000 KP950 KP900 KP800 KP700 KP600
KP1100 = exp(-DG1100*1000/(8.314*1373));
KP1000 = exp(-DG1000*1000/(8.314*1273));
KP950 = exp(-DG950*1000/(8.314*1223));
KP900 = exp(-DG900*1000/(8.314*1173));
KP800 = exp(-DG800*1000/(8.314*1073));
KP700 = exp(-DG700*1000/(8.314*973));
KP600 = exp(-DG600*1000/(8.314*873));
%-----------------------------------------------------------
C700 = 101325/(8.314*973); %mole/m3 CO2
C800 = 101325/(8.314*1073); %mole/m3 CO2
C900 = 101325/(8.314*1173); %mole/m3 CO2
C950 = 101325/(8.314*1223); %mole/m3 CO2
C1000 = 101325/(8.314*1273); %mole/m3 CO2
C1100 = 101325/(8.314*1373); %mole/m3 CO2

F = 1000*[FACCO2700C(1) - A*exp(-E/(8.314*973)) * ((1-psi*log(1-0.05))^0.5) * (C700)
    FACCO2700C(2) - A*exp(-E/(8.314*973)) * ((1-psi*log(1-0.1))^0.5) * (C700)
    FACCO2700C(3) - A*exp(-E/(8.314*973)) * ((1-psi*log(1-0.15))^0.5) * (C700)
    FACCO2700C(4) - A*exp(-E/(8.314*973)) * ((1-psi*log(1-0.2))^0.5) * (C700)

    FACCO2800C(1) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.05))^0.5) * (C800) %13total
    FACCO2800C(2) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.1))^0.5) * (C800)
    FACCO2800C(3) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.15))^0.5) * (C800)
    FACCO2800C(4) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.2))^0.5) * (C800)
    FACCO2800C(5) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.25))^0.5) * (C800)
    FACCO2800C(6) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.3))^0.5) * (C800)
    FACCO2800C(7) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.35))^0.5) * (C800)
    FACCO2800C(8) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.4))^0.5) * (C800)
    FACCO2800C(9) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.45))^0.5) * (C800)
    FACCO2800C(10) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.5))^0.5) * (C800)
    FACCO2800C(11) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.55))^0.5) * (C800)
    FACCO2800C(12) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.6))^0.5) * (C800)
    FACCO2800C(13) - A*exp(-E/(8.314*1073)) * ((1-psi*log(1-0.65))^0.5) * (C800)

    FACCO2900C(1) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.05))^0.5) * (C900)
    FACCO2900C(2) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.1))^0.5) * (C900)
    FACCO2900C(3) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.15))^0.5) * (C900)
    FACCO2900C(4) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.2))^0.5) * (C900)
    FACCO2900C(5) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.25))^0.5) * (C900)
    FACCO2900C(6) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.3))^0.5) * (C900)
```

```
FACCO2900C(7) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.35))^0.5) * (C900)
FACCO2900C(8) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.4))^0.5) * (C900)
FACCO2900C(9) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.45))^0.5) * (C900)
FACCO2900C(10) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.5))^0.5) * (C900)
FACCO2900C(11) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.55))^0.5) * (C900)
FACCO2900C(12) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.6))^0.5) * (C900)
FACCO2900C(13) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.65))^0.5) * (C900)
FACCO2900C(14) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.7))^0.5) * (C900)
FACCO2900C(15) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.75))^0.5) * (C900)
FACCO2900C(16) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.8))^0.5) * (C900)
FACCO2900C(17) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.85))^0.5) * (C900)
FACCO2900C(18) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.9))^0.5) * (C900)
FACCO2900C(19) - A*exp(-E/(8.314*1173)) * ((1-psi*log(1-0.95))^0.5) * (C900)

FACCO21000C(1) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.05))^0.5) * (C1000)
FACCO21000C(2) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.1))^0.5) * (C1000)
FACCO21000C(3) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.15))^0.5) * (C1000)
FACCO21000C(4) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.2))^0.5) * (C1000)
FACCO21000C(5) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.25))^0.5) * (C1000)
FACCO21000C(6) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.3))^0.5) * (C1000)
FACCO21000C(7) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.35))^0.5) * (C1000)
FACCO21000C(8) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.4))^0.5) * (C1000)
FACCO21000C(9) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.45))^0.5) * (C1000)
FACCO21000C(10) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.5))^0.5) * (C1000)
FACCO21000C(11) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.55))^0.5) * (C1000)
FACCO21000C(12) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.6))^0.5) * (C1000)
FACCO21000C(13) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.65))^0.5) * (C1000)
FACCO21000C(14) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.7))^0.5) * (C1000)
FACCO21000C(15) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.75))^0.5) * (C1000)
FACCO21000C(16) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.8))^0.5) * (C1000)
FACCO21000C(17) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.85))^0.5) * (C1000)
FACCO21000C(18) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.9))^0.5) * (C1000)
FACCO21000C(19) - A*exp(-E/(8.314*1273)) * ((1-psi*log(1-0.95))^0.5) * (C1000)];
```

B.1.4 particle sizing

```
%Andrew C. Lee
%Size distribution of 106-125micron raw fisher char
%02/16/09
% between 38.49 and 39.21micron is 216 particles
clear all;
format long;
Fishraw106125 =[3070 1774 1366 2153 4072 1987 573 434 353 345 294 317 290
260 250 231 180 221 227 207 217 211 207 184 174 183
204 191 197 193 207 220 206 192 222 191 188 210 172
219 202 205 228 214 197 216 204 220 204 215 205 183
212 180 205 198 201 211 193 192 179 176 158 182 169
157 155 176 182 157 168 183 211 160 175 194 178 209
175 184 164 179 189 178 176 162 147 157 124 157 156
156 139 149 153 131 135 140 115 125 99 116 111 103
107 95 77 73 61 64 64 45 33 50 37 38 25
14 17 20 27 14 10 9 10 8 5 11 7 3
8 4 1 3 5 1 1 2 2 4 2 4 2
2 3 3 1 0 2 2 2 0 1 2 2 0
1 0 0 0 0 2 1 0 1 0 1 1 0
0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
0 0 0 0 0 0 0 0 0 0 0 0 0];

N = 260;
Fish = zeros(1,N);
for i =1:20
    for j = 1:13
        m = (i-1)*13+j;
        Fish(m) = Fishraw106125(i,j);
    end
end
dsize = (189.6 - 6.263)/256
Fishsum = 0;
Fishvsum = 0;
for ii = 1:249
    LL(ii) = 6.263+dsize*(ii-1);
    UL(ii) = 6.263+dsize*(ii);

    Fishn(ii) = Fish(ii);
    Fishncum(ii) = Fishn(ii) + Fishsum;
    Fishv(ii) = Fishn(ii)*pi*(1/6)*((UL(ii)+LL(ii))*0.5)^3;
    Fishvcum(ii) = Fishv(ii) + Fishvsum;
    Fishsum=Fishncum(ii);
    Fishvsum = Fishvcum(ii);
end
%-----------------------------------------------------------
%Normalization of distibutions, number and volume
NR = sum(Fishn);
VR = sum(Fishv);
Fishnorm = Fishn/NR;
Fishcumnorm = Fishncum/NR;
FishnormV = Fishv/VR;
Fishvcumnorm = Fishvcum/VR;
%-----------------------------------------------------------
%Plot and calculation of number distribution
disp('Lower and upper bounds of distribution peak, number');
MLKR = max(Fishnorm);
IILKR = find(MLKR == Fishnorm);
lowerLKR=LL(IILKR)
upperLKR = UL(IILKR)
lr = num2str(lowerLKR);
ur = num2str(upperLKR);

figure(1);
[AX,H1,H2] = plotyy(LL,Fishnorm,LL,Fishcumnorm,'plot');
set(get(AX(1),'Ylabel'),'String','PDF');
set(get(AX(2),'Ylabel'),'String','CDF');
title('Size Characterization for Activated Carbon');
xlabel('LL (micron)');
%gtext(lr);
%gtext(ur);
%-----------------------------------------------------------
%Plot and calculation of volume distribution
disp('Lower and upper bounds of distribution peak, volume');
MLKRv = max(FishnormV);
IILKRv = find(MLKRv == FishnormV);
lowerLKRv =LL(IILKRv)
upperLKRv = UL(IILKRv)
lrv = num2str(lowerLKRv);
urv = num2str(upperLKRv);

figure(2);
[AX,H1,H2] = plotyy(LL,FishnormV,LL,Fishvcumnorm,'plot');
set(get(AX(1),'Ylabel'),'String','PDF');
set(get(AX(2),'Ylabel'),'String','CDF');
```

```
title('Size Characterization for Activated Carbon');
xlabel('LL (micron)');
%gtext(lrv);
%gtext(urv);
```

B.1.5 GC Peak Determination

Note that Y in the following code represents a column vector of the GC output signal taken at a frequency of 9.1743 Hz.

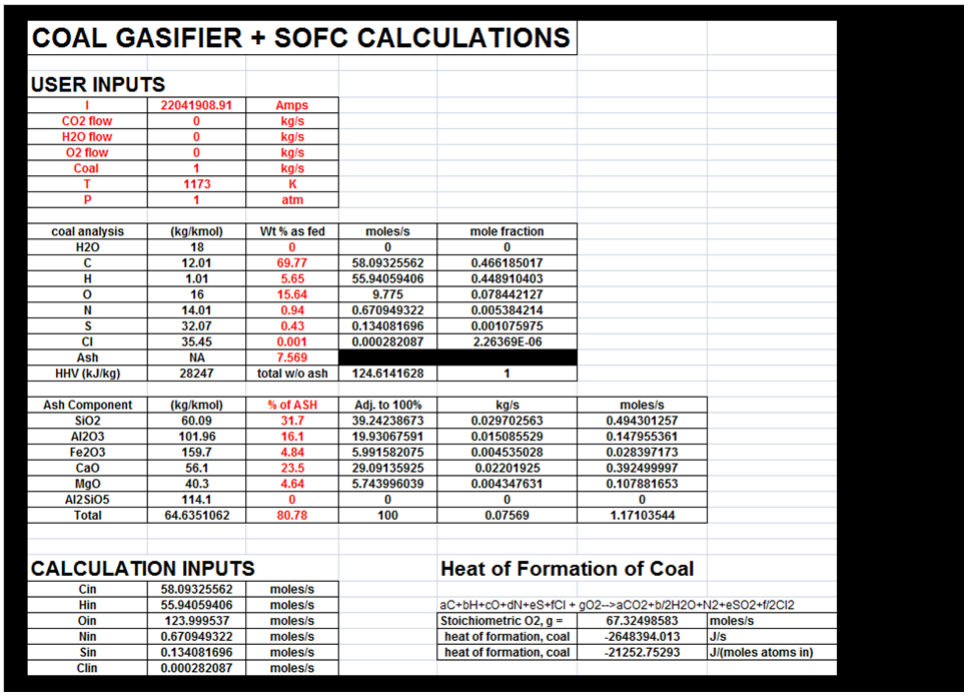
```
%Andrew Campbell Lee
%06/28/2006
%Numerical Integration of GC peaks
clear all;
format long;
Y = data;
N = length(Y);
dt = 0.109;
t = linspace(0,dt*(N-1),N);
plot(t,Y);title('reproduction of GC output');
xlabel('seconds');ylabel('signal');
index = input('Finished (2) or Refine (0)');
T = t;
y = Y;
while index == 0
LHS = input('Please enter the left limit');
RHS = input('Please enter the right limit');
LL = min(abs(t-LHS));
RR = min(abs(t-RHS));
LLL = find(t-LHS == -LL);
LLL2 = find(t-LHS == LL);
RRR = find(t-RHS == -RR);
RRR2 = find(t-RHS == RR);
if LLL2 > 1
LHindex = LLL2;
else
  LHindex = LLL;
end
if RRR2 > 1
RHindex = RRR2;
else
  RHindex = RRR;
end
clear T;
clear y;
for i = LHindex:RHindex
    T(i-LHindex+1) = t(i);
    y(i-LHindex+1) = Y(i);
end
plot(T,y,'o');title('reduced integration limits');
xlabel('seconds');ylabel('signal');
index = input('Finished (1) or Refine (0)');
end
NN = length(T);
slope = (y(NN)-y(1))/(T(NN)-T(1));
intercept = y(NN) - slope*T(NN);
for j = 1:NN
    y(j) = y(j) - (slope*T(j)+intercept);
end

Pk_heighht = max(y)
```

```
II = trapz(y);
Pk_Area = II*dt
```

B.2 Thermodynamic Calculations

The thermodynamic spreadsheet used to model the DCFC, IDG and ATG plants employs polynomial fits for $C_P(T)$. Shown in Figure B.1 is the input section of the spreadsheet, where the sample composition, operating temperature and pressure, current, and other in-streams are specified. The atomic flows are calculated for C, H, O, N, S, and Cl. Shown in Figure B.2 is a portion of the iterative element potential



COAL GASIFIER + SOFC CALCULATIONS

USER INPUTS

I	22041908.91	Amps
CO2 flow	0	kg/s
H2O flow	0	kg/s
O2 flow	0	kg/s
Coal	1	kg/s
T	1173	K
P	1	atm

coal analysis	(kg/kmol)	Wt % as fed	moles/s	mole fraction
H2O	18	0	0	0
C	12.01	69.77	58.09325562	0.466185017
H	1.01	5.65	55.94059406	0.448910403
O	16	15.64	9.775	0.078442127
N	14.01	0.94	0.670949322	0.005384214
S	32.07	0.43	0.134081696	0.001075975
Cl	35.45	0.001	0.000282087	2.26369E-06
Ash	NA	7.569		
HHV (kJ/kg)	28247	total w/o ash	124.6141628	1

Ash Component	(kg/kmol)	% of ASH	Adj. to 100%	kg/s	moles/s
SiO2	60.09	31.7	39.24238673	0.029702563	0.494301257
Al2O3	101.96	16.1	19.93067591	0.015085529	0.147955361
Fe2O3	159.7	4.84	5.991582075	0.004535028	0.028397173
CaO	56.1	23.5	29.09135925	0.02201925	0.392499997
MgO	40.3	4.64	5.743996039	0.004347631	0.107881653
Al2SiO5	114.1	0	0	0	0
Total	64.6351062	80.78	100	0.07569	1.17103544

CALCULATION INPUTS

Cin	58.09325562	moles/s
Hin	55.94059406	moles/s
Oin	123.999537	moles/s
Nin	0.670949322	moles/s
Sin	0.134081696	moles/s
Clin	0.000282087	moles/s

Heat of Formation of Coal

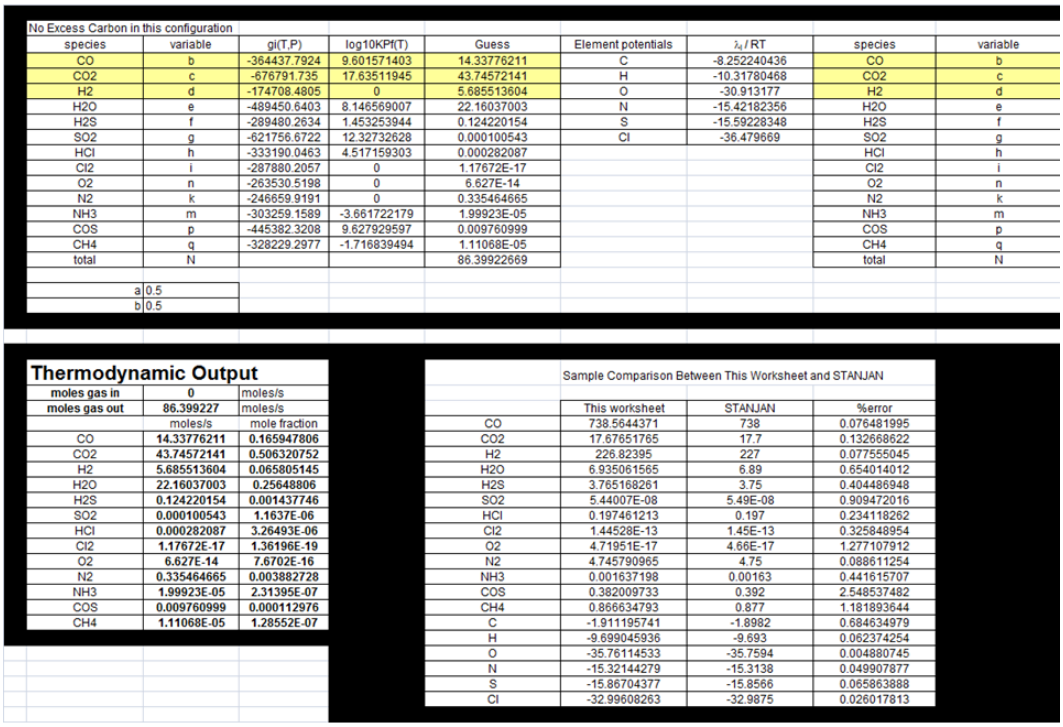
aC+bH+cO+dN+eS+fCl + gO2-->aCO2+b/2H2O+N2+eSO2+f/2Cl2

Stoichiometric O2, g =	67.32498583	moles/s
heat of formation, coal	-2648394.013	J/s
heat of formation, coal	-21252.75293	J/(moles atoms in)

Figure B.1: Sample screen shot of thermodynamic model inputs spreadsheet (for Wyodak coal in the DCFC).

approach to determining the equilibrium composition. The atomic flows from the previous figure, along with $g_i(T, P)$ are used as inputs. An initial guess of molar flows is made, and an intial estimate of the element potentials is made using the major

products assumption. The minor species are then solved for using these element potentials, and the major species are respecified from the atomic balances. This process repeats until the output and input agree to less than 0.001 % difference. This same equilibrium calculator is used for gasifiers, SOFC's and combustors. The two parameters a and b are damping coefficients used to maintain stable iteration. The



No Excess Carbon in this configuration

species	variable	gi(T,P)	log10KPf(T)	Guess	Element potentials	λ_i / RT	species	variable
CO	b	-364437.7924	9.601571403	14.33776211	C	-8.252240436	CO	b
CO2	c	-676791.735	17.63511945	43.74572141	H	-10.31780468	CO2	c
H2	d	-174708.4805	0	5.685513604	O	-30.913177	H2	d
H2O	e	-489450.6403	8.146569007	22.16037003	N	-15.42182356	H2O	e
H2S	f	-289480.2634	1.453253944	0.124220154	S	-15.59228348	H2S	f
SO2	g	-621756.6722	12.32732628	0.000100543	Cl	-36.479669	SO2	g
HCl	h	-333190.0463	4.517159303	0.000282087			HCl	h
Cl2	i	-287880.2057	0	1.17672E-17			Cl2	i
O2	n	-263530.5198	0	6.627E-14			O2	n
N2	k	-246659.9191	0	0.335464665			N2	k
NH3	m	-303259.1589	-3.661722179	1.99923E-05			NH3	m
COS	p	-445382.3208	9.627929597	0.009760999			COS	p
CH4	q	-328229.2977	-1.716839494	1.11068E-05			CH4	q
total	N			86.39922669			total	N

a	0.5
b	0.5

Thermodynamic Output

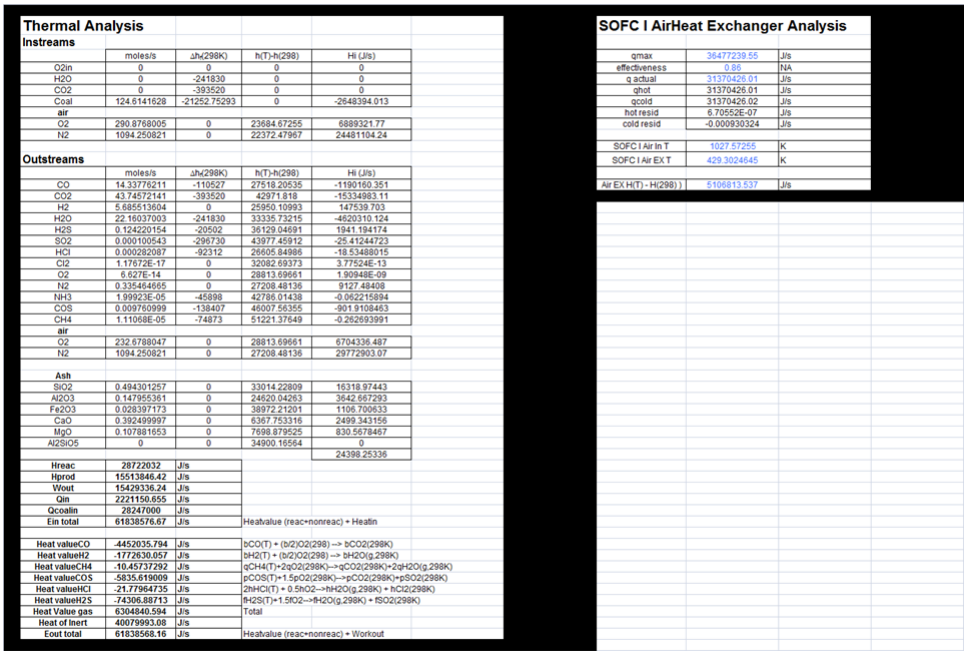
moles gas in	0	moles/s
moles gas out	86.399227	moles/s
	moles/s	mole fraction
CO	14.33776211	0.165947806
CO2	43.74572141	0.506320752
H2	5.685513604	0.065805145
H2O	22.16037003	0.25648806
H2S	0.124220154	0.001437746
SO2	0.000100543	1.1637E-06
HCl	0.000282087	3.26493E-06
Cl2	1.17672E-17	1.36196E-19
O2	6.627E-14	7.6702E-16
N2	0.335464665	0.003882728
NH3	1.99923E-05	2.31395E-07
COS	0.009760999	0.000112976
CH4	1.11068E-05	1.28552E-07

Sample Comparison Between This Worksheet and STANJAN

	This worksheet	STANJAN	%error
CO	738.5644371	738	0.076481995
CO2	17.67651765	17.7	0.132668622
H2	226.82395	227	0.077555045
H2O	6.935061565	6.89	0.654014012
H2S	3.765168261	3.75	0.404486948
SO2	5.44007E-08	5.49E-08	0.909472016
HCl	0.197461213	0.197	0.234118262
Cl2	1.44528E-13	1.45E-13	0.325848954
O2	4.71951E-17	4.66E-17	1.277107912
N2	4.745790965	4.75	0.088611254
NH3	0.001637198	0.00163	0.441615707
COS	0.382009733	0.392	2.548537482
CH4	0.866634793	0.877	1.181893644
C	-1.911195741	-1.8982	0.684634979
H	-9.699045936	-9.693	0.062374254
O	-35.76114533	-35.7594	0.004880745
N	-15.32144279	-15.3138	0.049907877
S	-15.86704377	-15.8566	0.065863888
Cl	-32.99608263	-32.9875	0.026017813

Figure B.2: Sample screen shot of thermodynamic model equilibrium calculator spreadsheet (for Wyodak coal in the DCFC).

outputs from the equilibrium calculation are then used to perform an energy balance to determine the heat interaction and the heating value of the exhaust stream (shown in Figure B.3). The thermodynamic data, in the form of polynomial fits for $C_P(T)$, are shown in tabular form in Figure B.4. This is a screenshot of the spreadsheet thermodynamic data section.



Thermal Analysis

Instreams

	moles/s	Δhf(298K)	h(T)-h(298)	Hi (J/s)
O2in	0	0	0	0
H2O	0	-241830	0	0
CO2	0	-393520	0	0
Coal	124.6141628	-21252.75293	0	-2648394.013
air				
O2	290.8768005	0	23684.67255	6889321.77
N2	1094.250821	0	22372.47967	24481104.24

Outstreams

	moles/s	Δhf(298K)	h(T)-h(298)	Hi (J/s)
CO	14.33776211	-110527	27518.20535	-1190160.351
CO2	43.74572141	-393520	42971.818	-15334983.11
H2	5.685513604	0	25950.10993	147539.703
H2O	22.16037003	-241830	33335.73215	-4620310.124
H2S	0.124220154	-20502	36129.04691	1941.194174
SO2	0.000100543	-296730	43977.45912	-25.41244723
HCl	0.000282087	-92312	26605.84986	-18.53488015
Cl2	1.17672E-17	0	32082.69373	3.77524E-13
O2	6.627E-14	0	28813.69661	1.90948E-09
N2	0.335464665	0	27208.48136	9127.48408
NH3	1.99923E-05	-45898	42786.01438	-0.062215894
COS	0.009760999	-138407	46007.56355	-901.9108463
CH4	1.11068E-05	-74873	51221.37649	-0.262693991
air				
O2	232.6788047	0	28813.69661	6704336.487
N2	1094.250821	0	27208.48136	29772903.07
Ash				
SiO2	0.494301257	0	33014.22809	16318.97443
Al2O3	0.147955361	0	24620.04263	3642.667293
Fe2O3	0.028397173	0	38972.21201	1106.700633
CaO	0.392499997	0	6367.753316	2499.343156
MgO	0.107881653	0	7698.879525	830.5678467
Al2SiO5	0	0	34900.16564	0
				24398.25336

Hreac	**28722032**	J/s
Hprod	**15513846.42**	J/s
Wout	**15429336.24**	J/s
Qin	**2221150.655**	J/s
Qcoalin	**28247000**	J/s
Ein total	**61838576.67**	J/s

Heatvalue (reac+nonreac) + Heatin

Heat valueCO	**-4452035.794**	J/s
Heat valueH2	**-1772630.057**	J/s
Heat valueCH4	**-10.45737292**	J/s
Heat valueCOS	**-5835.619009**	J/s
Heat valueHCl	**-21.77964735**	J/s
Heat valueH2S	**-74306.88713**	J/s
Heat Value gas	**6304840.594**	J/s
Heat of Inert	**40079993.08**	J/s
Eout total	**61838568.16**	J/s

bCO(T) + (b/2)O2(298) --> bCO2(298K)
bH2(T) + (b/2)O2(298) --> bH2O(g,298K)
qCH4(T)+2qO2(298K)-->qCO2(298K)+2qH2O(g,298K)
pCOS(T)+1.5pO2(298K)-->pCO2(298K)+pSO2(298K)
2hHCl(T) + 0.5hO2-->hH2O(g,298K) + hCl2(298K)
fH2S(T)+1.5fO2-->fH2O(g,298K) + fSO2(298K)
Total
Heatvalue (reac+nonreac) + Workout

SOFC I AirHeat Exchanger Analysis

qmax	36477239.55	J/s
effectiveness	0.86	NA
q actual	31370426.01	J/s
qhot	31370426.01	J/s
qcold	31370426.02	J/s
hot resid	6.70552E-07	J/s
cold resid	-0.000930324	J/s
SOFC I Air In T	1027.57255	K
SOFC I Air EX T	429.3024645	K
Air EX H(T) - H(298))	5106813.537	J/s

Figure B.3: Sample screen shot of thermodynamic model outputs spreadsheet (for Wyodak coal in the DCFC).

Specific heats: cp/R = a + bT + cT^2 + dT^3 + eT^4
enthalpy: h(T)/RT = a + (b/2)T + (c/3)T^2 + (d/4)T^3 + (e/5)T^4 + f/T
entropy: s(T)/R = a*ln(T) + (b)T + (c/2)T^2 + (d/3)T^3 + (e/4)T^4 + g

species	a	b	c	d	e	f	g	T (K)
C(s)	1.4902E+00	1.6621E-03	-6.6872E-07	1.2909E-10	-9.2053E-15	-7.0740E+02	-8.7178E+00	1000-5000 K
	-6.7057E-01	7.1815E-03	-5.6329E-06	2.1423E-09	4.1686E-13	-7.3395E+01	2.6016E+00	300- 1000 K
H2S	2.8831E+00	3.8278E-03	-1.4234E-06	2.4980E-10	-1.6603E-14	-3.4807E+03	7.2582E+00	1000-5000 K
	3.0710E+00	5.5783E-03	-1.0310E-05	1.2020E-08	-4.8384E-12	-3.5598E+03	5.9352E+00	300-1000K
HCl	2.7553E+00	1.4736E-03	-4.9713E-07	8.1087E-11	-5.0721E-15	-1.1918E+04	6.5151E+00	1000-5000 K
	3.3385E+00	1.2682E-03	-3.6669E-06	4.7040E-09	-1.8360E-12	-1.2132E+04	3.1936E+00	300-1000K
N2	2.9266E+00	1.4880E-03	-5.6848E-07	1.0097E-10	-6.7533E-15	-9.2280E+02	5.9805E+00	1000-5000 K
	3.2987E+00	1.4082E-03	-3.9632E-06	5.6415E-09	-2.4449E-12	-1.0209E+03	3.9504E+00	300- 1000 K
CO	3.0251E+00	1.4427E-03	-5.6308E-07	1.0186E-10	-6.9110E-15	-1.4268E+04	6.1082E+00	1000-5000 K
	3.2625E+00	1.5119E-03	-3.8818E-06	5.5819E-09	-2.4750E-12	-1.4311E+04	4.8489E+00	300-1000 K
H2	2.9914E+00	7.0006E-04	-5.6338E-08	-9.2316E-12	1.5828E-15	-8.3503E+02	-1.3551E+00	1000-5000 K
	3.2981E+00	8.2494E-04	-8.1430E-07	-9.4754E-11	4.1349E-13	-1.0125E+03	-3.2941E+00	300-1000 K
CO2	4.4536E+00	3.1402E-03	-1.2784E-06	2.3940E-10	-1.6690E-14	-4.8967E+04	-9.5540E-01	1000-5000 K
	2.2757E+00	9.9221E-03	-1.0409E-05	6.8667E-09	-2.1173E-12	-4.8373E+04	1.0188E+01	300- 1000 K
H2O	2.6721E+00	3.0563E-03	-8.7303E-07	1.2010E-10	-6.3916E-15	-2.9899E+04	6.8628E+00	1000-5000 K
	3.3868E+00	3.4750E-03	-6.3547E-06	6.9686E-09	-2.5066E-12	-3.0208E+04	2.5902E+00	300- 1000 K
NH3	2.4619E+00	6.0592E-03	-2.0050E-06	3.1360E-10	-1.9383E-14	-6.4933E+03	7.4721E+00	1000-5000 K
	2.2044E+00	1.0115E-02	-1.4653E-05	1.4472E-08	-5.3285E-12	-6.5255E+03	8.1271E+00	300-1000 K
O2	3.6976E+00	6.1352E-04	-1.2588E-07	1.7753E-11	-1.1364E-15	-1.2339E+03	3.1892E+00	1000-5000 K
	3.2129E+00	1.1275E-03	-5.7562E-07	1.3139E-09	-8.7686E-13	-1.0052E+03	6.0347E+00	300- 1000 K
SO2	5.2545E+00	1.9785E-03	-8.2042E-07	1.5764E-10	-1.1205E-14	-3.7569E+04	-1.1461E+00	1000-5000 K
	2.9114E+00	8.1030E-03	-6.9067E-06	3.3290E-09	-8.7771E-13	-3.6879E+04	1.1117E+01	300- 1000 K
Cl2	4.2746E+00	3.7173E-04	-1.8935E-07	5.3375E-11	-5.0576E-15	-1.3311E+03	2.2569E+00	1000-5000 K
	3.4396E+00	2.8708E-03	-2.3859E-06	2.8929E-10	2.9151E-13	-1.1318E+03	6.4714E+00	300- 1000 K
COS	5.1919E+00	2.5061E-03	-1.0244E-06	1.9439E-10	-1.3708E-14	-1.8462E+04	-2.8258E+00	1000-5000 K
	2.86E+00	9.52E-03	-8.88E-06	4.22E-09	-8.56E-13	-1.79E+04	9.08E+00	300- 1000 K
CH4	1.6835E+00	1.0237E-02	-3.8751E-06	6.7856E-10	-4.5034E-14	-1.0081E+04	9.6234E+00	1000-5000 K
	7.7874E-01	1.7477E-02	-2.7834E-05	3.0497E-08	-1.2239E-11	-9.8252E+03	1.3722E+01	300-1000 K

Figure B.4: Sample screen shot of thermodynamic model properties spreadsheet.

B.3 PTGA Method Files

The method files used to control the PTGA temperature and flow rates are edited in **.txt* format, with examples following for BET tests, isothermal reactivity tests, and temperature ramp tests. The general form of the method files consist of column 1 in which the method file is sectioned, column 2 in which the temperature is specified (in celsius), and column 3 in which the discreet times are specified for the flow rates given in column 4.

B.3.1 BET Test

The back pressure regulator on the PTGA exhaust stream must be set to 10 atm for the duration of this test.

```
1 25 720 14
1 150 500
2 150 0
1 240 300
2 240 200
1 330 200
2 330 300
1 420 150
2 420 350
1 510 100
2 510 400
1 600 0
2 600 500
1 690 500
2 690 0
```

B.3.2 Reactivity

This general form is used for all isothermal reactivity measurements, in which only the target temperature and time durations vary.

```
1 25 60 2
1   0   250
2   0   0
2 900 36 0
3 900 600 4
1   60 0
2   60 250
1   540 250
2   540 0
```

B.3.3 Temperature Ramp

The temperature ramp file is used for both cleaning chars and TPD measurements. The ramp rate can be adjusted by editing the ramp time interval, with the given arrangment giving a ramp rate of 24 $K \cdot min^{-1}$.

```
1 25 180 2
1   0   150
2   0   0
2 1100 45 0
```

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